Electrostatics

19/03/25 Wednesday

└ static means rest

properties of charge:

* charge is a scalar.
* charges are of two types.
  1) positive charge → deficiency of electrons
  2) negative charge → excess no. of e^-s.
* charge :
  S.I unit - Coulomb (C)
  C.G.S unit - esu (stat coulomb)
  emu (ab coulomb)

$1C = 3 \times 10^9$ estat coulomb.

$1C = \frac{1}{10}$ ab coulomb.

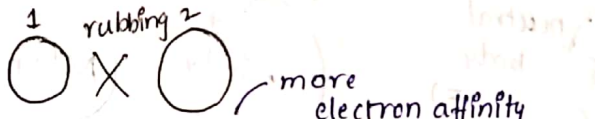
* charge is "invarient". i.e, charge doesn't change during "motion". where as mass is "varient". i.e, mass of body during motion is $\boxed{m = \frac{m_0}{\sqrt{1-(v^2/c^2)}}}$.

Where, m_0 – rest mass.

v – speed of body.

Methods To produce charged bodies:

1) Friction:



glass rod — silk cloth

⊕ — ⊖

$q = +ne$ — $q = -ne$

Here e = charge of electron

$= 1.6 \times 10^{-19}$ C

$m_e = 9.1 \times 10^{-31}$ kg

$n = 1, 2, 3, \cdots$

Charge on a body, $\boxed{q = \pm ne}$ — quantisation of charge.

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* No. of e^{-s} present in a body of charge 1C.

$$q = ne$$

$$\frac{1}{1.6 \times 10^{-19}} = n$$

$$\boxed{n = 6.25 \times 10^{18}}$$

* $1\,mC = 10^{-3}\,C$

$1\,\mu C = 10^{-6}\,C$

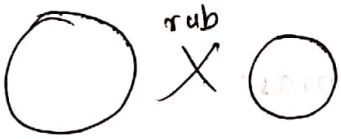
$1\,nC = 10^{-9}\,C$

$1\,pm = 10^{-12}\,C$

* charge is "conservative". i.e, total charge of isolated system is always constant. where as mass is not "conserved".

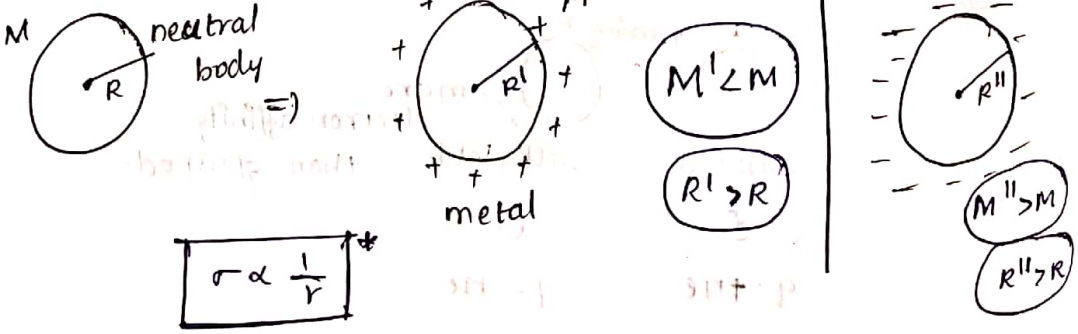
$$\boxed{E = mc^2}\ *; \quad c = 3 \times 10^8\ m/s$$

*



ebonite rod ⊖ fur (wool) ⊕

*



$$\boxed{\sigma \propto \frac{1}{r}}\ *$$

where σ - charge density

r - radius of curvature of surface

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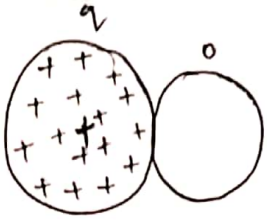
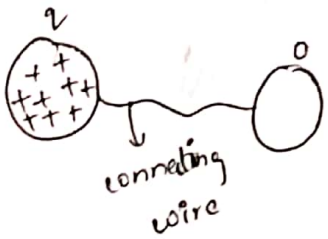
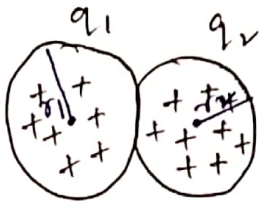
2) Conduction :
(contact)

* charge flow is done till they attain same potential.







Here energy is lost in the form of heat. (electrostatic pot. energy.)

heat, $H = mS\Delta\theta$

m = mass of wire
S = specific heat of wire
$\Delta\theta$ = rise in temp of wire

We have,

$$q = q_1 + q_2 \quad -(1)$$

$q_1 \propto r_1$ (if body is spherical)

$q_2 \propto r_2$

$$\frac{q_1}{q_2} = \frac{r_1}{r_2}$$

Finally, $q = q_1 + q_1 \frac{r_2}{r_1}$

$$= q_1 \left(1 + \frac{r_2}{r_1}\right)$$

$$q = \frac{q_1 (r_1 + r_2)}{r_1}$$

$$\boxed{q_1 = \frac{q r_1}{r_1 + r_2}}$$

Similarly, $\boxed{q_2 = \frac{q r_2}{r_1 + r_2}}$

If $r_1 = r_2 \Rightarrow q_1 = q_2 = q/2$

* Conduction preceeds "repulsion".

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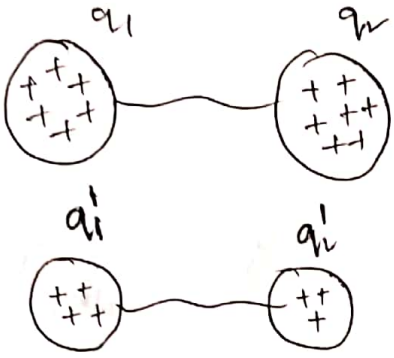
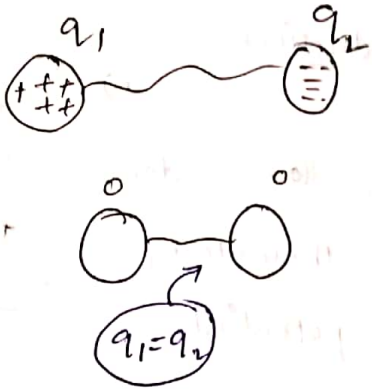
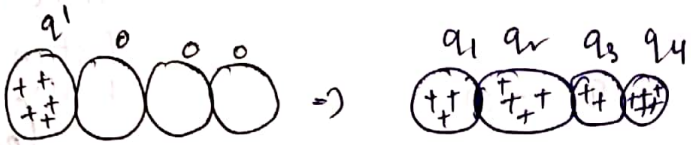




$q_1 + q_2 = q_1' + q_2'$ — ①

& $\frac{q_1'}{q_2'} = \frac{r_1}{r_2}$ — ②



$q = q_1 + q_2 + q_3 + q_4$ — ①

$q_1 : q_2 : q_3 : q_4 = r_1 : r_2 : r_3 : r_4$.

3) Induction :

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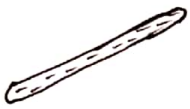
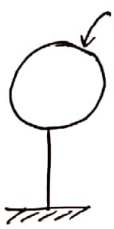
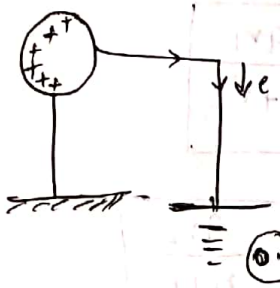
① 



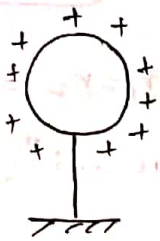
② inducing charge $-q$

(=) net charge = 0

③



④

⇒ 

If body is symmetric.

$q' = -q$

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* In case of dielectric,

$$q' = q\left(1 - \frac{1}{k}\right)$$

where, q' = induced charge

q = inducing charge

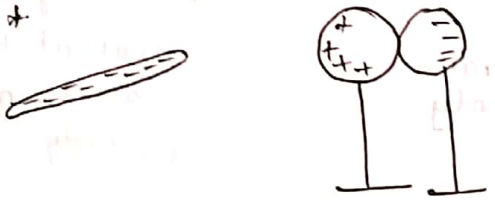
For metal $K = \infty$

for water $K = 80$

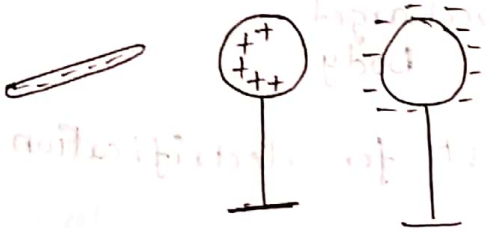
For air $K = 1$

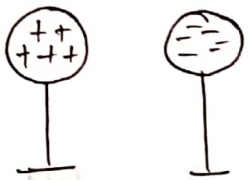
Ex:

*



* no charge accumulation at the junction.





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$q_1' + q_2' = q' \quad -(1)$

$\frac{q_1'}{q_2'} = \frac{r_1}{r_2} \quad -(2)$

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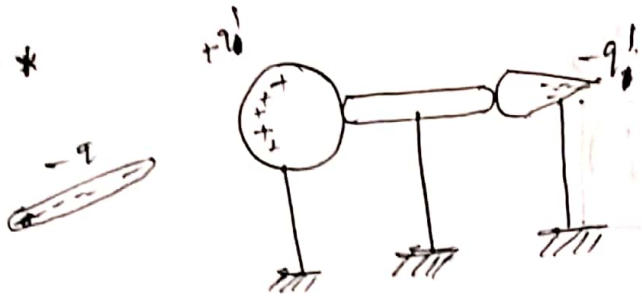

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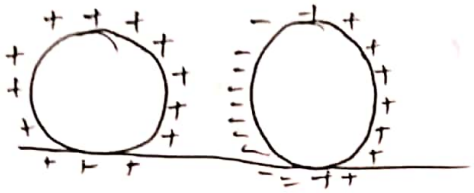
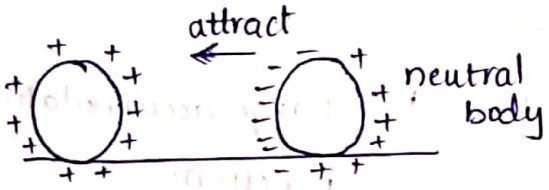


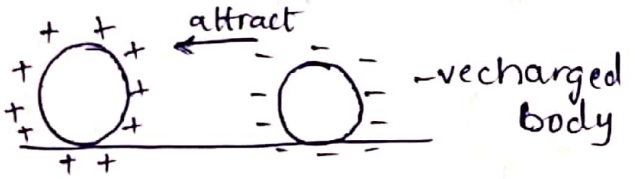
Induction preceeds attraction.

* Repulsion is the sure test for "electrification".
→ changing neutral body as charged body.

* (1)

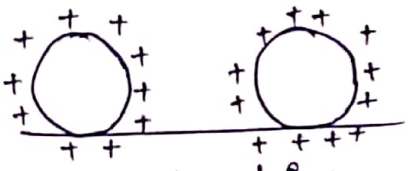


(2)



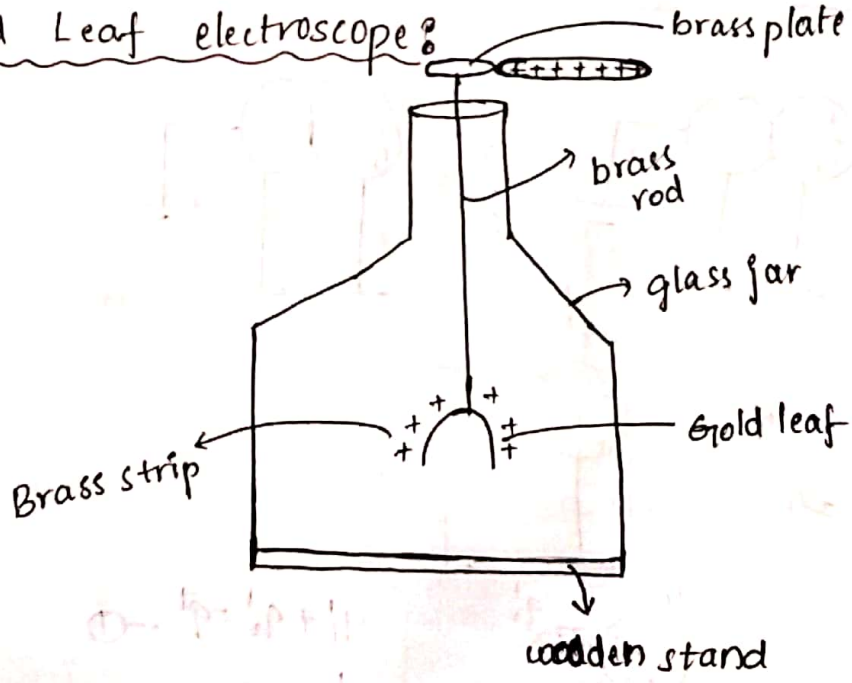
* attraction is not sure test for electrification.

*



repulsion.

Gold Leaf electroscope:



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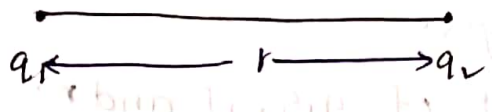
* charge produces electric field when it is at rest whereas it produces electric & magnetic fields when it is in motion.

* An accelerating charge radiates energy in the form of photons (or) light.

Coulomb's Inverse Square Law :-

q_1, q_2 are two point charges.



If $q_1 q_2 > 0$ then force is repulsive. $\boxed{F = +ve}$

(repulsive force is always positive force in physics)

If $q_1 q_2 < 0$ then force is attractive. $\boxed{F = -ve}$

It is observed that,

$$F \propto q_1 q_2 \quad -①$$

electrostatic force b/w two charges is directly proportional to product of their magnitudes.

$$F \propto \frac{1}{r^2} \quad -②$$

From eqns ① & ②,

$$F \propto \frac{q_1 q_2}{r^2}$$

$$\boxed{F = \frac{K_0 q_1 q_2}{r^2}} \quad -③$$

** The line of action of force is always along the line joining them.

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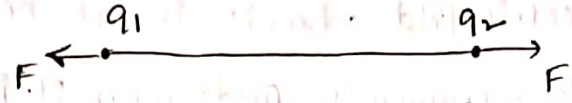


$q_1 q_2 > 0$

*

$q_1 q_2 < 0$

** The value of k_0 depends on

1) presence of medium between the charges.

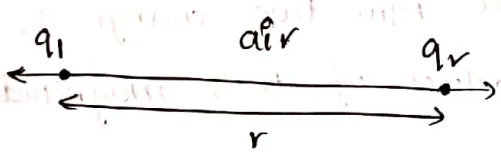
2) units of q_1, q_2, F and r.

* If the medium between the charges is air or vaccum & q_1, q_2, F, r are expressed in S.I units then $\boxed{K_0 = \frac{1}{4\pi\varepsilon_0} = 9 \times 10^9}$ * ($K_0 \rightarrow Nm^2/C^2$)

* ε_0 is called permitivity of air or vaccum.

$$\boxed{\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/Nm^2}$$ *.

Finally,



** $\boxed{F_{air} = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r^2} = 9 \times 10^9 \frac{q_1 q_2}{r^2}}$ *

* If the charges are placed in a "dielectric medium" then $\boxed{F_{med} = \frac{1}{4\pi\varepsilon}\frac{q_1 q_2}{r^2}}$ * ε = permitivity of medium

Dielectric constant: (also called relative permitivity of medium)

$$K = \varepsilon_r = \frac{F_{air}}{F_{medium}} = \frac{\frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r^2}}{\frac{1}{4\pi\varepsilon}\frac{q_1 q_2}{r^2}}$$

$$\boxed{K = \varepsilon_r = \frac{\varepsilon}{\varepsilon_0}}$$

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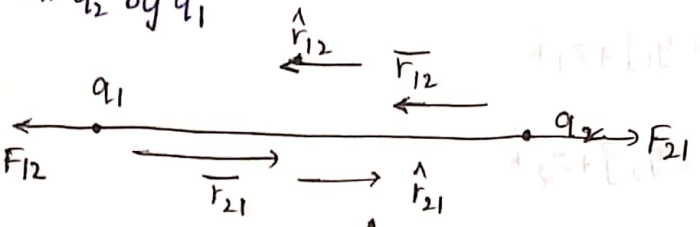
(or) $\boxed{\varepsilon = \varepsilon_0 K}$

(or) $\boxed{\varepsilon = \varepsilon_0 \varepsilon_r}$

Vector form:

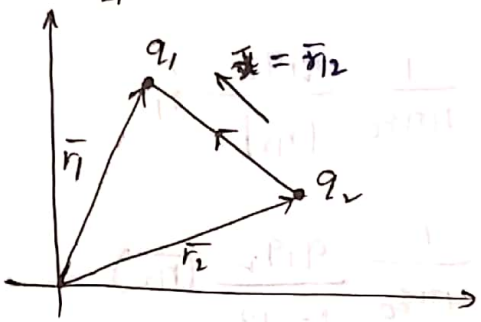
F_{12} = force on q_1 by q_2

F_{21} = force on q_2 by q_1



$\bar{r}_{12}$ = position vector of 1 w.r.to 2.
↳ from 2 to 1.

$\bar{r}_{21}$ = from 1 to 2.



$\bar{r}_2 + \bar{x} = +\bar{r}_1$

$\bar{x} = \bar{r}_1 \pm \bar{r}_2$

$\boxed{|\bar{x}| = |\bar{r}_{12}|}$

now,

$$\bar{F}_{12} = \frac{1}{4\pi\varepsilon_0} \frac{q_1 q_2}{|\bar{r}_{12}|^2} \hat{r}_{12}$$

&

$$\bar{F}_{21} = \frac{1}{4\pi\varepsilon_0} \frac{q_1 q_2}{|\bar{r}_{21}|^2} \hat{r}_{21}$$

From figure, $\bar{r}_{12} = -\bar{r}_{21}$ & $|\bar{r}_{12}| = |\bar{r}_{21}|$ & $\hat{r}_{12} = \frac{\bar{r}_{12}}{|\bar{r}_{12}|}$

$$\hat{r}_{21} = \frac{+\bar{r}_{21}}{|\bar{r}_{21}|} = \frac{-\bar{r}_{12}}{|r_{12}|} = -\hat{r}_{12} \quad \& \quad |\hat{r}_{12}| = |\hat{r}_{21}| = 1$$

$$\bar{F}_{12} = \frac{1}{4\pi\varepsilon_0} \frac{q_1 q_2}{|\bar{r}_{12}|^2} \hat{r}_{12}$$

$$= \frac{1}{4\pi\varepsilon_0} \frac{q_1 q_2}{|r_{12}|^2} (-\hat{r}_{21})$$

$$\boxed{\bar{F}_{12} = -\bar{F}_{21}} \Rightarrow \boxed{|\bar{F}_{12}| = |\bar{F}_{21}|}$$

↳ Newton's third law

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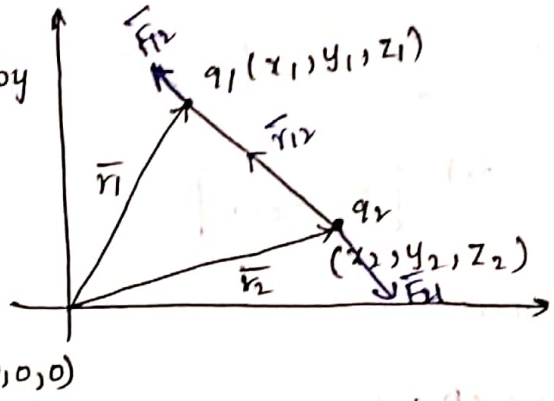
Ex:

Find force on q_1 by q_2.



Sol:

$$\bar{r}_1 = (x_1-0)\hat{i} + (y_1-0)\hat{j} + (z_1-0)\hat{k}$$

$$\bar{r}_1 = x_1\hat{i} + y_1\hat{j} + z_1\hat{k}$$

$$\bar{r}_2 = x_2\hat{i} + y_2\hat{j} + z_2\hat{k}$$

$$\bar{F}_{12} = \frac{1}{4\pi\varepsilon_0} \frac{q_1 q_2}{|\bar{r}_{12}|} \hat{r}_{12}$$

$$= \frac{1}{4\pi\varepsilon_0} \frac{q_1 q_2}{|\bar{r}_{12}|^3} (\bar{r}_{12})$$

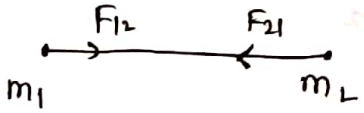
$$\boxed{\bar{F}_{12} = \frac{1}{4\pi\varepsilon_0} \frac{q_1 q_2}{|\bar{r}_1 - \bar{r}_2|^3} (\bar{r}_1 - \bar{r}_2)}$$

★★★ & $\bar{F}_{21} = -\bar{F}_{12}$

* In gravitation:

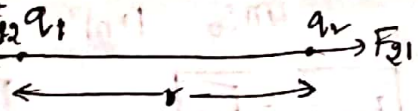
$$F_g = \frac{G m_1 m_2}{r^2} \quad \text{(always attractive)}$$

$G = 6.67 \times 10^{-11}\ Nm^2/kg^2$ (does not change with medium)



In electrostatics:

$$F_e = \frac{1}{4\pi\varepsilon_0} \frac{q_1 q_2}{r^2}$$



$m \rightarrow q$

$G \rightarrow \left(\frac{1}{4\pi\varepsilon_0}\right) \rightarrow$ change with medium

$$\boxed{K = \frac{F_{air}}{F_{med}}}$$

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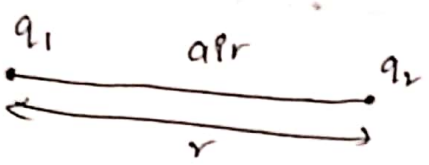
Imp points:

① 

• q – charge is split into 2 parts q_1, q_2.
$q = q_1 + q_2$

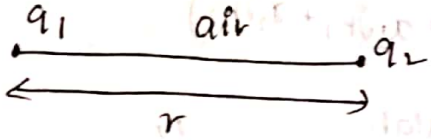
$q_2 = q - q_1$

$$F = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r^2}$$

$$F = \frac{1}{4\pi\varepsilon_0}\frac{q_1(q-q_1)}{r^2}$$

$F = F_{max} \Rightarrow \frac{dF}{dq_1} = 0 \Rightarrow \boxed{q_1 = q_2}$

$$\boxed{q_1 = q_2 = q/2}$$

② 

$$F_{air} = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r^2}$$

$$F_{med} = \frac{1}{4\pi\varepsilon_0 K}\frac{q_1 q_2}{r^2} = \frac{F_{air}}{K}$$

(glass, paper etc.)

now, medium is replaced by air and sepelation between the charges increases to r_1 to keep

$F_{air} = F_{med}$

$$\frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r_1^2} = \frac{1}{4\pi\varepsilon_0 K}\frac{q_1 q_2}{r^2}$$

$$r_1^2 = Kr^2$$

$$r_1 = r\sqrt{K}$$

$$\boxed{r_{air} = r_{med}\sqrt{K}} \quad **$$

(or) $\boxed{r_{med} = \frac{r_{air}}{\sqrt{K}}}$

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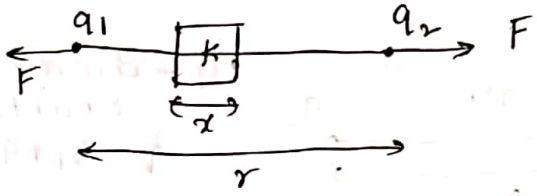
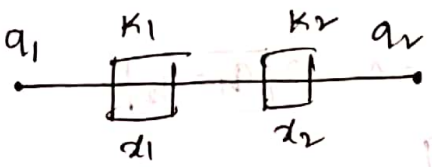
Ex:



$$F = \frac{1}{4\pi\varepsilon_0} \frac{q_1 q_2}{((r-x)+x_{med})^2}$$

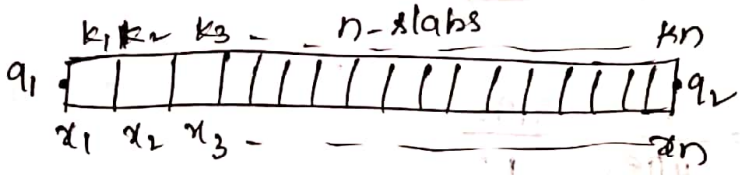
$$\boxed{F = \frac{1}{4\pi\varepsilon_0} \frac{q_1 q_2}{((r-x)+x\sqrt{K})^2}}$$

Ex:



$$F = \frac{1}{4\pi\varepsilon_0} \frac{q_1 q_2}{((r-x_1-x_2)+x_1\sqrt{K_1}+x_2\sqrt{K_2})^2}$$

Ex:



clearly $r = x_1 + x_2 + x_3 + \dots + x_n$

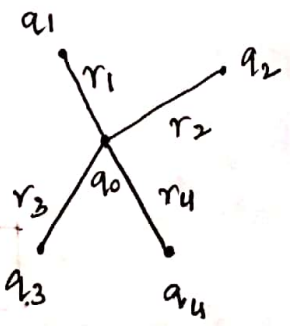
$$F = \frac{1}{4\pi\varepsilon_0} \frac{q_1 q_2}{(x_1\sqrt{K_1} + x_2\sqrt{K_2} + \dots x_n\sqrt{K_n})^2}$$

Principle of Superposition

22|3|25| Saturday

The net force acting on any charge is equal to vector sum of individual forces acting on same charge. i.e,

$$\bar{F}_0 = \bar{F}_{01} + \bar{F}_{02} + \bar{F}_{03} + \dots$$



$$\bar{F}_{01} = \frac{Kq_1q_0}{r_1^2}(\hat{r}_1)$$

$$\bar{F}_{02} = \frac{Kq_2q_0}{r_2^2}(\hat{r}_2)$$

and so on...

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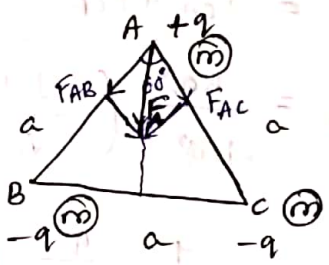
Ex:

① Find i) net force on A & B.

ii) Immediate acceleration of A & B.



Sol:

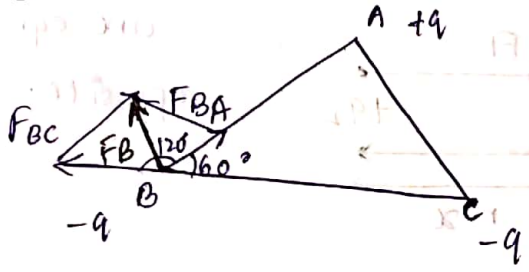
① $F_{AB} = F_{AC} = \frac{Kq^2}{a^2}$

$\overline{F_A} = \overline{F_{AB}} + \overline{F_{AC}}$ & $|\overline{F_A}| = 2F_{AB}\cos\left(\frac{60}{2}\right)$

$\Rightarrow \overline{F_A} = 2\left(\frac{Kq^2}{a^2}\right)\left(\sqrt{3}/2\right)$

$$\boxed{\overline{F_A} = \frac{\sqrt{3}Kq^2}{a^2}}$$ & $a_A = \frac{F_A}{m_A}$

②



$F_{BA} = F_{BC} = \frac{Kq^2}{a^2}$

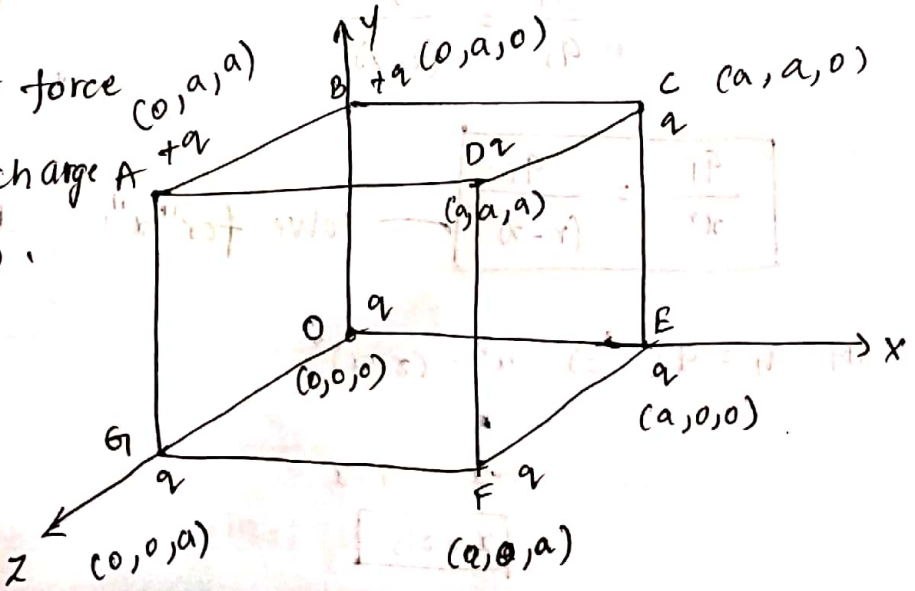
$F_B = 2F_{BC}\cos\left(\frac{120^\circ}{2}\right)$

$= 2\frac{Kq^2}{a^2}(1/2)$

$$\boxed{F_B = Kq^2/a^2}$$ & $a_B = \frac{F_B}{m_B}$.

Ex:

Find net force acting on charge A at origion.



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Sol:

$$\vec{F_o} = \vec{F_{oA}} + \vec{F_{oB}} + \vec{F_{oC}} + \vec{F_{oD}} + \vec{F_{oE}} + \vec{F_{oF}} + \vec{F_{oG}}$$

$$\vec{F_o} = \frac{Kq^2}{(\sqrt{2}a)^2}\left(\frac{-\hat{j}-\hat{k}}{\sqrt{2}}\right) + \frac{Kq^2}{a^2}(-\hat{j}) + \frac{Kq^2}{(\sqrt{2}a)^2}\left(\frac{-\hat{i}-\hat{j}}{\sqrt{2}}\right)$$

$$+ \frac{Kq^2}{(\sqrt{2}a)^2}\left(\frac{-\hat{i}-\hat{j}-\hat{k}}{\sqrt{3}}\right) + \frac{Kq^2}{a^2}(-\hat{i}) + \frac{Kq^2}{(\sqrt{2}a)^2}\left(\frac{\hat{i}-\hat{j}}{\sqrt{2}}\right)$$

$$+ \frac{Kq^2}{a^2}(-\hat{k})$$

Equilibrium:

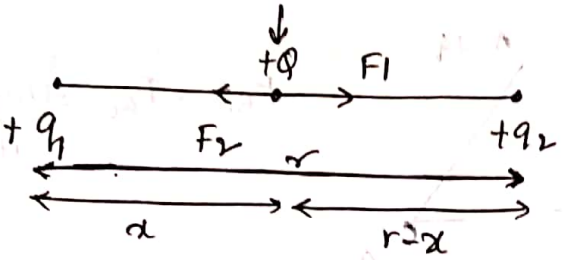
The charge is in equilibrium when net force on charge & net torque on charge is equal to zero. i.e,

$$F_{net} = 0 \;,\; \tau_{net} = 0$$

$\vec{F_1} + \vec{F_2} = 0 \Rightarrow \vec{F_1} = -\vec{F_2}$ & $|\vec{F_1}| = |\vec{F_2}|$ clearly F_1 & F_2 are equal & opposite.

Ex:

①



In eqm, $F_1 = F_2$

$$\frac{Kq_1Q}{x^2} = \frac{Kq_2Q}{(r-x)^2}$$

$$\frac{q_1}{q_2} = \frac{x^2}{(r-x)^2}$$

$$\boxed{\frac{q_1}{x^2} = \frac{q_2}{(r-x)^2}}$$ — solve for "x".

* The pt at which net force on charge is zero then it is called null point.
* eqm depends on q_1, q_2, r & does not depends on magnitude of Q.

If $q_1 = q_2 \Rightarrow x^2 = (r-x)^2$

$$x = r - x$$

$$\boxed{x = r/2}$$

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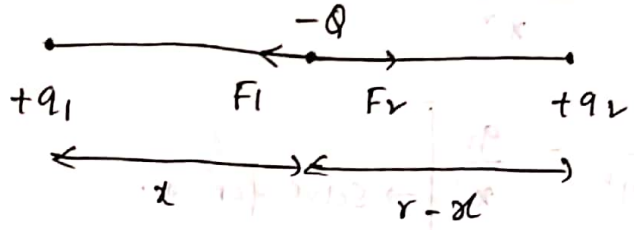
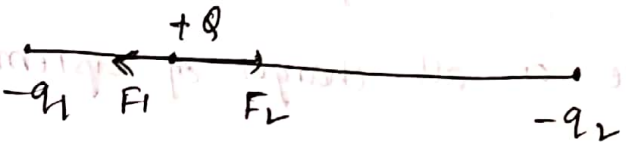
* usually eqm position is obtained at points closer to weaker charge (lesser magnitude charge).

(2)

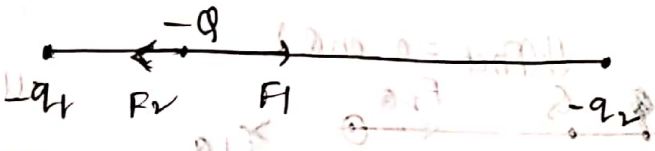


$$\frac{q_1}{x^2} = \frac{q_2}{(r-x)^2}$$

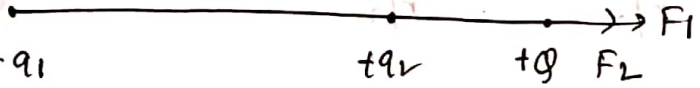
(3)



(4)



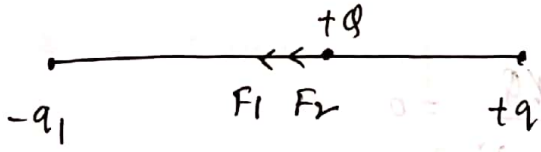
(5)



$F_{net} \neq 0$

no eqm position at all points out side line joining q_1 & q_2 (like charges).

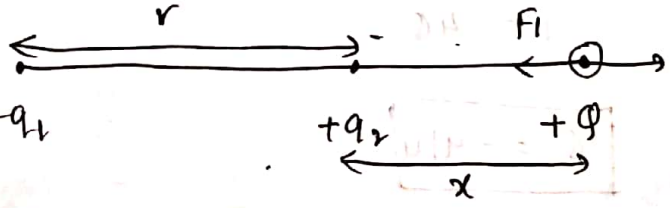
(6)



$F_{net} \neq 0$

no eqm position at all pts inside line joining unlike charges.

(7)



In eqm,

$$\frac{q_1}{(r+x)^2} = \frac{q_2}{x^2}$$

$|q_1| > |q_2|$

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In eqm, $F_1 = F_2$

$$\frac{kq_1Q}{(r+x)^2} = \frac{kq_2Q}{x^2}$$

$$\boxed{\frac{q_1}{(r+x)^2} = \frac{q_2}{x^2}} \rightarrow \text{Solve for } x.$$

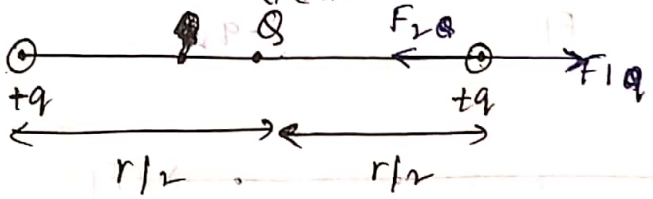
* **Eqm of system of charges**

* All charges present in system must be in eqm.

* Here net force on all charges of system = 0.

Ex:

(1)



// (Fnet = 0, on Q) → all charges must be collinear.

for system in eqm,

$F_{net, on\ q} = 0$ & $F_{net, on\ Q} = 0$.

$F_{1q} + F_{2q} = 0$

$$\frac{kqq}{(r)^2} + \frac{kqQ}{(r/2)^2} = 0$$

$$\frac{q}{r^2} + \frac{4Q}{r^2} = 0$$

$$q = -4Q$$

$$\boxed{Q = -q/4}$$

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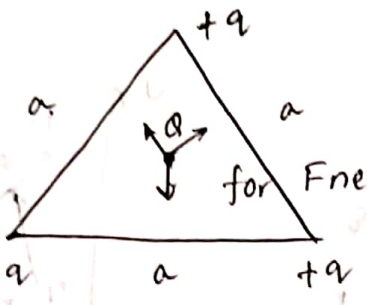
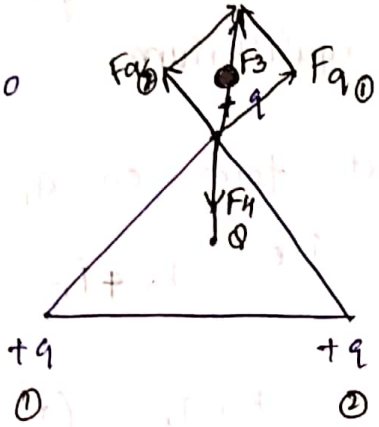
(2)



for $F_{net, on\ Q} = 0$ (Q must be at centroid)

For system in eqm,

$F_{net, on\ q} = 0$



$F_3 = 2\ F_1 \cos\left(\frac{60}{2}\right)$

$F_3 = \frac{\sqrt{3}\, k q^2}{a^2}$

$F_3 + F_4 = 0$

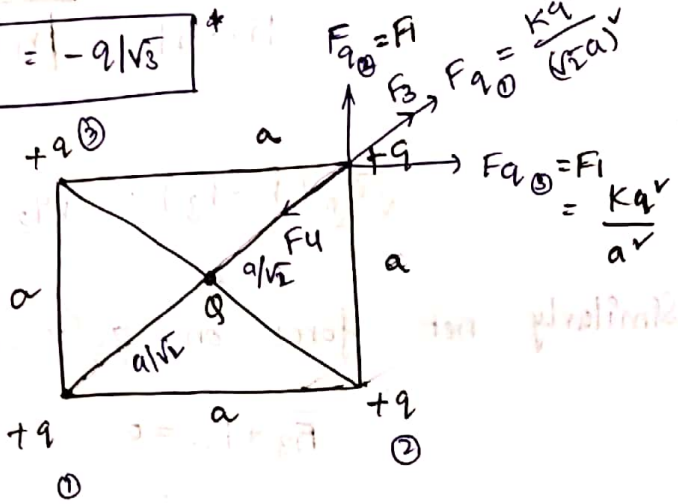
$\frac{\sqrt{3} k q^2}{a^2} + \frac{k q Q}{(a/\sqrt{3})^2} = 0$

$\frac{\sqrt{3} q}{a^2} + \frac{3Q}{a^2} = 0$

$\boxed{Q = -q/\sqrt{3}}$ *

(3)

Q must be at centre of square.



$F_3 = \sqrt{2} F_1$

$= \frac{\sqrt{2} k q^2}{a^2}$

In eqm, $F_{net} = 0 \Rightarrow F_3 + F_{q①} + F_H = 0$

$\frac{kq^2}{(\sqrt{2}a)^2} + \frac{(\sqrt{2})kq^2}{a^2} + \frac{kqQ}{(a/\sqrt{2})^2} = 0$

$\frac{q}{2a^2} + \frac{\sqrt{2}q}{a^2} + \frac{2Q}{a^2} = 0 \Rightarrow 2Q = -q\left(\frac{1}{2} + \sqrt{2}\right)$

$\boxed{Q = -\frac{q(1+2\sqrt{2})}{4}}$

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Ex:

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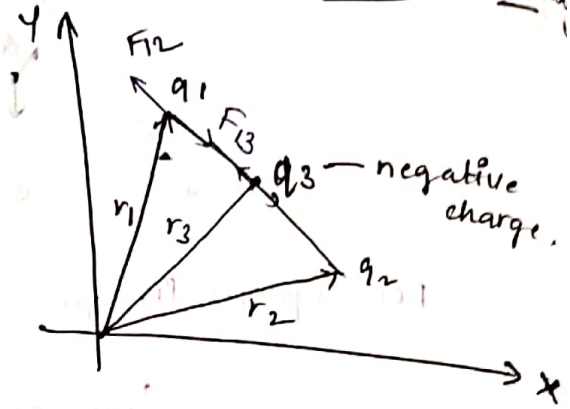
Find the value of negative charge q_3 & q_3's position vector r_3 so that all the charges are in equilibrium.



Soln net force on q_1 is zero.

i.e, $\bar{F}_{12} + \bar{F}_{13} = 0$

$$\frac{Kq_1q_2}{|\bar{r}_1-\bar{r}_2|^3}\frac{(\bar{r}_1-\bar{r}_2)}{|\bar{r}_1-\bar{r}_2|} - \frac{Kq_1q_3}{|\bar{r}_1-\bar{r}_3|^3}\frac{(\bar{r}_1-\bar{r}_3)}{|\bar{r}_1-\bar{r}_3|} = 0$$

clearly, $\hat{r}_{12} = \hat{r}_{13}$

$$\frac{\bar{r}_1-\bar{r}_2}{|\bar{r}_1-\bar{r}_2|} = \frac{\bar{r}_1-\bar{r}_3}{|\bar{r}_1-\bar{r}_3|}$$

$$\sqrt{q_2}(\bar{r}_1-\bar{r}_3) = \sqrt{q_3}(\bar{r}_1-\bar{r}_2) \quad -(a)$$

Similarly net force on q_3 is zero.

$$\bar{F}_{13} + \bar{F}_{23} = 0$$

$$\frac{Kq_1q_3}{|\bar{r}_1-\bar{r}_3|^3}\frac{(\bar{r}_1-\bar{r}_3)}{|\bar{r}_1-\bar{r}_3|} - \frac{Kq_2q_3}{|\bar{r}_3-\bar{r}_2|^3}\frac{(\bar{r}_3-\bar{r}_2)}{|\bar{r}_3-\bar{r}_2|} = 0$$

$$\sqrt{q_1}(\bar{r}_3-\bar{r}_2) = \sqrt{q_2}(\bar{r}_1-\bar{r}_3)$$

$$(\sqrt{q_1}+\sqrt{q_2})\bar{r}_3 = \sqrt{q_2}\bar{r}_1 + \sqrt{q_1}\bar{r}_2$$

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$$\bar{r}_3 = \frac{\sqrt{q_2}\,\bar{r}_1 + \sqrt{q_1}\,\bar{r}_2}{\sqrt{q_1} + \sqrt{q_2}}$$

from (a) :-

$$\sqrt{q_2}\left(\frac{\sqrt{q_1}\,\bar{r}_1 + \sqrt{q_2}\,\bar{r}_1 - \sqrt{q_2}\,\bar{r}_1 - \sqrt{q_1}\,\bar{r}_2}{\sqrt{q_1} + \sqrt{q_2}}\right)$$

$$= \sqrt{q_3}\,(\bar{r}_1 - \bar{r}_2)$$

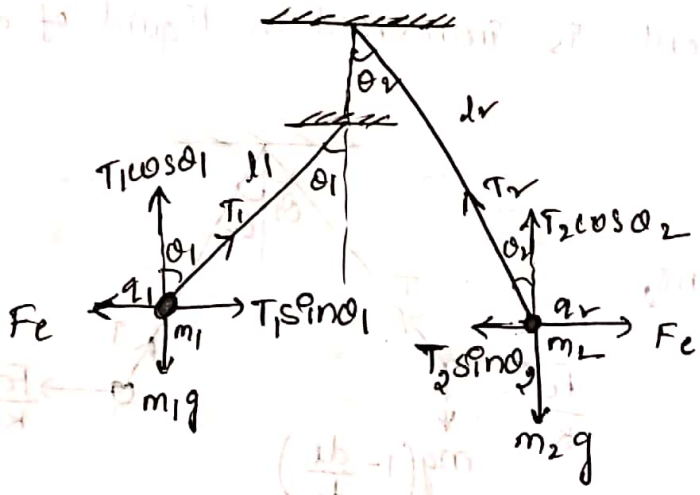
$$\sqrt{q_2} \times \sqrt{q_1}\,(\bar{r}_1 - \bar{r}_2) = \sqrt{q_3}\,(\sqrt{q_1} + \sqrt{q_2})(\bar{r}_1 - \bar{r}_2)$$

$$\sqrt{q_3} = \frac{\sqrt{q_1 q_2}}{\sqrt{q_1} + \sqrt{q_2}}$$

$$q_3 = \frac{-q_1 q_2}{(\sqrt{q_1} + \sqrt{q_2})^2}$$

Charged Pendulums :-

①



In equilibrium,

$$T_1\cos\theta_1 = m_1 g \qquad T_2\cos\theta_2 = m_2 g$$

$$T_1\sin\theta_1 = Fe \qquad T_2\sin\theta_2 = Fe$$

$$\tan\theta_1 = \frac{Fe}{m_1 g} \quad -① \qquad \tan\theta_2 = \frac{Fe}{m_2 g} \quad -②$$

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From eqns ① & ②,

$$\boxed{\frac{\tan\theta_1}{\tan\theta_2} = \frac{m_2}{m_1}}$$

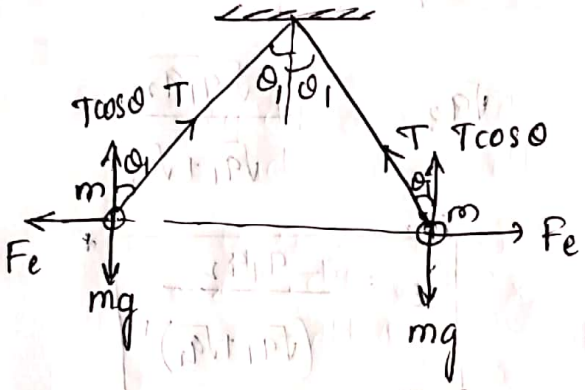
Points:-

i) $m_2 > m_1 \longrightarrow \theta_1 > \theta_2$

ii) $m_2 < m_1 \longrightarrow \theta_2 > \theta_1$

3) $m_1 = m_2 \longrightarrow \theta_1 = \theta_2$

Identical charged pendulums:

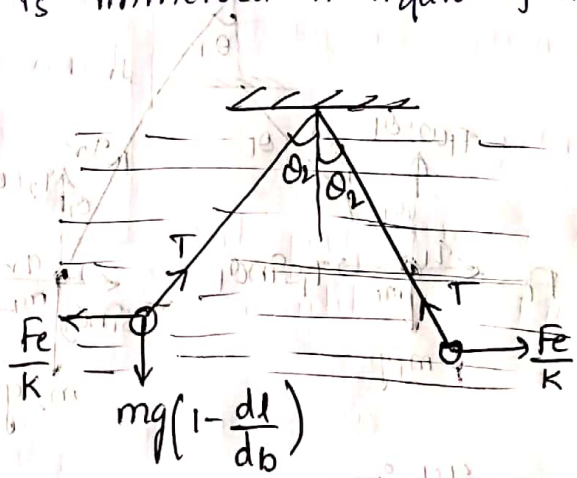


* mass decides θ but not charge.

clearly, $\boxed{Fe = mg\tan\theta_1}$ *

*If the arrangement is immersed in liquid of dielectric const 'K'.

$$\frac{Fe}{K} = mg\left(1-\frac{d_l}{d_b}\right)\tan\theta_2$$



If $\theta_1 = \theta_2$

$$\tan\theta_1 = \tan\theta_2$$

$$\frac{Fe}{mg} = \frac{Fe}{K(mg)\left(1-\frac{d_l}{d_b}\right)}$$

$$\boxed{K = \frac{d_b}{d_b - d_l}}$$ *

** the electrostatic force exerted by the liquid on one of the balls $= Fe - \frac{Fe}{K}$

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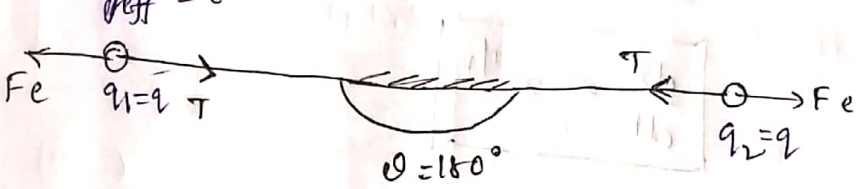
* If arrangement is placed in the region where $q = 0$ (or) $q_{eff} = 0$.

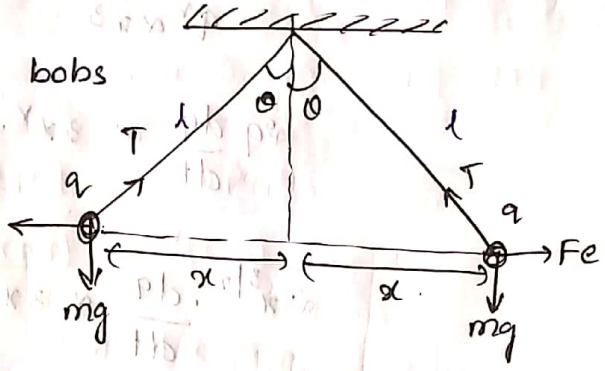


Tension in each string,

$$T = F_e = \frac{Kq_1q_2}{r^2} = \frac{Kq^2}{(2d)^2}$$

$$\boxed{T = \frac{Kq^2}{4d^2}}$$

ex: If the charge on the bobs leaks at a constant rate i.e, $\frac{dq}{dt} = \text{const.}$ Find the relation b/w velocity of approach of bob & distance b/w bobs.



Soln

at any intermediate position,

$$F_e = \frac{1}{4\pi\varepsilon_0}\frac{q^2}{(2x)^2}$$

we have, $F_e = mg\tan\theta$ (θ is very small, $\sin\theta = \tan\theta = \frac{x}{d}$)

$$\frac{1}{4\pi\varepsilon_0} \times \frac{q^2}{4x^2} = mg\left(\frac{x}{d}\right)$$

clearly $q^2 \propto x^3$

on differentiating,

$$2q\frac{dq}{dt} \propto 3x^2\frac{dx}{dt}$$

$$\frac{dx}{dt} \propto \frac{q}{x^2}$$

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$$\frac{dx}{dt} \propto \frac{x^{3/2}}{x^2}$$

$$\boxed{\frac{dx}{dt} \propto x^{-1/2}}$$

$$\Rightarrow \boxed{v \propto x^{-1/2}}$$

*In another way, given $v = \frac{a}{\sqrt{x}}$ = velocity of approach of each bob.

Find $\frac{dq}{dt} =$ _____.

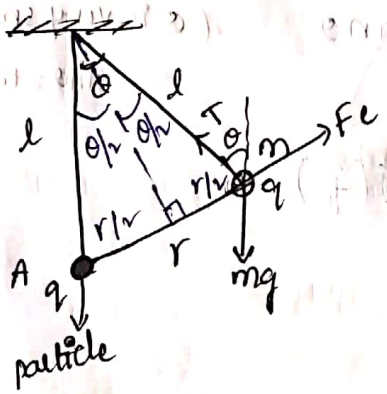
$$q^2 \propto x^3$$

$$\therefore 2q\frac{dq}{dt} \propto 3x^2 \cdot \frac{dx}{dt}$$

$$2x^{3/2} \cdot \frac{dq}{dt} \propto 3x^2 \times \frac{a}{x^{1/2}}$$

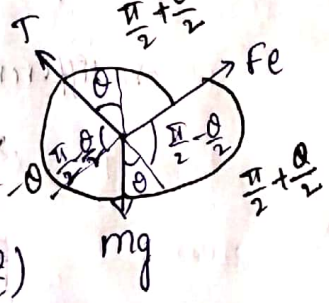
$$\frac{dq}{dt} \propto \frac{3a}{2}$$

$$\boxed{\frac{dq}{dt} = \text{const}}$$

Eg: 

Here pendulum is in equilibrium, Find θ.

Sol: From lami's theorem,



$$\frac{Fe}{\sin(\pi-\theta)} = \frac{mg}{\sin\left(\frac{\pi}{2}+\frac{\theta}{2}\right)} = \frac{T}{\sin\left(\frac{\pi}{2}+\frac{\theta}{2}\right)}$$

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$$F_e = \frac{mg\sin\theta}{\cos\theta/2}$$

$$\frac{Kq^2}{r^2} = \frac{mg(2\sin\theta/2\cos\theta/2)}{\cos\theta/2}$$

From diagram,

$$\sin\frac{\theta}{2} = \frac{r}{2d}$$

$$r = 2d\sin\theta/2$$

$$\frac{Kq^2}{4d^2\sin^2\theta/2} = 2mg\sin\theta/2$$

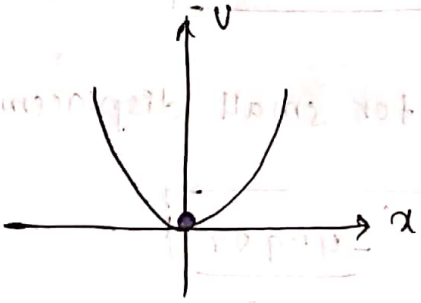
$$\sin^3\theta/2 = \frac{Kq^2}{8d^2mg}$$

$$\sin\theta/2 = \left(\frac{Kq^2}{8mgd^2}\right)^{1/3}$$

$$\boxed{\theta = 2\sin^{-1}\left(\frac{Kq^2}{8mgd^2}\right)^{1/3}}$$

Types of eqm^{n}

① Stable eqm^{n}



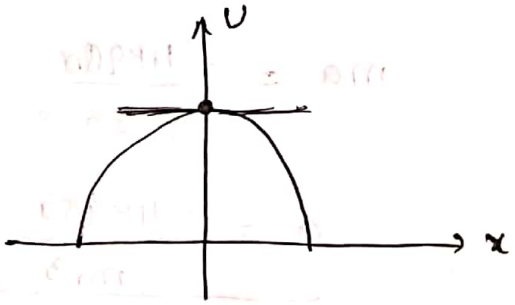
(1) $V = V_{min}$

(2) $F = -\frac{dV}{dx}$

$= 0$

(3) Here, $\frac{d^2V}{dx^2} = +ve$.

② ~~Unst~~ Unstable eqm^{n}



(1) $V = V_{max}$

(2) $F = -\frac{dV}{dx} = 0$

(3) Here, $\frac{d^2V}{dx^2} = -ve$.

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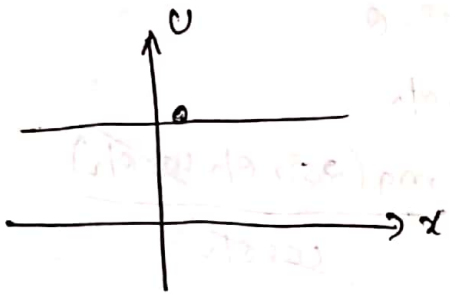
③ Neutral eqm :



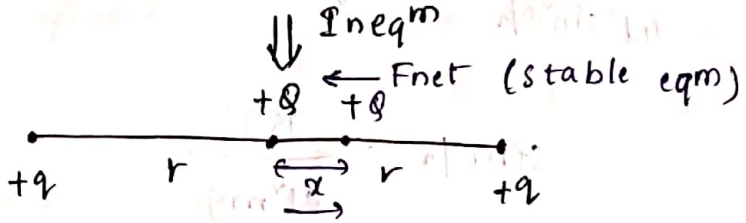
(i) $U = \text{const}$

(ii) $\frac{dU}{dx} = 0$

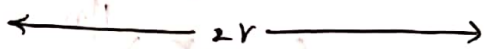
(iii) $\frac{d^2U}{dx^2} = 0$

Ex:

①



At any position,



net force on Q is,

$$F = \frac{KqQ}{(r-x)^2} - \frac{KqQ}{(r+x)^2}$$

$$\boxed{F = \frac{-4rx\,KqQ}{(r^2-x^2)^2}}$$ — oscillatory motion.

If $x \ll\ll r$ (for small displacement)

$$\boxed{F = \frac{-4KqQx}{r^3}}$$

It is simple harmonic motion.

$$ma = -\frac{4KqQx}{r^3}$$

$$a = \frac{-4KqQx}{mr^3}$$

$$\boxed{\omega = \sqrt{\frac{4KqQ}{mr^3}}}$$

$K = \frac{1}{4\pi\varepsilon_0}$

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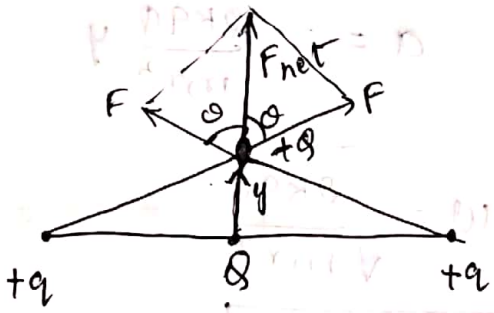
$$T = \frac{2\pi}{\omega}$$

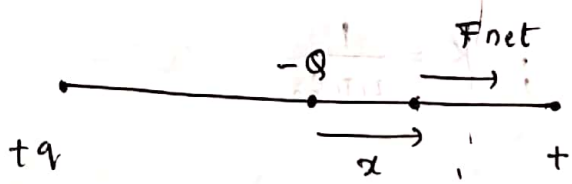
$$T = 2\pi\sqrt{\frac{mr^3}{4kqQ}} \quad *$$

(2)

Unstable eqm.

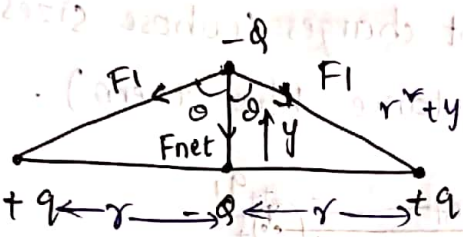


(3)



unstable eqm.

(4)



$$F_{net} = 2F_1\cos\theta$$

$$= \frac{2kqQ}{(r^2+y^2)} \times \frac{y}{\sqrt{r^2+y^2}}$$

$$F_{net} = \frac{-2kqQy}{(r^2+y^2)^{3/2}}$$

$\Downarrow$

oscillatory motion

If $y \ll r$ (for small displacements)

$$F_{net} = -\frac{2kqQ}{r^3}y$$

$\Downarrow$

It is an SHM.

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$$ma = -\frac{2Kq_0q}{r^3}$$

$$a = -\frac{2Kq_0}{mr^3}y$$

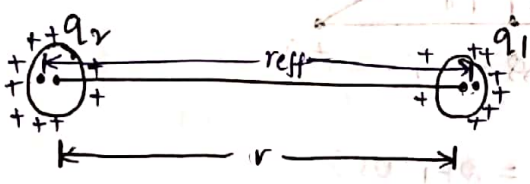
$$\omega = \sqrt{\frac{2Kq_0}{mr^3}}$$

$$\boxed{T = 2\pi\sqrt{\frac{mr^3}{2Kq_0}}} \quad ; \quad K = \frac{1}{4\pi\varepsilon_0}$$

Limitations of Coulomb's Law :

* It is valid only for pt charges (whose sizes are very small compared to distance blw them).

①



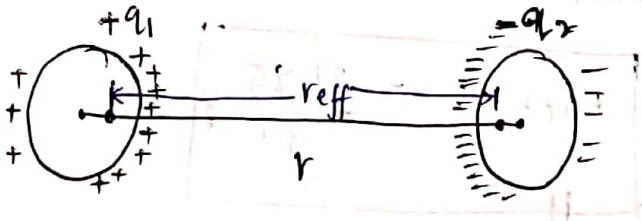
$\boxed{r_{eff} > r}$ $\qquad F_c = \frac{1}{4\pi\varepsilon_0}\frac{q_1q_2}{r^2}$

Actual force between the charges,

$$F_{actual} = \frac{1}{4\pi\varepsilon_0}\frac{q_1q_2}{r_{eff}^2}$$

clearly, $\boxed{F_{actual} < F_c}$

②



$r_{eff} < r$, $F_c = \frac{1}{4\pi\varepsilon_0}\frac{q_1q_2}{r^2}$, $F_{actual} = \frac{1}{4\pi\varepsilon_0}\frac{q_1q_2}{r_{eff}^2}$

clearly, $\boxed{F_c < F_{actual}}$

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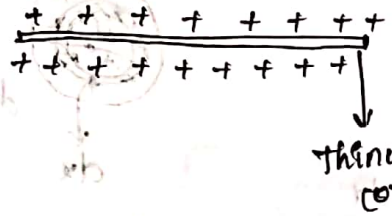
* It is valid only for "static charges".

$$\bar{F} = \bar{F}_e + \bar{F}_M$$

↓ net force on the charge.

Types of charge densities:

01 linear charge density:

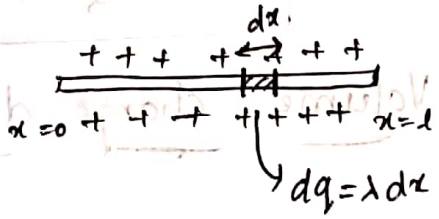


$$\lambda = \frac{Q_T}{l}$$

$$\boxed{\lambda = \frac{dq}{dl}}$$

Ex: For a given wire, $\lambda = \lambda_0 \left(1 + \frac{x}{l}\right)$. Find Total charge on the wire.

Sol:



$$dq = \lambda\, dx$$

$$dq = \lambda_0 \left(1 + \frac{x}{l}\right) \cdot dx$$

$$Q_T = \int dq = \int_0^l \lambda_0 \left(1 + \frac{x}{l}\right) \cdot dx$$

$$= \lambda_0 \left(x + \frac{x^2}{2l}\right)_0^l$$

$$= \lambda_0 \left(l + \frac{l^2}{2l}\right)$$

$$\boxed{Q_T = \frac{3\lambda_0 l}{2}}$$

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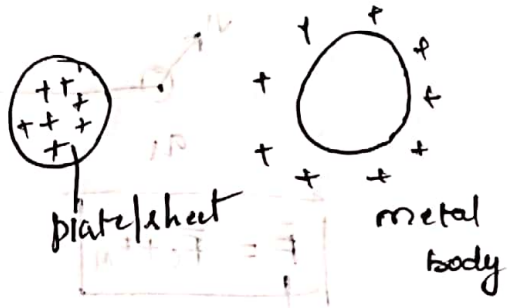
(2) Surface charge density:

$$\sigma = \frac{Q_T}{A_T} \quad C/m^2$$

$$\sigma = \frac{dq}{dA}$$

$$Q = \int dq = \int \sigma \cdot dA$$

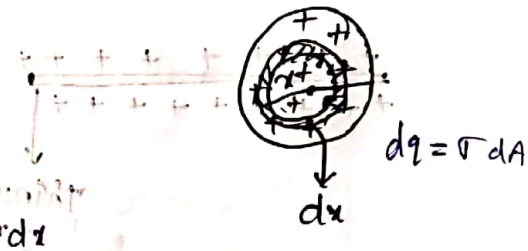


Ex: For circular plate, $\sigma = \sigma_0\left(\frac{x}{R}\right)$. Find total charge on plate.

Sol:

$$dq = \sigma (2\pi x) \cdot dx$$

$$dq = \sigma_0 \left(\frac{x}{R}\right)(2\pi x) \cdot dx$$



$$Q = \int dq = \frac{2\pi\sigma_0}{R} \int_{x=0}^{x=R} x^2 \cdot dx$$

$$Q = \frac{2\pi\sigma_0}{R} \cdot \frac{R^3}{3}$$

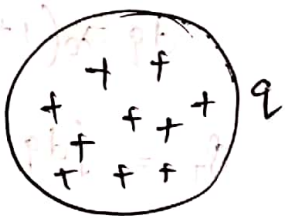
$$\boxed{Q = \frac{2\pi\sigma_0 R^2}{3}}$$

(3) Volume charge density:

$$\rho = \frac{Q_T}{V_T} \quad C/m^3$$

$$\rho = \frac{dq}{dv}$$

$$Q_T = \int dq = \int \rho \cdot dv$$



non-conducting bodies

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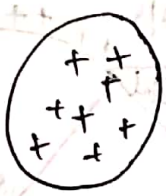
Ex! given $\rho = \rho_0\left(1 - \frac{r}{R}\right)$.

Find total charge in sphere.



$$Q = \int \rho \cdot dv$$

$$= \int \rho_0\left(1 - \frac{r}{R}\right) \cdot 4\pi r^2 \cdot dr$$

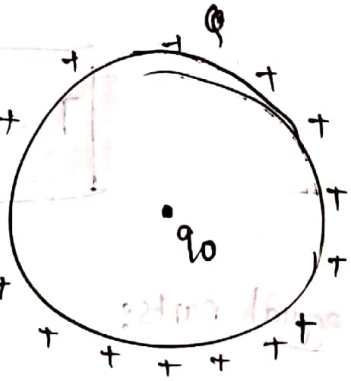
$$= 4\pi\rho_0 \int_{r=0}^{r=R} r^2 - \frac{r^3}{R} \cdot dr$$

$$= 4\pi\rho_0 \left[\frac{R^3}{3} - \frac{R^4}{4R}\right]$$

$$\boxed{Q = \frac{\pi \rho_0 R^3}{3}}$$

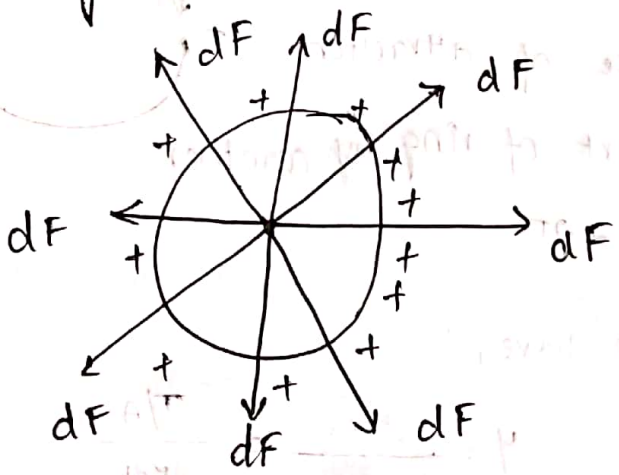
Ex 6

A charge q_0 is placed at the centre of circular ring. Find increment in the tension of ring.



Sol 6

① Net force on the ring is zero.



②

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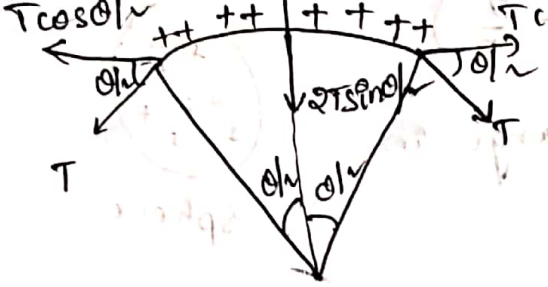


$dq = \lambda dx$

$= \frac{q_0}{2\pi R} \cdot dx$

In eqm,

$F_e = 2T\sin\theta/2$

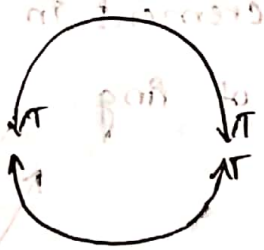
$\frac{1}{4\pi\varepsilon_0} \cdot \frac{q_0\, dq}{R^2} = 2T(\theta/2)$

$\frac{q_0}{4\pi\varepsilon_0 R^2} \times \frac{q}{2\pi R} \cdot dx = T\theta$

$$\boxed{T = \frac{qq_0}{8\pi^2\varepsilon_0 R^2}}$$

$$\boxed{T = \frac{1}{4\pi\varepsilon_0}\frac{qq_0}{R^2} \times \frac{1}{2\pi}}$$

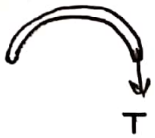
(3) Two equal parts:



Force of attraction on part of ring by another part $= 2T$

(4) We have,

$$Y = \frac{\text{stress}}{\text{strain}} = \frac{T/A}{\frac{2\pi dR}{2\pi R}}$$



A = area of cross-sectio of materi of ring

$$\boxed{\frac{dR}{R} = \frac{T}{AY}}$$ fractional increase in radius.

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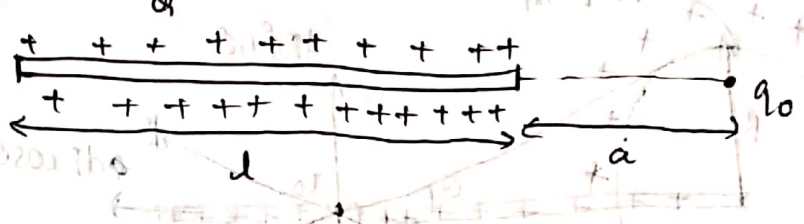
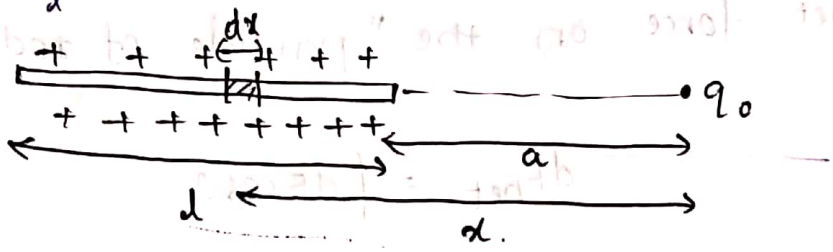
Ex 6



Find force on rod due to point charge.

Sol^n

$$\lambda = \frac{Q}{l}$$



$$F = \int dF = \int_{x=a}^{x=a+l} \frac{1}{4\pi\varepsilon_0} \frac{q_0 dq}{x^2}$$

$$F = \frac{1}{4\pi\varepsilon_0} q_0 \lambda \int_{x=a}^{x=a+l} \frac{1}{x^2} \cdot dx$$

$$F = \frac{q_0 \lambda}{4\pi\varepsilon_0} \left[\frac{1}{a} - \frac{1}{a+l}\right]$$

$$= \frac{q_0 \lambda (a+l-a)}{4\pi\varepsilon_0 (a)(a+l)}$$

$$F = \frac{1}{4\pi\varepsilon_0} \frac{q_0 Q}{a(a+l)}$$

$$\boxed{F = \frac{K q_0 Q}{a(a+l)}} \quad *$$

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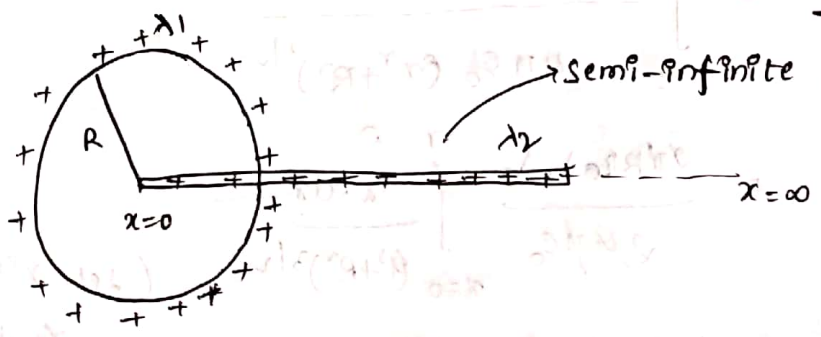
Ex 8



Find the force of interaction b/w the bodies.

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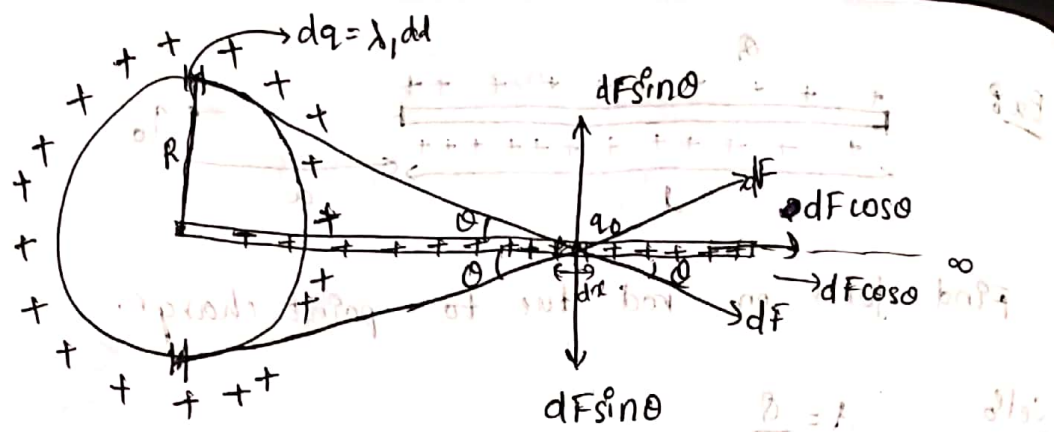
Sol:



* net force on the "particle of rod due to ring.

$$dF_{net} = \int dF\cos\theta$$

$$= \frac{1}{4\pi\varepsilon_0}\int \frac{q_0\,dq}{x^2+R^2}\cdot\frac{x}{\sqrt{x^2+R^2}}$$

$$= \frac{q_0 x}{4\pi\varepsilon_0 (x^2+R^2)^{3/2}}\int \lambda_1\,dl$$

$$= \frac{q_0 x \lambda_1 (2\pi R)}{4\pi\varepsilon_0 (x^2+R^2)^{3/2}}$$

$$dF_{net} = \frac{2\pi R q_0 x \lambda_1}{4\pi\varepsilon_0 (x^2+R^2)^{3/2}}$$

$(q_0 = \lambda_2\, dx)$

Finally, net force on the entire rod due to ring.

$$F = \int dF_{net}$$

$$= \int_{x=0}^{x=\infty} \frac{(\lambda_2 dx)(\lambda_1)(2\pi R)(x)}{4\pi\varepsilon_0 (x^2+R^2)^{3/2}}$$

$$= \frac{2\pi R \lambda_1 \lambda_2}{2\cdot 4\pi\varepsilon_0}\int_{x=0}^{x=\infty} \frac{x\cdot dx}{(x^2+R^2)^{3/2}}$$

(let $x^2+R^2 = t^2$

$2x\cdot dx = 2t\cdot dt$

$$F = \frac{\lambda_1\lambda_2 R}{2\varepsilon_0}\int \frac{t\cdot dt}{t^3}$$

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$$F = \frac{\lambda_1 \lambda_2 R}{2\varepsilon_0}\left[\frac{-1}{\sqrt{x^2+R^2}}\right]_{x=0}^{x=\infty}$$

$$F = \frac{\lambda_1 \lambda_2 R}{2\varepsilon_0}\left[-\frac{1}{\infty} + \frac{1}{R}\right]$$

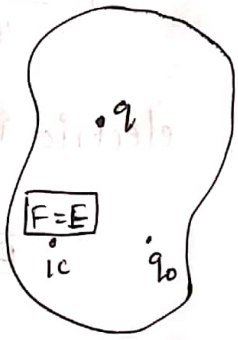
$$F = \frac{\lambda_1 \lambda_2 R}{2\varepsilon_0 R}$$

$$\boxed{F = \frac{\lambda_1 \lambda_2}{2\varepsilon_0}}$$

If $\lambda_1 = \lambda_2$ then $\boxed{F = \lambda^2/2\varepsilon_0}$.

Electric Field:

It is the region or space that surrounds the charge upto which its influence is felt & called its electric field.



1) Intensity of electric Field: (E)

The force acting on unit positive charge when placed in the electric field is called intensity of electric field at that particular point.

$q_0 \rightarrow F$

test charge (Small magnitude)

$1 \rightarrow \boxed{\frac{F}{q_0} = E}$

$$\boxed{\vec{\varepsilon} = \frac{\vec{F}}{q_0}}$$

units: N/c or volt/m.

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* Force acting on the charge q placed at a point in field of intensity $\bar{E}$ is,

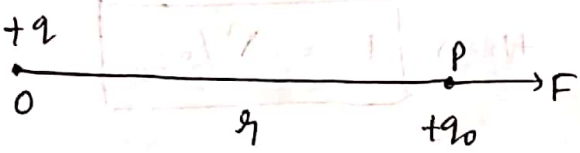
$$\boxed{\bar{F} = q\bar{E}}$$

* If q = +ve then $\bar{F} \parallel \bar{E}$

* If q = -ve then $\bar{F} \parallel -\bar{E}$

Electric intensity at a point due to point charge.

*.



Force on charge 'q_0' is,

$$F = \frac{1}{4\pi\varepsilon_0} \frac{qq_0}{r^2}$$

now, electric intensity at 'P' is,

$$\bar{E} = \frac{F}{q_0}$$

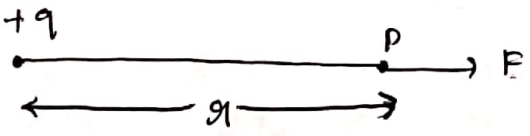
$$\boxed{\bar{E} = \frac{1}{4\pi\varepsilon_0} \frac{q}{r^2}}$$

* In another medium of dielectric const K,

$$\bar{E} = \frac{1}{4\pi\varepsilon_0 K} \cdot \frac{q}{r^2}$$

$$\boxed{\bar{E}_{med} = \frac{\bar{E}_{air}}{K}}$$

*



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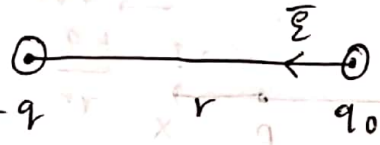
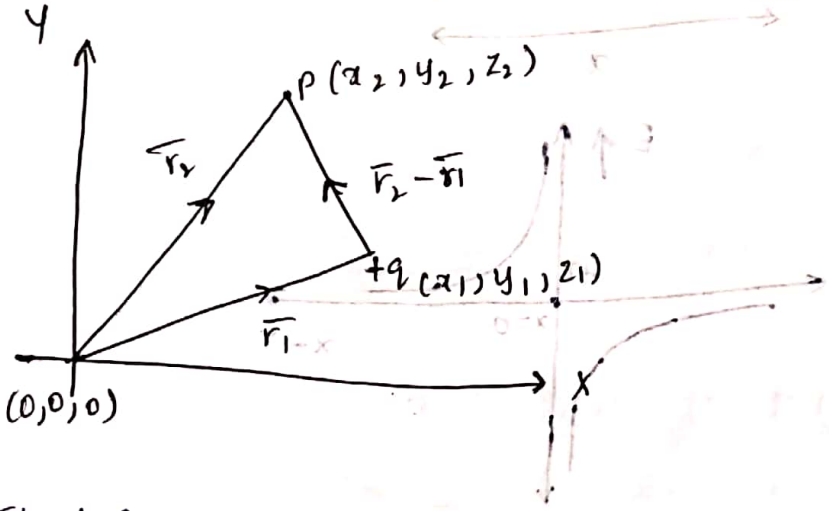


$$\mathcal{E} = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}$$

Ex:



Electric intensity at P is,

$$E = \frac{1}{4\pi\varepsilon_0}\frac{q}{|\bar{r}_2 - \bar{r}_1|^2}\frac{(\bar{r}_2 - \bar{r}_1)}{|\bar{r}_2 - \bar{r}_1|}$$

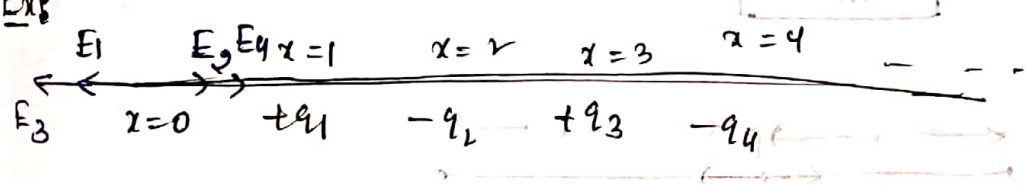
$$\boxed{\bar{E} = \frac{1}{4\pi\varepsilon_0}\frac{q(\bar{r}_2 - \bar{r}_1)}{|\bar{r}_2 - \bar{r}_1|^3}}$$

Principle of Superposition:

Net electric field intensity at pt P is,

$$\bar{\mathcal{E}} = \bar{\mathcal{E}}_1 + \bar{\mathcal{E}}_2 + - - - -$$

Ex:



Net electric field at $x = 0$ is ____.

Sol:

$$\bar{\mathcal{E}} = \bar{\mathcal{E}}_1 + \bar{\mathcal{E}}_2 + \bar{\mathcal{E}}_3 + - - -$$

$$\bar{\mathcal{E}} = \frac{Kq_1^2}{(1)^2}(-\hat{\imath}) + \frac{Kq_2}{(2)^2}(\hat{\imath}) + \frac{Kq_3}{(3)^2}(-\hat{\imath}) + - - - -$$

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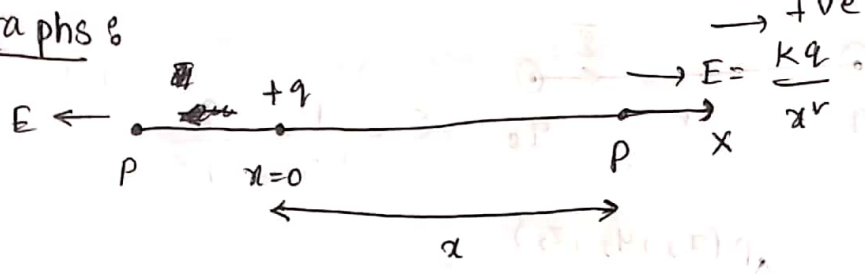
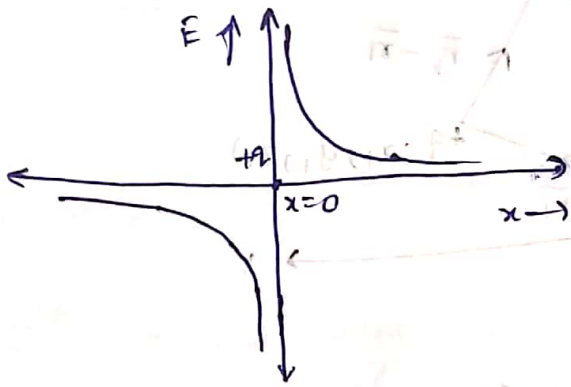
Graphs :

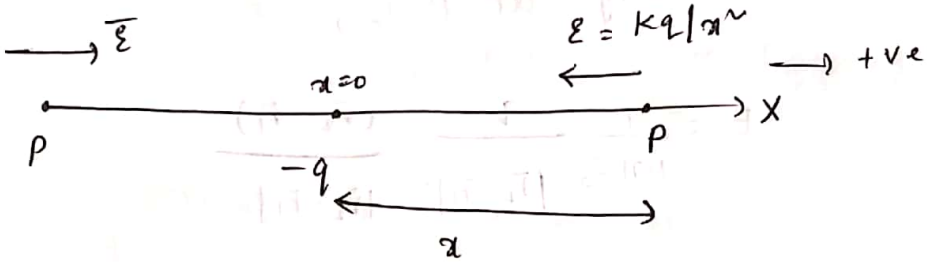
①



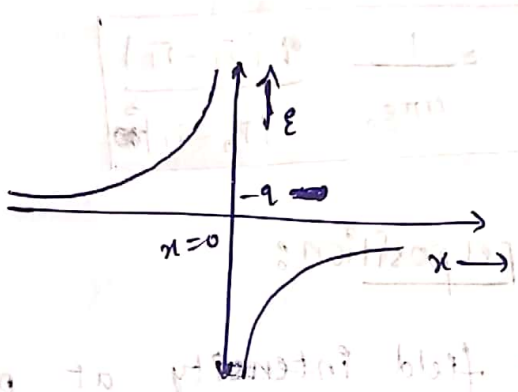
graph –



②



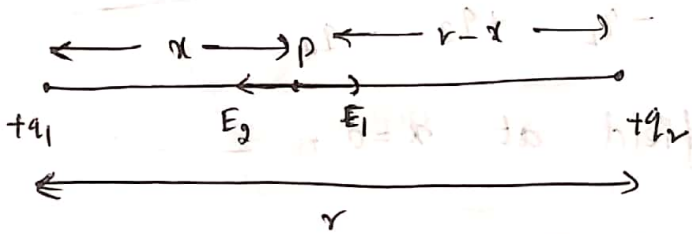
graph –



Null point :

$\bar{\varepsilon}_{net} = 0$

①



Net field, $\bar{\varepsilon} = \bar{\varepsilon}_1 + \bar{\varepsilon}_2$

at null point $\bar{\varepsilon} = 0$

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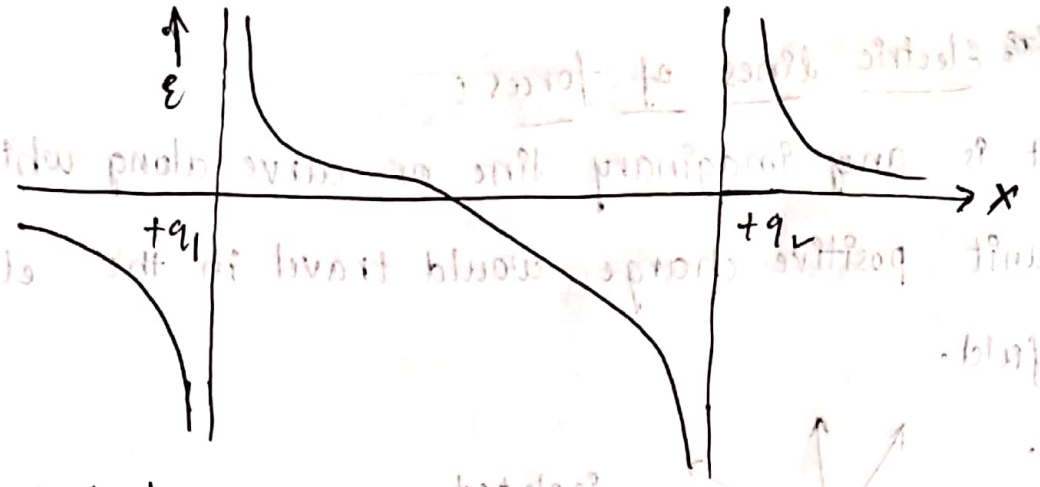
$\vec{E_1} + \vec{E_2} = 0 \Rightarrow \vec{E_1} = -\vec{E_2} \Rightarrow |\vec{E_1}| = |\vec{E_2}|$

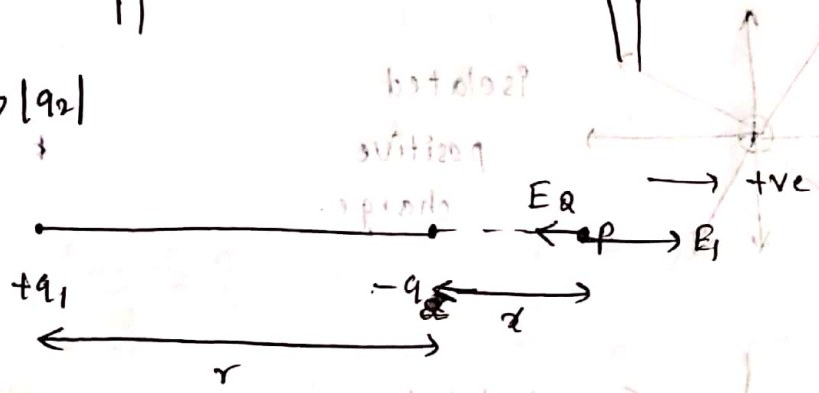
$$\frac{Kq_1}{x^2} = \frac{Kq_2}{(r-x)^2}$$

$$\boxed{\frac{q_1}{x^2} = \frac{q_2}{(r-x)^2}}$$ — solve for x.

graphe



Ex: $|q_1| > |q_2|$



At null point,

$$\frac{Kq_1}{(r+x)^2} = \frac{Kq_2}{x^2}$$

$$\boxed{\frac{q_1}{(r+x)^2} = \frac{q_2}{r^2}}$$

solve for x.

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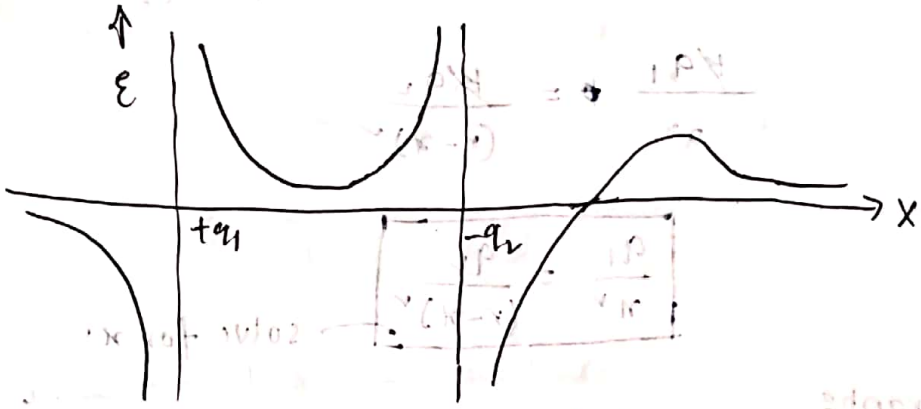
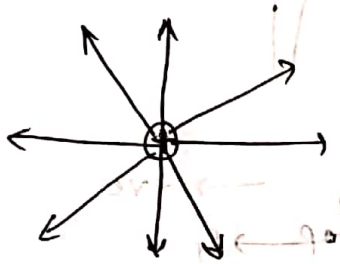
graph:



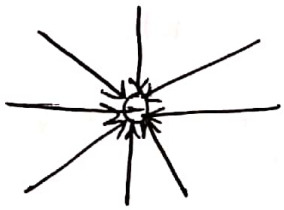
Electric lines of forces:

It is any imaginary line or curve along which an isolated unit positive charge would travel in the electric field.

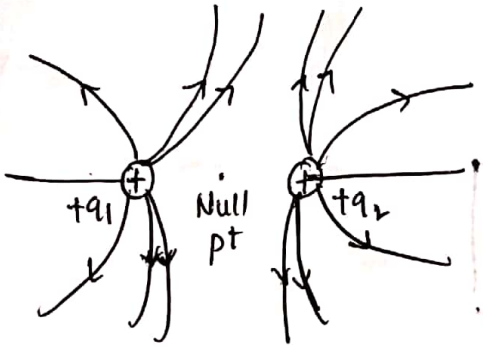
1. 

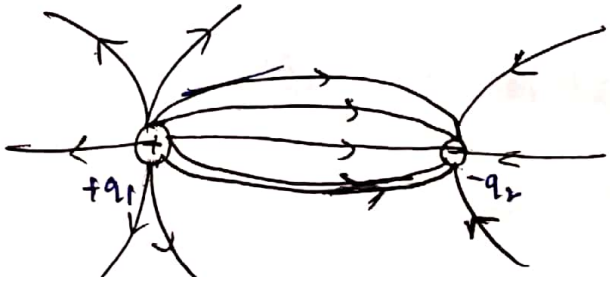
Isolated positive charge.

* pt charge ↓ non-uniform electric field.

2. 

Isolated negative charge.

3. 

4. 

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5. Electric field lines are called "open loop curves".

6. tangent ⟶ direction of electric field.

7. Number of lines passing through unit gives magnitude of electric field.

8. we have,

$$E = \frac{1}{4\pi\varepsilon_0} \times \frac{q}{r^2}$$

$$E \times 4\pi r^2 = q/\varepsilon_0$$

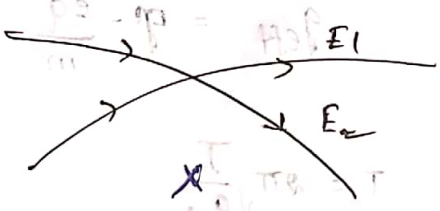
Total no of lines passing through an area ($4\pi r^2$) is called electric flux.

i.e, $N = q/\varepsilon_0$

$$N \propto q$$

$$\therefore \frac{N_1}{N_2} = \frac{q_1}{q_2}$$

9. No two electric lines can intersect with each other.



10. In uniform electric field,

$\bar{E}$ = const at all pts.

uniform electric field

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Charged Pendulum in Electric Field:

$T = 2\pi\sqrt{\frac{l}{g}}$

(with out electric field)

* when electric field is switched on -

①

$T + \varepsilon q = mg$

$T = mg - \varepsilon q$

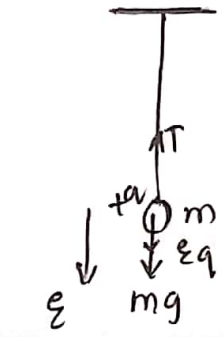
$mg_{eff} = mg - \varepsilon q$

$g_{eff} = g - \frac{\varepsilon q}{m}$

$T = 2\pi\sqrt{\frac{l}{g_{eff}}}$

$$\boxed{T = 2\pi\sqrt{\frac{l}{g - \frac{\varepsilon q}{m}}}}$$

②



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$T = Eq + mg$

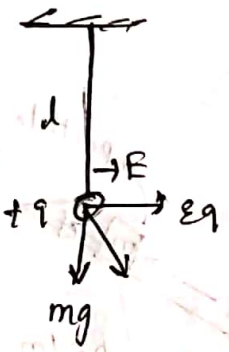
$mg_{eff} = Eq + mg$

$$\boxed{g_{eff} = g + \frac{Eq}{m}}$$

$$T = 2\pi \sqrt{\frac{l}{g_{eff}}}$$

$$\boxed{T = 2\pi \sqrt{\frac{l}{g + \frac{Eq}{m}}}}$$

(3)



$$mg_{eff} = \sqrt{(mg)^2 + (Eq)^2}$$

$$\boxed{g_{eff} = \sqrt{g^2 + \left(\frac{Eq}{m}\right)^2}}$$

$$T = 2\pi \sqrt{\frac{l}{g_{eff}}}$$

$$\boxed{T = 2\pi \sqrt{\frac{l}{\sqrt{g^2 + \frac{E^2q^2}{m^2}}}}}$$

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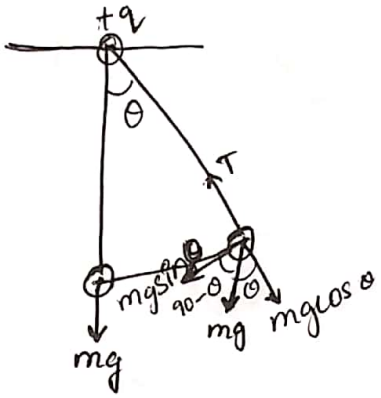
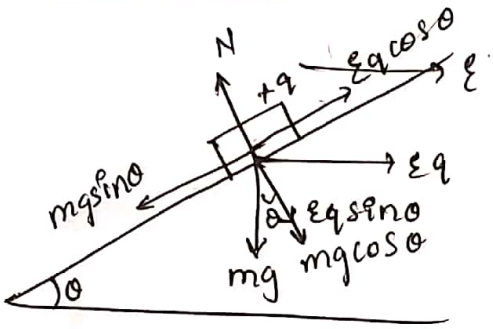
(4)

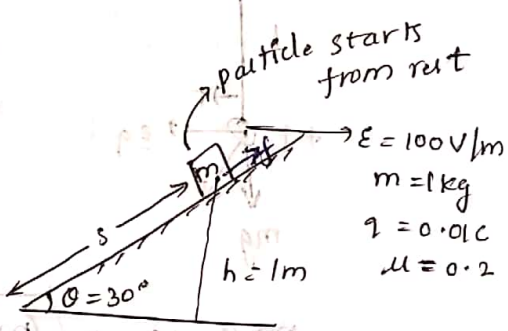


$$T = 2\pi\sqrt{\frac{l}{g}}$$

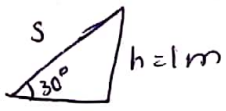
*



Ex: If the particle slides down from rest from a height of 1m. Find the time taken by the particle to reach the bottom of the inclined plane.



Sol:



$$\sin 30^\circ = \frac{h}{s}$$

$$\frac{1}{2} = \frac{h}{s} \Rightarrow s = 2h = 2\,m$$

$$N = \varepsilon q\sin\theta + mg\cos\theta$$

$$f = \mu N = \mu(\varepsilon q\sin\theta + mg\cos\theta)$$

$$f = 0.2 \times \left(\frac{1}{2} + 1 \times \frac{4.9}{9.8} \times \frac{\sqrt{3}}{2}\right)$$

$$f = 0.1 + 0.98 \times \sqrt{3} = 1.6974 + 0.1 = 1.7974$$

$$a = ma + \varepsilon q\cos\theta + f = mg\sin\theta$$

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$-(\frac{1.73}{2} + 1.70074) + 4.9 = a$

$a =$

$S = \frac{1}{2} a t^2$

$ma = mg\sin\theta - Eq\cos\theta - f$

$a = \frac{1}{2} \times 9.8 - \frac{\sqrt{3}}{2} - 1.7074$

$a = 4.9 - 0.866 - 1.7074$

$\boxed{a = 2.32}$

$S = \frac{1}{2} a t^2$

$\frac{9 \times 2}{2.32} = t^2$

$t = \sqrt{\frac{4}{2.32}}$

$\boxed{t = 1.34 \text{ sec}}$

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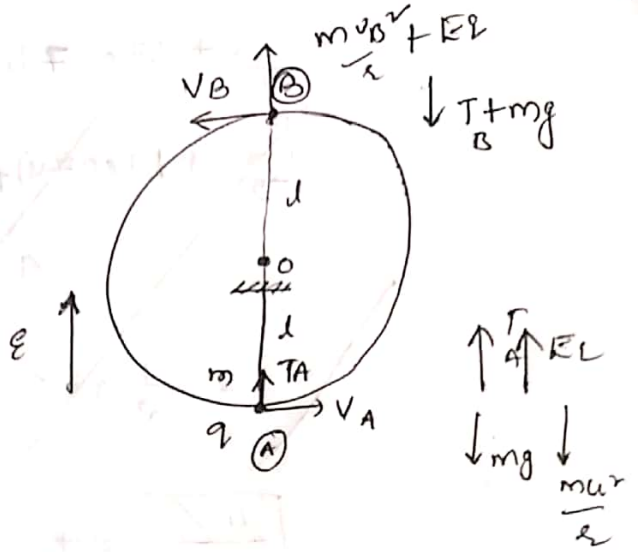
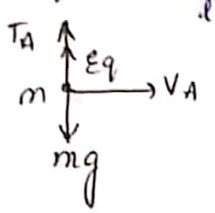
Vertical Circular Motion :



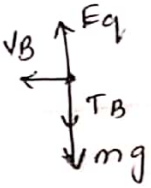
At bottom most point,

$$T_A + Eq - mg = \frac{mv_A^2}{l} \quad -(1)$$



At top-most point,

$$T_B + mg - Eq = \frac{mv_B^2}{l} \quad -(2)$$



* charge will not suffer force in its own field.

From $w-\varepsilon$ theorem,

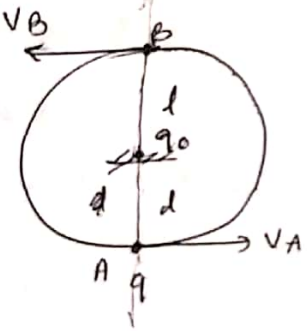
$$w_g + w_e + w_T = \Delta K$$

$$-mg(2l) + Eq(2l) + 0 = \frac{1}{2}mv_B^2 - \frac{1}{2}mv_A^2 \quad -(3)$$

* Just to complete the circular loop,

$T_B = 0$

At A -



$$T_A - mg - F_c = \frac{mv_A^2}{l} \quad -(1)$$

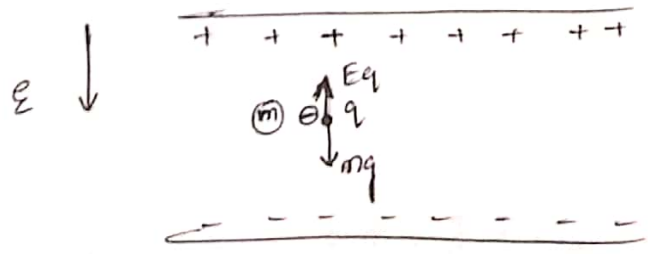
At B -

$$T_B + mg - F_c = \frac{mv_B^2}{l} \quad -(2)$$

$w-\varepsilon$ Theorem -

$$-mg(2l) + 0 + 0 = \frac{1}{2}mv_B^2 - \frac{1}{2}mv_A^2 \quad -(3)$$

Motion of a charged particle in a Uniform Electric field :



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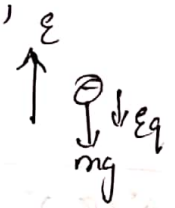
If charge is at rest,

$$Eq = mg.$$

Suddenly if field is reversed,

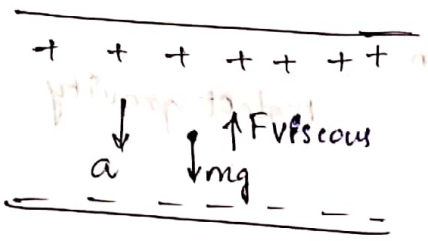


$$ma = mg + Eq$$

$$ma = mg + mg$$

$$\boxed{a = 2g} \ *$$

*



without electric field,

$$F_{net} = mg - F_v$$

$$ma = mg - F_v$$

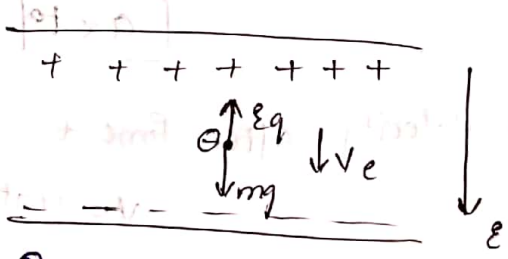
at terminal velocity, $\boxed{a = 0}$

$$mg = F_v$$

$$\frac{4}{3}\pi r^3 \rho g (s - \sigma) g = 6\pi\eta r V_g$$

$$\boxed{V_g = \frac{2}{9\eta}(s-\sigma) g r^2} \ *$$

when electric field is switched on,



$$mg - Eq = 6\pi\eta r V_e$$

$$\frac{4}{3}\pi r^3 (s-\sigma) g - Eq = 6\pi\eta r V_e \quad -②$$

$$6\pi\eta r V_g - Eq = 6\pi\eta r V_e$$

$$6\pi\eta r (V_g - V_e) = Eq$$

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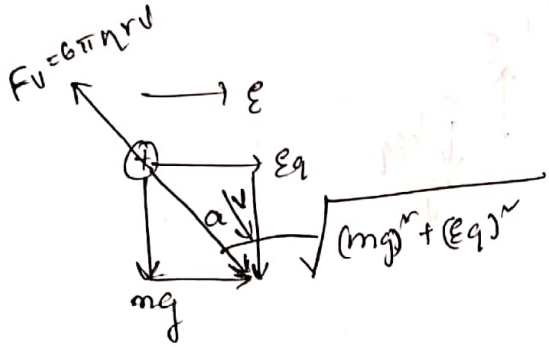
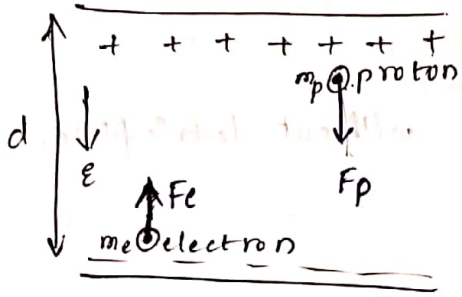
$$q = \frac{6\pi\eta r(v_g - v_e)}{\mathcal{E}}$$

*.



Ex:



neglect gravity.

equal $\begin{cases} F_e = \mathcal{E}q \\ F_p = \mathcal{E}q \end{cases}$ $(q = 1.6 \times 10^{-19} C)$

* $a_e = \frac{F_e}{m_e}$, $a_p = \frac{F_p}{m_p}$

$$\frac{a_e}{a_p} = \frac{m_p}{m_e} > 1$$

$$a_e \gg a_p$$ *

& acceleration is directly proportional to t^0.

$$a \propto t^0$$

* Velocity after time t:

$$v = u + at$$

$$v = 0 + \frac{F}{m}t$$

$$v = \frac{Ft}{m}$$

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$\Rightarrow \boxed{v \propto t}$ *

now, $v_e = \frac{F_e}{m_e} t$ & $v_p = \frac{F_e}{m_p} t$.

$$\frac{v_e}{v_p} = \frac{m_p}{m_e} > 1$$

$\boxed{v_e >> v_p}$ *.

* displacement of particle at any time t,

$$s = ut + \frac{1}{2} a t^2$$

$$s = \frac{1}{2} \times \frac{F}{m} t^2$$

$\Rightarrow \boxed{s \propto t^2}$

now, $s_e = \frac{1}{2} \times \frac{F_e}{m_e} t^2$ & $s_p = \frac{1}{2} \times \frac{F_e}{m_p} t^2$

$$\frac{s_e}{s_p} = \frac{m_p}{m_e} > 1$$

$\boxed{s_e >> s_p}$ *

For same s,

$$t = \sqrt{\frac{2s}{a}}$$

$$\Rightarrow \frac{t_e}{t_p} = \sqrt{\frac{a_p}{a_e}} < 1$$

$\boxed{t_e < t_p}$ *

* electron wins the race *

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Finally,

momentum,

$$P = mv$$

$$P \propto v$$

$$v \propto t$$

$$\Rightarrow \boxed{P \propto t}$$

Kinetic energy,

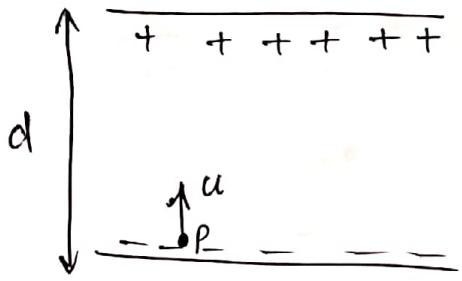
$$KE = \frac{P^2}{2m}$$

$$P \propto t$$

$$\Rightarrow \boxed{K.E \propto t^2}$$

* Both the particles move on straight line path.
(If they are released from rest in same uniform electric field; though forces are same, a, v, s, t are different)

Ex:



Particle P just hit the upper plate. (cover max height)

$$d = H = \frac{u^2}{2g_{eff}}$$

$$mg + qE = mg_{eff}$$

$$g_{eff} = -\left(g + \frac{qE}{m}\right)$$

$$\bar{F} = \bar{F}_g + \bar{F}_e$$

$$m\bar{g}_{eff} = \bar{F}_g + \bar{F}_e$$

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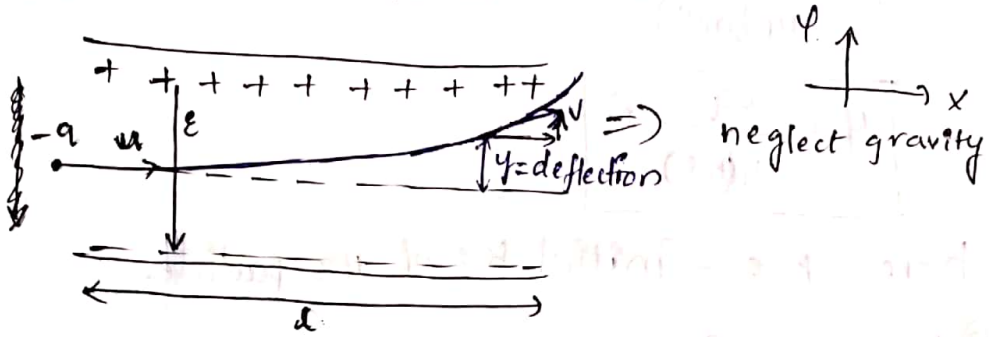
projection velocity is perpendicular to electric field :



$\Rightarrow$ neglect gravity

$u_x = u$, $u_y = 0$

$a_x = 0$, $a_y = \frac{Eq}{m}$

Velocity of particle at any time 't',

$\bar{v} = \bar{v}_x + \bar{v}_y$

$$\boxed{\bar{v} = u\hat{i} + \frac{Eq}{m} t\hat{j}}$$

* a charge at rest in electric field has st. line path. (If free to move)

Displacement of particle after a time 't' :

$\bar{s} = \bar{x} + \bar{y}$

$\bar{x} = \bar{u}_x t + \frac{1}{2} \bar{a}_x t^2$ ($\bar{a}_x = 0$)

$$\boxed{x = ut} \quad \text{— (1)}$$

& $\bar{y} = \bar{u}_y t + \frac{1}{2} \bar{a}_y t^2$ ($\bar{u}_y = 0$)

$$\boxed{y = \frac{1}{2} \frac{Eq}{m} t^2}$$

now, $y = \frac{1}{2} \frac{Eq}{m} \left(\frac{x}{u}\right)^2$

$$\boxed{y = \frac{Eq}{2u^2 m} x^2}$$

$\therefore$ path is parabola.

* a charge is projected then, in the direction of field - st. line and in ⊥er direction of field - parabola. (paths of charge)

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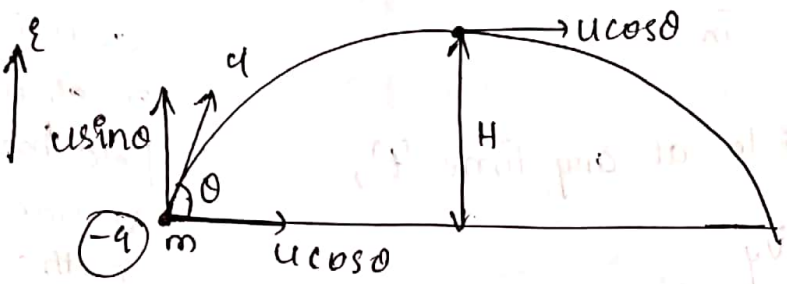
$$y = \frac{Eq}{4\left(\frac{1}{2}mu^2\right)} x^2$$

$$\boxed{y = \frac{Eq}{4(K.E)} x^2}$$

here K.E – initial K.E of the particle.

* (2) Oblique projection : (neglect gravity)



$$a_y = \frac{Eq}{m} (\downarrow)$$

(1) $v_y^2 - u_y^2 = 2a_y \cdot y$

at $v_y = 0$, $y = H$

$$+u_y^2 = +2\frac{Eq}{m}(H)$$

$$H = \frac{u_y^2}{2\left(\frac{Eq}{m}\right)}$$

$$\boxed{H = \frac{u^2 \sin^2\theta}{\frac{2Eq}{m}}}$$

(2) $T = \frac{2u_y}{a_y}$

(3) $R = \frac{2u_x u_y}{a_y} = u_x T$.

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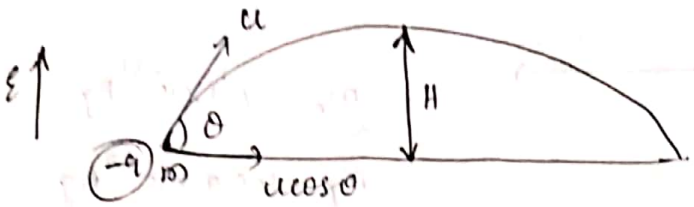
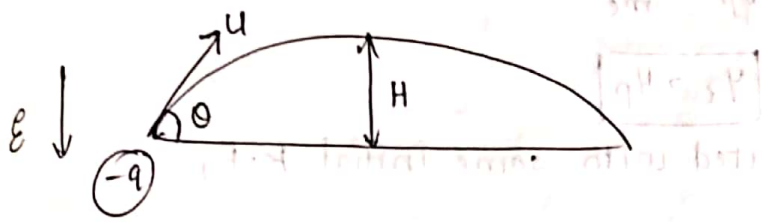
* By Considering Gravity:



$$mg_{eff} = mg + Eq$$

$$\boxed{g_{eff} = \left(g + \frac{Eq}{m}\right)(\downarrow)}$$



$$mg_{eff} = mg - Eq$$

$$\boxed{g_{eff} = g - \frac{Eq}{m}}$$

i.e, in general,

$$\boxed{g_{eff} = g \pm \frac{Eq}{m}}$$

1) $H = \dfrac{u^2 \sin^2\theta}{2 g_{eff}}$

2) $T = \dfrac{2u\sin\theta}{g_{eff}}$

3) $R = u\cos\theta\, T$

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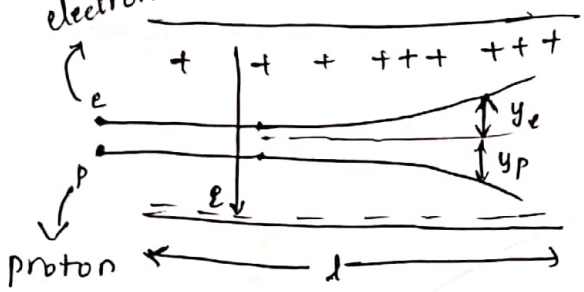
Ex: electron



$m_e = 9.1 \times 10^{-31}$ kg

$m_p = 1.67 \times 10^{-27}$ kg

we have,

$$y = \frac{Eq}{2u^2 m} l^2$$

① If both are projected with same speed.

then, $y \propto \frac{1}{m}$

$$\frac{y_e}{y_p} = \frac{m_p}{m_e} > 1$$

$$\boxed{y_e > y_p}$$

② If both are projected with same initial K.E,

$$y = \frac{Eq}{4(K.E)} l^2$$

$$\frac{y_e}{y_p} = 1 \Rightarrow \boxed{y_e = y_p}$$

③ If both are projected with same initial momentum,

$$y = \frac{Eq}{2A\left(\frac{p^2}{m}\right)} l^2 = \frac{mEql^2}{2p^2}$$

$y \propto m$

$$\frac{y_e}{y_p} = \frac{m_e}{m_p} < 1$$

$$\boxed{y_e < y_p}$$

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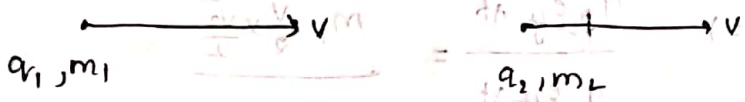
Ex: Two balls of charges q_1 & q_2 are initially have a velocity v with having same magnitude & direction. After a uniform electricfield has been applied for a certain time interval, the direction of 1st ball changes by 60° and velocity magnitude reduced to half. The direction of velocity of second ball changes by 90°. what is the velocity magnitude of second ball.

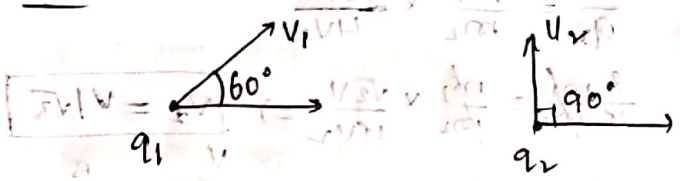
Sol:

case-1:
(without electrifield)



[initial condn]

case-2:



Let $\vec{E} = E_x\hat{\imath} + E_y\hat{\jmath}$ Δt = time duration

From impulse-momentum theorem,

impulse = change in momentum

For first ball,

$$q_1(E_x\hat{\imath} + E_y\hat{\jmath})\Delta t = m_1\left(\frac{v}{2}\cos 60^\circ\hat{\imath} + \frac{v}{2}\sin 60^\circ\hat{\jmath}\right) - m_1 v\hat{\imath}$$

Comparing coefficients,

$$q_1 E_x \Delta t = m_1\left(\frac{v}{2}\cos 60^\circ - v\right) \quad -①$$

& $$q_1 E_y \Delta t = m_1 \frac{v}{2}\sin 60^\circ \quad -②$$

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For second ball,

$$q_2(E_x\hat{i} + E_y\hat{j})\,dt = m_2 v_2\hat{j} - m_2 v\hat{i}$$

$$q_2 E_x\,dt = -m_2 v \quad -③$$

$$q_2 E_y\,dt = m_2 v_2 \quad -④$$

$$\frac{\text{eqn } ①}{\text{eqn } ③} \Rightarrow \frac{q_1 E_x\,dt}{q_2 E_x\,dt} = \frac{m_1\left(\frac{v}{4} - v\right)}{-m_2 v}$$

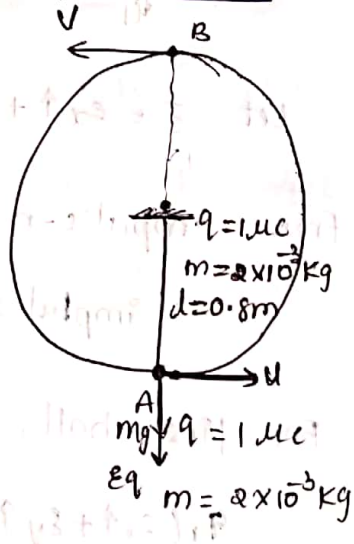
$$\frac{q_1}{q_2} = \frac{3m_1}{4m_2}$$

$$\frac{\text{eqn } ②}{\text{eqn } ④} \Rightarrow \frac{q_2 E_y\,dt}{q_2 E_y\,dt} = \frac{m_1 \frac{v}{2} \times \frac{\sqrt{3}}{2}}{m_2 v_2}$$

$$\frac{q_1}{q_2} = \frac{m_1}{m_2} \times \frac{\sqrt{3}v}{4v_2}$$

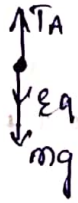
$$\frac{3m_1}{4m_2} = \frac{m_1}{m_2} \times \frac{\sqrt{3}v}{4v_2} \Rightarrow \boxed{v_2 = v/\sqrt{3}}$$

Ex! Find min horizontal velocity given to the lower ball so that it can make complete revolution.



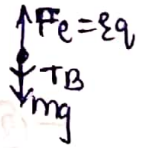
Sol⁸

At A –



$$T_A - Eq - mg = \frac{mu^2}{l} \quad -①$$

At B –



$$T_B + mg - Eq = \frac{mv^2}{l} \quad -②$$

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W-E Theorem:

$$-mg(2d) = \frac{1}{2} m (u^2 - v^2)$$

Ex: A particle of mass m & charge q moves rectilinearly under the action of an electric field $E = A - Bx$ where A, B are positive constants & (x) is distance from the point where the particle was initially at rest. Calculate:

1) distance travelled by the particle till it comes to rest.
2) acceleration at that moment.

Sol:

Force on the particle,

$$F = Eq$$

$$F = q(A - Bx)$$

$$a = \frac{q}{m}(A - Bx)$$

$$F = q(A - Bx)$$

$$mv \cdot \frac{dv}{dx} = q(A - Bx)$$

$$\int_{v=0}^{0} v \cdot dv = \frac{q}{m} \int_{x=0}^{x=x_{max}} (A - Bx) \cdot dx$$

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$$0 = \frac{q}{m}\left(A x_{max} - \frac{B x_{max}}{2}\right)$$

$$x_{max}\left(A - \frac{B}{2}x_{max}\right) = 0$$

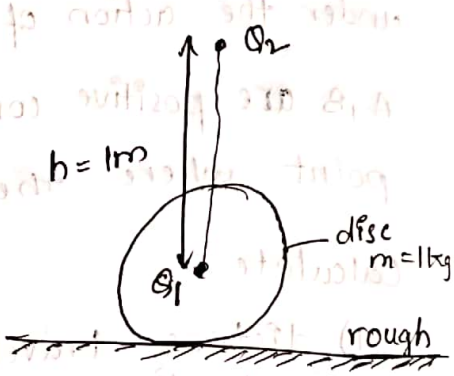
$$\boxed{x_{max} = \frac{2A}{B}}$$

* At this moment;

$$a = \frac{q}{m}\left(A - B\left(\frac{2A}{B}\right)\right)$$

$$\boxed{a = -\frac{qA}{m}}^{*}$$

Ex! A pt charge $Q_1 = -125\mu C$ is fixed at the centre of an insulated disc of mass 1kg. The disc rests on a rough horizontal plane. Another charge $Q_2 = +125\mu C$ is fixed vertically above the centre of the disc at a height of $h = 1m$. after the disc is displaced slightly in horizontal direction, friction is sufficient to prevent the slipping. Find time period of oscillation of disc.

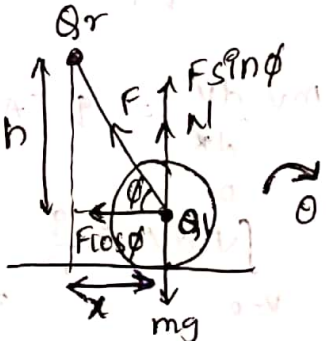


Sol: let R = radius of disc

θ = angle of rotation of disc (small)

then $x = R\theta$

* apply torque about the bottom most point,



~~$I\alpha = -R(F\cos\phi)$~~

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$$\tau = F\cos\phi\,(R)$$

$$I\alpha = R\,\frac{KQ_1Q_2}{x^2+h^2}\cos\phi$$

$$= \frac{RKQ_1Q_2}{(x^2+h^2)} \times \frac{x}{\sqrt{x^2+h^2}}$$

$$I\alpha = \frac{RKQ_1Q_2x}{h^3}$$

$$\left(\frac{mR^2}{2} + mR^2\right)\alpha = -\frac{RKQ_1Q_2}{h^3}(R\theta)$$

$$\frac{3mR^2}{2}\alpha = -\frac{KQ_1Q_2}{h^3}R^2\theta$$

$$\alpha = \frac{-2KQ_1Q_2}{3mh^3}\theta$$

$$\omega = \sqrt{\frac{2KQ_1Q_2}{3mh^3}}$$

$$T = 2\pi\sqrt{\frac{3mh^3}{2KQ_1Q_2}}$$

$$= 2\pi\sqrt{\frac{3\times1\times1}{2\times9\times10^9\times(125)^2}}$$

$$T = \frac{2\pi\times1\sqrt{3}}{125\times10^4}\sqrt{\frac{3}{20}}$$

$$= \frac{2\pi\times1}{3\times125\times10^4}\times\sqrt{\frac{3}{5}}$$

$$T = \frac{\pi\sqrt{0.6}}{375\times10^4}$$

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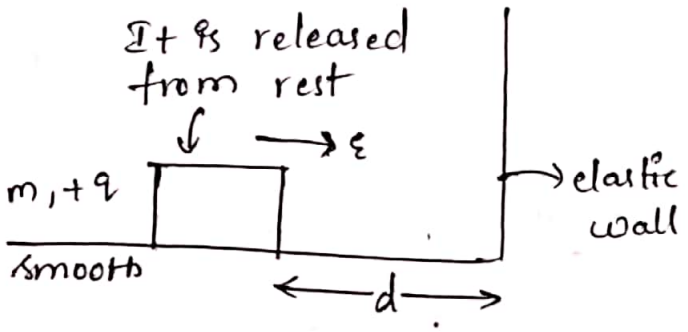
Electrostatics (continuation)

Ex: Find time period of oscillation of block



Sol:

① $F_e = Eq$

$\Rightarrow a = \frac{F_e}{m} = \frac{Eq}{m} = \text{const}$

Velocity of block just before hitting the wall,

$v^2 - u^2 = 2as$

$$\boxed{v = \sqrt{\frac{2Eqd}{m}}}$$

After collision it retards with same speed and stops at initial position.

∴ Time period,

$T = 2t$

$= 2\sqrt{\frac{2dm}{Eq}}$

$$\boxed{T = \sqrt{\frac{8dm}{Eq}}}$$

where $t = \sqrt{\frac{2s}{a}}$

$= \sqrt{\frac{2d}{Eq/m}}$

$$\boxed{t = \sqrt{\frac{2dm}{Eq}}}$$

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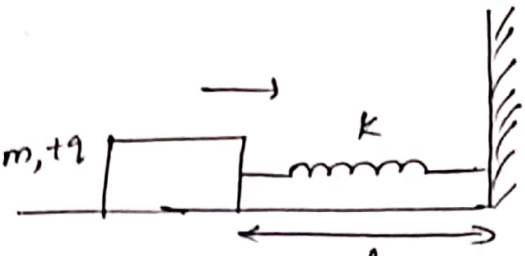
Ex: Block is released at rest



① Eqm position,

$F_{net} = 0$

$Eq - kx = 0$

$Eq = kx$

$$\boxed{x = \frac{Eq}{k}}$$

② Velocity of block at mean position,

ω-ε theorem,

$W_e + W_{sp} + \cancel{W_N}^{0} + \cancel{W_{mg}}^{0} = \Delta K$

$Eqx - \frac{1}{2}kx^2 = \frac{1}{2}mv^2$

$Eq\left(\frac{Eq}{k}\right) - \frac{1}{2}k\left(\frac{Eq}{k}\right)^2 = \frac{1}{2}mv^2$

$\frac{E^2q^2}{k} - \frac{1}{2}\frac{Eq^2}{k} = \frac{1}{2}mv^2$

$\frac{E^2q^2}{2k} = \frac{1}{2}mv^2$

$$\boxed{v = \frac{Eq}{\sqrt{mk}}}$$ max speed.

* Angular frequency does not depend on external-factor Only depends on mass.

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③ Maximum compression:

$$W_e + W_{sp} = \cancel{0}\vec{K}^{0}$$

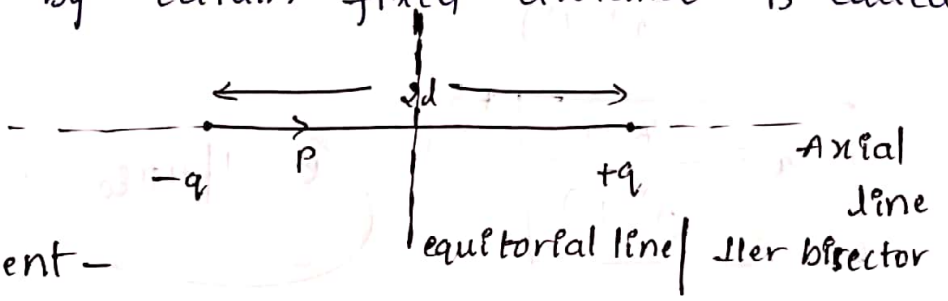
$$Eq\, x_{max} - \frac{1}{2} K x_{max}^{2} = 0$$

$$x_{max} = 2\frac{Eq}{K}$$

$$\boxed{x_{max} = 2x}$$

Electric Dipole:

The arrangement of two equal and opposite charges seperated by certain fixed distance is called dipole.



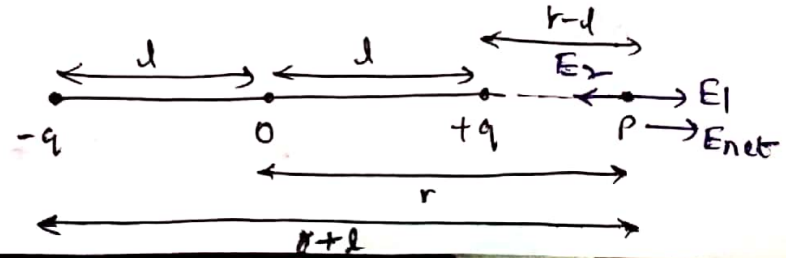
Dipole moment-

Product of magnitude of one of the charge & distance b/w the charges.

$$\boxed{P = q \times 2d}$$ unit → c-m

* It is vector.
* It's direction is from '-q' to '+q' along the line joining them.

Calculation of Electric Intensity at a point on the axial line of dipole:



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Net field at 'P' is,

$$E_a = E_1 - E_2$$

$$= \frac{Kq}{(r-l)^2} - \frac{Kq}{(r+l)^2}$$

$$= Kq\left(\frac{(r+l)^2-(r-l)^2}{(r^2-l^2)^2}\right)$$

$$= \frac{4Kqrl}{(r^2-l^2)^2}$$

$$E_a = \frac{2K(q\times 2l)r}{(r^2-l^2)^2}$$

$$\boxed{E_a = \frac{2Kpr}{(r^2-l^2)^2}} \quad *$$

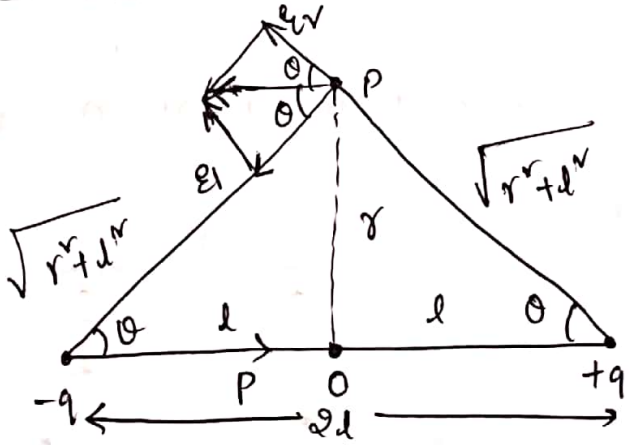
$K = 1/4\pi\varepsilon_0$

For short dipole, $l \ll r$

$$\boxed{E_a = \frac{2Kp}{r^3}}$$

clearly, $\boxed{\bar{\varepsilon} \parallel \bar{p}}$ **

Electric intensity at a point on the equitorial line :



$$E_1 = E_2$$

$$= \frac{Kq}{r^2+l^2}$$

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Resultant field at 'P',

$$\mathcal{E}_{net} = 2\mathcal{E}_1 \cos\theta$$

$$= \frac{2Kq}{r^2+d^2} \times \frac{l}{\sqrt{r^2+d^2}}$$

$$\mathcal{E}_{net} = \frac{2Kql}{(r^2+d^2)^{3/2}}$$

$$\boxed{\mathcal{E}_{net} = \frac{K(q \times 2l)}{(r^2+d^2)^{3/2}}}$$

$$K = 1/4\pi\varepsilon_0$$

$$\boxed{\mathcal{E}_{net} = \frac{Kp}{(r^2+d^2)^{3/2}}} \quad *$$

For short dipole,

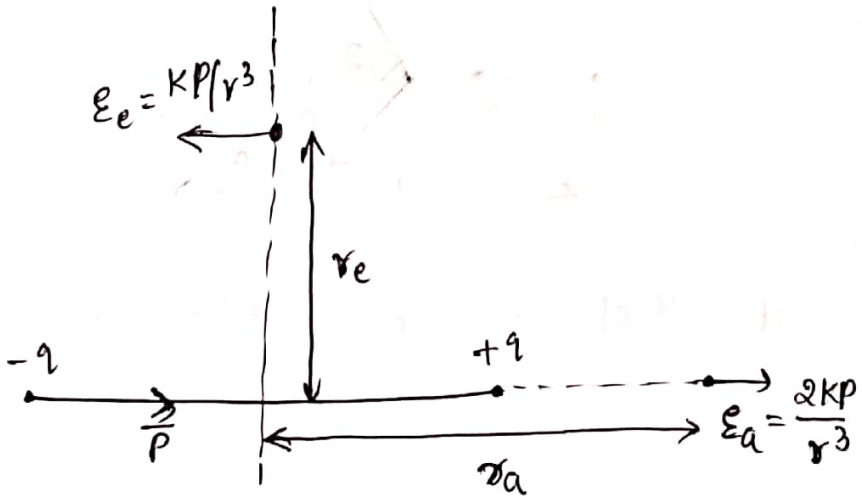
$$l << r$$

$$\boxed{\mathcal{E}_e = \frac{Kp}{r^3}} \quad *$$

Clearly, $\boxed{\overline{\mathcal{E}_e} \parallel -\overline{p}}$.

Notes

For short dipole :-



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(1) For $\varepsilon_a = \varepsilon_e$

$$\frac{2Kp}{r_a^3} = \frac{Kp}{r_e^3}$$

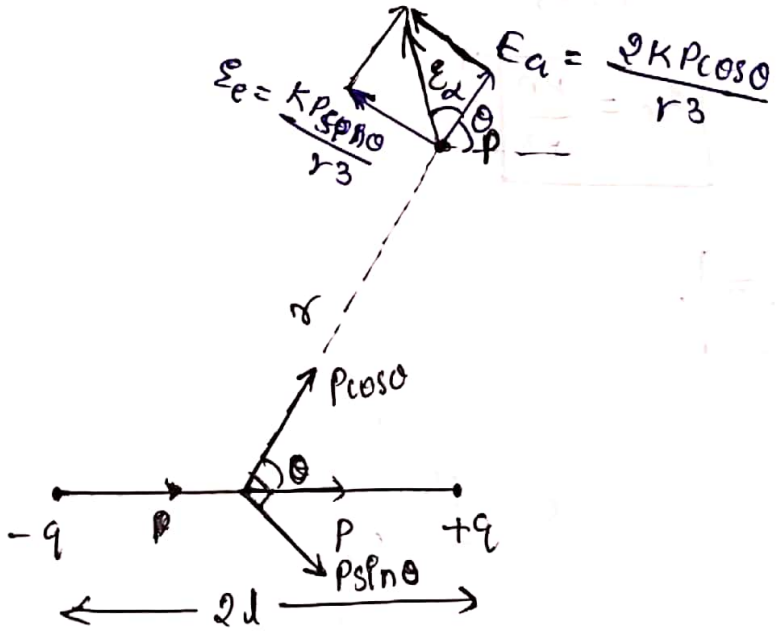
$$\boxed{r_a = 2^{1/3} r_e}$$

(2) For $r_a = r_e$

$$\frac{\varepsilon_a}{\varepsilon_e} = \frac{\frac{2Kp}{r^3}}{\frac{Kp}{r^3}}$$

$$\boxed{\varepsilon_a = 2\varepsilon_e}$$

Electric Intensity at a general point of dipole:
(For short dipole)



Let dipole can be considered as 2 ller dipoles.

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Net field at P is,

$$\varepsilon = \sqrt{\varepsilon_a^2 + \varepsilon_e^2}$$

$$= \frac{KP}{r^3}\sqrt{\sin^2\theta + 4\cos^2\theta}$$

$$\boxed{\varepsilon = \frac{KP}{r^3}\sqrt{1+3\cos^2\theta}}$$

$K = 1/4\pi\varepsilon_0$

Points:

1) on axial line,

$$\theta = 0^\circ$$

$$\varepsilon_a = \frac{2KP}{r^3}$$

2) on equitorial line,

$$\theta = 90^\circ$$

$$\boxed{\varepsilon_e = \frac{KP}{r^3}}$$

3) direction of ε is

$$\tan\alpha = \frac{\varepsilon_e}{\varepsilon_a} = \frac{\tan\theta}{2}$$

$$\boxed{\alpha = \tan^{-1}\left(\frac{1}{2}\tan\theta\right)}$$ with ε_a.

4) If $\alpha + \theta = \pi/2$.

$$\tan\left(\pi/2 - \theta\right) = \frac{\tan\theta}{2}$$

$$\cot\theta = \frac{\tan\theta}{2}$$

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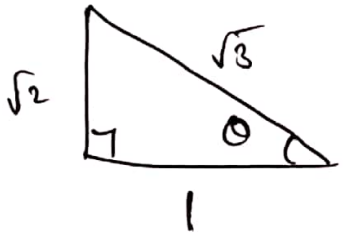
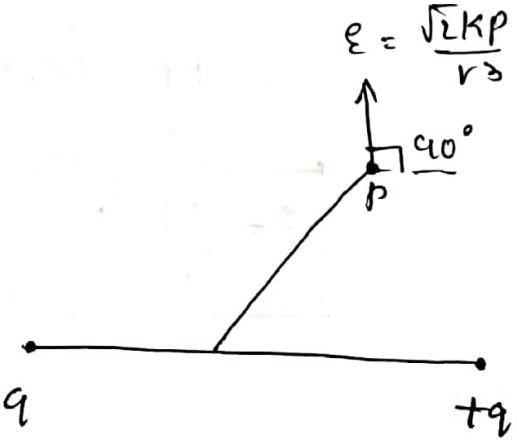
$\tan^2\theta = 2$

$\tan\theta = \sqrt{2}$

$$\boxed{\theta = \tan^{-1}\sqrt{2}}$$



$\cos\theta = 1/\sqrt{3}$



$$\varepsilon = \frac{kp}{r^3}\sqrt{1+3\cos^2\theta}$$

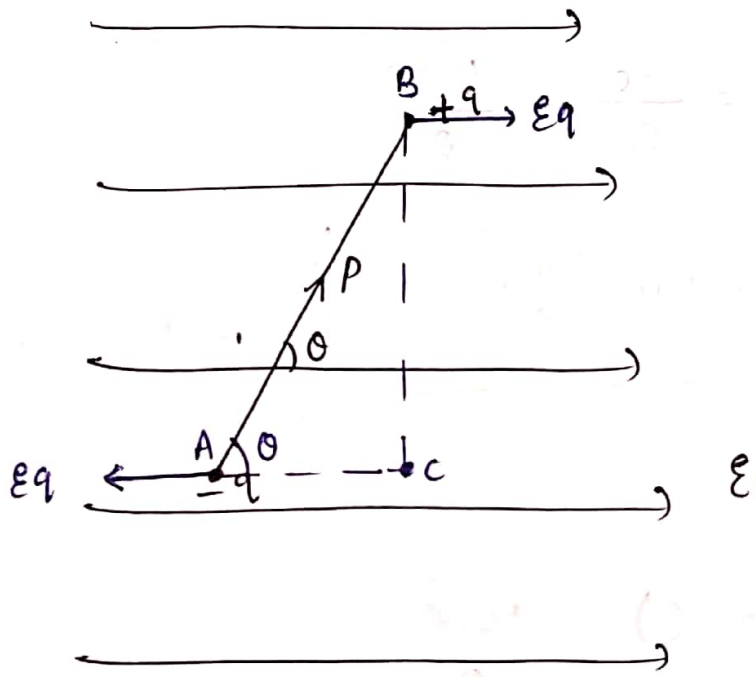
$$= \frac{kp}{r^3}\sqrt{1+(\sqrt{3})^2 \times 1}$$

$$\boxed{\varepsilon = \frac{\sqrt{2}kp}{r^3}}$$

Torque acting on the dipole placed in uniform electric Field:

$AB = 2l$

$BC = 2l\sin\theta$



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(1) Net force on the dipole = 0 but $\tau_{net} \neq 0$.

(because forces are non-collinear) ← [$\tau_{net} = 0$ for collinear forces]

(2) Here two equal, opposite and non-collinear forces acting on the dipole constitute the torque.

Torque (τ) = F × ⊥ler distance between the forces.

$= Eq \times BC$

$= Eq \times 2d \sin\theta$

$$\boxed{\tau = pE\sin\theta}$$

In vector form, $\boxed{\vec{\tau} = \vec{p} \times \vec{E}}$

* Torque try to rotate dipole to the direction of E. (τ will be min)

* Again τ will be max with $\vec{p}$ & $\vec{E}$ are perpendicular.

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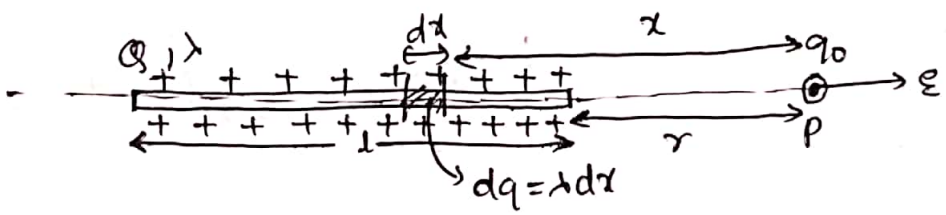
Electric field intensity for Extended charged bodies

(1) Thin rod:

a) on the axis of rod:



$\lambda = \frac{Q}{l}$ = const

uniformly charge is distributed

* $dE = \frac{K\,dq}{x^2} = \frac{K\lambda\,dx}{x^2}$

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* Total electric field,

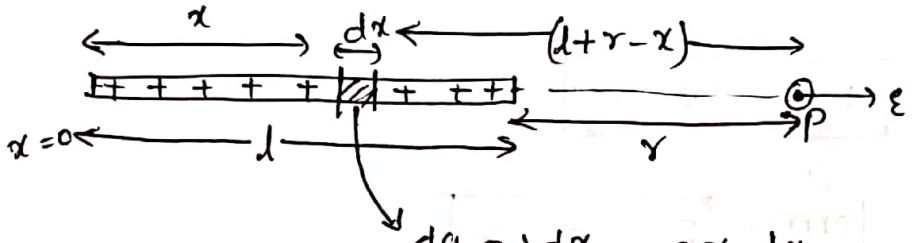
$$\mathcal{E} = \int d\mathcal{E} = \int_{x=r}^{x=r+l} \frac{K\lambda \cdot dx}{x^2} = -K\lambda \left[\frac{1}{x}\right]_{x=r}^{x=r+l}$$

$$\mathcal{E} = K\lambda \left[-\frac{1}{r+l} + \frac{1}{r}\right],$$

$$\boxed{\mathcal{E} = \frac{K\lambda l}{r(r+l)}}$$

b) For non-uniform distribution :

For ex, $\lambda = cx$; c = const



$$dq = \lambda dx = cx \cdot dx$$

now, $$d\mathcal{E} = \frac{K dq}{(l+r-x)^2} = \frac{K(cx)\cdot dx}{(l+r-x)^2}$$

net field,

$$\mathcal{E} = \int_{x=0}^{x=l} d\mathcal{E} = Kc \int_{x=0}^{x=l} \frac{x \cdot dx}{(l+r-x)^2}$$

let $l+r-x = t$

$-dx = dt$

$$\mathcal{E} = -kc \int_{t=l+r}^{t=r} \frac{(l+r-t)\cdot dt}{t^2}$$

limits: when $x=0$, $t = l+r$

when $x=l$, $t=r$

$$= -Kc \int_{t=l+r}^{t=r} \frac{(l+r)\cdot dt}{t^2} + Kc \int_{t=l+r}^{t=r} \frac{t}{t^2} \cdot dt$$

$$= \frac{Kcl}{r(l+r)}(l+r) + Kc[\ln r - \ln(l+r)]$$

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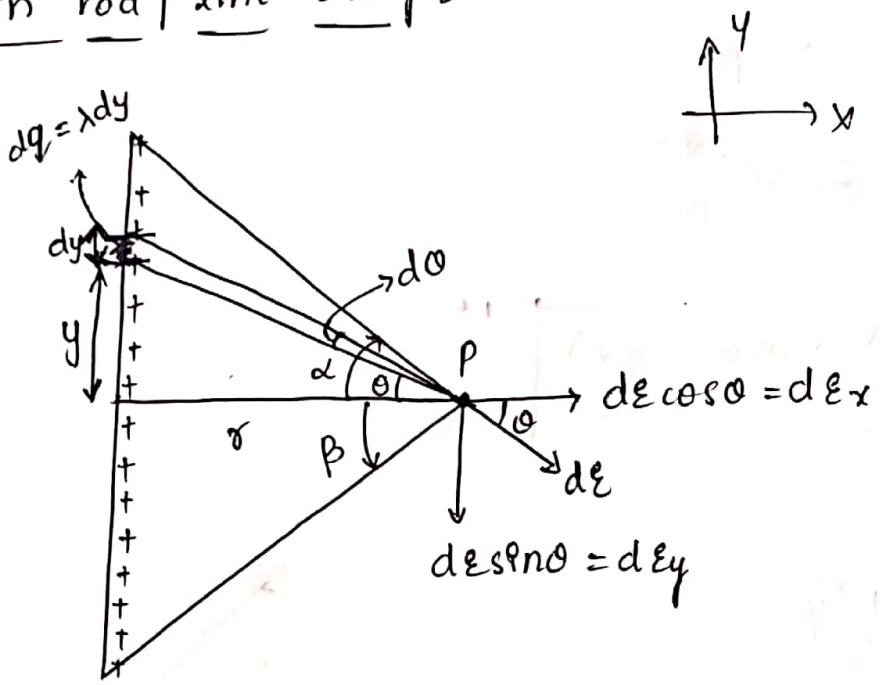
$$\varepsilon = \frac{Kql}{r} + Kc \ln\left(\frac{r}{d+r}\right)$$

c) Thin rod / line charge:-



$\lambda = \frac{q}{l}$

now, $d\varepsilon = \frac{K \cdot dq}{(\sqrt{r^2+y^2})^2} = \frac{K dq}{r^2+y^2}$

From figure, $\tan\theta = \frac{y}{r}$

$$y = r\tan\theta$$

$$dy = r\sec^2\theta \cdot d\theta$$

$$d\varepsilon = \frac{K\lambda r \sec^2\theta \cdot d\theta}{r^2 + r^2\tan^2\theta}$$

$$\boxed{d\varepsilon = \frac{K\lambda}{r} \cdot d\theta}$$

now, $d\varepsilon_x = d\varepsilon \cos\theta$

$$\varepsilon_x = \int_{\theta=-\beta}^{\theta=\alpha} d\varepsilon_x = \frac{K\lambda}{r}\int_{-\beta}^{\alpha} \cos\theta \cdot d\theta = \frac{K\lambda}{r}(\sin\theta)_{-\beta}^{\alpha}$$

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$$\boxed{E_x = \frac{K\lambda}{r}(\sin\alpha + \sin\beta)} \quad **$$

&

$$E_y = \int dE_y = \int_{\theta=-\beta}^{\theta=\alpha} \frac{K\lambda}{r}\sin\theta \cdot d\theta$$

$$E_y = \frac{K\lambda}{r}(-\cos\theta)_{-\beta}^{\alpha}$$

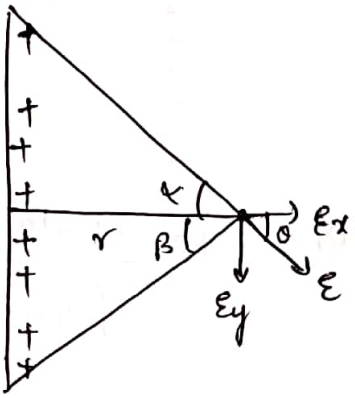
$$\boxed{E_y = \frac{K\lambda}{r}(\cos\beta - \cos\alpha)} \quad **$$

Finally,

$$\bar{E} = \bar{E_x} + \bar{E_y}$$

$$\bar{E} = E_x(\uparrow) + E_y(-\hat{j})$$

$$|\bar{E}| = \sqrt{E_x^2 + E_y^2}$$



& direction, $\theta = \tan^{-1}\left(\frac{E_y}{E_x}\right)$ with E_x.

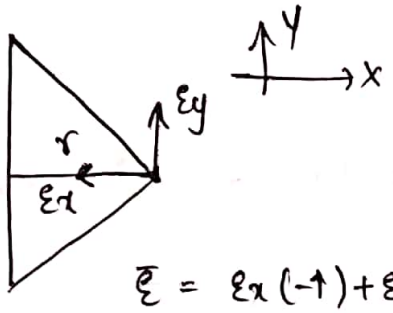
* on perpendicular bisector,

$$\boxed{\alpha = \beta} \Rightarrow E_y = 0$$

& $$E_x = \frac{2K\lambda}{r}\sin\alpha$$

where $$\sin\alpha = \frac{d/2}{\sqrt{r^2 + d^2/4}}$$

* Gravitation:



$$E_x = \frac{G\lambda}{r}(\sin\alpha + \sin\beta)$$

$$E_y = \frac{G\lambda}{r}(\cos\beta - \cos\alpha)$$

$$\bar{E} = E_x(-\uparrow) + E_y(\uparrow)$$

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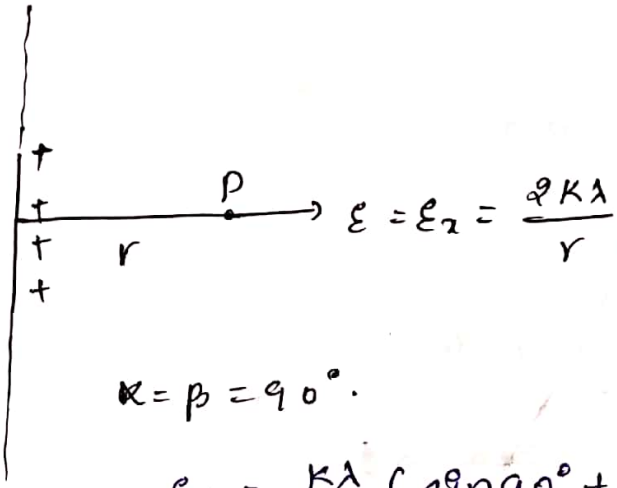
Points:

1) For infinite line charge:-



$\alpha = \beta = 90°$.

$$\varepsilon_x = \frac{K\lambda}{r}(\sin 90° + \sin 90°)$$

$$\boxed{\varepsilon_x = \frac{2K\lambda}{r}}$$

$$\varepsilon_y = \frac{K\lambda}{r}(\cos 90 - \cos 90)$$

$$\boxed{\varepsilon_y = 0}$$

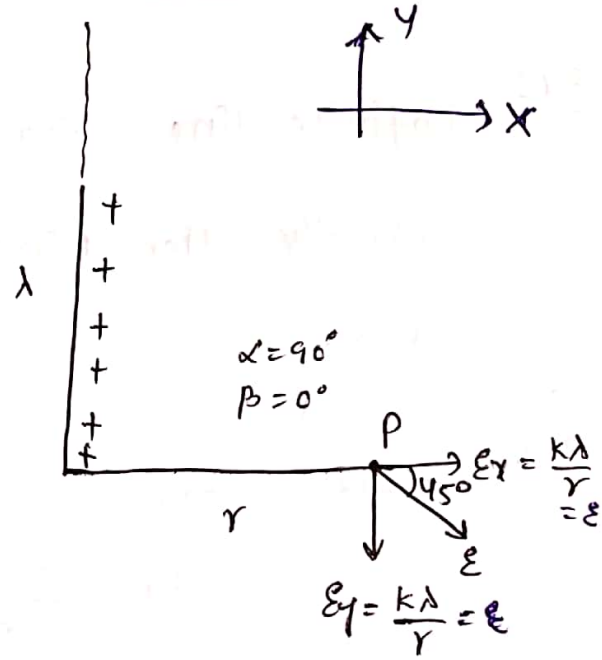
2) At one end of semi-infinite line charge:-

$$\varepsilon_x = \frac{K\lambda}{r}(\sin 90 + \sin 0)$$

$$\boxed{\varepsilon_x = \frac{K\lambda}{r}}$$

$$\varepsilon_y = \frac{K\lambda}{r}(\cos 0 - \cos 90)$$

$$\boxed{\varepsilon_y = \frac{K\lambda}{r}}$$



$$\bar{\varepsilon} = \frac{K\lambda}{r}(\hat{i}) + \frac{K\lambda}{r}(-\hat{j}) \Rightarrow \boxed{|\bar{\varepsilon}| = \sqrt{2}\,\varepsilon}$$

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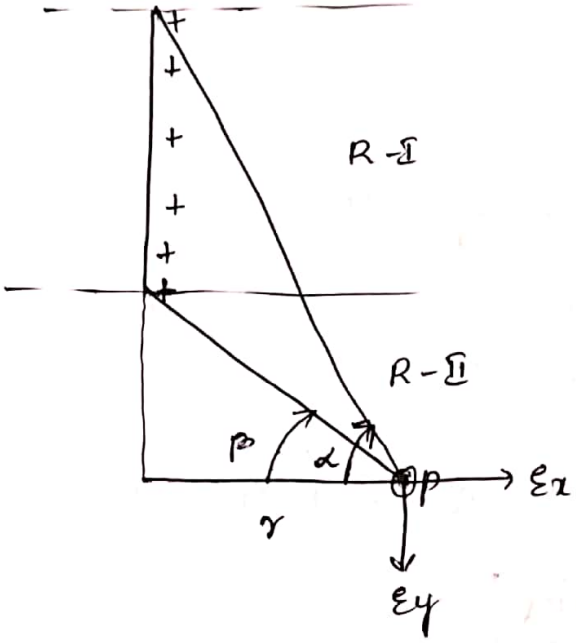
3) Point is not infront of wire :

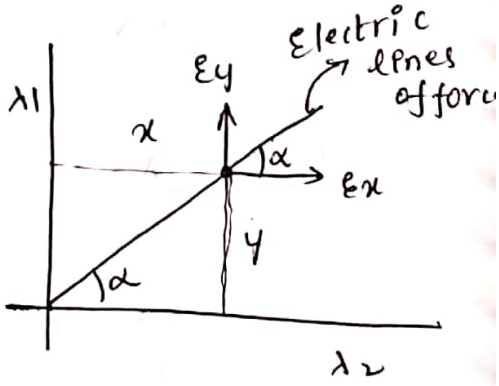


Replace β by $-\beta$.

$$\mathcal{E}_x = \frac{K\lambda}{r}(\sin\alpha - \sin\beta)$$

$$\mathcal{E}_y = \frac{K\lambda}{r}(\cos\beta - \cos\alpha)$$

Ex: Infinite line charged wires are initially ller. Find $\frac{\lambda_1}{\lambda_2}$ in terms of α.



Sol: we have,

$$\mathcal{E}_x = \frac{2K\lambda_1}{x} = \frac{\lambda_1}{2\pi\varepsilon_0 x}$$

$$\mathcal{E}_y = \frac{2K\lambda_2}{y} = \frac{\lambda_2}{2\pi\varepsilon_0 y}$$

now, $\tan\alpha = \frac{\mathcal{E}_y}{\mathcal{E}_x}$

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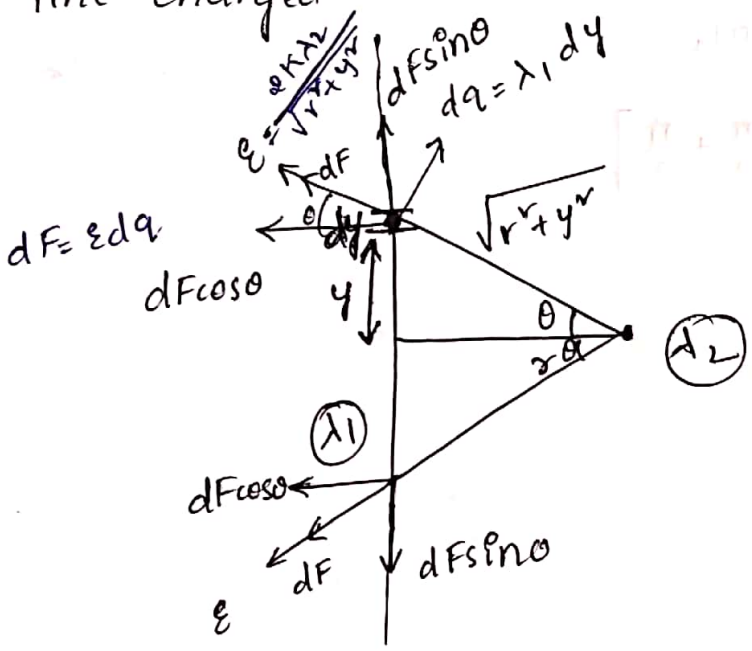
$$\tan\alpha = \frac{\lambda_2}{\lambda_1} \times \frac{x}{y}$$

$$\tan\alpha = \frac{\lambda_2}{\lambda_1} \cot\alpha$$

$$\boxed{\frac{\lambda_1}{\lambda_2} = \cot^2\alpha}$$

Ex: Find the force of interaction between two infinite line charged wires as shown.



* $\varepsilon = \frac{2K\lambda_2}{\sqrt{r^2+y^2}}$

* $dF = \varepsilon dq$

$$= \frac{2K\lambda_2}{\sqrt{r^2+y^2}} \times \lambda_1 dy$$

$$= \frac{2K\lambda_1\lambda_2 dy}{\sqrt{r^2+y^2}}$$

Net force of interaction is,

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$$F = \int_{\theta=-\pi/2}^{\theta=\pi/2} dF\cos\theta$$

$$F = 2K\lambda_1\lambda_2 r \int_{\theta=-\pi/2}^{\theta=\pi/2} \frac{dy}{r^2+y^2}$$

$$= 2K\lambda_1\lambda_2 \int_{\theta=-\pi/2}^{\theta=\pi/2} \frac{r\sec^2\theta \cdot d\theta}{r^2\sec^2\theta}$$

from figure,

$y = r\tan\theta$

$dy = r\sec^2\theta \cdot d\theta$

$$= 2K\lambda_1\lambda_2 \left[\frac{\pi}{2} + \frac{\pi}{2}\right]$$

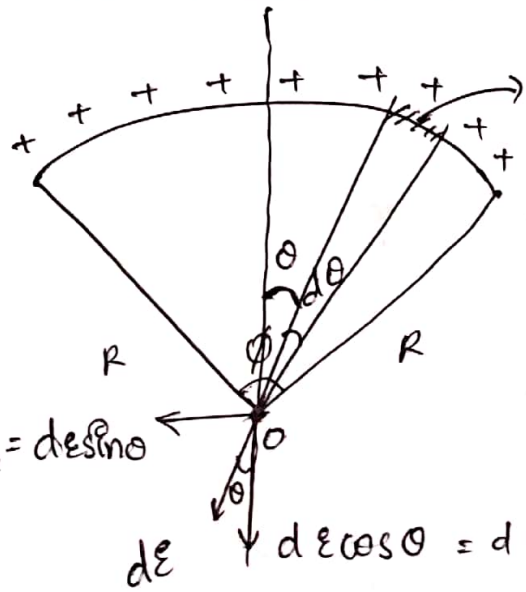
$$= 2\pi K\lambda_1\lambda_2$$

$$= \frac{\lambda_1\lambda_2}{\frac{1}{2\pi\varepsilon_0}}$$

$$\boxed{F = \frac{\lambda_1\lambda_2}{2\varepsilon_0}}$$

Arc shaped charged wire / ring :

Field at the centre of circle :-



$dq = \lambda dl = \frac{Q}{R\phi} dl = \frac{Q}{R\phi} R\,d\theta$

$dq = \frac{Q}{\phi} \cdot d\theta$

ϕ – radians $dE_x = dE\sin\theta$

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$$d\mathcal{E} = \frac{k\,dq}{R^2} = \frac{kQ}{\phi R^2}\,d\theta$$

$$\mathcal{E}_x = \int d\mathcal{E}_x = \int_{\theta=-\phi/2}^{\theta=\phi/2} d\mathcal{E}\sin\theta = \frac{kQ}{\phi R^2}\int_{\phi=-\phi/2}^{\theta=\phi/2} \sin\theta\,d\theta$$

$$\mathcal{E}_x = \frac{kQ}{\phi R^2}(-\cos\theta)\Big|_{-\phi/2}^{\phi/2}$$

$$\boxed{\mathcal{E}_x = 0}$$

$$\mathcal{E}_y = \int d\mathcal{E}_y = \int_{\theta=-\phi/2}^{\theta=\phi/2} d\mathcal{E}\cos\theta$$

$$= \frac{kQ}{\phi R^2}\int_{\theta=-\phi/2}^{\theta=\phi/2} \cos\theta$$

$$\boxed{\mathcal{E}_y = \frac{2kQ}{\phi R^2}\sin(\phi/2)}$$

∴ net field,

$$\vec{\mathcal{E}} = \cancel{\vec{\mathcal{E}}_x}^{\,0} + \vec{\mathcal{E}}_y$$

$$\boxed{\mathcal{E} = \frac{2kQ}{\phi R^2}\sin(\phi/2)}$$

It is along bisector line.

(or)

$$\boxed{\mathcal{E} = \frac{2k\lambda}{R}\sin(\phi/2)}$$

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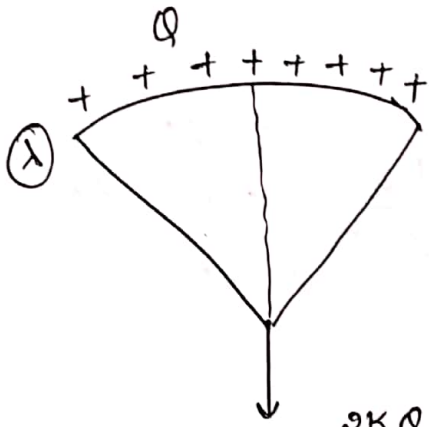
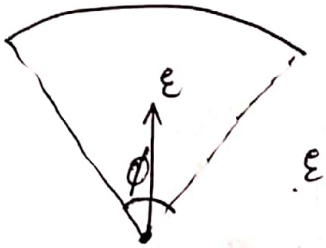
i.e,



$$\varepsilon = \frac{2KQ}{\phi R^2} \sin(\phi/2)$$

$$= \frac{2K\lambda}{R} \sin(\phi/2)$$

*Gravitation:



$$\varepsilon = \frac{2GM}{\phi R^2} \sin(\phi/2)$$

$$\varepsilon = \frac{2G\lambda}{R} \sin(\phi/2)$$

Points:

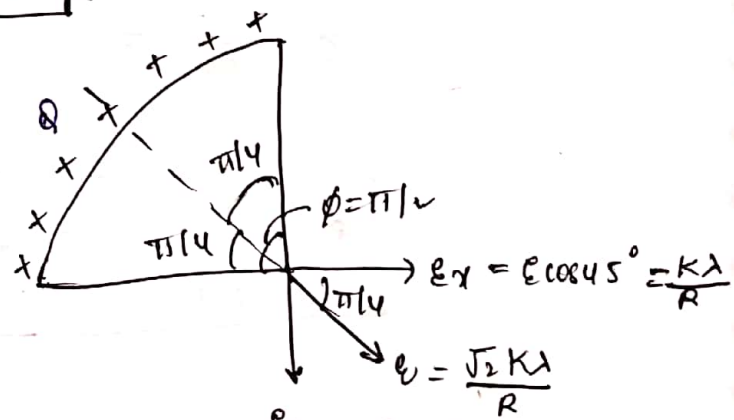
1) For quater shaped charged ring:

$$\lambda = \frac{Q}{\frac{\pi R}{2}} = \frac{2Q}{\pi R}$$

$$\varepsilon = \frac{2K\lambda}{R} \sin\phi/2$$

$$= \frac{2K\lambda}{R} \sin\pi/4$$

$$\boxed{\varepsilon = \frac{\sqrt{2}K\lambda}{R}}$$



$$\bar{\varepsilon} = \varepsilon_x \rightarrow + \varepsilon_y (\uparrow)$$

$$= \frac{K\lambda}{R} \rightarrow - \frac{K\lambda}{R} \uparrow$$

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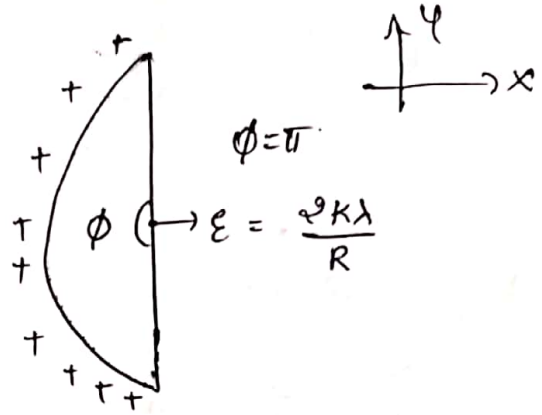
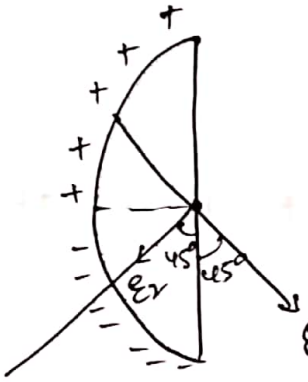
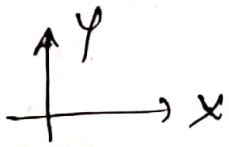
2) For semi-circular ring :

$$\lambda = \frac{Q}{\pi R}$$



Ex:



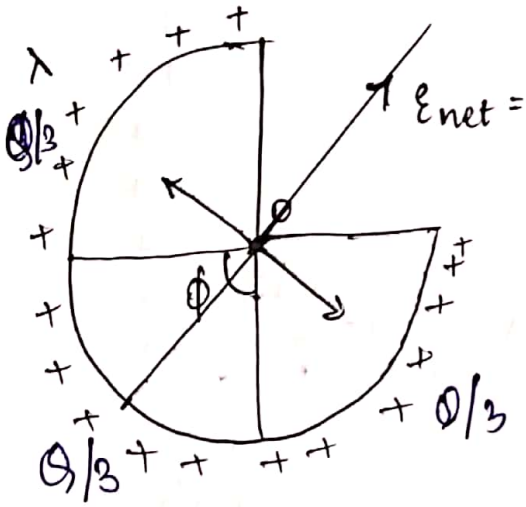


$$\varepsilon_1 = \varepsilon_2 = \frac{\sqrt{2}K\lambda}{R}$$

$$\varepsilon = \sqrt{\varepsilon_1^2 + \varepsilon_2^2} = \sqrt{2}\,\varepsilon_1 = \sqrt{2}\left(\frac{\sqrt{2}K\lambda}{R}\right)$$

$$\boxed{\varepsilon = \frac{2K\lambda}{R}(-\hat{j})}$$

Ex:



$$\varepsilon_{net} = \varepsilon = \frac{\sqrt{2}K\lambda}{R} = \frac{\sqrt{2}K\left(\frac{2Q}{3\pi R}\right)}{R} = \frac{2\sqrt{2}KQ}{3\pi R^2}$$

(or)

$$\varepsilon = \frac{2KQ'}{\phi R^2}\sin(\phi/2)$$

$$= \frac{2K(Q/3)}{\frac{\pi}{2}R^2}\sin(\pi/4)$$

$$\boxed{\varepsilon = \frac{2\sqrt{2}\,KQ}{3\pi R^2}}$$

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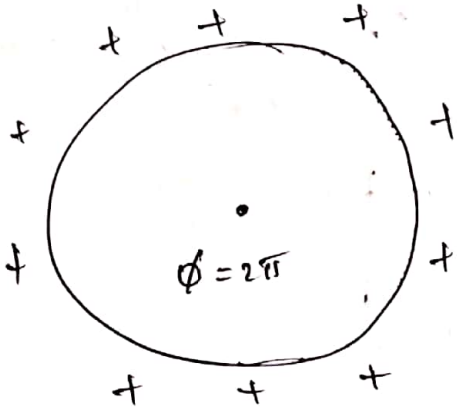
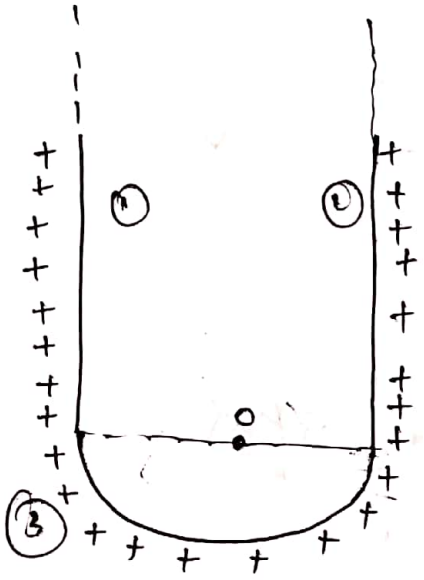
3) For complete Ring:



$$E_c = \frac{2K\lambda}{R}\sin\left(\frac{2\pi}{2}\right)$$

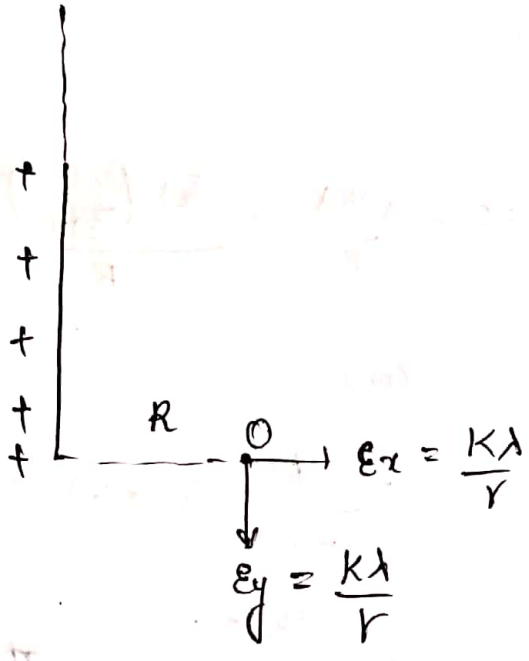
$$\boxed{E_c = 0}$$

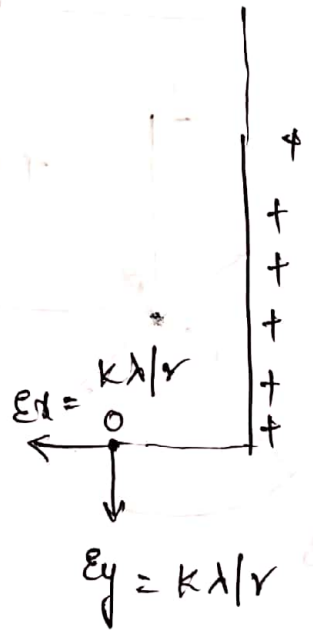
Ex:

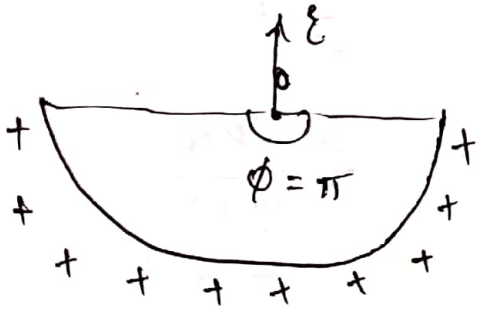


Find net field at point O.

Sol:







$$E = \frac{2K\lambda}{R}\sin(\pi/2)$$

$$= \frac{2K\lambda}{R}.$$

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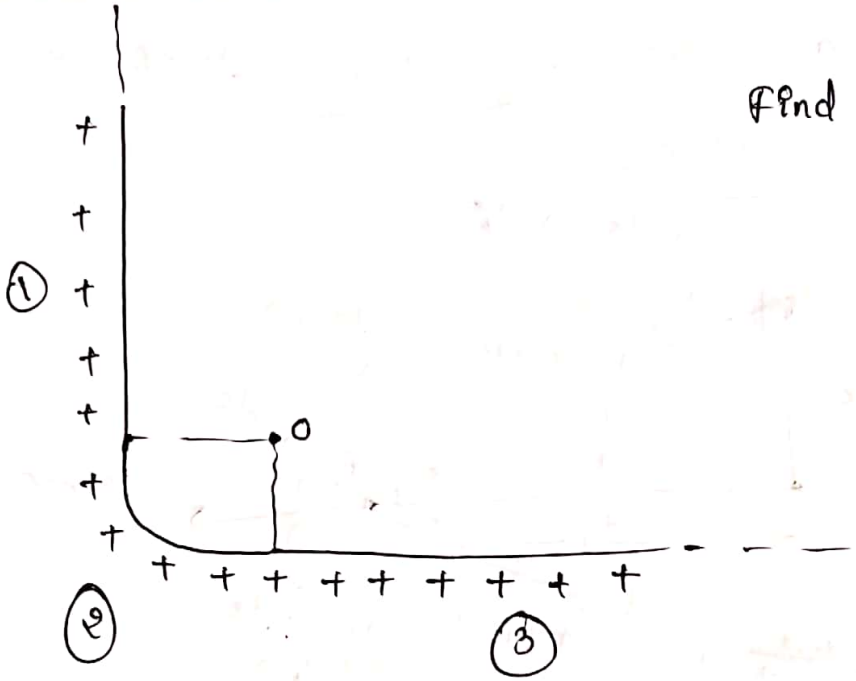
Eg:



Find Net field at point O.

Sol:

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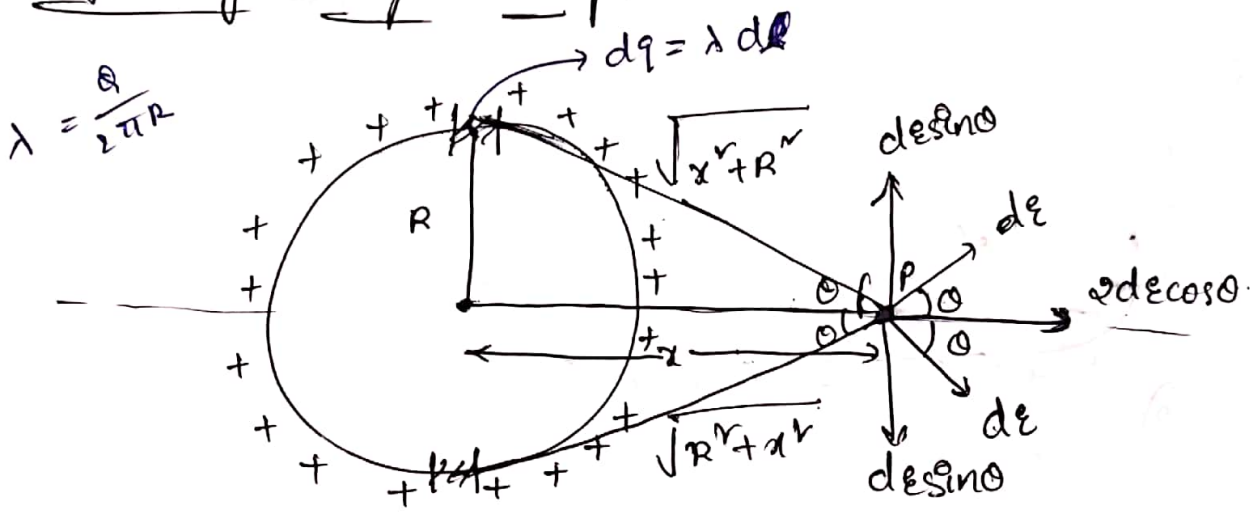
Electric field at a point on the axis of a uniformly charged ring :



Net field,

$$\varepsilon = \int d\varepsilon\cos\theta = \int \frac{k\,dq}{(x^2+R^2)} \times \frac{x}{(x^2+R^2)^{1/2}}$$

$$= \frac{kx\lambda}{(x^2+R^2)^{3/2}} \int dl$$

$$= \frac{kx\lambda\,(2\pi R)}{(x^2+R^2)^{3/2}}$$

$$\boxed{\varepsilon = \frac{kQx}{(x^2+R^2)^{3/2}}} \quad *$$

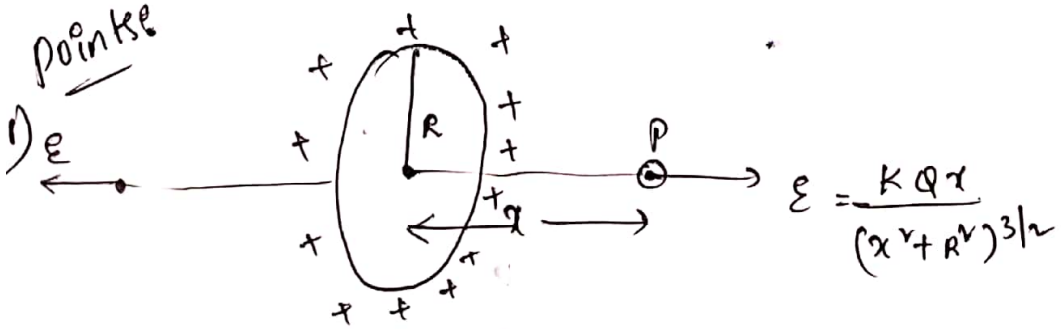
$$k = \frac{1}{4\pi\varepsilon_0}.$$

At centre of ring, $x = 0 \Rightarrow \varepsilon = 0$.

At ∞, $x = \pm\infty \Rightarrow \varepsilon = 0$.

Pointe

1) ε



$$\varepsilon = \frac{kQx}{(x^2+R^2)^{3/2}}$$

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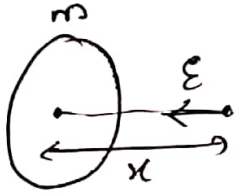
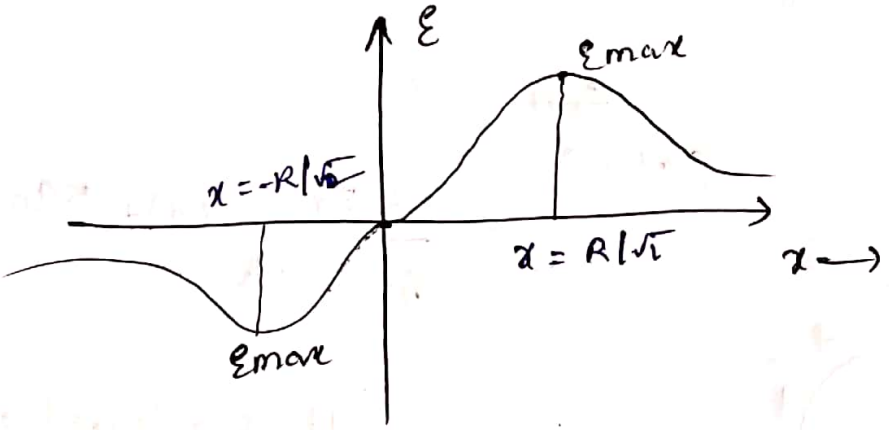
*Gravitation:



$$E = \frac{GMx}{(x^2+R^2)^{3/2}}$$

2).



At max, $E = E_{max}$, $\frac{dE}{dx} = 0$

$$\frac{d}{dx}\frac{x}{(x^2+R^2)^{3/2}} = 0$$

$$\frac{(x^2+R^2)^{3/2} - x \times \frac{3}{2}(x^2+R^2)^{1/2}(2x)}{(x^2+R^2)^3} = 0$$

$$\boxed{x = \pm R/\sqrt{2}}$$

$$E_{max} = \frac{KQ\, R/\sqrt{2}}{\left(\frac{R^2}{2}+R^2\right)^{3/2}} = \frac{KQR}{\sqrt{2}\, R^3 \times \left(\frac{3}{2}\right)^{3/2}}$$

$$= \frac{2KQ}{R^2 \times 3^{3/2}}$$

$$\boxed{E_{max} = \frac{2KQ}{3\sqrt{3}R^2}}$$

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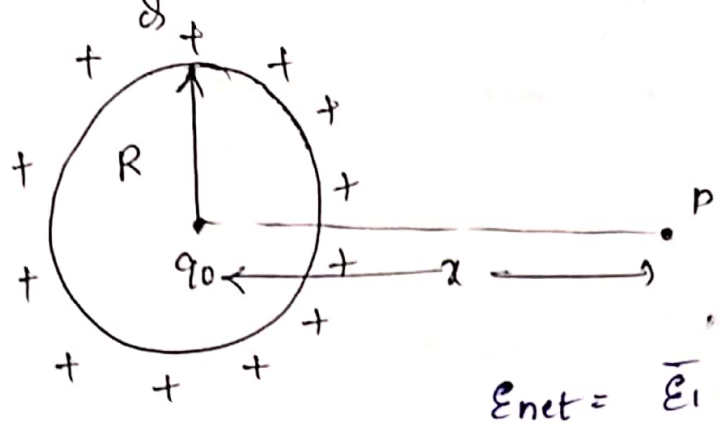
3)

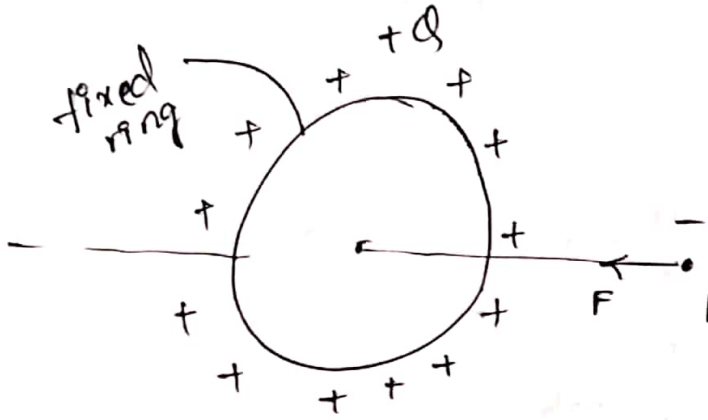


$$\mathcal{E}_{net} = \bar{\mathcal{E}}_1 + \bar{\mathcal{E}}_2$$

(at P)

$$= \frac{kQx}{(x^2+R^2)^{3/2}}(\uparrow) + \frac{kq_0}{x^2}(\uparrow)$$

4)



It is released from rest.

Point charge

Force on charge,

$$F = q\mathcal{E} = \frac{-kQxq}{(x^2+R^2)^{3/2}}$$

For small displacements, $x \ll R$

x^2 can be neglected.

$$F = \frac{-kQqx}{R^3}$$

$$ma = \frac{-kqQx}{R^3}$$

$$a = -\frac{kqQx}{R^3 m}$$

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$$\boxed{a = -\omega^2 x}$$

motion is SHM.

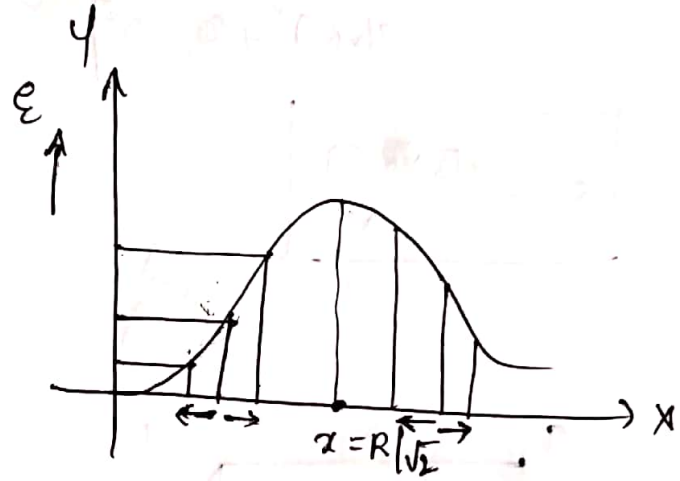
$$\boxed{\omega = \sqrt{\frac{KQq}{mR^3}}}$$

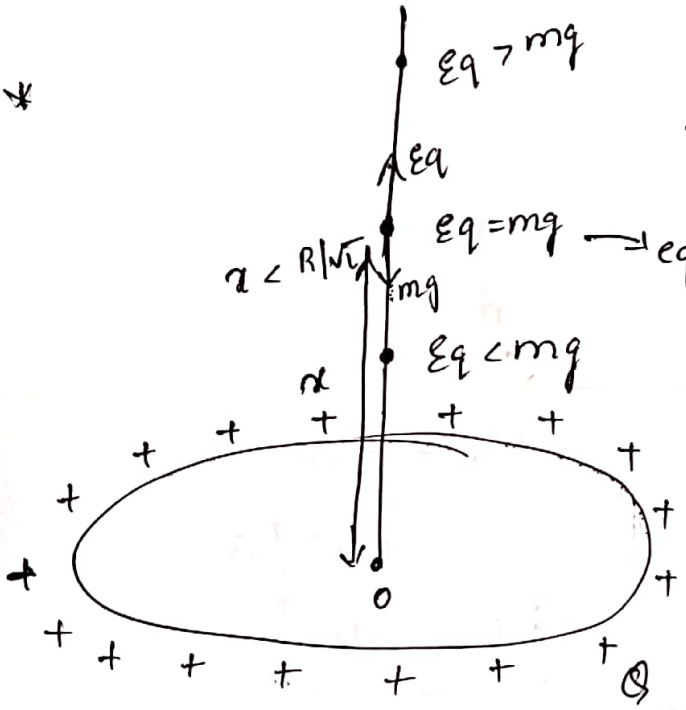
$$T = \frac{2\pi}{\omega} \Rightarrow \boxed{T = 2\pi\sqrt{\frac{mR^3}{KQq}}}$$

***

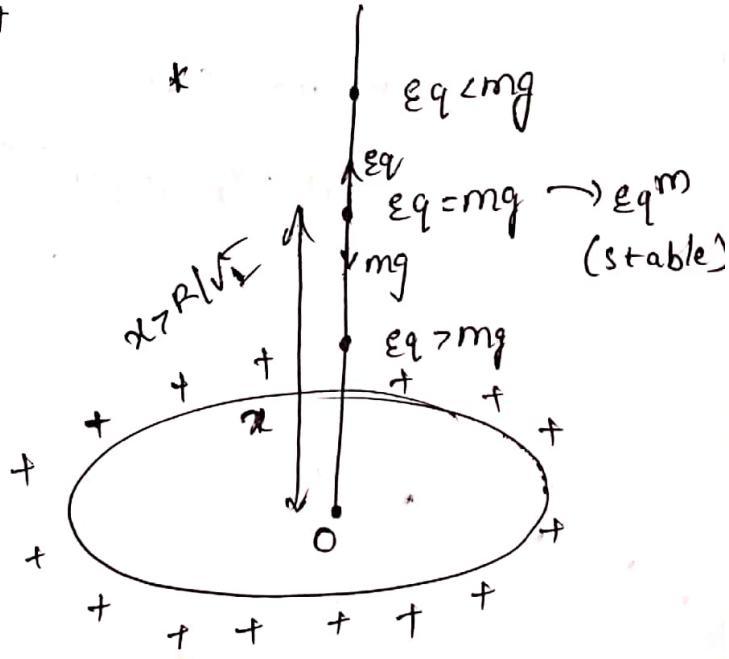
Equilibrium :

$$\boxed{F_{net} = 0}$$





Eq = mg → eq^m (unstable)



Eq = mg → eq^m (stable)

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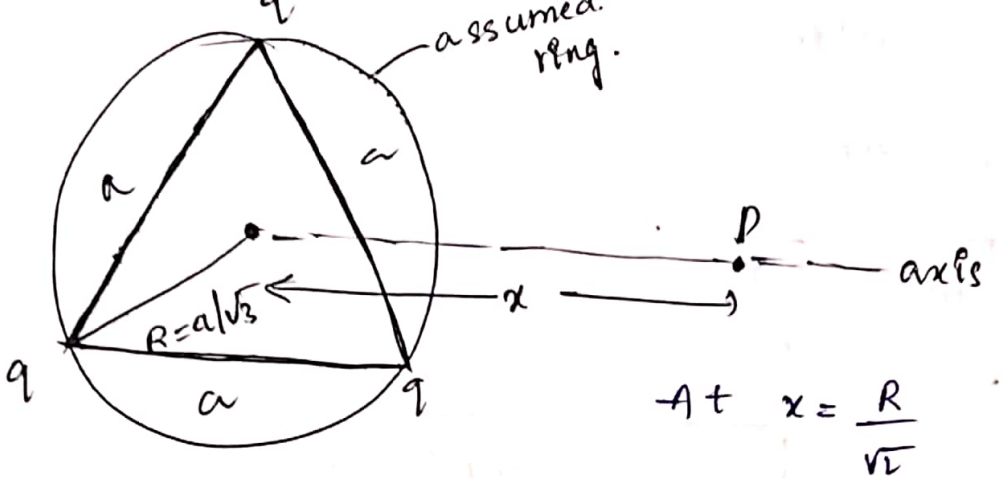
Ex 6



At $x = \frac{R}{\sqrt{2}}$

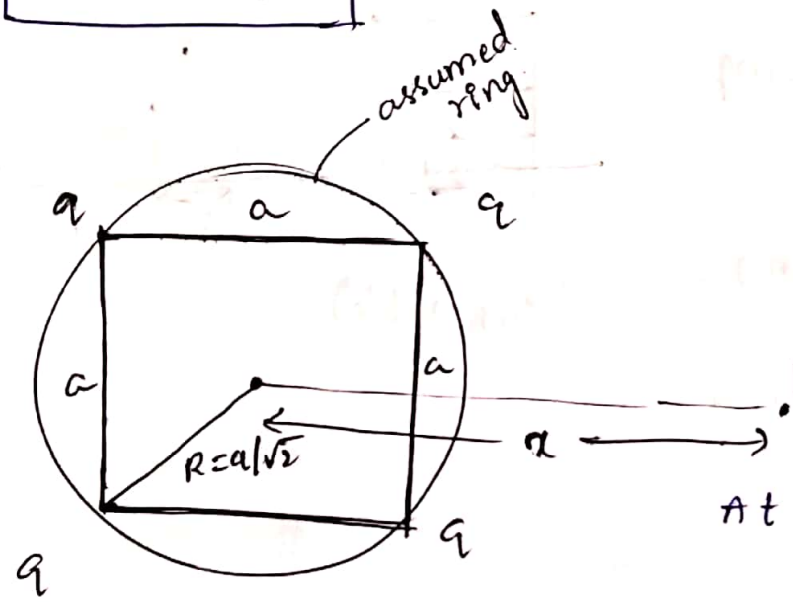
$= \frac{a/\sqrt{3}}{\sqrt{2}}$

$x = a/\sqrt{6}$

$$\mathcal{E} = \frac{K(3q)(a/\sqrt{6})}{\left[(a/\sqrt{6})^2 + (a/\sqrt{3})^2\right]^{3/2}}$$

$$\mathcal{E} = \frac{2\sqrt{3}\,Kq}{a^2}$$

Ex 6



At $x = R/\sqrt{1}$

$= a/\sqrt{1}/\sqrt{2}$

$x = a/2$

$$\mathcal{E} = \frac{K(4q)(a/2)}{\left[(a/2)^2 + (a/\sqrt{1})^2\right]^{3/2}}$$

=

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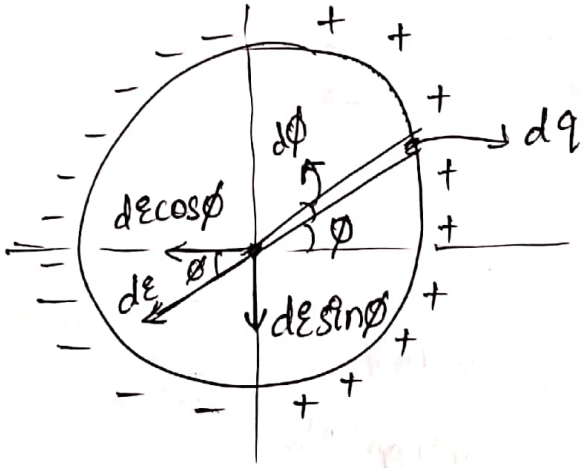
Non-Uniform charge distribution

* $\lambda = \lambda_0 \cos\phi$

Find field at the centre of ring.



Net field at centre,

$$E_1 = \int_{\phi=-\pi/2}^{\phi=\pi/2} dE_1 \cos\phi$$

$$= \int_{\phi=-\pi/2}^{\phi=\pi/2} \frac{K\,dq}{R^2} \cos\phi$$

$$E_1 = \frac{K\lambda_0}{R^2} \int_{\phi=-\pi/2}^{\phi=\pi/2} \cos^2\phi \cdot d\phi$$

$$= \frac{K\lambda_0}{R} \int_{\phi=-\pi/2}^{\phi=\pi/2} \left(\frac{1+\cos 2\phi}{2}\right) \cdot d\phi$$

$$\boxed{E_1 = \frac{\pi K \lambda_0}{2R}}$$

Net field at centre (O) due to both the half rings

$$\boxed{E = 2E_1}.$$

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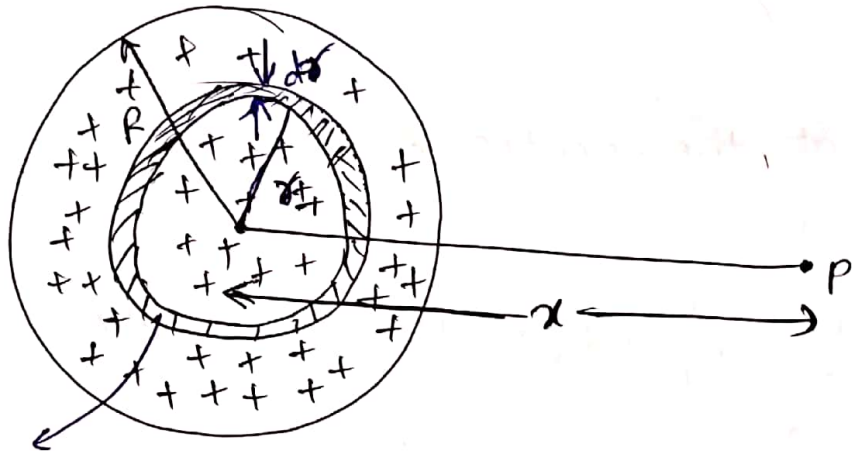
Uniformly charged Non-conducting circular Disc

$\sigma = \frac{Q}{\pi R^2}$



$dq = \sigma dA$

$= \sigma (2\pi r dr)$

now,

$$d\varepsilon = \frac{K dq x}{(x^2+r^2)^{3/2}} \quad \text{(ring)}$$

net field,

$$\varepsilon = \int d\varepsilon$$

$$= \int \frac{K\sigma(2\pi r \cdot dr)\, x}{(x^2+r^2)^{3/2}}$$

$$= 2\pi\sigma x K \int_{r=0}^{r=R} \frac{r \cdot dr}{(x^2+r^2)^{3/2}}$$

let $x^2+r^2 = t^2$

$2r \cdot dr = 2t \cdot dt$

$$= 2\pi\sigma K x \int \frac{t \cdot dt}{t^3}$$

$$= \frac{2\pi\sigma}{4\pi\varepsilon_0} x \left[-\frac{1}{t}\right]$$

$$= \frac{\sigma x}{2\varepsilon_0} \left[\frac{-1}{\sqrt{x^2+r^2}}\right]_{r=0}^{r=R}$$

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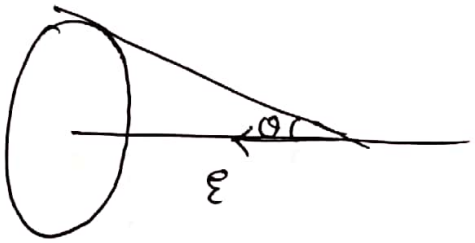
$$E = \frac{\sigma x}{2\varepsilon_0}\left[\frac{1}{x} - \frac{1}{\sqrt{x^2+R^2}}\right]$$

$$\boxed{E = \frac{\sigma}{2\varepsilon_0}\left[1 - \frac{x}{\sqrt{x^2+R^2}}\right]}$$

$$\boxed{E = \frac{\sigma}{2\varepsilon_0}(1-\cos\theta)} \quad *$$

* Gravitation

$$\sigma = \frac{m}{\pi R^2}$$

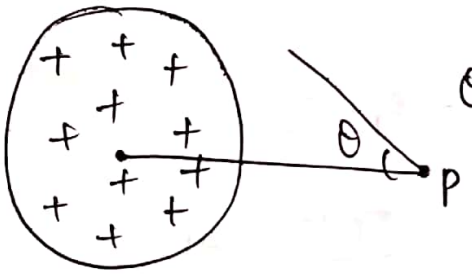


$$\frac{1}{4\pi\varepsilon_0} = G$$

$$\frac{1}{\varepsilon_0} = 4\pi G$$

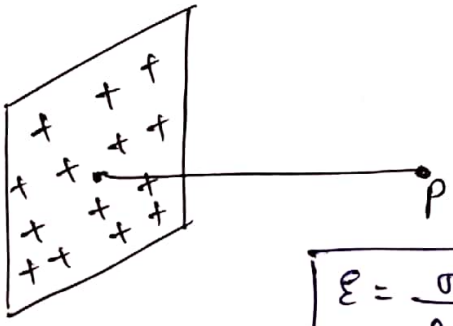
$$\boxed{E = 2\pi\sigma G(1-\cos\theta)}$$

* If the point is very close to disc (i.e. infinite charged disc) :



$\theta = 90°$.

$$\boxed{E = \frac{\sigma}{2\varepsilon_0}} \quad *$$



$$\boxed{E = \frac{\sigma}{2\varepsilon_0}}$$

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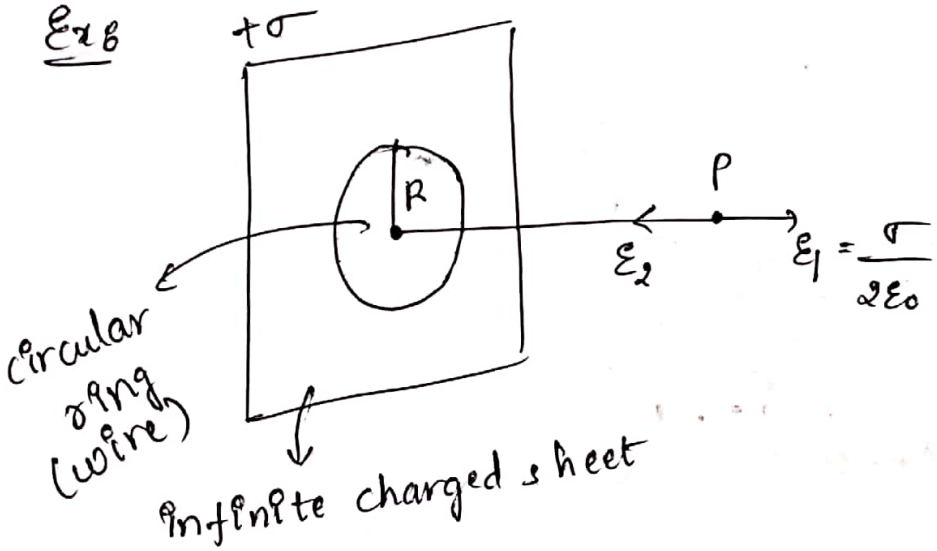
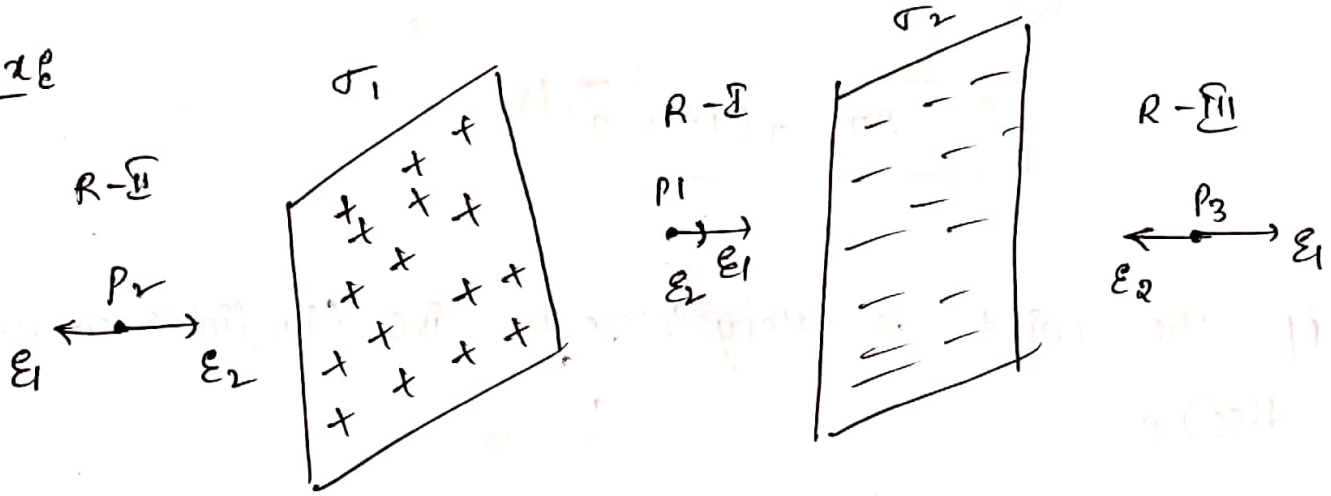
Ex 6



$$E_2 = \frac{\sigma}{2\varepsilon_0}\left(1 - \frac{x}{\sqrt{x^2+R^2}}\right)$$

$$\Rightarrow \boxed{E = E_1 - E_2}$$

Ex 8



$$E_{P_2} = (E_2 - E_1)(\uparrow)$$

$$= \frac{\sigma_2 - \sigma_1}{2\varepsilon_0}(\uparrow)$$

$$E_{P_1} = E_1 + E_2$$

$$= \frac{\sigma_1 + \sigma_2}{2\varepsilon_0}(\uparrow)$$

$$E_{P_3} = (E_1 - E_2)(\uparrow)$$

$$= \left(\frac{\sigma_2 - \sigma_2}{2\varepsilon_0}\right)(\uparrow)$$

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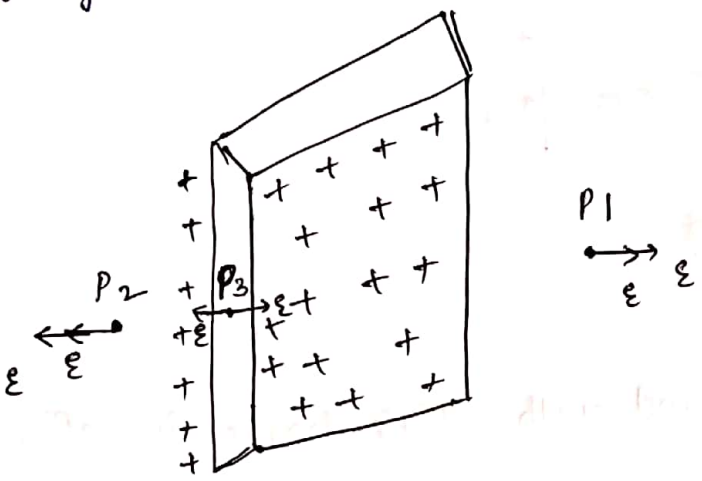
28 | 03 | 25 | Friday

Conducting sheet :

* It is the combination of two thin "non-conducting sheets".
* charge resides on it's outer surface.
* charge inside the sheet = 0.

Ex1:



P_3 is inside pt.

σ = surface charge density of each sheet.

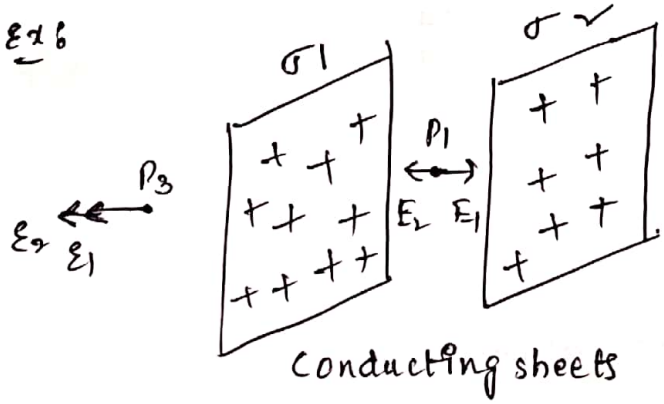
then $\mathcal{E}_{P_1} = 2\mathcal{E} = \frac{\sigma}{\varepsilon_0}$ (→)

$\mathcal{E}_{P_2} = 2\mathcal{E} = \frac{\sigma}{\varepsilon_0}$ (←)

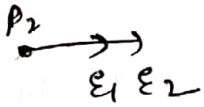
$\boxed{\mathcal{E}_{P_3} = 0}$

i.e, net field at any point inside the conductor = 0.

Ex 2:



Conducting sheets



$\mathcal{E}_1 = \frac{\sigma_1}{\varepsilon_0}$

$\mathcal{E}_2 = \frac{\sigma_2}{\varepsilon_0}$

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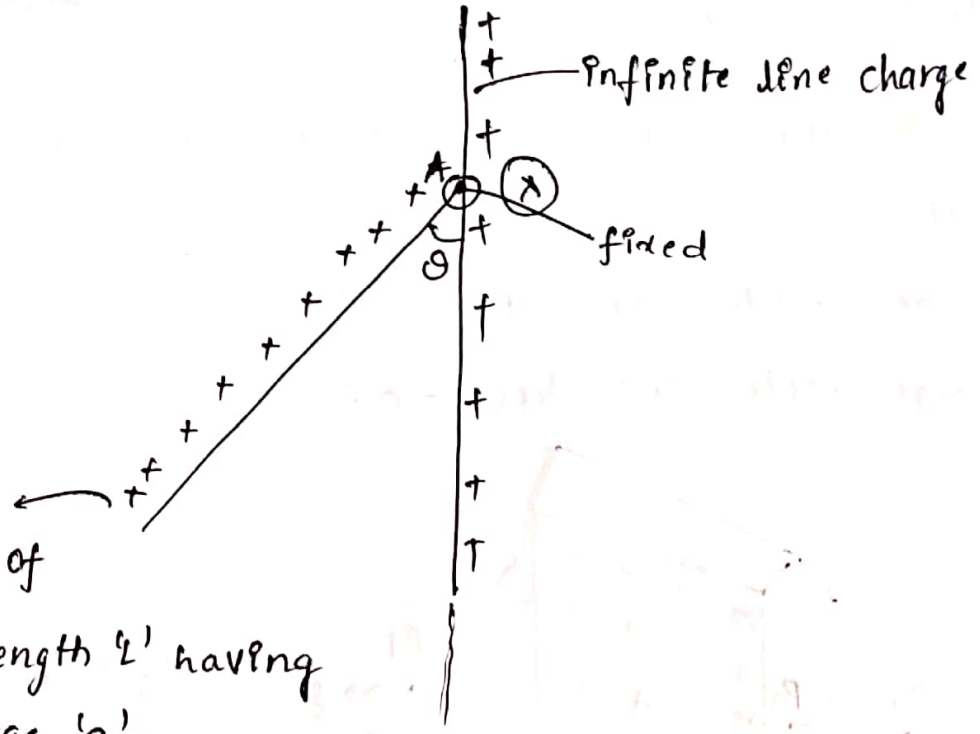
Ex:



mass 'm', length 'l' having total charge 'q'.

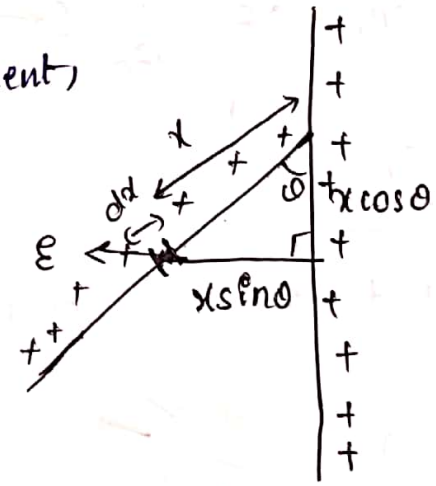
If θ = angle made by rod with line charge in eqm. Find the charge on the rod.

Sol:

In eqm of rod,

It is a non-uniform electric field.

$$\tau_{elec} = \tau_g \text{ (about A)}$$



electric field at the element,

$$E = \frac{2K\lambda}{x\sin\theta}$$

Force on the element,

$$dF = E(dq)$$

$$dF = \frac{2K\lambda}{x\sin\theta} \cdot \frac{q}{l} \cdot dx$$

now, torque on the element about A

$$d\tau_e = (dF)(x\cos\theta)$$

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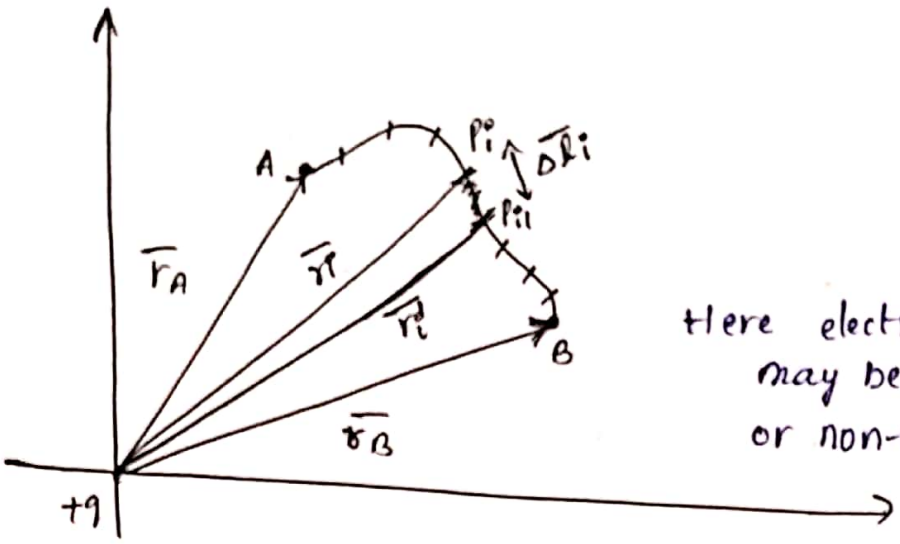
$$q = \frac{2\pi\varepsilon_0 \tan\theta}{\sigma}$$

Electrostatic field is a conservative field :

*



Here electric field may be uniform or non-uniform.

From figure,

$$\overline{r_i} + \overline{\Delta l_i} = r_i' \quad \text{— (1)}$$

$$\overline{r_i'} \cdot \overline{r_i'} = (\overline{r_i} + \overline{\Delta l_i})(\overline{r_i} + \overline{\Delta l_i})$$

$$(r_i')^2 = r_i^2 + (\Delta l_i)^2 + 2\Delta l_i r_i$$

$$\overline{r_i} \cdot \overline{\Delta l_i} = \frac{(r_i')^2 - (r_i)^2}{2}$$

$$\overline{r_i} \cdot \overline{\Delta l_i} = \frac{1}{2}(r_i + r_i')(r_i' - r_i)$$

$$\overline{r_i} \cdot \overline{\Delta l_i} = r_i (r_i' - r_i) \qquad \left[\because r_i = \frac{r_i + r_i'}{2}\right]$$

now,

$$\overline{E_i} \cdot \overline{\Delta l_i} = \frac{kq}{r_i^3}(\overline{r_i}) \cdot \overline{\Delta l_i}$$

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$$= \frac{Kq}{r_i \cdot r} (r_i) \quad (r_i' - r_i)$$

$$\bar{E}_1 \cdot \Delta\bar{l}_1 = \frac{Kq}{r_i \cdot r} (r_i' - r_i)$$

$$= \frac{Kq}{r_i \cdot r_i'} (r_i' - r_i)$$

$$\boxed{\bar{E}_1 \cdot \Delta\bar{l}_i = Kq \left(\frac{1}{r_i} - \frac{1}{r_i'} \right)}$$

how,

$$\bar{E}_1 \cdot \Delta\bar{l}_1 + \bar{E}_2 \Delta\bar{l}_2 + \text{- - - - +} \bar{E}_n \Delta\bar{l}_n$$

$$= Kq \left[\left(\frac{1}{r_1} - \frac{1}{r_2} \right) + \left(\frac{1}{r_2} - \frac{1}{r_3} \right) + \text{- - -} + \left(\frac{1}{r_{n-1}} - \frac{1}{r_n} \right) \right]$$

$$= Kq \left(\frac{1}{r_1} - \frac{1}{r_n} \right)$$

$$\boxed{\sum_{i=1}^{n} \bar{E}_i \Delta\bar{l}_i = Kq \left(\frac{1}{r_1} - \frac{1}{r_n} \right)}$$

* $n \longrightarrow \infty$ then $\Delta\bar{l}_i \longrightarrow 0$

$$\text{It } \lim_{n \to \infty} \sum_{i=1}^{n} \bar{E}_i \cdot \Delta\bar{l}_i = \int_A^B \bar{E} \cdot d\bar{l}$$

$$\boxed{\int_A^B \bar{E} \cdot (d\bar{l}) = Kq \left(\frac{1}{r_A} - \frac{1}{r_B} \right)} \; *$$

work done in moving a electric charge from A to B.

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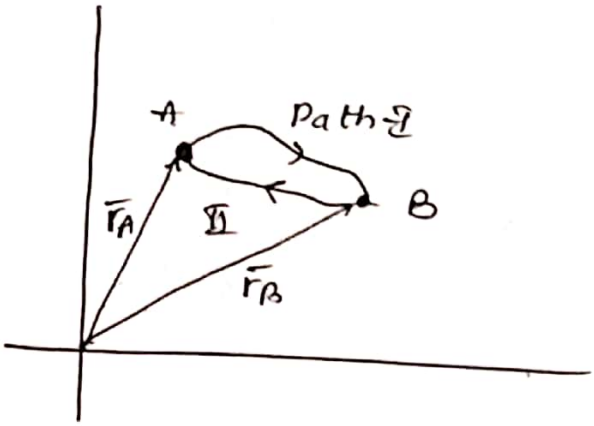
* Line integral of electric field doesnot depends on path followed but depends only on initial and final positions.

*

For path-I :

$$\int_A^B \bar{E}\cdot d\bar{l} = Kq\left(\frac{1}{r_A} - \frac{1}{r_B}\right)$$



For path-II :

$$\int_B^A \bar{E}\cdot d\bar{l} = Kq\left(\frac{1}{r_B} - \frac{1}{r_A}\right)$$

For closed path,

$$\oint_{ABA} \bar{E}\cdot d\bar{l} = Kq\left(\frac{1}{r_A} - \frac{1}{r_B}\right) + Kq\left(\frac{1}{r_B} - \frac{1}{r_A}\right)$$

$$\boxed{\oint \bar{E}\cdot d\bar{l} = 0}$$

∴ Electrostatic field is a conservative field.

⇒ Electrostatic force is a conservative force.

* For a conservative force,

* $\boxed{F_e = -\frac{dU}{dr}}$ * U = electrostatic potential energy.

$\frac{dU}{dr}$ – gradient of potential energy.

$$d\bar{U} = -\bar{F_e}\cdot d\bar{r}$$

$$\int_i^f dU = -\int_i^f \bar{F_e}\cdot d\bar{r}$$

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$$U_f - U_i = -W_{elec}$$

$$\boxed{W_{elec} = -\Delta U}^{*} \qquad (\Delta U = U_f - U_i)$$

Electrostatic potential energy :

* P.E - stored energy.
* K.E - utilized energy.

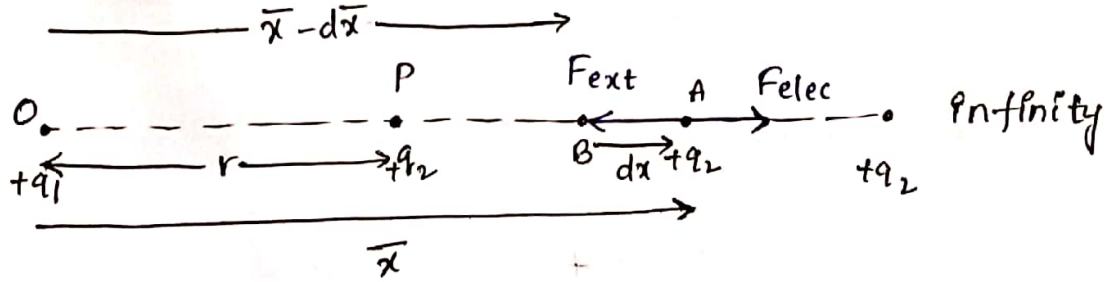
1) Interaction P.E 2) Self energy.

*only conservative forces
|
law of conservation of energy
[(or) non-conservative doing no work]
* W-E Theorem - for all

1) Interaction P.E :

It is the amount of work done in bringing the charges from infinity to certain seperation between them against field force is called interaction P.E.

Derivation :



Here 1) $F_{ele} = \frac{Kq_1q_2}{x^2}$

2) Net force on the charge = 0 (moved very slowly so a = 0)

$$\bar{F}_{ele} + \bar{F}_{ext} = \bar{0}$$

$$\bar{F}_{ext} = -\bar{F}_{ele}$$

$$\bar{F}_{ext} = -\frac{Kq_1q_2}{x^2} \Rightarrow F_{ext} = \frac{Kq_1q_2}{x^2}$$

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3) Work done by $\bar{F}_{ext}$ in moving charge from A to B.

$$dw_{ext} = \bar{F}_{ext} \cdot d\bar{x}$$

$$= +\frac{kq_1q_2}{x^2} \cdot dx \cos 180^\circ$$

$$dw_{ext} = -\frac{kq_1q_2}{x^2} \cdot dx$$

∴ Total work done,

$$w_{ext} = \int_{x=\infty}^{x=r} dw_{ext}$$

$$= \int_{x=\infty}^{x=r} -\frac{kq_1q_2}{x^2} \cdot dx$$

$$= -kq_1q_2 \int_{x=\infty}^{x=r} \frac{1}{x^2} \cdot dx$$

$$\boxed{w_{ext} = \frac{kq_1q_2}{r}}$$

From work-Energy Theorem,

$$w_{ext} + w_{ele} = \Delta k^{0}$$

$$w_{ext} = -w_{ele}$$

$$= -(-\Delta U)$$

$$\frac{kq_1q_2}{r} = \Delta U$$

$$U_f - U_i = \frac{kq_1q_2}{r}$$

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$$U_f = U_i + \frac{Kq_1q_2}{r}$$ * — Increase in P.E.

* If $U_i = 0 \Rightarrow U_f = \frac{Kq_1q_2}{r}$ (at ∞)

$$U = \frac{Kq_1q_2}{r}$$ * — Increase in P.E → here it is absolute P.E.

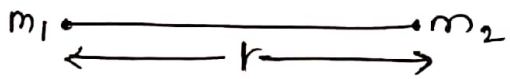
[take signs of q_1 & q_2]

* It is defined for a pair of charges.

* If U = +ve, like charges are taken, field force is repulsive.

* If U = -ve, unlike charges are taken, field force is attractive.

* Gravitation:



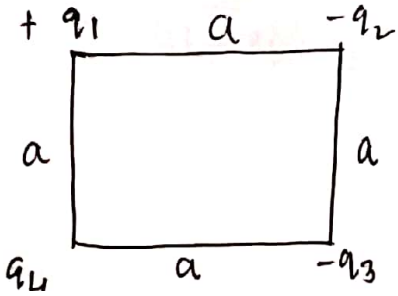
$$U_f - U_i = -\frac{Gm_1m_2}{r}$$

Principle of Superposition:

U_{system} = algebraic sum of P.E.s of individual pairs of system.

For n-particle system,

No. of pairs = $\frac{n(n-1)}{2}$.

Ex: 

Find potential energy of system.

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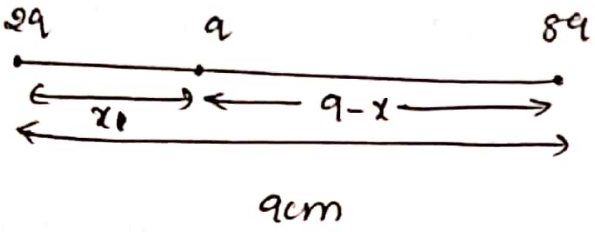
$$U_{system} = -\frac{kq_1q_2}{a} - \frac{kq_1q_3}{\sqrt{2}a} + \frac{kq_1q_4}{a} + \frac{kq_2q_3}{a} - \frac{kq_2q_4}{\sqrt{2}a} + \frac{kq_1q_4}{a}$$

Ex: Three charges $q, 2q, 8q$ are placed on the line of length 9cm. find the seperation between the charges so that potential energy of system is min. And also find net force acting on charge q in this position.

Sol:



9cm

$$U_{system} = \frac{2kq^2}{x} + \frac{8kq^2}{9-x} + \frac{16kq^2}{9}$$

$$\frac{dU}{dx} = 0$$

$$\frac{-2}{x^2} + \frac{8}{(9-x)^2} = 0$$

$$2(9-x)^2 = 8x^2$$

$$9-x = 2x \quad \Rightarrow \quad 3x = 9$$

$$x = 3$$

$$(U_{system})_{min} = \frac{2kq^2}{3} + \frac{8kq^2}{83} + \frac{16kq^2}{9}$$

$$= \frac{26kq^2}{8} + \frac{16kq^2}{9} = \frac{(18+16)kq^2}{9}$$

$$= \frac{34}{9} kq^2 \times 100$$

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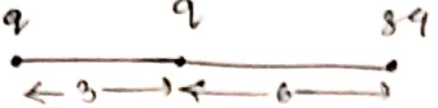


$$F_q = \frac{2kq^2}{9} - \frac{8kq^2}{36}$$

$$F_q = \frac{2kq^2}{q} - \frac{2kq^2}{q}$$

$$\boxed{F_q = 0}$$

net force on charge $q = 0$.

Note:

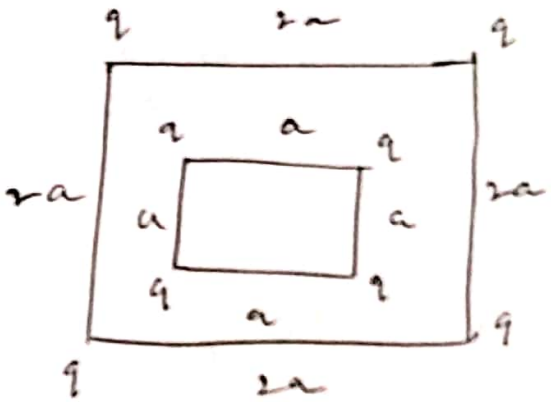
* we have,

$$\boxed{W_{ext} = U_f - U_i}$$

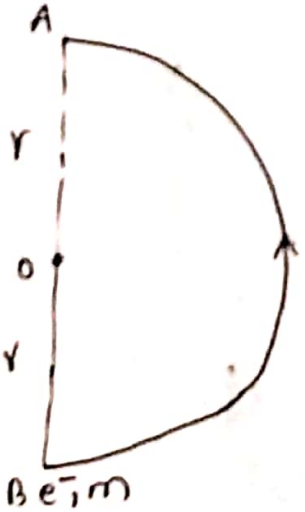
* $W_{ele} + W_{ext} = \Delta K$

$$-\Delta U + W_{ext} = \Delta K$$

$$\boxed{W_{ext} = \Delta U + \Delta K}$$



Ex:



Find work done.

Sol: $W_{ele} = U_f - U_i^\circ = 0$

$W_g = -mg(2r)$

$W_{ele}^{\,0} + W_g + W_{ext} = \Delta K^{\,0}$

$W_{ext} = -W_g$

$$\boxed{W_{ext} = 2mgr}$$

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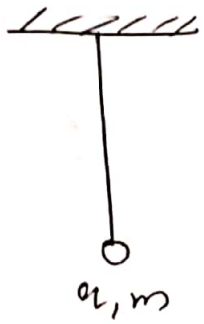
Ex: given $\mathcal{E} = 10^6$ v/m (↓)

$m = 0.01$ kg

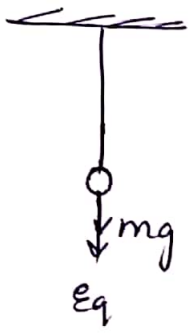
$q = 10^{-6}$ c

$l = 1$ m



Find Time period.

Sol:



$$mg_{eff} = mg + Eq$$

$$g_{eff} = g + \frac{Eq}{m} = \frac{10^6 \times 10^{-6}}{0.01} + 10 = 110$$

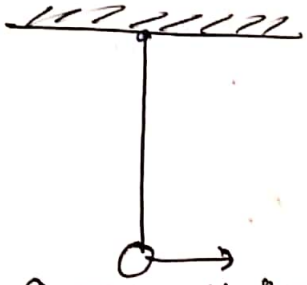
$$T = 2\pi\sqrt{\frac{l}{g_{eff}}} = 2\pi\sqrt{\frac{1}{110}}$$

$$T = 2\pi \times 0.095 = 0.19\pi = 0.599$$

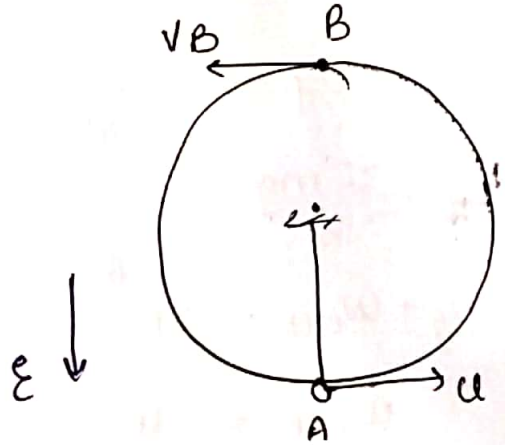
$$\boxed{T = 0.6 \text{ sec}}$$

Ex:

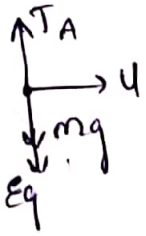
Find T_{max}, T_{min}, U_{min} from above given data.



Sol:



At A:



$$T_A - mg - Eq = \frac{mu^2}{l} \quad -(a)$$

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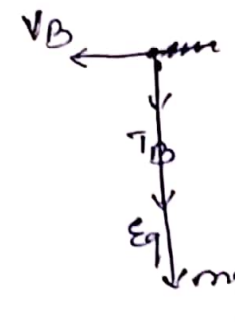
At B,



$$T_B + Eq + mg = \frac{mv_B^2}{l} \quad \text{— (b)}$$

To complete vertical circular motion, $T_B = 0$

eqn (b) becomes,

$$Eq + mg = \frac{mv_B^2}{l} \quad \text{— (1)}$$

W-E theorem,

$$-mg(2l) - Eq(2l) = \frac{1}{2}mv_B^2 - \frac{1}{2}mu^2$$

$$-2l(mg + Eq) = \frac{1}{2}mv_B^2 - \frac{1}{2}mu^2 \quad \text{(from (1))}$$

$$\frac{1}{2}mu^2 = \frac{1}{2}l(Eq + mg) + 2l(mg + Eq)$$

$$\frac{1}{2}mu^2 = \frac{5}{2}l(Eq + mg) \Rightarrow mu^2 = 5l(Eq + mg) \quad \text{— (2)}$$

$$u^2 = \frac{5l(Eq + mg)}{m}$$

$$= \frac{5 \times 1 \times (1 + 0.01 \times 10)}{0.001}$$

$$u^2 = 500 \times 1.1$$

$$u^2 = 550$$

$$\boxed{u = 23.45 \text{ m/s}}$$

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Substitute eqn ⑤ in ④,

$$T_A - (mg + \varepsilon q) = \frac{5\,d(mg+\varepsilon q)}{t}$$

$$T_A = 6(mg + \varepsilon q)$$

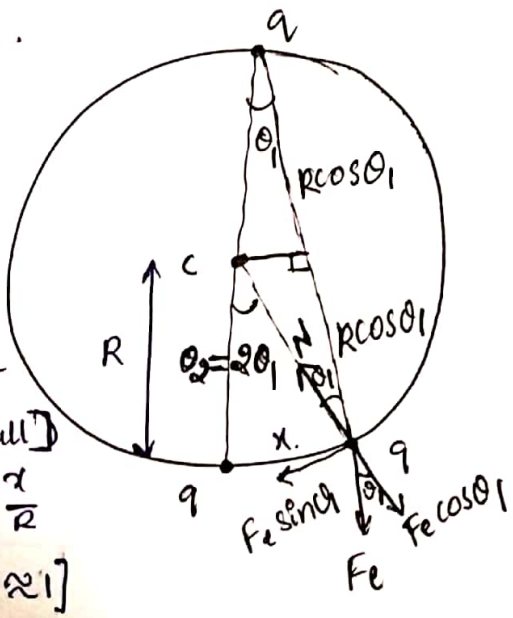
$$= 6(0.01 \times 10 + 1)$$

$$= 6(1.1)$$

$$\boxed{T_A = 6.6\ N}$$

$T_{max} = 6.6N$, $T_{min} = 0$, $u_{min} = 23.45 m/s$.

Ex: Two beads each with charge q and mass m are on horizontal, frictionless, non-conducting, circular hoop of radius R. One of the beads is glued to the hoop at some point while the other one performs small oscillations about it's eqm position along the hoop. The square of the angular frequency of the small oscillations is given by ——.



Sol: Restoring force $= -F_e \sin\theta_1$

$$F = -F_e \sin(\theta_2/2)$$

$$F = \frac{-Kq^2}{(2R\cos\theta_1)^2} \cdot \frac{\theta_2}{2}$$

$$ma = \frac{-Kq^2}{8R^2\cos^2\theta_1} \times \frac{x}{R}$$

$\sin\theta_2/2 \approx \theta_2/2$ [$\because \theta_2$ very small]

$\theta_2 = \frac{x}{R}$

$[\cos\theta_1 \approx 1]$

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$$ma = \frac{-kq^2}{8R^3}x$$

$$a = \frac{-kq^2}{8R^3m}x$$

$$\boxed{a \propto -x}$$

∴ The motion is SHM.

$$a = -\omega^2 x$$

$$\omega^2 = \frac{kq^2}{8R^3m}$$

$$= \frac{1}{4\pi\varepsilon_0} \cdot \frac{q^2}{8R^3m}$$

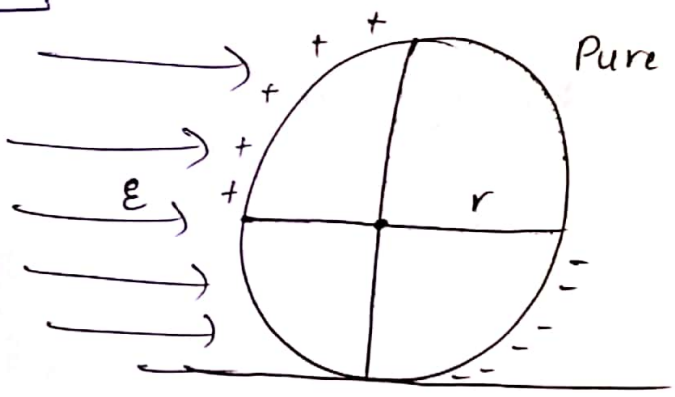
$$\boxed{\omega^2 = \frac{q^2}{32\pi\varepsilon_0 mR^3}}$$

Ex:

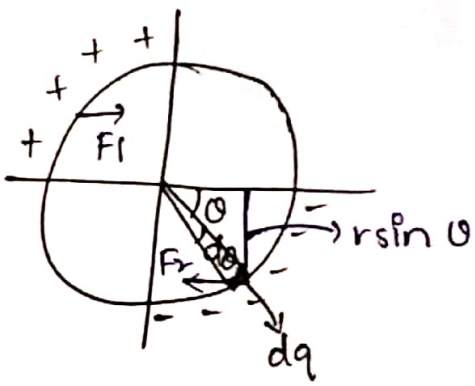
Find

1) Total Torque about centre of mass.

2) Friction force.



Sol: 1)



F_1, F_2 are equal and opposite forces.

$$dF = \varepsilon dq$$

$$d\tau = dF\,(r\sin\theta)$$

$$= \varepsilon dq\,(r\sin\theta)$$

$$dq = \lambda dl = \lambda r \cdot d\theta$$

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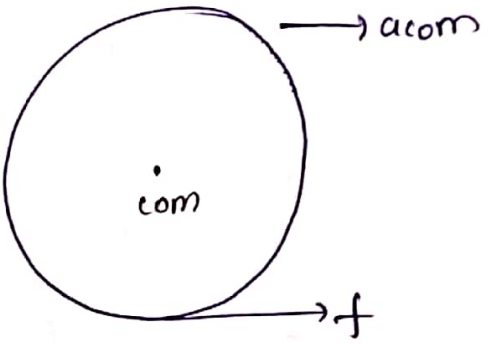
$$d\tau = \varepsilon \lambda r^2 \sin\theta \cdot d\theta$$

$$\tau = \int d\tau = \varepsilon \lambda r^2 \int_{\theta=0}^{\theta=\pi/2} \sin\theta \cdot d\theta$$

$$\tau = \varepsilon \lambda r^2 (1)$$

Total torque $= \Theta\tau = 2\varepsilon\lambda r^2$.

2)



$$f = ma_{com} \quad -①$$

$$a_{com} = r\alpha \quad -②$$

$$f = mr\alpha$$

$$\alpha = \frac{f}{mr} \quad -③$$

$$2\tau - rf = I\alpha \quad -④$$

$$2\tau - rf = \cancel{m}r\cancel{r}\cdot\frac{f}{\cancel{m}\cancel{r}}$$

$$2\tau = 2rf$$

$$rf = \tau$$

$$f = \frac{\tau}{r}$$

$$\boxed{f = \varepsilon\lambda r}$$ *

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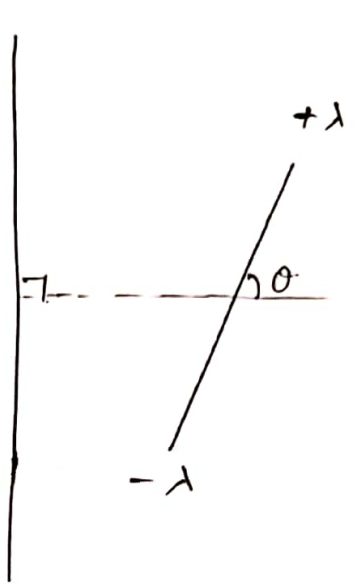
Ex: A large sheet carries uniform surface charge density σ. A rod of length $2l$ has a linear charge density λ on one half and $-\lambda$ on the other half. The rod is hinged at midpoint 'O' and makes an angle 'θ' with normal to the sheet. The torque experienced by the rod is



Sol:

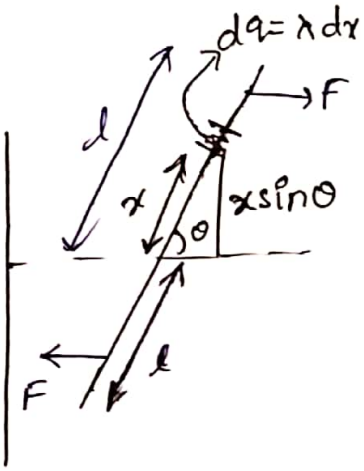
$$d\tau = \frac{\sigma}{2\varepsilon_0}\, dq \cdot x\sin\theta$$

$$= \frac{\sigma \sin\theta}{2\varepsilon_0} \cdot \lambda x \cdot dx$$

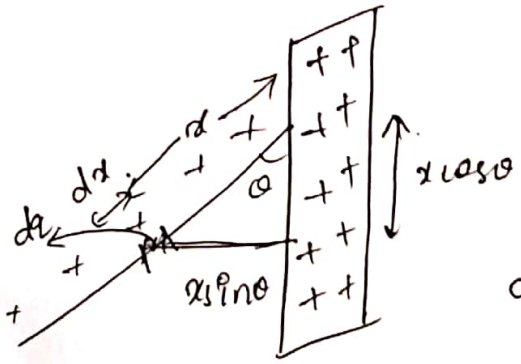
$$\tau = \int d\tau = \frac{\sigma\lambda\sin\theta}{2\varepsilon_0}\int_0^l x \cdot dx$$

$$\tau = \frac{\sigma\lambda\sin\theta}{2\varepsilon_0}\,\frac{l^2}{2}$$

$$\text{Total torque} = 2\tau = \frac{\sigma\lambda l^2}{2\varepsilon_0}\sin\theta$$



Ex: Find charge on rod.



Sol: M-II: $d\tau = dF \cdot x\cos\theta$

$$d\tau = \frac{\sigma}{2\varepsilon_0} \cdot \frac{q}{l} \cdot dx \cdot x\cos\theta \qquad dq = \frac{q}{l} \cdot dx$$

$$\int d\tau = \frac{\sigma q x\cos\theta}{2l\varepsilon_0}\int_{x=0}^{x=l} x \cdot dx \;\Rightarrow\; \tau = \frac{\sigma q l}{4\varepsilon_0}\cos\theta$$

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$$\tau_e = \tau_g$$

$$\frac{\sigma q x}{4\pi\varepsilon_0} \cos\theta = \frac{mgx}{2} \sin\theta$$

$$q = \frac{2\pi\varepsilon_0 mg}{\sigma} \tan\theta$$

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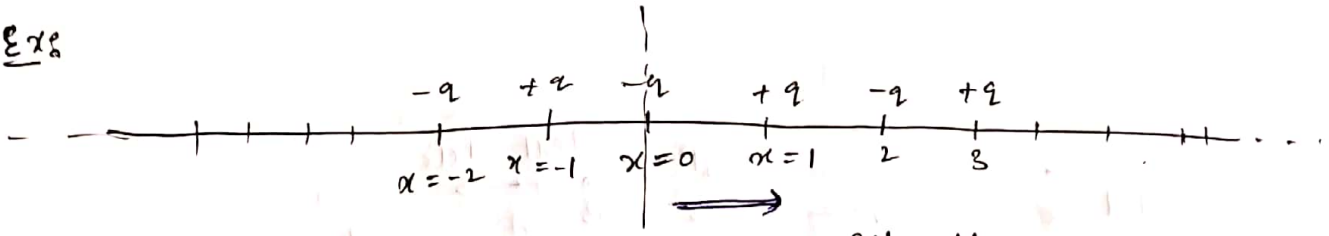
Ex:



Find potential energy of one charge with all other charges.

Sol:

$$U_1 = \frac{Kq^2}{1} - \frac{Kq^2}{2} + \frac{Kq^2}{3} - - - -$$

$$= Kq^2\left(1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + - - - -\right)$$

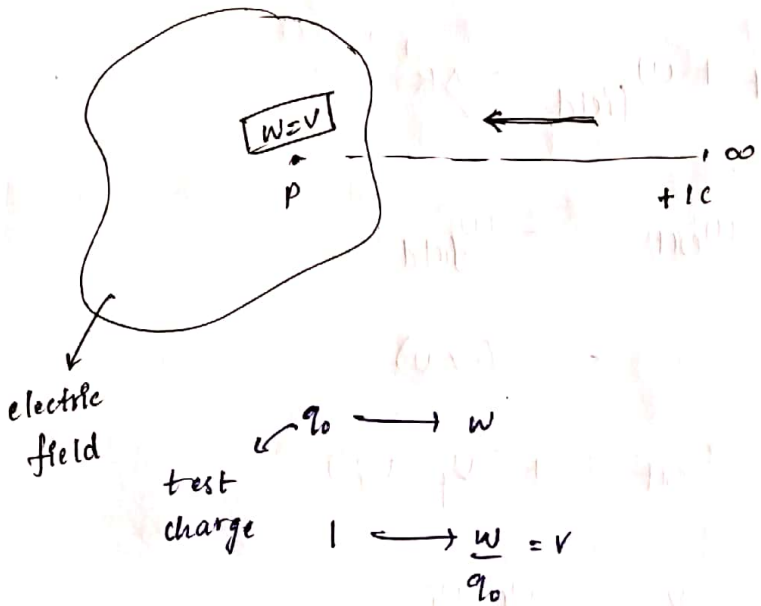
$$= Kq^2 \ln 2$$

* net potential energy,

$$U_{net} = 2U \Rightarrow \boxed{U_{net} = 2Kq^2 \ln 2}$$

Electric Potential (V):

It is the amount of workdone in bringing a unit positive charge slowly from infinity to a point in the field and against to it is called potential at that point.



$$\boxed{V = \frac{W}{q_0}}$$ unit: J/C (or) Volt

* It is a scalar.

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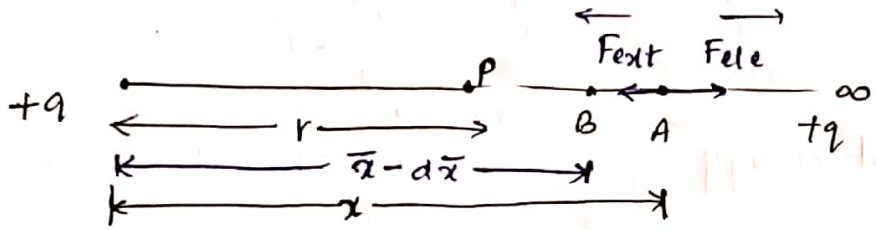
Electric Potential at a point due to point charges



* electric field force do -ve work.
* external force do +ve work.

$$dw_{ext} = \bar{F}_{ext} \cdot d\bar{x}$$

$$\Rightarrow \quad w = \int_{x=\infty}^{x=r} dw_{ext}$$

$$\boxed{w = \frac{kqq_0}{r}}$$

Finally potential at P, $V = \frac{w}{q_0}$

$$\boxed{V = \frac{kq}{r}} \quad **$$

we have,

$$w_{ext} + w_{field} = \cancel{\Delta K}^{\,0}$$

$$w_{ext} = -w_{field}$$

$$= -(-\Delta U)$$

$$w_{ext} = +(U_f - U_i)$$

$$V = \frac{U_f - \cancel{U_i}^{\,0}}{q_0}$$

$$\boxed{V = U/q_0}$$

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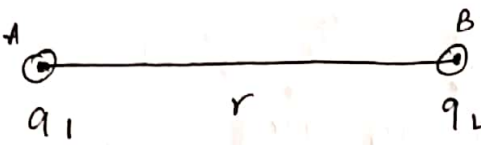
i.e,



$$V_A = \frac{\frac{Kq_1q_2}{r}}{q_1}$$

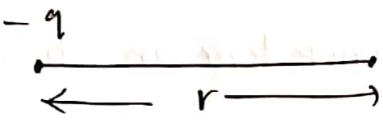
$$V_B = \frac{\frac{Kq_1q_2}{r}}{q_2}$$

$$\boxed{V_A = \frac{Kq_2}{r}}$$

$$\boxed{V_B = \frac{Kq_1}{r}}$$

*

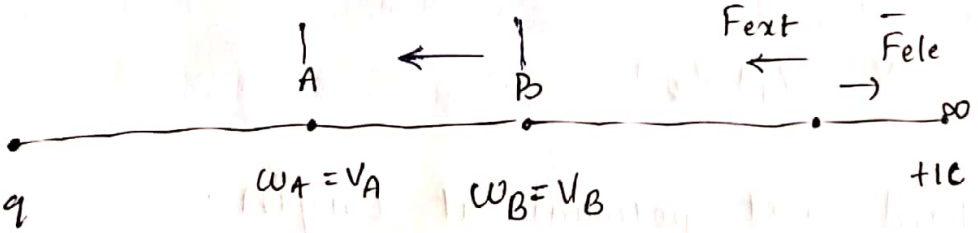
$$V = \frac{Kq}{r}$$

* 

$$V = -\frac{Kq}{r}$$

* * Neutral body produces zero potential in it's surrounding.

Potential difference between two points in a field:

* 

$W_A = V_A$ $\quad W_B = V_B$

(or) $V_A = \frac{W_A}{q_0}$ $\quad$ (or) $V_B = \frac{W_B}{q_0}$

Clearly $W_A > W_B$.

$V_A > V_B$

* $\boxed{V_{AB} = V_A - V_B = +ve}$

$$V_{AB} = \frac{W_A - W_B}{q_0} = \frac{W}{q_0}$$

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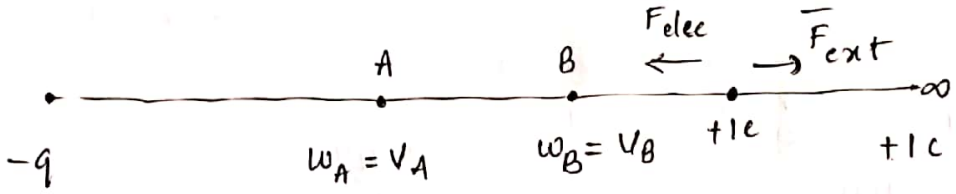
*



clearly, $|W_A| > |W_B|$

but $V_A < V_B$

* electric field force do +ve work

* ext force do -ve work

* $\boxed{V_{AB} = V_A - V_B = -ve}$

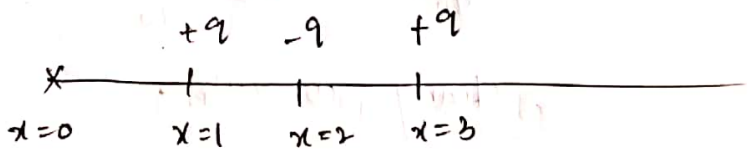
$$V_{AB} = \frac{W}{q_0}.$$

* Definition :- It is the amount of work done in bringing a unit positive charge between 2 pts in the field and against to it is called potential difference b/w those points.

Principle of Superpositions

$$V = V_1 + V_2 + - - -$$

Ex: Find net potential at Origin.



Sol:

$$V = \frac{Kq}{1} - \frac{Kq}{2} + \frac{Kq}{3} - \frac{Kq}{4} + - - -$$

$$\boxed{V = Kq \ln 2 \, (V)}$$

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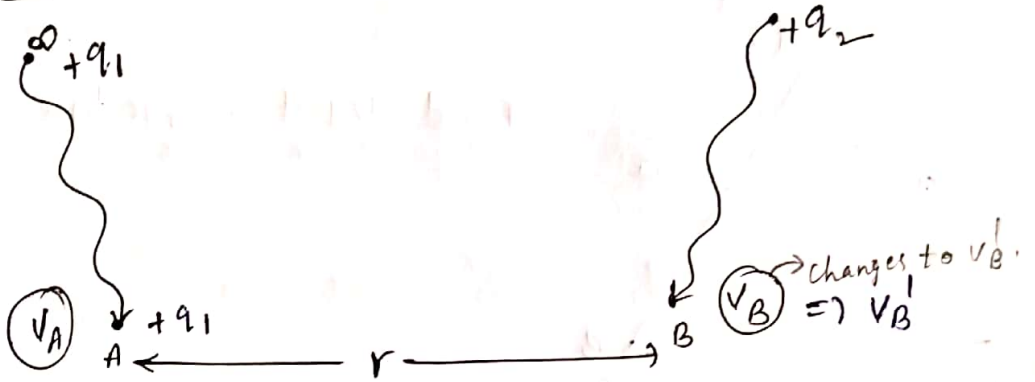
Electrostatic potential Energy of two particle system placed in external field :



work done in moving the charge 'q1' from infinity to point A is,

$$W_A = q_1 (V_A - \cancel{V_\infty}^{0}) = q_1 V_A$$

now, potential at point B,

$$V_B' = V_B + \frac{Kq_1}{r}$$

work done in moving the charge 'q2' from infinity to point B is,

$$W_B = q_2 (V_B' - \cancel{V_\infty}^{0})$$

$$W_B = q_2 \left[V_B + \frac{Kq_1}{r}\right]$$

$$W_B = V_B q_2 + \frac{Kq_1q_2}{r}$$

Finally, Total work done,

$$W = \text{P.E of system}$$

$$= W_A + W_B$$

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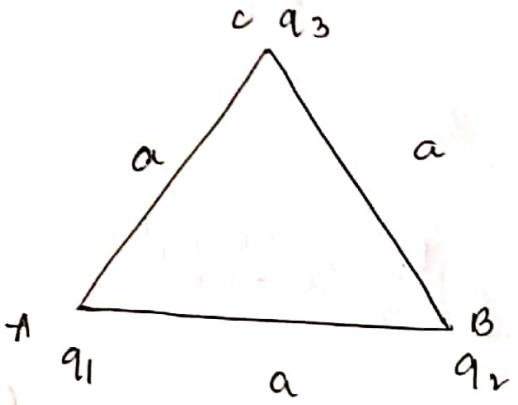
$$W_{system} = q_1 V_A + q_2 V_B + \frac{K q_1 q_2}{r}$$

Ex:



Find total work done.

Sol:

(1) $W_A = q_1 V_A = q_1 (0) = 0.$

(2) potential at B is,

$$V_B = \frac{K q_1}{a}$$

(3) $W_B = q_2 \left(\frac{K q_1}{a} \right) = q_2 (V_B - \cancel{V_\infty}^{0})$

$$W_B = \frac{K q_1 q_2}{a}$$

(4) potential at C is,

$$V_C = \frac{K q_1}{a} + \frac{K q_2}{a}$$

(5) $W_C = q_3 (V_C - \cancel{V_\infty}^{0})$

$$= q_3 \left(\frac{K q_1}{a} + \frac{K q_2}{a} \right) = \frac{K q_1 q_3}{a} + \frac{K q_2 q_3}{a}.$$

(6) Total work done $= W_A + W_B + W_C$

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$$W = \frac{Kq_1q_2}{a} + \frac{Kq_1q_3}{a} + \frac{Kq_2q_3}{a}$$

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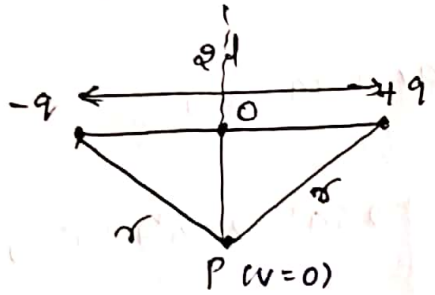
Zero potential points :



locus = st. line *

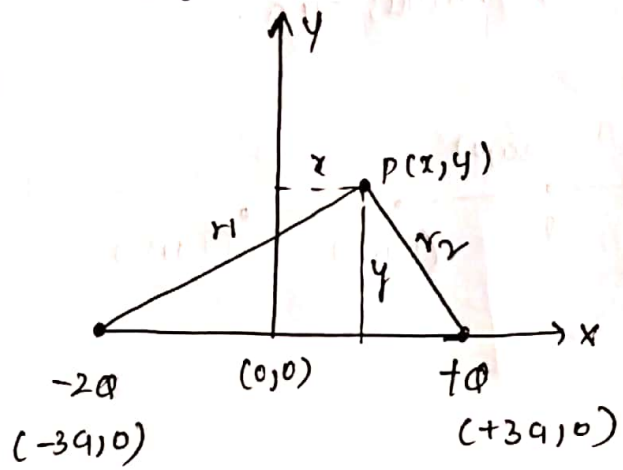
(V=0 but $\mathcal{E} \neq 0$)

$V_0 = V_- + V_+$ $\qquad \mathcal{E} = \frac{KP}{r^3}$

$$V_0 = \frac{-Kq}{d} + \frac{Kq}{d}$$

$$= 0$$

Ex: Find the locus of points where net potential is zero.



From figure,

$$r_1 = \sqrt{(3a+x)^2 + y^2}$$

$$r_2 = \sqrt{(3a-x)^2 + y^2}$$

Since $V_P = 0$

$$V_- + V_+ = 0 \Rightarrow \frac{-2Kq}{r_1} + \frac{Kq}{r_2} = 0$$

$$2r_2 = r_1$$

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$$4\left((3a-x)^2+y^2\right) = (3a+x)^2+y^2$$

$$4\left\{9a^2+x^2-6ax+y^2\right\} = 9a^2+x^2+6ax+y^2$$

$$27a^2+3x^2-30ax+3y^2=0$$

$$x^2+y^2-10ax+9a^2=0$$

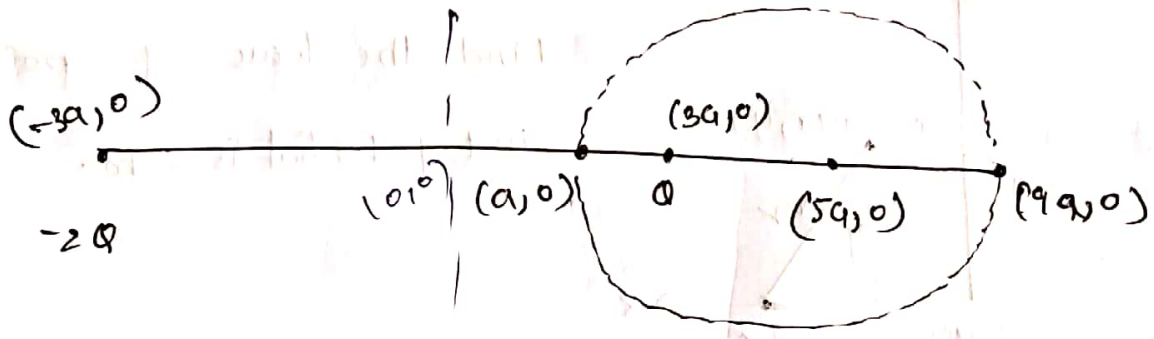
$$x^2+y^2-2(5a)x+25a^2-25a^2+9a^2=0$$

$$\boxed{(x-5a)^2+y^2=16a^2}$$

⟶ circle

with centre $(5a,0)$ & radius $=4a$.

*



At a general point,

$$V_{P(x,y)} = \frac{-2KQ}{\sqrt{(3a+x)^2+y^2}} + \frac{KQ}{\sqrt{(3a-x)^2+y^2}}$$

At any point on the axis, $y=0$

$$\boxed{V_{(x)} = \frac{-2KQ}{|3a+x|} + \frac{KQ}{|3a-x|}}$$

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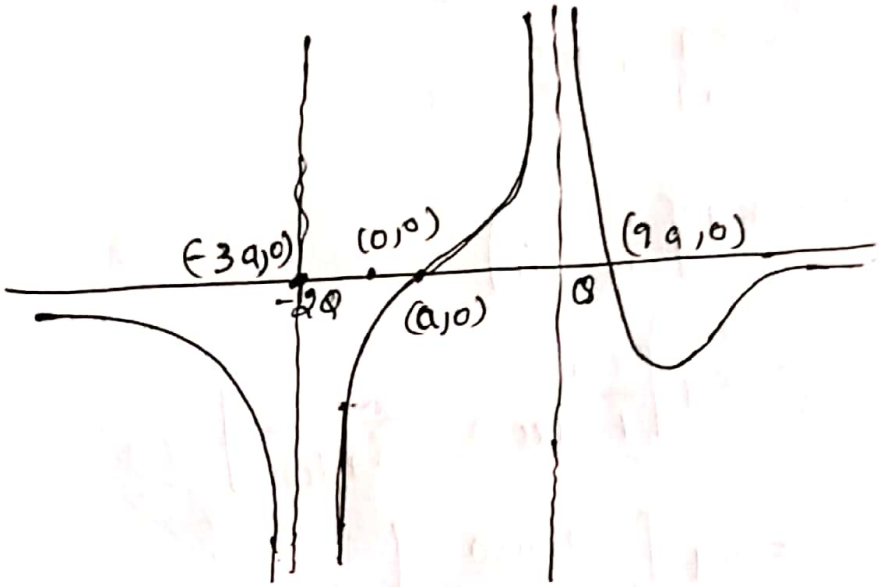
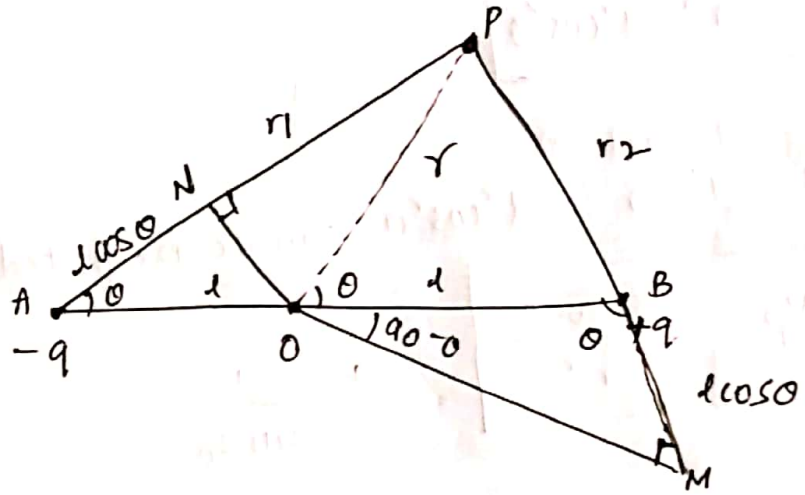
V_x (vs) x graph:



Potential at a point due to dipole:



$NP = OP = r$

$AP = r_1 = AN + NP$

$$\boxed{r_1 = l\cos\theta + r}$$

Again,

$OP = PM = r$

$BP = r_2 = MP - BM$

$$\boxed{r_2 = r - l\cos\theta}$$

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$$V_p = V_- + V_+$$

$$V_p = -\frac{kq}{r_1} + \frac{kq}{r_2}$$

$$= kq\left[\frac{1}{r_2} - \frac{1}{r_1}\right]$$

$$= kq\left[\frac{1}{r - d\cos\theta} - \frac{1}{r + d\cos\theta}\right]$$

$$= kq\left[\frac{2d\cos\theta}{r^2 - d^2\cos^2\theta}\right]$$

$$\boxed{V_p = \frac{kp\cos\theta}{r^2 - d^2\cos^2\theta}}$$

for short dipole $d \ll r$ $d^2\cos^2\theta$ can be neglected.

$$\boxed{V_p = \frac{kp\cos\theta}{r^2}}$$

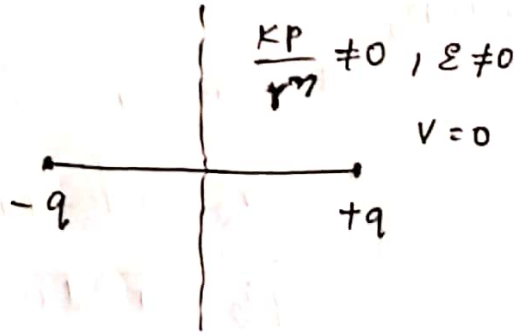
$$k = \frac{1}{4\pi\varepsilon_0}$$

① on axial line $\cos\theta = \pm 1$

$$\boxed{V_p = \frac{kp}{r^2}}$$

② on equitorial line, $\theta = 90°$

$$\boxed{V = 0}$$



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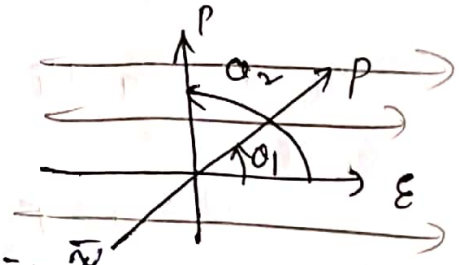
Workdone in rotating the dipole slowly:



we have, $\tau_{net} = 0$

$$\tau_{ext} + \tau_{elec} = 0 \Rightarrow \vec{\tau}_{ext} = -\vec{\tau}_{ele}$$

$$\boxed{\tau_{ext} = \tau_{elec}}$$

⇒ In intermediate state

$$\tau_{ext} = \tau_{elec} = pE\sin\theta, \quad \theta_1 \leq \theta < \theta_2$$

Total work done,

$$W = \int dW_{ext} = \int \vec{\tau}_{ext} \cdot d\theta = \int_{\theta_1}^{\theta_2} pE\sin\theta \cdot d\theta$$

$$W = pE\,[-\cos\theta]_{\theta_1}^{\theta_2}$$

$$\boxed{W = pE(\cos\theta_1 - \cos\theta_2)}$$

Potential Energy stored in the dipole:

we have, $W_{ext} + W_{ele} = \Delta K^{0}$ (slowly)

$$W_{ext} - \Delta U = 0$$

$$W_{ext} = \Delta U = U_f - U_i$$

let us take at $\theta = 90°$, Potential Energy stored in dipole is zero.

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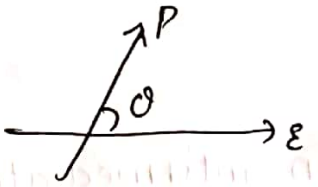
$$U_f - U_i = pE[\cos\theta_1 - \cos\theta_2]$$

$\theta_1 = 90°$ & $U_i = 0$ & $U_f = U$

$$U = pE[-\cos\theta]$$

$$\boxed{U = -pE\cos\theta}$$

In vector form, $\boxed{U = -\vec{p}\cdot\vec{E}}$



In general, $\vec{\tau} = \vec{p} \times \vec{E}$

$$\boxed{U = -\vec{p}\cdot\vec{E}}$$

Equilibrium:

$\tau_{net} = 0$, $F_{net} = 0$

① Stable equilibrium:

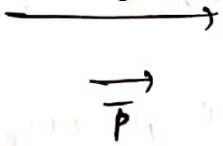
$F_{net} = 0$, $\tau_{net} = 0$

$pE\sin\theta = 0$

$\sin\theta = 0$

$\theta = 0°, \pi$

✓ ✗



② $U = U_{min}$

$-pE\cos\theta = U$

$U = -pE$ $(\theta = 0°)$ (min)

$U = pE$ $(\theta = 180°)$ (max)

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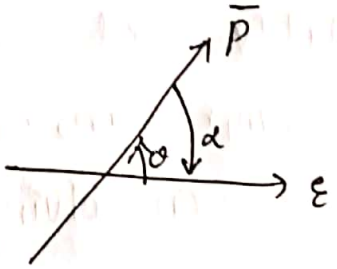
Time period of oscillation of dipole:

Let the dipole is displaced or turned through small angle 'θ' from eqm position and released.



Restoring torque,

$$\tau = -P\mathcal{E}\sin\theta$$

$$I\alpha = -P\mathcal{E}\theta$$

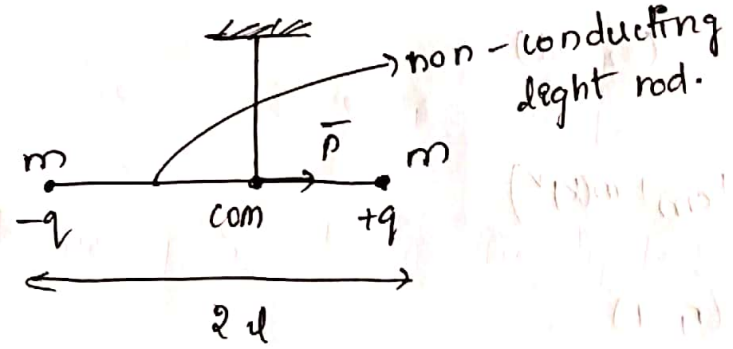
$$\alpha = -\frac{P\mathcal{E}}{I}\theta \Rightarrow \boxed{\alpha = -\omega^2\theta}$$

Angular SHM

$$\omega = \sqrt{\frac{P\mathcal{E}}{I}}$$

$$\boxed{T = 2\pi\sqrt{\frac{I}{P\mathcal{E}}}}$$

(i)



$$I = md^2 + md^2 = 2md^2$$

$$P = q(2d)$$

(ii) If rod has mass M.

$$I = md^2 + md^2 + \frac{M(2d)^2}{12}$$

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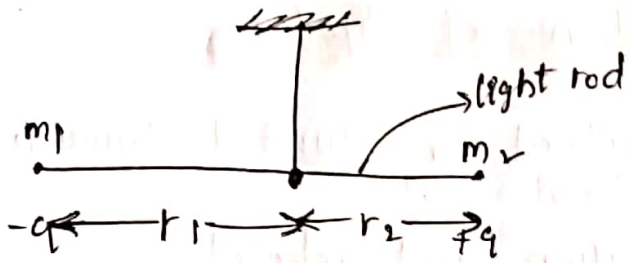
(2)



$r_1 + r_2 = 2d$

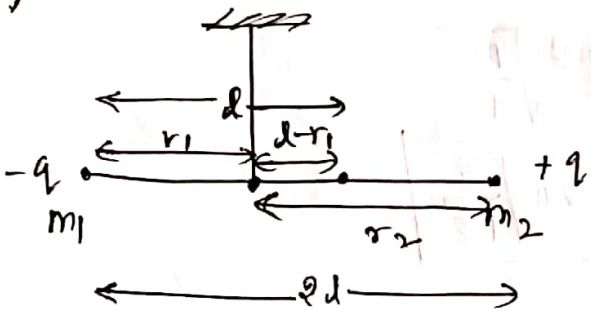
$m_1 r_1 = m_2 r_2$

On solving, $r_1 = \frac{2dm_2}{m_1+m_2}$, $r_2 = \frac{m_1(2d)}{m_1+m_2}$

$\boxed{P = q(2d)}$

$I = m_1 r_1^2 + m_2 r_2^2$

(3)



$I = m_1 r_1^2 + m_2 r_2^2 + (I_{cm} + m(k)^2)$

$k = (d - r_1)$ (or) $(r_1 - d)$

$x_{cm} = \frac{m_1(0) + m_2(2d) + M(d)}{m_1 + m_2 + M}$

Unstable eqm:

(1) $\tau_{net} = 0$, (2) $F_{net} = 0$

$\theta = 0°$ (or) $180°$

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(3) $U = U_{max} = +P\cdot E$

$\theta = 180^\circ$

Now, the dipole is slightly disturbed & released from unstable eqm, when it comes to stable eqm position,

$W_{ext}^{\,0} + W_{elec} = \Delta K$

$U_i - U_f = \Delta K$

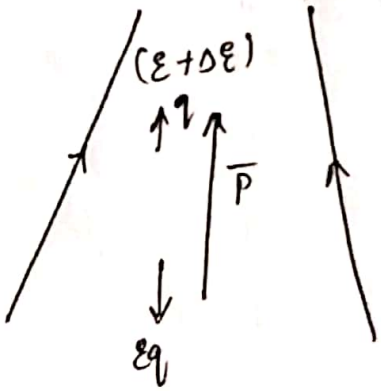
$-PE\cos 180^\circ - (-PE\cos 0^\circ) = \Delta K$

$\boxed{\Delta K = 2PE}$ → max KE

$\frac{1}{2}$ rm $\frac{1}{2} I\omega_{max}^2 = 2PE$

$$\boxed{\omega_{max} = 2\sqrt{\frac{PE}{I}}}$$

Force acting on the dipole placed in non-uniform electric field:



~~clearly~~

clearly $F_{net} \neq 0$

$F = q(E+\Delta E) - qE$

$F = q\Delta E$

$F = q\left[\frac{\Delta E}{\Delta x}\right](\Delta x)$

Here, $q(\Delta x) = P$

$\Rightarrow F = P\left(\frac{\Delta E}{\Delta x}\right)$

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For $\Delta x \to 0 \Rightarrow F = p\left(\frac{d\varepsilon}{dx}\right)$

In vector form,

$$\boxed{F = \bar{p} \cdot \frac{d\bar{\varepsilon}}{dx}} \quad *$$

II Method

$$F = -\frac{dU}{dr} \Rightarrow F = \frac{-d}{dr}(-\bar{p}\cdot\bar{\varepsilon})$$

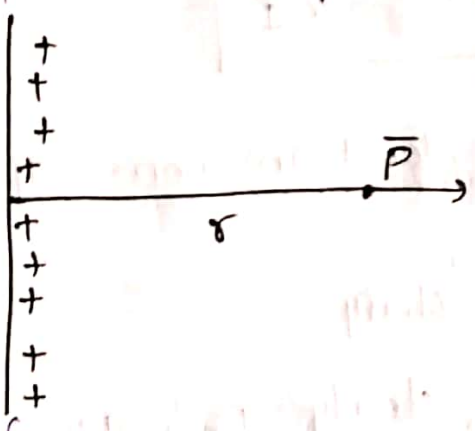
$$\boxed{F = \bar{p} \cdot \frac{d\bar{\varepsilon}}{dr}}$$

If F is +ve (repulsive force)

F = -ve (attractive force)

Eg

1)



$$\tau = \bar{p} \times \bar{\varepsilon} = 0$$

Force on the dipole,

$$\bar{F} = \bar{p} \cdot \frac{d\bar{\varepsilon}}{dr}$$

$$= p(\hat{i}) \cdot \frac{d}{dr}\left(\frac{2k\lambda}{r}(\hat{i})\right)$$

$$= p\left(\frac{-2k\lambda}{r^2}\right)(\hat{i}\cdot\hat{i})$$

$$\boxed{F = -\frac{2kp\lambda}{r^2}} \quad \text{(attractive force)}$$

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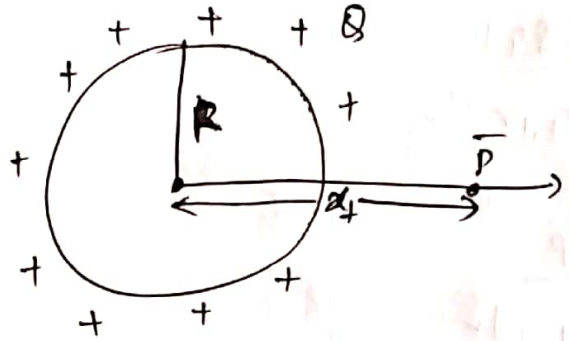
2)



Force acting on dipole,

$$F = \bar{P} \cdot \frac{d\bar{E}}{dx}$$

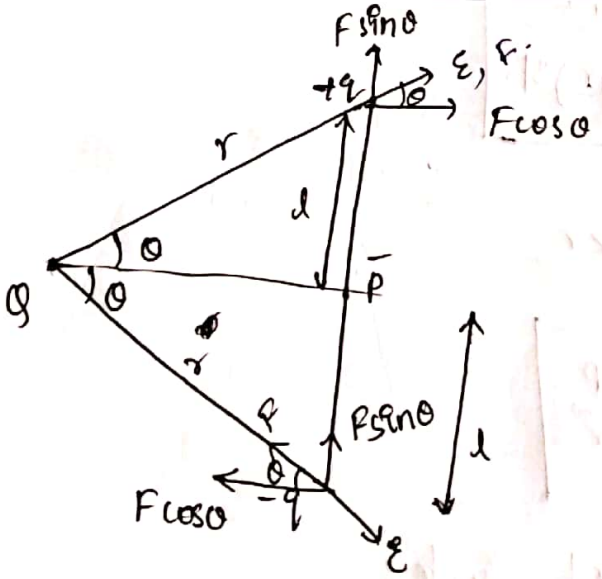
$$= P(\uparrow) \frac{d}{dx} \left(\frac{KQx}{(x^2+R^2)^{3/2}} \right) (\uparrow)$$

$$= P K Q \left[\frac{(x^2+R^2)^{3/2} - x\left(\frac{3}{2}\right)(x^2+R^2)^{1/2}(2x)}{(x^2+R^2)^3} \right]$$

at $x = R/\sqrt{1} \Rightarrow E_{max}$:

$$\frac{dE}{dx} = 0. \qquad \boxed{\bar{F} = \bar{P} \cdot \frac{d\bar{E}}{dx} = 0} \qquad \boxed{\bar{\tau} = \bar{P} \times \bar{E} = 0}$$

3)



∴ force on dipole $F_{net}^{1} = 2F\sin\theta\ (\uparrow)$

$$F_{net} = 2\left(\frac{KQq}{r^2+d^2}\right)\frac{l}{(r^2+d^2)^{1/2}}$$

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$$F_{net} = \frac{2KQqd}{(r^2+d^2)^{3/2}}$$

$$F_{net} = \frac{KQp}{(r^2+d^2)^{3/2}}$$

Force on the charge due to dipole ($\downarrow$)

$$F = \frac{KQp}{(r^2+d^2)^{3/2}} \quad (\downarrow)$$

Torque on the dipole.

$$\tau = \bar{r} \times \bar{F}$$

$$= 2dF\cos\theta$$

$$= 2d \times \frac{KQq}{r^2+d^2} \times \frac{r}{(r^2+d^2)^{3/2}}$$

$$\boxed{\tau = \frac{KQpr}{(r^2+d^2)^{3/2}}}$$

for short dipole, $d \ll r$

$$\boxed{\tau = \frac{KQp}{r^2}}$$

Dipole moment of distributed dipole s

↓

at least one extended body.

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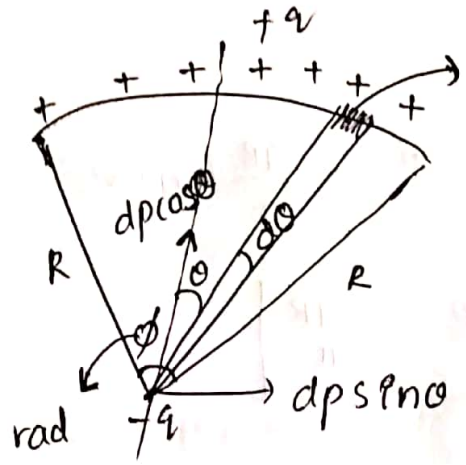
Ex:



$dp = dq\,(R)$

$= \frac{q}{\phi} R\,d\theta$

$d\bar{P} = dp\sin\theta\,\hat{\imath} + dp\cos\theta\,\hat{\jmath}$

$d\bar{P} = \frac{q}{\phi} R\sin\theta\,d\theta\,\hat{\imath} + \frac{q}{\phi} R\cos\theta\cdot d\theta\,\hat{\jmath}$

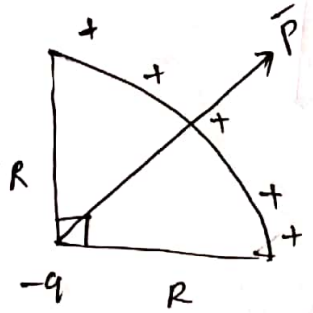
$$\bar{P} = \int_{\theta=-\phi/2}^{\theta=\phi/2} d\bar{P} = \frac{q}{\phi} R\left[-\cos\theta\right]_{-\phi/2}^{\phi/2}\hat{\imath} + \frac{q}{\phi} R\left[\sin\theta\right]_{-\phi/2}^{\phi/2}\hat{\jmath}$$

$$= \frac{q}{\phi} R\left[-\cos\phi/2 + \cos\phi/2\right]\hat{\imath} + \frac{qR}{\phi}\left[2\sin\phi/2\right]\hat{\jmath}$$

$$\boxed{\bar{P} = \frac{2qR}{\phi}\sin\phi/2}$$

Cases:

①



$\phi = \pi/2$

$$P = \frac{2qR}{\pi/2}\sin\pi/4$$

$$\boxed{P = \frac{2\sqrt{2}qR}{\pi}}$$

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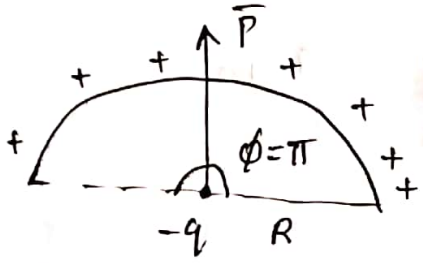
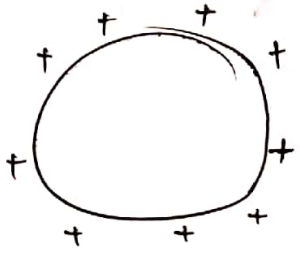
② for semi circular ring:



$$\bar{P} = \frac{2qR}{\pi} \sin\pi/2$$

$$\boxed{P = \frac{2qR}{\pi}}$$

③ for complete ring:

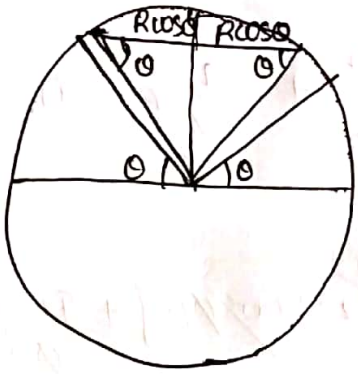


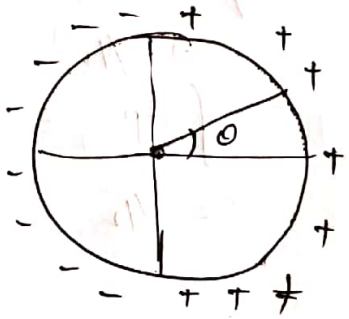
$$\phi = 2\pi$$

$$P = \frac{2qR}{2\pi} \sin\frac{2\pi}{2}$$

$$\boxed{P=0}$$

④ for non-uniform charge density:





$\lambda = \lambda_0 \cos\theta$

$$dq = \lambda R\, d\theta$$

$$= \lambda_0 R\cos\theta \cdot d\theta$$

$$dp = dq\,(2R\cos\theta)$$

$$= \lambda_0 \cos\theta\, R \cdot d\theta\,(2R\cos\theta)$$

$$dp = 2\lambda_0 R^2 \cos^2\theta \cdot d\theta$$

$$P_{net} = \int_{-\pi/2}^{\pi/2} dp = \int_{-\pi/2}^{\pi/2} 2\lambda_0 R^2 \cos^2\theta \cdot d\theta$$

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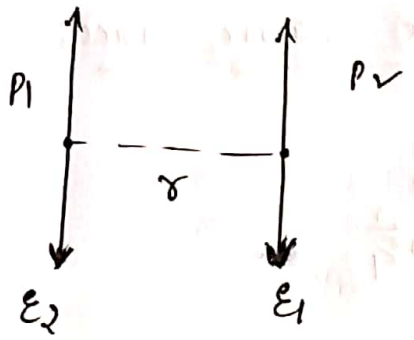
$$= 2\lambda_0 R^2 \int_{-\pi/2}^{\pi/2} \left(\frac{1+\cos 2\theta}{2}\right) \cdot d\theta$$

$$= \lambda_0 R^2 [\pi + 0]$$

$$\boxed{P_{net} = \pi \lambda_0 R^2}$$

Interaction force b/w the short dipoles

① dipoles are arranged paallel to each others



$$E_1 = \frac{KP_1}{r^3}$$

$$U_2 = P_2\left(\frac{KP_1}{r^3}\right) = \frac{KP_1P_2}{r^3}$$

Now, potential energy stored on 2nd dipole,

$$U_2 = -\vec{P_2} \cdot \vec{E_1}$$

$$= -\vec{P}_2 \cdot \vec{E_1} \cos 180°$$

$$\boxed{U_2 = P_2 E_1}$$

force acting on dipole 2 is

$$F = -\frac{dU}{dr} = -\left(\frac{d}{dr}\frac{KP_1P_2}{r^3}\right)$$

$$\boxed{F = \frac{3KP_1P_2}{r^4}} \rightarrow \text{repulsive}$$

$$\boxed{f \propto \frac{1}{r^4}}$$

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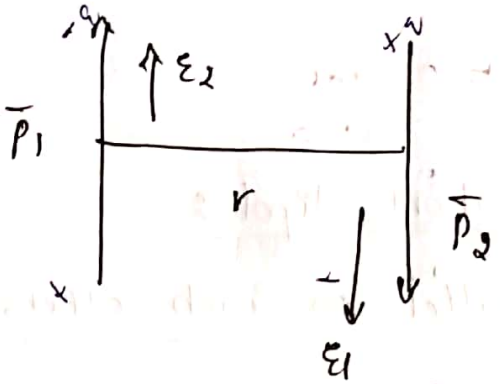
Torque on each dipole,

$$\tau_1 = \bar{p} \times \bar{\varepsilon} \qquad \bar{\tau}_2 = \bar{p}_2 \times \bar{\varepsilon}_1$$

$$= \bar{p}_1 \times \bar{\varepsilon}_2 = 0 \qquad = 0 .$$

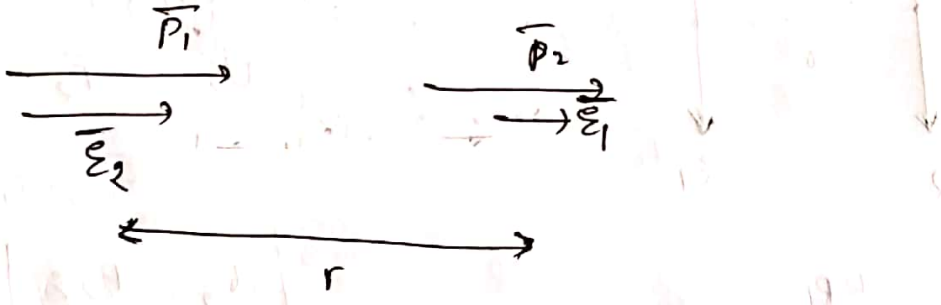
*.



$$F = \frac{-3K p_1 p_2}{r^4} \Rightarrow \text{attractive}$$

$\tau = 0$

(2) dipoles are placed on the same axes



$$U_2 = -\bar{p}_2 \cdot \bar{\varepsilon}_1$$

$$U_2 = -p_2 \varepsilon_1 \cos 0^\circ = -p_2 \varepsilon_1$$

$$U_2 = -p_2 \times \frac{2Kp_1}{r^3}$$

$$\boxed{U_2 = -\frac{2K p_1 p_2}{r^3}}$$

$$F = -\frac{dU_2}{dr} = 2Kp_1p_2\left(\frac{-3}{r^4}\right)$$

$$\boxed{F = -\frac{6Kp_1p_2}{r^4}} \Rightarrow \text{attractive} \qquad \boxed{F \propto \frac{1}{r^4}}$$

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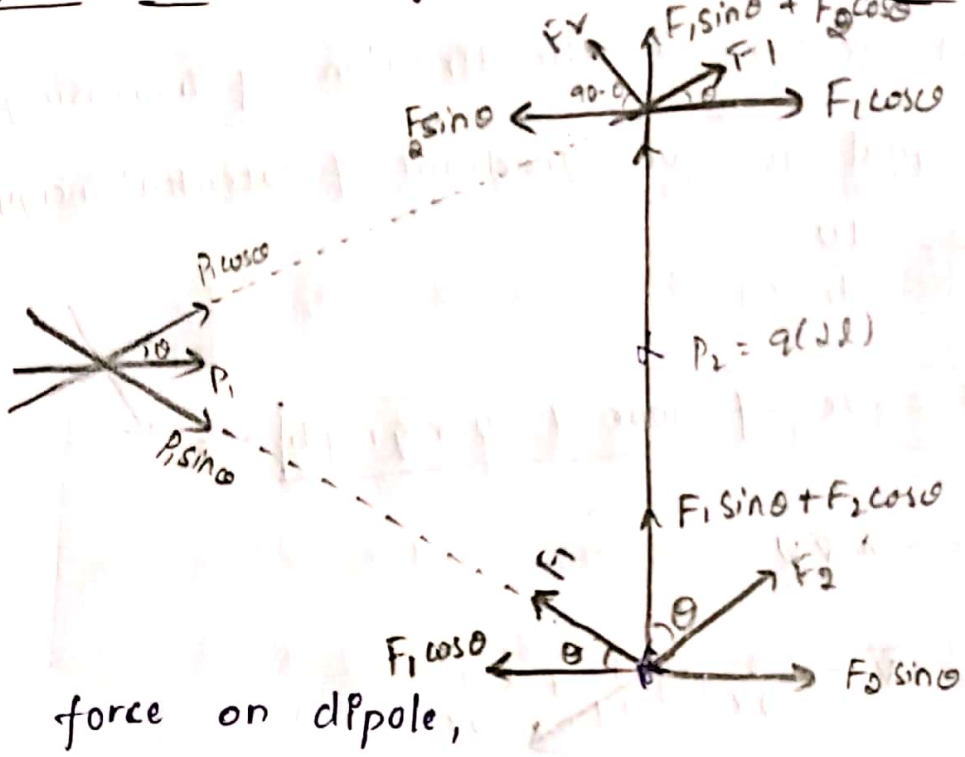
Torque on each dipole = 0

$$\tau = \bar{P} \times \bar{E} = 0$$

③ dipoles are arranged perpendicular to each other



Net force on dipole,

$$f = 2F_1 \sin\theta + 2F_2 \cos\theta$$

$$= 2\left[\frac{2KP_1\cos\theta}{(r^2+d^2)^{3/2}}\right]\sin\theta + 2\left[\frac{KP_1\sin\theta}{(r^2+d^2)^{3/2}}\right]\cos\theta$$

$$= \frac{\sin 2\theta\,(KP_1)}{(r^2+d^2)^{3/2}} \quad (3)$$

$$= \frac{3KP_1P_2}{r^4}$$

$$F = \frac{6KP_1Q}{r^3}$$

& $\tau = (F_1\cos\theta - F_2\sin\theta)\,2d \neq 0$.

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Relation b/w $\bar{F}$ & V   21/04/25 | wednesday

For a conservative force,

$$\boxed{\bar{F} = -\frac{dU}{dr}\hat{r}}$$

$\hat{r}$ = unit vector in the direction of increasing r.

i.e, 'F' is equal to -ve gradiant of Potential Energy.

& $F = -\frac{dU}{dr}$

$$\boxed{F = -ve \text{ of slope of } U\text{-}r \text{ graph}}$$

$\therefore \bar{F} = -\nabla U$

where, $\bar{\nabla} = i\frac{\partial}{\partial x} + j\frac{\partial}{\partial y} + k\frac{\partial}{\partial z}$

$$F_x\hat{i} + F_y\hat{j} + F_z\hat{k} = -\hat{i}\frac{\partial U}{\partial x} - \hat{j}\frac{\partial U}{\partial y} - k\frac{\partial U}{\partial z}$$

$$\boxed{\begin{aligned} F_x &= -\frac{\partial U}{\partial x} \\ F_y &= -\frac{\partial U}{\partial y} \\ F_z &= -\frac{\partial U}{\partial z} \end{aligned}}$$

finally,

$$\bar{F} = F_x\hat{i} + F_y\hat{j} + F_z\hat{k}$$

and $dU = -\bar{F}\cdot d\bar{r}$

$$\int_{U_i}^{U_f} dU = -\int_{i}^{f} \bar{F}\cdot d\bar{r}$$

$$\boxed{U_f - U_i = -\int_{i}^{f} \bar{F}\cdot d\bar{r}}$$

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Ex6 $U = x^2yz$. Then find conservative force at this point.

Sol6
$$F_x = -\frac{\partial U}{\partial x} = -2xyz$$

$$F_y = -\frac{\partial U}{\partial y} = -x^2z$$

$$F_z = -\frac{\partial U}{\partial z} = -x^2y$$

Finally, $\bar{F} = -2xyz\hat{i} - x^2z\hat{j} - x^2y\hat{k}$

Ex! $F = x^2\hat{i} + y\hat{j} + z^3\hat{k}$. Find change in PE blw the points $(1,-2,3)$ & $(2,2,4)$.

Sol6
$$U_f - U_i = -\int_i^f \bar{F}\cdot d\bar{r}, \quad d\bar{r} = dx\hat{i} + dy\hat{j} + dz\hat{k}$$

$$U_f - U_i = -\int_{(1,-2,3)}^{(2,2,4)} (x^2\hat{i} + y\hat{j} + z^3\hat{k})\cdot(dx\hat{i} + dy\hat{j} + dz\hat{k})$$

$$U_f - U_i = -\int_i^f x^2\,dx - \int_i^f y\,dy - \int z^3\,dz$$

$$= -\left[\frac{x^3}{3}\right]_i^f - \left[\frac{y^2}{2}\right]_i^f - \left[\frac{z^4}{4}\right]_i^f$$

$$= -\left(\frac{7}{3} + 0 + \frac{256-81}{4}\right)$$

$$=$$

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* we have,

$$F = -\frac{dV}{dr}$$

$$qE = -\frac{dV}{dr}$$

$$E = -\frac{dV}{q\,dr}$$

$$\boxed{E = -\frac{dV}{dr}} \quad (\text{or}) \quad \boxed{\vec{E} = -\frac{dV}{dr}\hat{r}}$$

clearly, $\vec{E}$ = -ve of slope of V-r graph.

& -ve sign indicates, electric intensity is always in the direction of "decreasing potential".

$$\vec{E} = -\vec{\nabla} V$$

$$E_x\hat{i} + E_y\hat{j} + E_z\hat{k} = -\hat{i}\frac{\partial V}{\partial x} - \hat{j}\frac{\partial V}{\partial y} - \hat{k}\frac{\partial V}{\partial z}$$

$$\boxed{\begin{aligned} E_x &= -\frac{\partial V}{\partial x} \\ E_y &= -\frac{\partial V}{\partial y} \\ E_z &= -\frac{\partial V}{\partial z} \end{aligned}}$$

& $dV = -\vec{E}\cdot d\vec{r}$

$$\int_i^f dV = -\int_i^f \vec{E}\cdot d\vec{r}$$

$$\boxed{V_f - V_i = -\int_i^f \vec{E}\cdot d\vec{r}}$$

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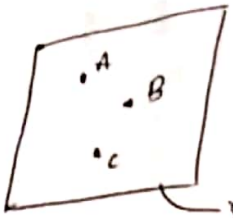
Equipotential Surface :

These are the surfaces on which potential is same at each & every point.

i.e, $V_A = V_B = V_C$

i.e, potential diff b/w any two pts = 0



Equipotential Surface

* work done in moving the charge between any two points on the surface = 0.

i.e, $W = 0$

$$\bar{F} \cdot d\bar{r} = 0$$

$$q\bar{E} \cdot d\bar{r} = 0$$

$$\bar{E} \cdot d\bar{r} = 0$$

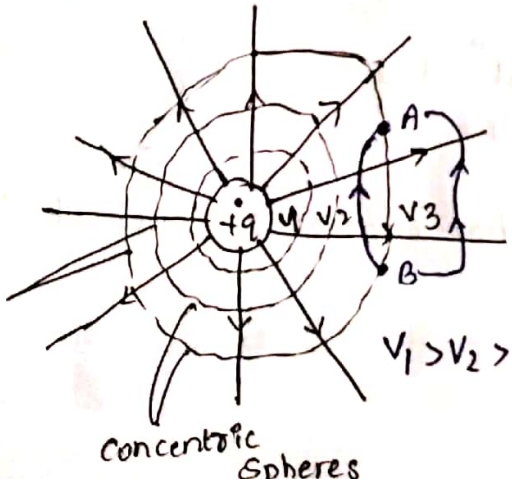
$$E \cdot dr \cdot \cos\theta = 0$$

$$\cos\theta = 0$$

$$\boxed{\theta = 90^\circ}$$

clearly Electric field lines are always perpendicular to equipotential Surface.

①



$V_A = V_B$

$W = \frac{\Delta V}{q} = 0$

EPS

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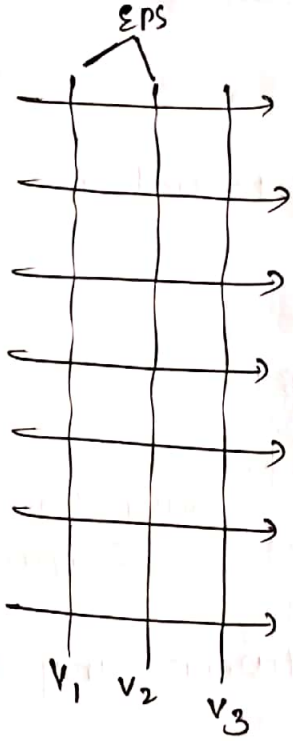
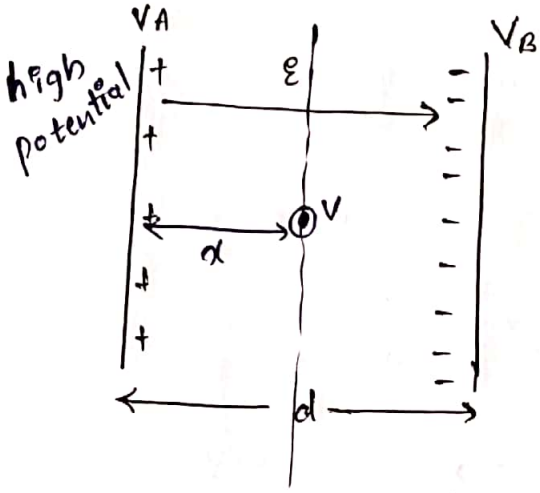
(2)



uniform $\mathcal{E}$

$$V_1 > V_2 > V_3$$

*.

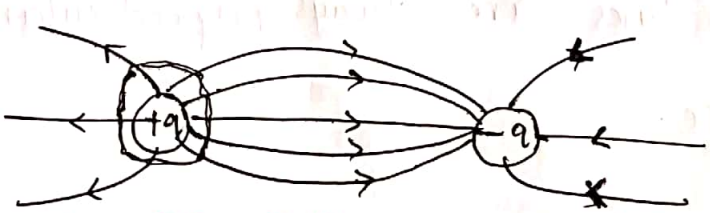


$$\mathcal{E} = \frac{V_A - V_B}{d} = \text{const}$$

(only when direction is fixed)

$$V = V_A - \left(\frac{V_A - V_B}{d}\right)x$$ *

(3)



Here. E.F Pattern disturbed due to $-q$ so, not perfect sphere.

For 'q' only we will get Exact sphere

Problems:

1) $\bar{E} = (x\hat{i} - 2y\hat{j} + 2\hat{k})$ V/m . Find potential difference (V_{AB}) b/w the points A(2,1,0) & B(0,2,4) m

Sol:

$$\int_{V_B}^{V_A} dV = -\int_{B}^{A} \bar{E}.dr$$

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$$V_A - V_B = -\int_B^A (x\hat{i} - 2y\hat{j} + z\hat{k}) \cdot (dx\hat{i} + dy\hat{j} + dz\hat{k})$$

$$= -\int_B^A x \cdot dx + \int_B^A 2y \cdot dy - \int_B^A z \cdot dz$$

$$V_{AB} = -\left[\frac{x^2}{2}\right]_B^A + 2\left[\frac{y^2}{2}\right]_B^A - \left[\frac{z^2}{2}\right]_B^A$$

$$= -\frac{1}{2}(4) + 1(1-4) - \frac{1}{2}(0-16)$$

$$= -2 - 3 + 8$$

$$\boxed{V_{AB} = 3\,V}$$

2) $\bar{E} = (2\hat{i} + 3\hat{j} + 4\hat{k})$ v/m . Find V_{AB} where $\bar{r}_A = (\hat{i} - 2\hat{j} + 4\hat{k})$ m

$\bar{r}_B = (2\hat{i} + \hat{j} - 2\hat{k})$ m

Sol:

$$V_A - V_B = -[2x]_B^A - [3y]_B^A - [4z]_B^A$$

$$= -2(1-2) - 3(-2-1) - 4(4+2)$$

$$= 2 + 9 - 24 = -13$$

3) Find electric field strength at this point.

i) $V = x(y^2 - 4x^2)$

ii) $V = a(x^2 - y^2)$

iii) $V = axy$

Sol: i) $E_x = \frac{-\partial V}{\partial x} = -\frac{\partial}{\partial x}(y^2 x - 4x^3) = -y^2 + 12x^2$

$$E_y = \frac{-\partial V}{\partial y} = \frac{-\partial}{\partial y}(y^2 x - 4x^3) = -2xy$$

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$$\bar{\varepsilon} = \varepsilon_x \hat{i} + \varepsilon_y \hat{j}$$

$$\boxed{\bar{\varepsilon} = (2x^2 - y^2)\hat{i} - 2xy\hat{j}}$$

ii) $V = a(x^2 - y^2)$

$$\varepsilon_x = -\frac{\partial V}{\partial x} = -2ax$$

$$\varepsilon_y = -\frac{\partial V}{\partial y} = +2ay$$

$$\bar{\varepsilon} = \varepsilon_x \hat{i} + \varepsilon_y \hat{j}$$

$$\boxed{\bar{\varepsilon} = -2ax\hat{i} + 2ay\hat{j}}$$

iii) $V = axy$

$$\varepsilon_x = -\frac{\partial V}{\partial x} = -ay$$

$$\varepsilon_y = -\frac{\partial V}{\partial y} = -ax$$

$$\boxed{\bar{\varepsilon} = -ay\hat{i} - ax\hat{j}}$$

4) $\bar{\varepsilon} = (4axy\sqrt{z})\hat{i} + (2ax^2\sqrt{z})\hat{j} + \frac{ax^2y}{\sqrt{z}}\hat{k}$ where $a = \text{const.}$

$$\int_{V_B}^{V_A} dV = -\int_i^f \bar{\varepsilon} \cdot d\bar{r}$$

$$0 = -\int_i^f \bar{\varepsilon} \cdot dr$$

$$\int \varepsilon \cdot dr = \int 4axy\sqrt{z} \cdot dx + \int 2ax^2\sqrt{z} \cdot dy + \int a x^2 y \times \frac{1}{\sqrt{z}} \cdot dz$$

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$$0 = \frac{4}{2}ay\sqrt{z}\cdot\frac{x^2}{2} + 2ax^2\sqrt{z}\cdot y + 2a^0x^2y\sqrt{z} + c$$

$$0 = 2ax^2y\sqrt{z} + 2ax^2y\sqrt{z} + 2a^0x^2y\sqrt{z} + c$$

$$0 = \frac{4}{2}ax^2y\sqrt{z} + 2a^0x^2y\sqrt{z} + c$$

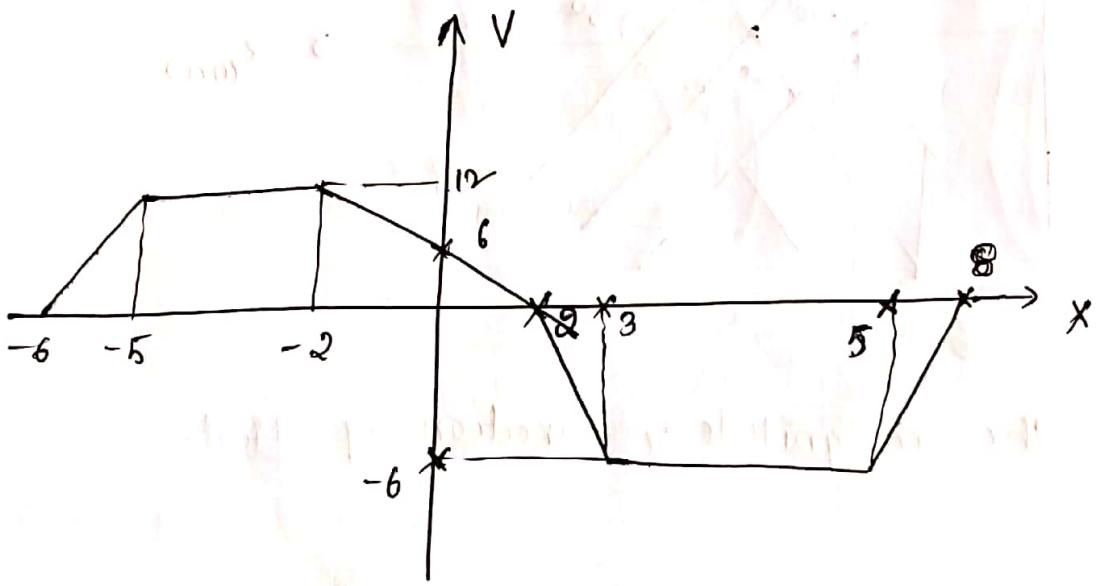
$$0 = 3x^2y\sqrt{z} + c$$

$$\frac{-c}{3} = x^2y\sqrt{z}$$

$$\frac{k}{x^2y} = \sqrt{z}$$

$$z = \frac{\text{const}}{x^4y^2}$$

Ex 6



Draw ε-x graph and find ε-values at i) $x = -5.5m$

ii) $x = -3m$

iii) $x = 6m$

iv) $x = 0m$

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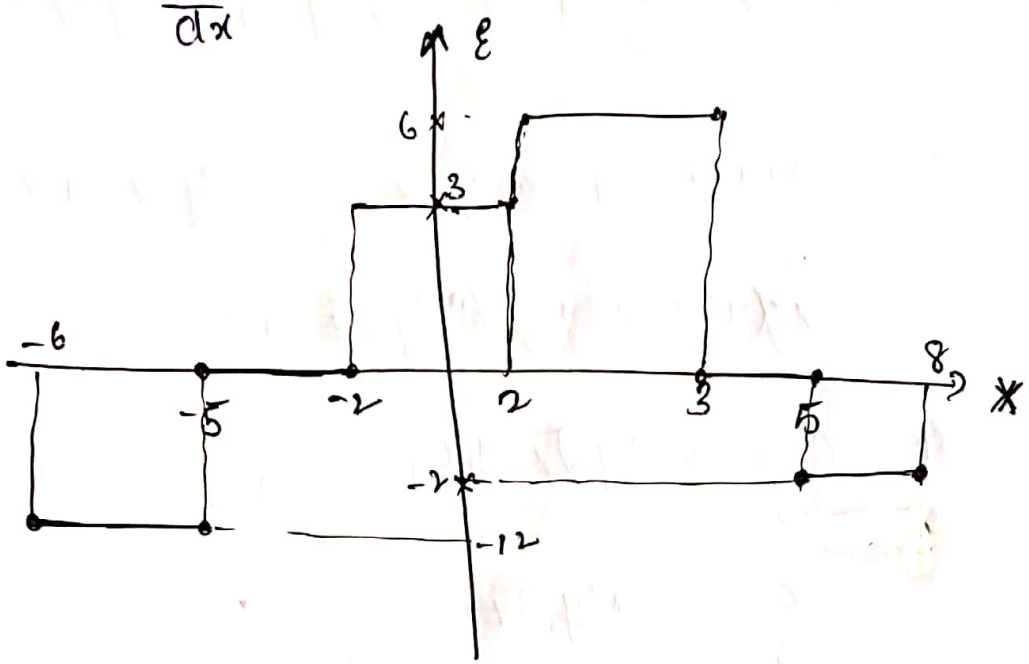
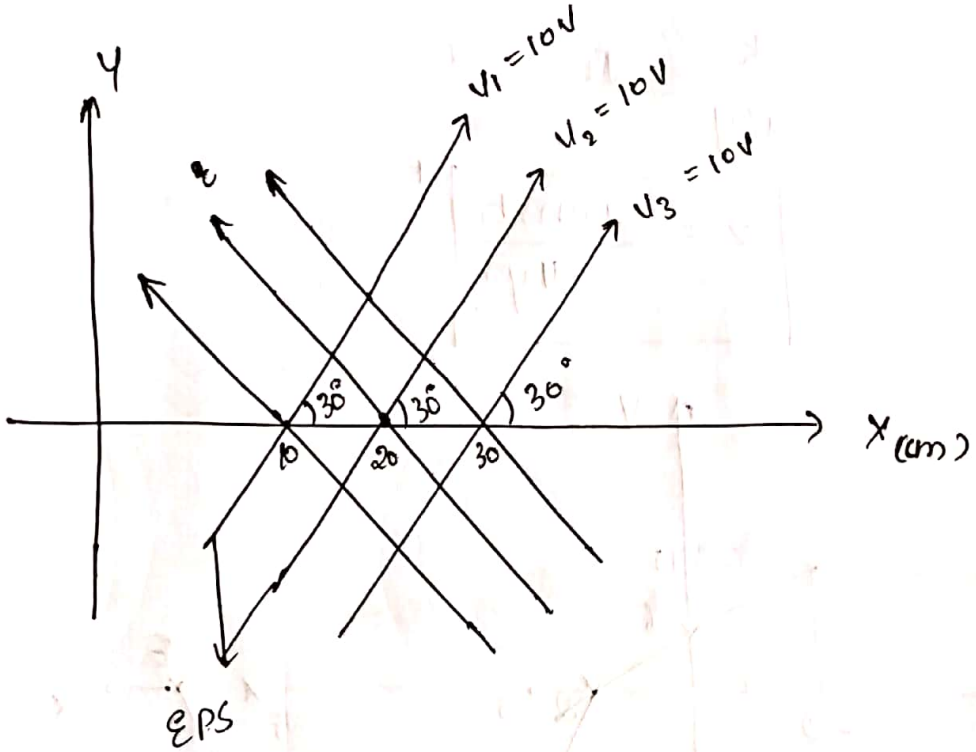
Sol^{n} $E = -\frac{dV}{dx}$



Ex^{n}



Find the magnitude & direction of field.

Sol^{n} $E = \frac{20-10}{10\sin 30^\circ}$ V/cm

$= 2$ V/cm $= 200$ V/m at an angle 120° with $+x$-axis in anticlockwise direction.

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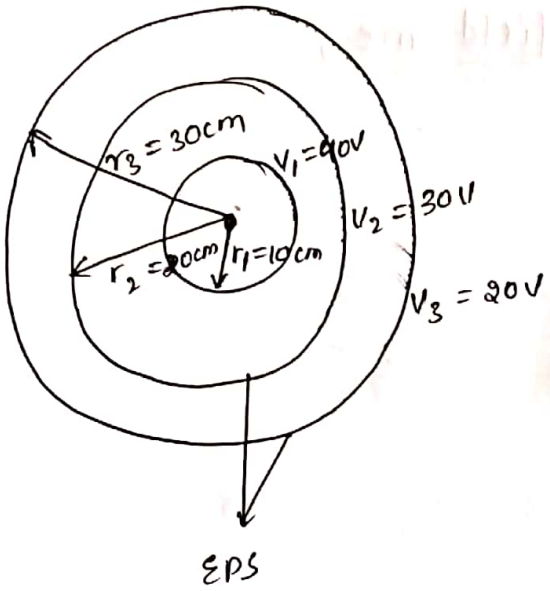
Ex:



This position is due to isolated +ve charge.

$$V = \frac{kq}{r} \Rightarrow kq = Vr = const.$$

$$kq = 60 \times 10 = V\text{-}cm$$

$$\boxed{kq = 6\ V\text{-}m}$$

now,

$$E = \frac{kq}{r^2} = \frac{6}{r^2} = \frac{6}{0.01}\ V/m$$

$$\boxed{E = 600\ V/m}$$

Ex: For a given potential $V = -kxy$ $(k > 0)$. Find the field strength at this point and draw field pattern.

Sol:

$$V = -kxy$$

$$E_x = -\frac{\partial V}{\partial x} = +ky$$

$$E_y = \frac{-\partial V}{\partial y} = +kx$$

$$\boxed{\vec{E} = ky\uparrow + kx\uparrow}$$

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Slope of the tangent of field line,

$$\frac{dy}{dx} = \frac{E_y}{E_x}$$

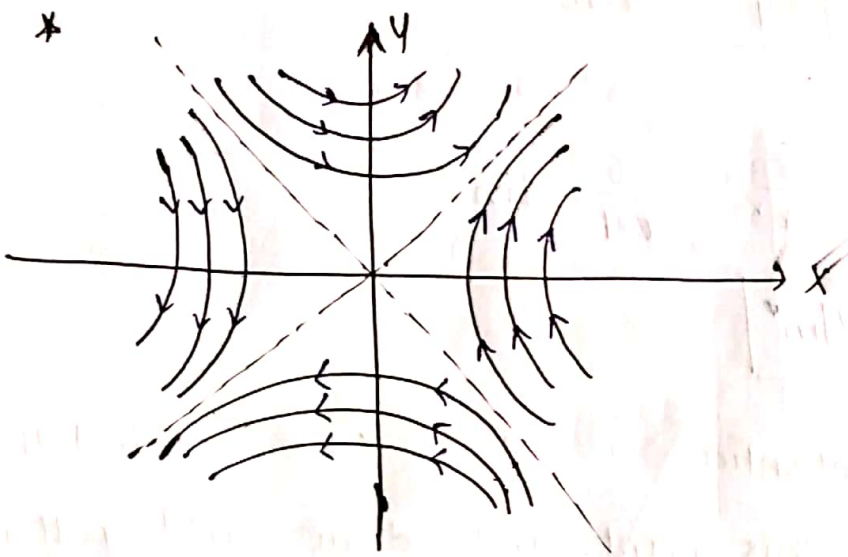
$$\frac{dy}{dx} = \frac{kx}{ky} = \frac{x}{y}$$

$$y\,dy = x\,dx$$

$$\frac{y^2}{2} = \frac{x^2}{2} + c$$

$$\boxed{x^2 - y^2 = \text{const}}$$

(rectangular) hyperbola.



Ex: Electric field at a point P in a region whose co-ordinates are x, y, z is given by $\boxed{\vec{E} = E_0 \frac{(-y\hat{i} + x\hat{j})}{x^2 + y^2}}$ where E_0 is const.

Draw field pattern.

Sol^n

$$\frac{dy}{dx} = \frac{E_y}{E_x} = -\frac{x}{y}$$

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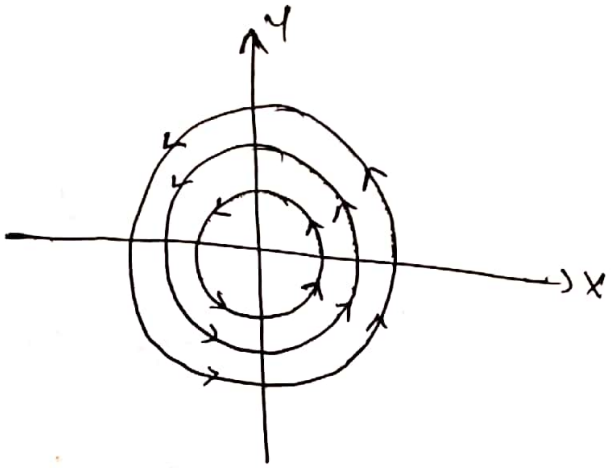
$$y \cdot dy = -x \cdot dx$$

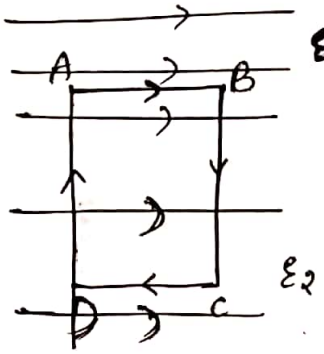
$$\frac{y^2}{2} = \frac{-x^2}{2} + c$$

$$\boxed{x^2 + y^2 = \text{const}}$$

circle.



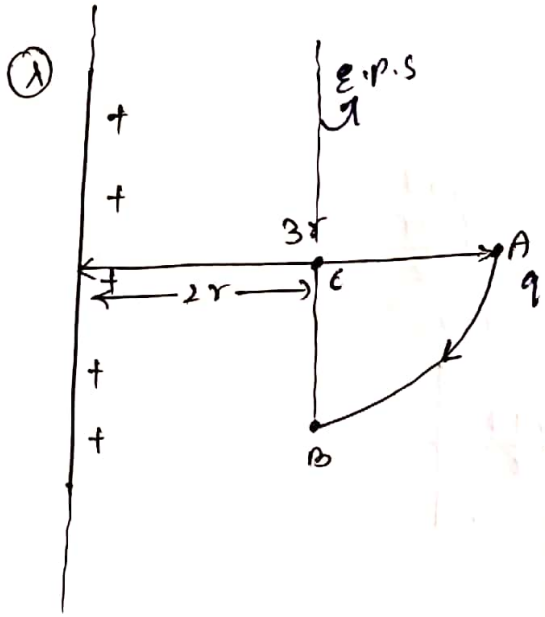
Ex6



work done is not zero.

So, Such electric field lines doesnot exist.

Ex6



Find workdone in moving charge from A to B in circular path.

Sol6

$$V_B = V_C \quad (\varepsilon \cdot p \cdot s)$$

$$W_{ext} + W_{ele} = \cancel{K}^{0}$$

$$W_{ext} = -W_{ele}$$

$$= -\int_{3r}^{2r} \bar{F}_e \cdot d\bar{r}$$

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$$W_{ext} = q \int_{2r}^{3r} \frac{2K\lambda}{x}\, dx$$

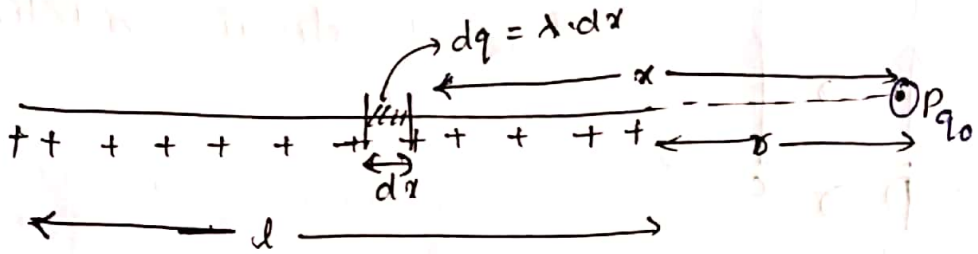
$$= 2Kq\lambda \left[\ln x\right]_{2r}^{3r}$$

$$\boxed{W_{ext} = 2Kq\lambda \ln(1.5)}$$

** Potential Due to Extended Charged Bodies:

① Thin finite length rod:

(a)



① $dV_P = \frac{K\,dq}{x} = \frac{K\lambda\, dx}{x}$

② $V_P = \int_{x=r}^{r+l} dV_P = K\lambda \int_{x=r}^{r+l} \frac{dx}{x}$

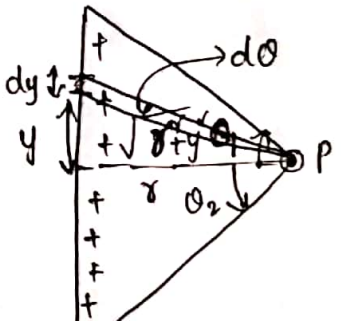
$$\boxed{V_P = K\lambda \ln\left(\frac{r+l}{r}\right)}$$

(b) when q_0 is at pt P.

P.E stored in the charge q_0,

$$U = Vq_0 = K\lambda q_0 \ln\left(\frac{r+l}{r}\right)$$

(b)



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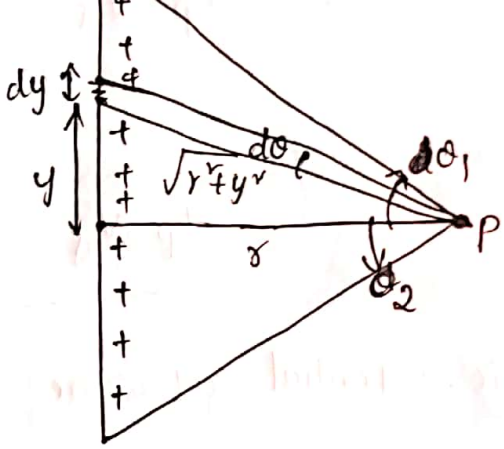
① $dV_P = \frac{K dq}{\sqrt{r^2+y^2}}$



$\theta = \theta_1$

② $V_P = \int dV_P$

$\theta = -\theta_2$

$\theta = \theta_1$

$= \int \frac{K\, r \cdot d\theta (\sec^2\theta)}{\sqrt{r^2 + r^2\tan^2\theta}}$

$\theta = -\theta_2$

$\theta = \theta_1$

$= K \int \frac{r\sec^2\theta}{r\sec\theta} \cdot d\theta$

$\theta = -\theta_2$

$\theta = \theta_1$

$= K \int \sec\theta \cdot d\theta$

$\theta = -\theta_2$

$= K \left[\ln|\sec\theta + \tan\theta|\right]_{-\theta_2}^{\theta_1}$

From figure,

$\tan\theta = \frac{y}{r}$

$y = r\tan\theta$

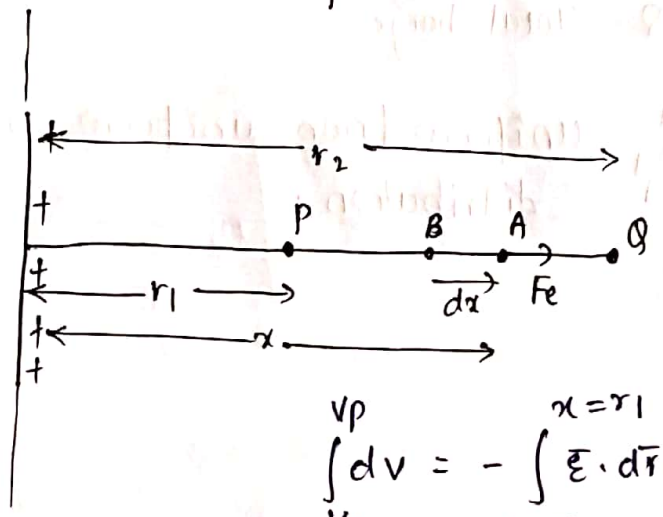
$dy = r^2\sec^2\theta \cdot d\theta$

$$\boxed{V_P = K \ln\left|\frac{\sec\theta_1 + \tan\theta_1}{\sec\theta_2 - \tan\theta_2}\right|}$$

Infinite line charge:

*Absolute potential at a pt is undefined.

*Here we can take potential diff blw 2 points.



$$\int_{V_Q}^{V_P} dV = -\int_{x=r_2}^{x=r_1} E \cdot d\bar{r}$$

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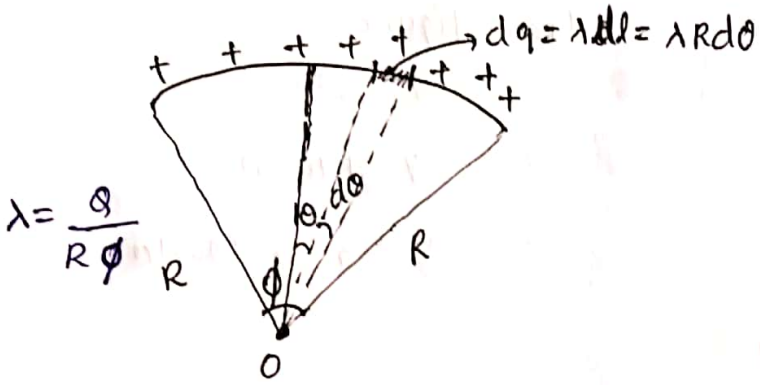
$$V_P - V_Q = \int_{x=r_1}^{x=r_2} \frac{2k\lambda}{x} \cdot dx$$

$$\boxed{V_P - V_Q = 2k\lambda \ln(r_2/r_1)}$$

Electric potential at the centre of the Arc shape charged wires



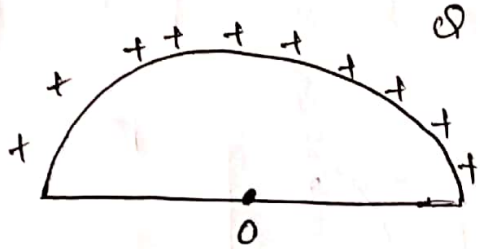
(1) $dV_\theta = \frac{k\,dq}{R} = \frac{k\lambda\phi \cdot d\theta}{R}$

(2) $V_0 = \int_{\theta=-\phi/2}^{\theta=+\phi/2} dV_0 = k\lambda \int_{-\phi/2}^{\phi/2} d\theta$

$$V_0 = k\lambda\phi = \frac{kQ}{R}$$

$$\boxed{V_0 = \frac{kQ}{R}}$$

(i)



Q — Total charge

uniform / non-uniform charge distribution.

$$V_0 = \frac{kQ}{R}$$

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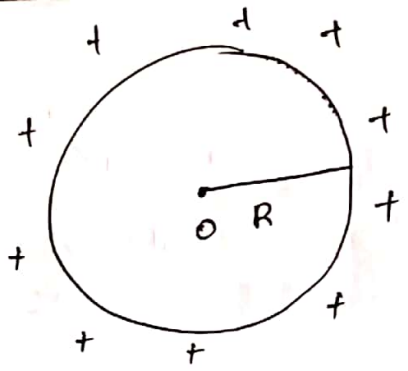
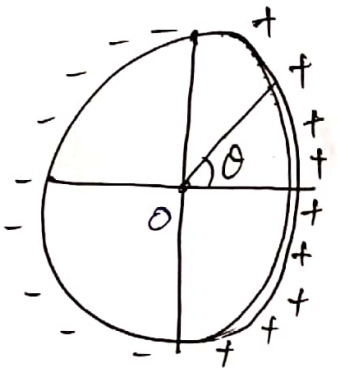
(2)



$V_0 = \frac{KQ}{R}$

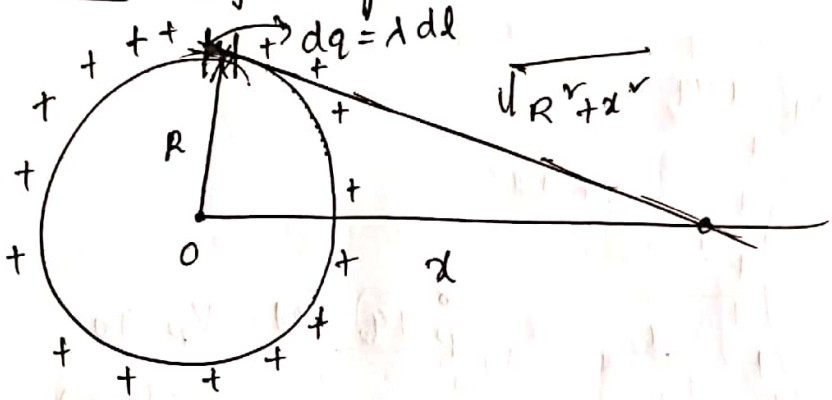
(3)



$\lambda = \lambda_0 \cos\theta$

$\boxed{V_0 = 0}$

On the Axis of ring:



$\lambda = \frac{Q}{2\pi R}$

M-I:

(1) $dV = \frac{K dq}{\sqrt{R^2+x^2}} \Rightarrow dV = \frac{K\lambda\, dl}{\sqrt{R^2+x^2}}$

(2) $V = \int dV = \frac{K\lambda}{\sqrt{R^2+x^2}} \int dl$

$$\boxed{V = \frac{2\pi K R \lambda}{\sqrt{R^2+x^2}}} \Rightarrow \boxed{V = \frac{KQ}{\sqrt{R^2+x^2}}}$$

M-II:

$$\int_{V_\infty}^{V_P} dV = -\int_{x=\infty}^{x=x} \bar{E} \cdot d\bar{r}$$

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$$V_p - V_\infty = -\int_{x=\infty}^{x=x} \frac{KQx}{(x^2+R^2)^{3/2}} \cdot dx$$

$a^2+R^2=t^2$

$2x \cdot dx = 2t \cdot dt$

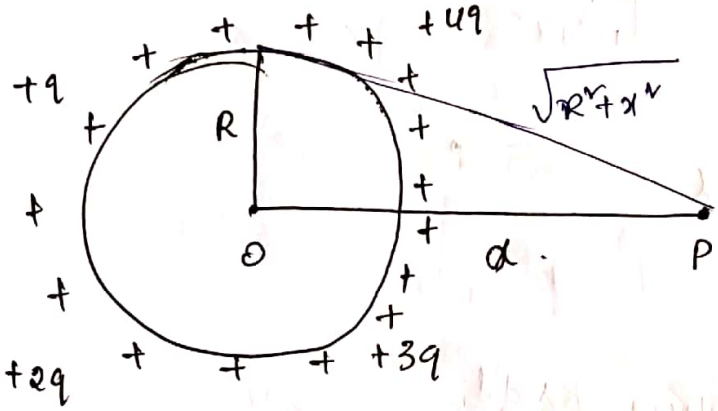
$$V_p = -\int KQ \cdot \frac{t \cdot dt}{t^3}$$

$$V_p = KQ \left[\frac{1}{\sqrt{R^2+x^2}}\right]_\infty^x$$

$$= KQ \left[\frac{1}{\sqrt{x^2+R^2}} - 0\right]$$

$$\boxed{V_p = \frac{KQ}{\sqrt{R^2+x^2}}}$$

Ex 6

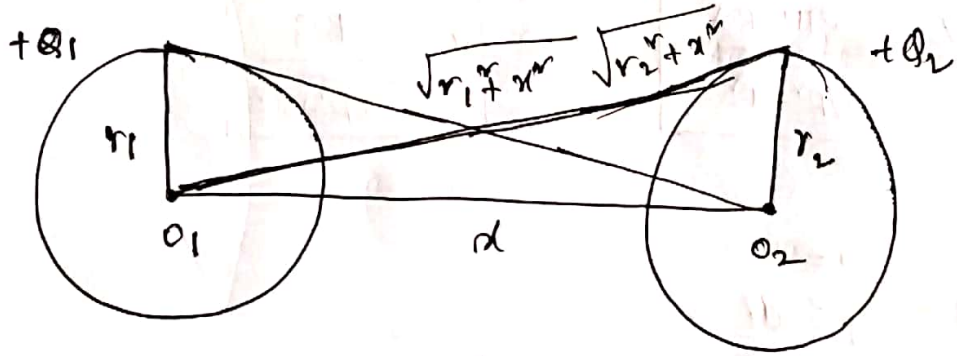


$V = V_p \cdot q_0$

$$U = \frac{K(10q) \cdot q_0}{\sqrt{R^2+x^2}}$$

$$V_0 = \frac{K(10q)}{R}, \quad V_p = \frac{K(10q)}{\sqrt{R^2+x^2}}$$

Combination of Rings :



$$V_{O_1O_2} = V_{O_1} - V_{O_2} = \left(\frac{KQ_1}{R_1} + \frac{KQ_2}{\sqrt{R_2^2+d^2}}\right) - \left(\frac{KQ_2}{R_2} + \frac{KQ_1}{\sqrt{R_1^2+x^2}}\right)$$

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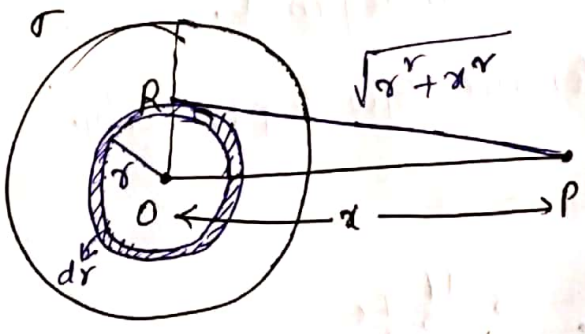
Potential Due to charged disc



$$dq = \sigma dA$$

$$= \sigma (2\pi r \cdot dr)$$

(1) $$dV_p = \frac{K dq}{\sqrt{r^2+x^2}}$$

$$dV_p = \frac{K(2\pi r)(r \cdot dr)}{\sqrt{r^2+x^2}}$$

(2) $$V_p = \int_{r=0}^{r=R} dV_p = 2\pi K\sigma \int_{r=0}^{r=R} \frac{r}{\sqrt{r^2+x^2}} \cdot dr$$

$r^2+x^2 = t^2$

$2r \cdot dr = 2t \cdot dt$

$$= 2\pi K\sigma \int \frac{t \cdot dt}{t}$$

$$= 2\pi K\sigma \left(\sqrt{r^2+x^2}\right)_{r=0}^{r=R}$$

$$= 2\pi K\sigma \left(\sqrt{R^2+x^2} - x\right)$$

$$\boxed{V_p = \frac{\sigma}{2\varepsilon_0}\left(\sqrt{R^2+x^2} - x\right)}$$

At centre of disc, $x = 0$

$$\boxed{V_p = \frac{\sigma R}{2\varepsilon_0}}$$

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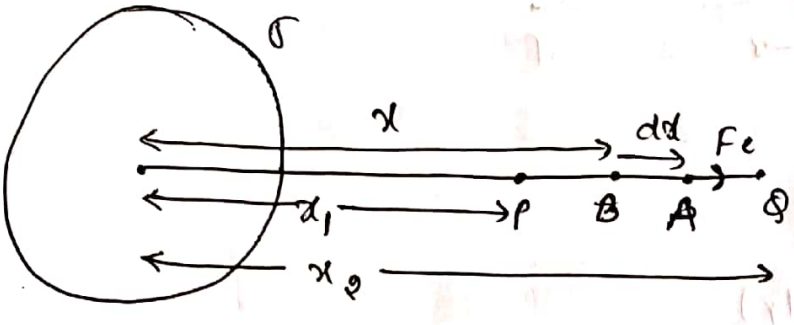
For infinite size charged sheet

Absolute potential at a pt is undefined.

we can calculate potential difference b/w 2 pts only.



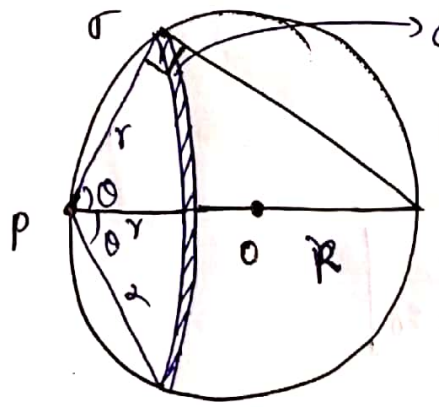
$$\int_{V_Q}^{V_P} dV = -\int_{x=x_2}^{x=x_1} \vec{E} \cdot dr$$

$$V_P - V_Q = -\int_{x=x_2}^{x=x_1} E \cdot dx$$

$$V_P - V_Q = +\int_{x=x_1}^{x=x_2} \frac{\sigma}{2\varepsilon_0} \cdot dx$$

$$\boxed{V_P - V_Q = \frac{\sigma}{2\varepsilon_0}(x_2 - x_1)}$$

Potential at one edge/rim of disc



$$dq = \sigma dA$$

$$= \sigma (2r \cdot d\theta)(dr)$$

$$(1)\ dV = \frac{K\,dq}{r}$$

$$dV = 2K\sigma\theta \cdot dr$$

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Total potential,

$$V = \int_{\theta=\pi/2}^{\theta=0} dV$$

$$= -2\sigma k \int_{\theta=0}^{\theta=\pi/2} -2R\sin\theta \cdot \theta \cdot d\theta$$

$$= 4\sigma RK \int_{\theta=0}^{\theta=\pi/2} \theta \sin\theta \cdot d\theta$$

$$V = 4K\sigma R \quad (1)$$

$$V = 4K\sigma R$$

$$\boxed{V = \frac{\sigma R}{\pi \varepsilon_0}}$$

from figure,

$$\cos\theta = \frac{r}{2R}$$

$$-2R\sin\theta \cdot d\theta = dr$$

ILATE rule:

$$\int u \cdot dv = uv - \int v \cdot du$$

$u = t$, $V = -\cos t$

$\int t \cdot \sin t \cdot dt$ $\quad dv = \sin t \cdot dt$

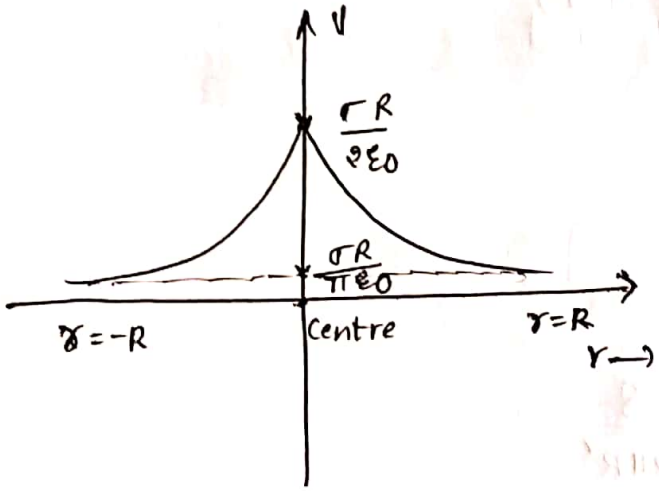
$$= -t\cos t - \int -\cos t \cdot dt$$

$$= -t\cos t + \sin t$$

$$\int_{t=0}^{\pi/2} t \cdot \sin t \cdot dt$$

$$= \left(-\frac{\pi}{2}(0) + 1\right) - (0 + 0)$$

$$= 1.$$



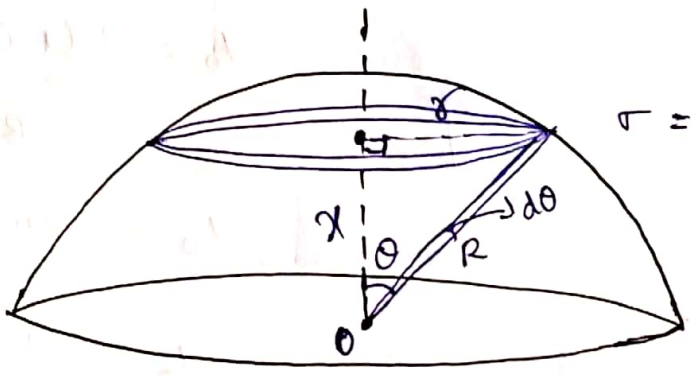
Field & potential at the centre of charged hemi-spherical shells

Electric Intensity:-

$$dq = \sigma dA$$

$$= \sigma(2\pi r \cdot dr)$$

$$= 2\pi\sigma(R\sin\theta) \cdot R d\theta$$



$$\sigma = \frac{q}{2\pi R^2}$$

from fig, $\sin\theta = \frac{r}{R} \Rightarrow r = R\sin\theta \Rightarrow dr = R d\theta$

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$$d\varepsilon_p = \frac{k\,dq\,x}{(r^2+x^2)^{3/2}} \qquad \frac{x}{R} = \cos\theta$$

$$x = R\cos\theta$$

$$= \frac{k(2\pi r R^2 \sin\theta \cdot d\theta)(R\cos\theta)}{(R^2\sin^2\theta + R^2\cos^2\theta)^{3/2}}$$

$$= \frac{2\pi r k R^3 \sin\theta\cos\theta \cdot d\theta}{(R^2)^{3/2}}$$

$$d\varepsilon_p = \pi r k (\sin 2\theta)\cdot d\theta$$

$$\varepsilon_p = \int_{\theta=0}^{\theta=\pi/2} d\varepsilon_p = \pi k r \left[\frac{-\cos 2\theta}{2}\right]_0^{\pi/2}$$

$$\varepsilon_p = \frac{\pi r}{4\pi\varepsilon_0}\left(\frac{-1}{2}\right)(-1-1)$$

$$\varepsilon_p = \frac{r}{4\varepsilon_0}\left(\frac{-1}{2}\right)(-2)$$

$$\boxed{\varepsilon_p = \frac{r}{4\varepsilon_0}}$$

Potentials

$$r = \frac{Q}{2\pi R^2}$$

$$V_0 = \frac{kQ}{R}$$

$$V_0 = \frac{k(r(2\pi R^2))}{R}$$

$$V_0 = 2\pi k r R$$

$$V_0 = \frac{2\pi r R}{4\pi\varepsilon_0} \Rightarrow \boxed{V_0 = \frac{rR}{2\varepsilon_0}}$$

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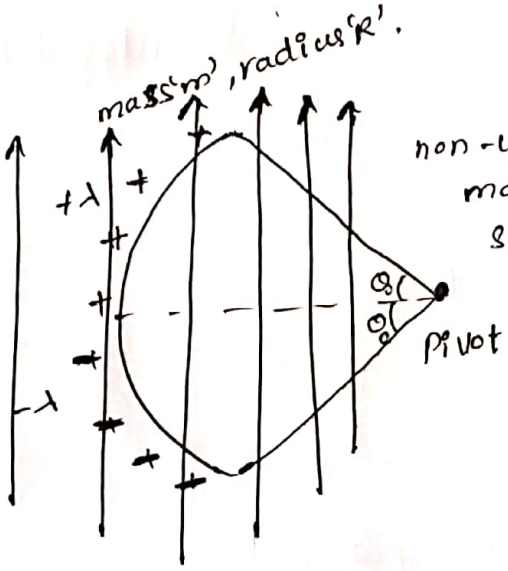
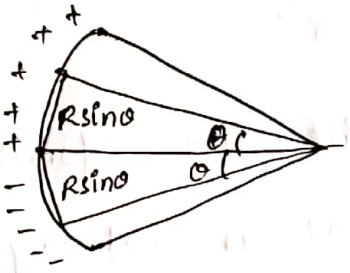
3|02|25| Thursday

Exc



non-conducting mass less system.

Show that for a small angular displacement, the system will perform SHM. determine its time period. considering the size of system to be very small determine the work required to turn it 180°. (neglect motion of charge)



Sol:

① $T = 2\pi\sqrt{\frac{I}{P\mathcal{E}}}$

Calaulation of 'P' :-

$dp = dq\,(2R\sin\theta)$

$= 2\lambda R\sin\theta\,(dl)$

$= 2\lambda R\sin\theta\,(R\,d\theta)$

$dp = 2\lambda R^2 \sin\theta \cdot d\theta$

$\theta = \theta_0$

$P = \int dP$

$\theta = -\theta_0$

$= 2\lambda R^2 \int_{-\theta_0}^{\theta_0} \sin\theta \cdot d\theta$

$= 2\lambda R^2 \left(-\cos\theta\right)_{-\theta_0}^{\theta_0}$

$\boxed{P = 2\lambda R^2 (1 - \cos\theta_0)}$

$P = 4\lambda R^2 \sin^2\theta/2$

$\boxed{I = mR^2}$

$\boxed{T = 2\pi\sqrt{\frac{mR^2}{2\lambda R^2(1-\cos\theta)}}}$

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work done

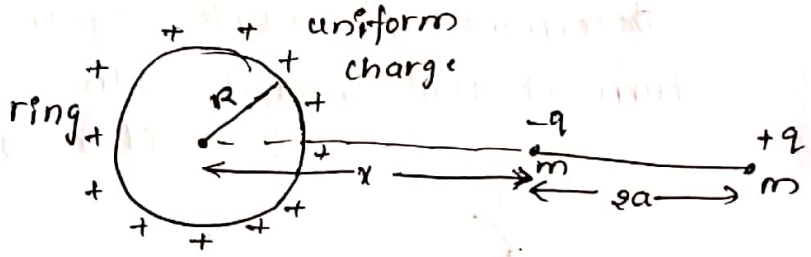
$$W = pE(\cos\theta_1 - \cos\theta_2)$$

Here $\theta_1 = 0°$ & $\theta_2 = 180°$

$$W = pE(\cos 0° - \cos 180°)$$

$$\boxed{W = 2pE}$$

Ex!



① Find the net force acting on the dipole.

② work done in rotating the dipole through 180°.

③ The dipole is slightly displaced and it's eqm position and released. Find the period of oscillation.

Sol⁸ ① $\bar{F} = \bar{p} \cdot \frac{d\bar{E}}{dx}$

$$\bar{F} = (q \times 2a)(\hat{i}) \cdot \frac{d}{dx}\left(\frac{kQx}{(x^2+R^2)^{3/2}}\hat{i}\right)$$

$$\boxed{F = \frac{aQq}{2\pi\varepsilon_0}\left[\frac{R^2-2x^2}{(R^2+x^2)^{3/2}}\right]}$$

② work done,

$$W = pE(\cos\theta_1 - \cos\theta_2)$$

here $\theta_1 = 0°$, $\theta_2 = 180°$

$$\boxed{W = 2pE}$$

③ $$T = 2\pi\sqrt{\frac{I}{pE}}$$

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$I = ma^2 + ma^2 = 2ma^2$ | $P = q(2a)$ | $\varepsilon = \frac{kqx}{(x^2+R^2)^{3/2}}$

Conservative laws:

① Law of conservation of Energy:

* Here only conservative forces are doing work.

⇓

gravitational,
electrostatic,
spring, etc..

Here $T.E = const$

$$\boxed{U_i + K_i = U_f + K_f}$$

* If external / non-conservative forces do work then we use W-ε theorm.

$$\boxed{W_c + W_{nc} + W_{ext} = \Delta K}$$

$-\Delta U$

$$W_{nc} + W_{ext} = \Delta K + \Delta U$$

$$\boxed{W_{nc} + W_{ext} = \Delta E}$$

② Law of conservation of linear momentum:

If $\Sigma F_{ext} = 0$ then $\frac{d\bar{P}}{dt} = 0$

$$\bar{P}_{sys} = const$$

i.e, $\boxed{\bar{P}_i = \bar{P}_f}$

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(3) Law of conservation of Angular Momentum

$$\Sigma \tau_{ext} = 0$$

$$\frac{d\bar{L}}{dt} = 0$$

$$\boxed{\bar{L} = \text{const}}$$

i.e, $\boxed{\bar{L_i} = \bar{L_f}}$

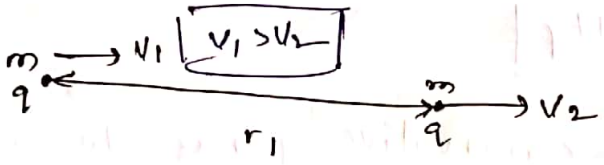
$$\boxed{\bar{L} = m(\bar{r} \times \bar{v})}$$

(or)

$$\boxed{\bar{L} = I\omega}$$

Ex:

Protons

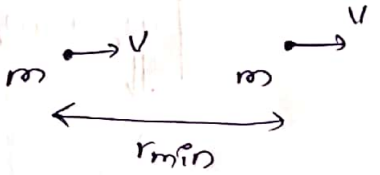


[neglect gravity]

Find min seperation b/w the charges.

Sol:

① L.C.L.M,



$$mv_1 + mv_2 = 2mV$$

$$v_1 + v_2 = 2V \quad -①$$

② L.C.E,

$$U_i + K_i = U_f + K_f$$

$$\frac{Kq^2}{r} + \frac{1}{2}mv_1^2 + \frac{1}{2}mv_2^2 = \frac{Kq^2}{r_{min}} + \frac{1}{2}mV^2 + \frac{1}{2}mV^2 \quad -②$$

solve ① & ②.

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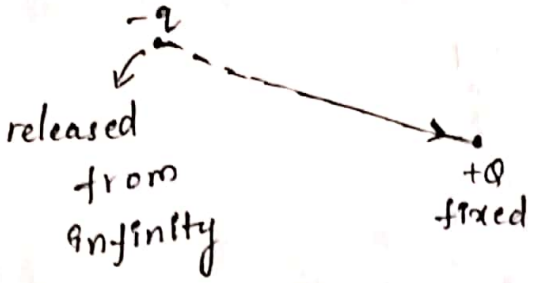
Ex:

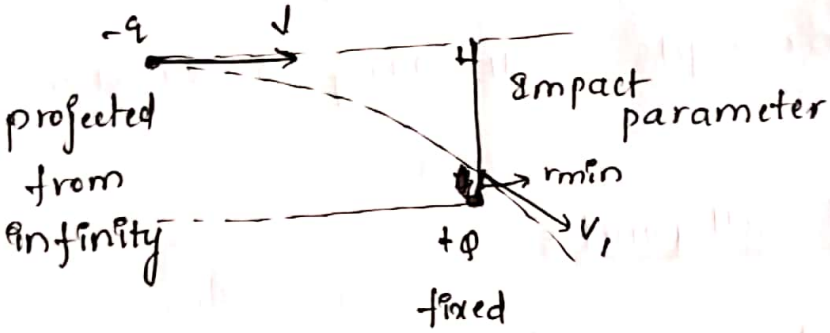


Energy = 0

Angular momentum = 0

Law of conservation of linear momentum is not valid & also angular momentum is also not conserved.

Ex:



L·C·L·M ✗

L·C·E ✓

L·C·A·M ✓

① L·C·A·M :

$mrv = mr_{min}v_1$

$rv = r_{min}v_1$

② L·C·E :

$0 + \frac{1}{2}mv^2 = -\frac{kQq}{r_{min}} + \frac{1}{2}mv_1^2$

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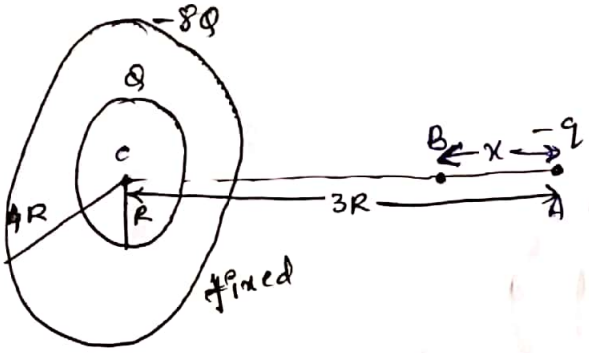
Ex 6



Find min velocity with which a point charge of mass 'm' & charge -q should be projected so that it will reach the centre of the rings.

Sol6

Initially -q charge experiences repulsive force due to -8Q. and at near pts to the ring it experiences attractive force.

*Calculation of null point:

At null point - $E_{net} = 0$.

$$\frac{K\cancel{Q}\cancel{x}}{(x^2+R^2)^{3/2}} = \frac{K(8Q)x}{(x^2+16R^2)^{3/2}}$$

$$\& \quad (x^2+R^2)^{3/2} = (x^2+16R^2)^{3/2}$$

$$4(x^2+R^2) = x^2+16R^2$$

$$4x^2 + 4R^2 = x^2+16R^2$$

$$3x^2 = 12R^2$$

$$x^2 = 4R^2$$

$$\boxed{x = 2R}$$

Min velocity is sufficient to send the particle from A to B.

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From L·C·E,

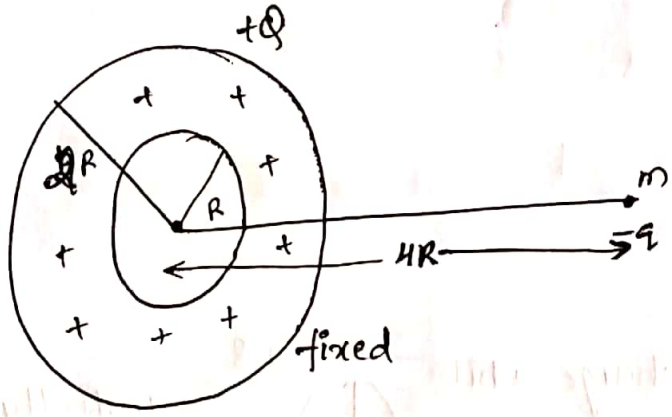
$$U_A + K_A = U_B + K_B \quad (K_B \to 0)$$

$$\frac{-KQq}{\sqrt{R^2+(3R)^2}} + \frac{K(8Q)q}{\sqrt{(4R)^2+(3R)^2}} + \frac{1}{2}mv_{min}^2 = \frac{-KQq}{\sqrt{R^2+(2R)^2}} + \frac{K(8Q)q}{\sqrt{(4R)^2+(2R)^2}} + 0$$

*To find speed of particle at centre of ring:

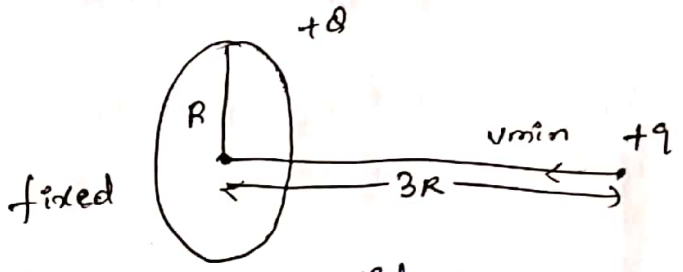
$$(T\cdot E)_B = (T\cdot E)_C$$

Ex:


+Q
+
+
+

Ex6 Find the min speed required for the charged particle so that it never turns back.

Sol:

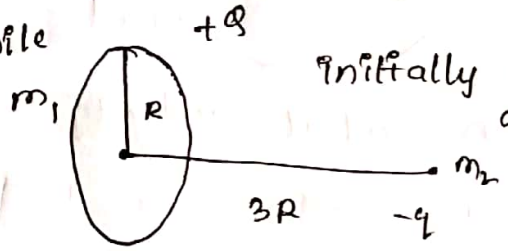


Min speed is sufficient for the particle to reach the centre of the ring.

$$U_1 + K_1 = U_2 + K_2$$

$$\frac{KQq}{\sqrt{(3R)^2+R^2}} + \frac{1}{2}mv_{min}^2 = \frac{KQq}{R} + 0$$

Ex8 Find velocity of each charge while crossing each other.



Initially both are at rest.

Sol:

L.C.L.M,

$$0 = m_1v_1 - m_2v_2$$

$$m_1v_1 = m_2v_2$$

L.C.E,

$$U_1 + \cancel{K_1}^{0} = U_2 + K_2$$

$$\frac{-K(Q)q}{\sqrt{(3R)^2+R^2}} + 0 = \frac{-KQq}{R} + \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2$$

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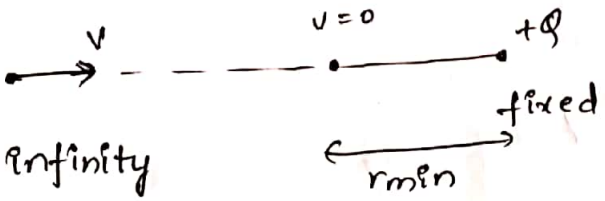
Ex1:

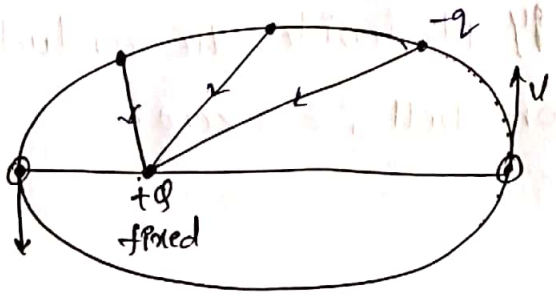


$$\cancel{U_1}^{0} + K_1 = U_2 + \cancel{K_2}^{0}$$

$$\frac{1}{2}mv^2 = \frac{K(2q)Q}{r_{min}} + 0$$

$$\boxed{\frac{1}{2}mv^2 = \frac{K(2q)Q}{r_{min}}} \Rightarrow \boxed{r_{min} = \frac{4KQq}{mv^2}}$$

Ex:



$\tau = 0$

angular momentum is conserved.

But $J \cdot \varepsilon = const.$

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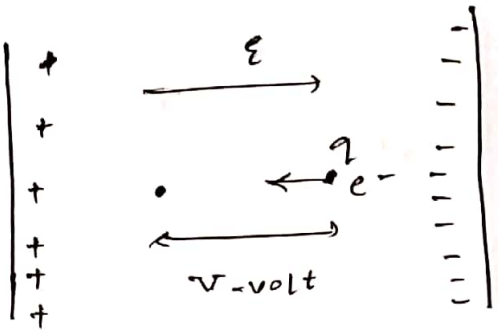
Electron Volt

It is the energy aquired by an electron when accelerated through a potential difference of 1Volt.

$W_{ele} = \Delta K$

$$\boxed{Vq = \Delta K}$$



If $V = 1$ volt

$q = 1C$

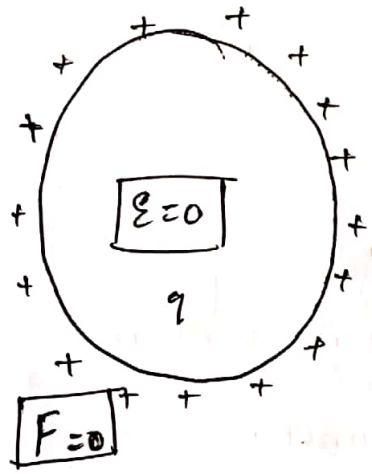
then $\Delta K = 1.6 \times 10^{-19} \times 1$ V-C

$$\boxed{1eV = 1.6 \times 10^{-19} J}$$

Electric Intensity & Potential at a point due to conducting Sphere/shell

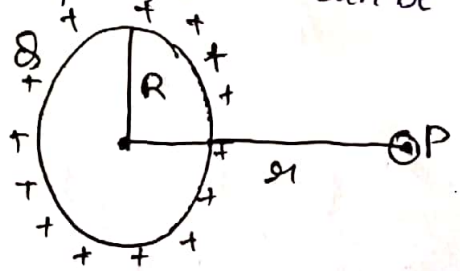
Shell Theorems

(1)



At every pt inside the conducting sphere (or shell, E = zero.

(2) For outside pts, shell can be treated as pt charge.



$$\boxed{E_p = \frac{KQ}{r^2}}$$

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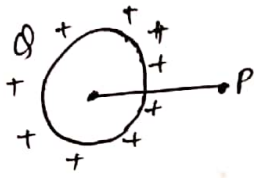
Points

① For $r < R$, $\boxed{E = 0}$



② For $r > R$, $\boxed{E = \frac{KQ}{r^2}}$

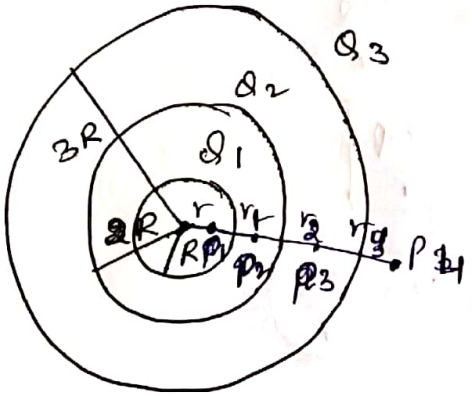
③ on the surface, $r = R$

$$\boxed{E = \frac{KQ}{R^2}}$$

$$E = \frac{1}{4\pi\varepsilon_0} \times \frac{\sigma (4\pi R^2)}{R^2}$$

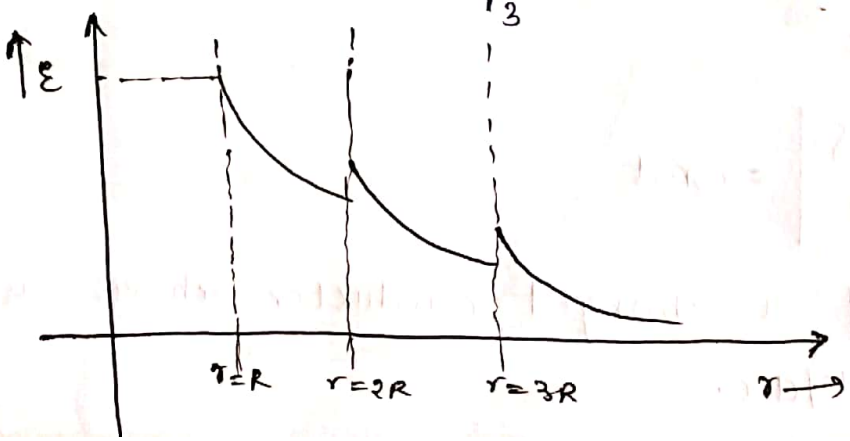
$$\boxed{E = \frac{\sigma}{\varepsilon_0}} \Rightarrow \text{For any "conductor" (Just outside)}$$

Ex:



$$E_{P_1} = 0 \qquad E_{P_2} = \frac{KQ_1}{r_1^2} \qquad E_{P_3} = \frac{K(Q_1+Q_2)}{r_2^2}$$

$$E_{P_4} = \frac{K(Q_1+Q_2+Q_3)}{r_3^2}$$



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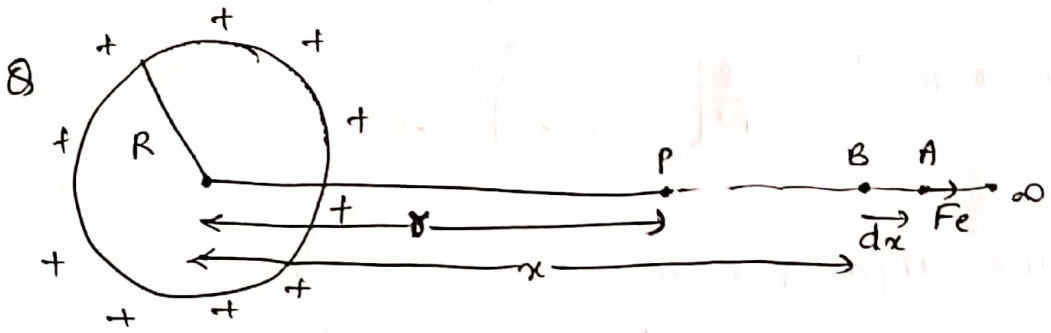
Potentials

(1) point lies outside the shell :



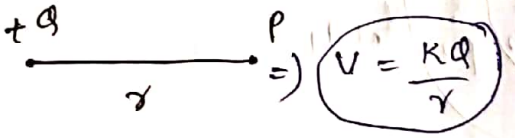
$$\int_{V_\infty}^{V_p} dV = -\int_{\infty}^{r} \vec{E}\cdot d\vec{x}$$

$$V_p - V_\infty = -\int_{\infty}^{r} E\cdot dx$$

$$V_p - V_\infty = -\int_{\infty}^{r} \frac{KQ}{x^2}$$

$$V_p - \cancel{V_\infty}^{\,0} = \frac{KQ}{r}$$

$$\boxed{V_p = \frac{KQ}{r}}$$



$\Rightarrow$ $\boxed{V = \frac{KQ}{r}}$

* clearly shell behaves as point charge when we calculating potential at out side points.

(2) point lies on the surface :

$r = R$

$$\boxed{V = \frac{KQ}{R}} = \text{const}$$

i.e, surface of a charged conductor behaves as equipotential surface.

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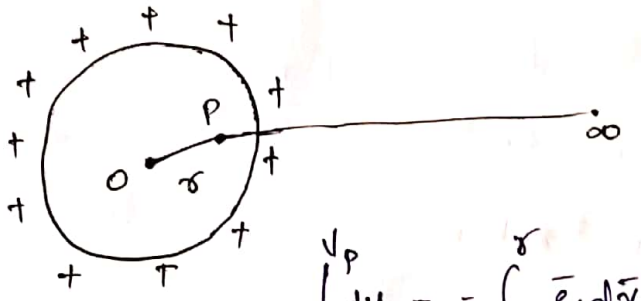
i.e, field lines are perpendicular to conducting surface.

③ point lies inside the shell :

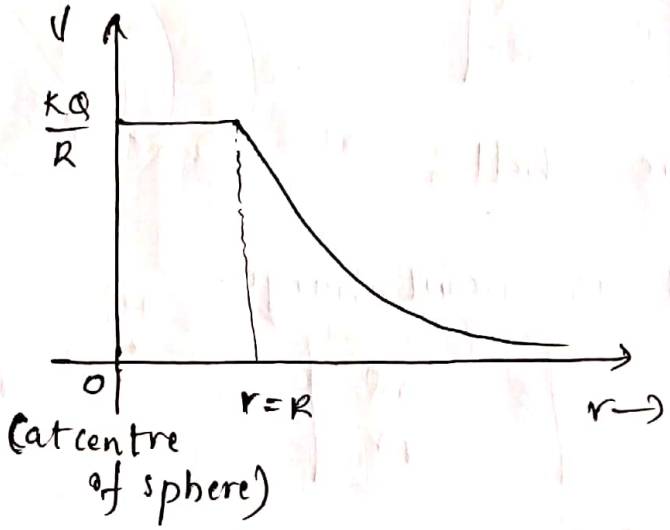


$$\int_{V_\infty}^{V_p} dV = -\int_{\infty}^{r} \bar{E}\cdot d\bar{x}$$

$$V_p - V_\infty = -\int_{\infty}^{R} E\,dx - \int_{R}^{r} E\,dx = \frac{kQ}{R}$$

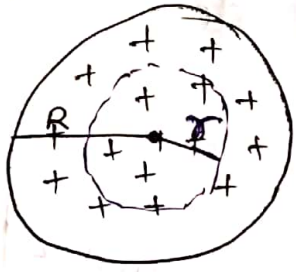
$$\boxed{V_p = \frac{kQ}{R}}$$

* potential at every pt inside the conducting shell is same/uniform and is equal to potential on it's surface.



Uniformly charged non-conducting solid sphere :

① point lies inside the shell :



$$S = \frac{Q}{\frac{4}{3}\pi R^3} = \text{const}$$

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$$E_p = \frac{kq}{r^2} \qquad q = \rho \times \frac{4}{3}\pi R^3$$

$$\varepsilon_p = \frac{1}{4\pi\varepsilon_0} \times \rho \times \frac{4}{3}\pi r^3 \times \frac{1}{r^2}$$

$$\boxed{\varepsilon_p = \frac{\rho r}{3\varepsilon_0}} \Rightarrow (\varepsilon \propto r)$$

(or) $$\varepsilon_p = \frac{\frac{4}{3}\pi k \rho r}{3\varepsilon_0 \times \frac{4}{3}\pi R^3}$$

$$\boxed{\varepsilon_p = \frac{kQr}{R^3}}$$

At centre, $r = 0 \Rightarrow \boxed{\varepsilon = 0}$

② On the surface :

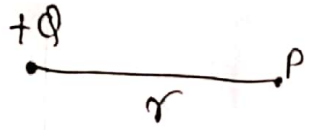
$r = R$

$$\boxed{\varepsilon_p = \frac{kQ}{R^2}}$$

③ Out side the shell :

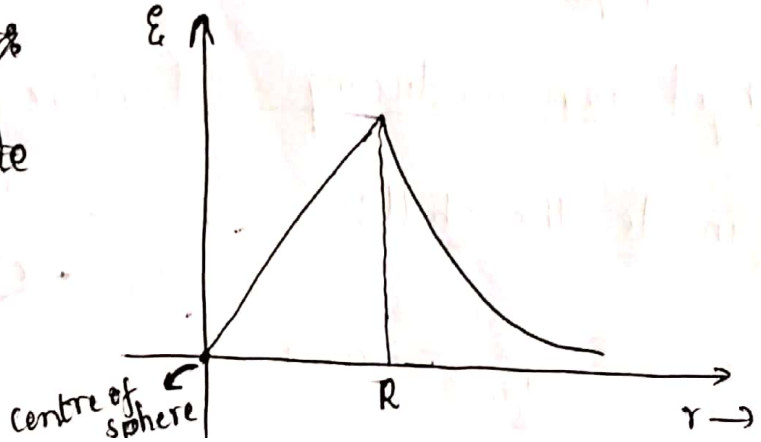
Considered as point charge.

$$\boxed{\varepsilon = \frac{kQ}{r^2}}$$



~~Potential :~~

~~① Outside~~



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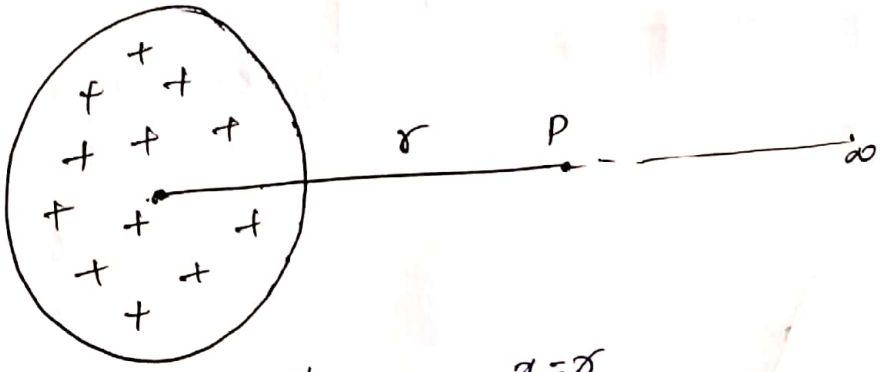
Potentials

(1) Outside point :



$$\int_{V_\infty}^{V_p} dV = -\int_{x=\infty}^{x=r} \bar{E}\cdot d\bar{x}$$

$$V_p - V_\infty = -\int_{x=\infty}^{x=r} \frac{KQ}{x^2}\cdot dx$$

$$\boxed{V_p = \frac{KQ}{r}}$$

(2) on the surface :

$$\boxed{V = \frac{KQ}{R}} \qquad r=R$$

(3) Inside point :

$$\int_{V_\infty}^{V_p} dV = -\int_{\infty}^{r} \bar{E}\cdot d\bar{x} = -\int_{\infty}^{R} E_1\cdot dx - \int_{R}^{r} E_2\cdot dx$$

$$V_p - \cancel{V_\infty}^{0} = -\int_{\infty}^{R} \frac{KQ}{x^2}\cdot dx - \int_{R}^{r} \frac{KQx}{R^3}\cdot dx$$

$$\boxed{V_p = \frac{KQ}{2R^3}(3R^2 - r^2)}$$

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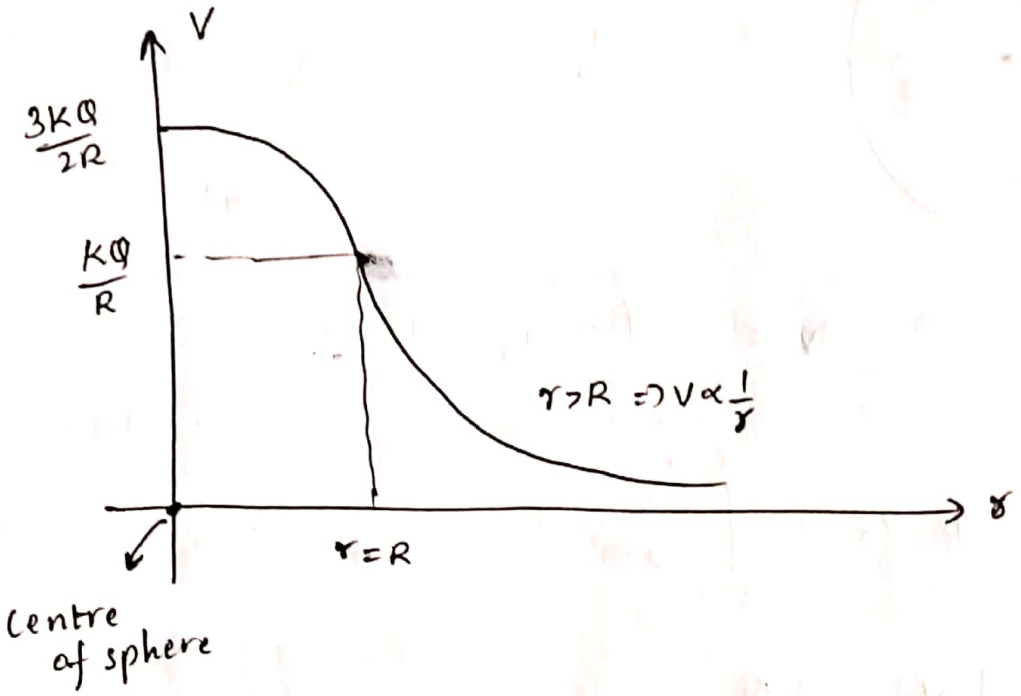
at centre,

$r = 0$

$$V_p = \frac{3KQ}{2R} = \frac{3}{2} V_{surface}$$



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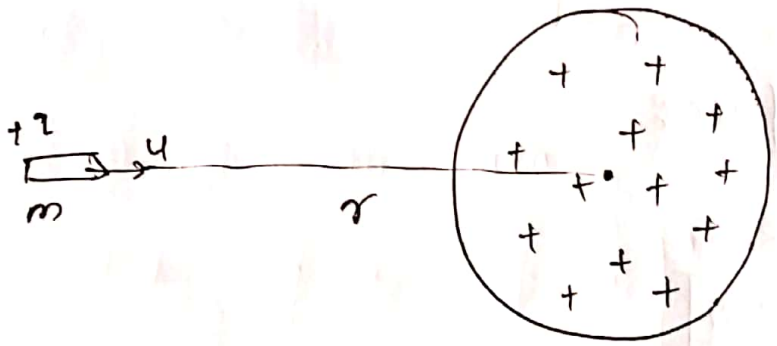

$$\frac{KQq}{r} + \frac{1}{2}mv^2 = \frac{3KQq}{2R} + 0$$

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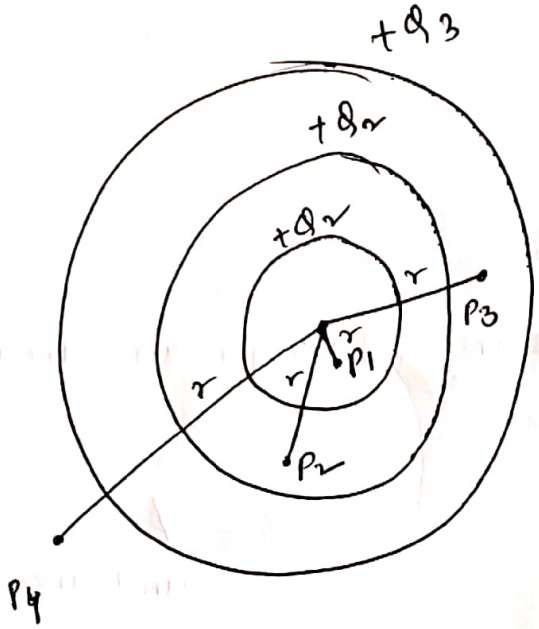
4|04|25| Friday

Ex 8



$$E_{P_1} = 0 \quad ; \quad V_{P_1} = \frac{KQ_1}{R_1} + \frac{KQ_2}{R_2} + \frac{KQ_3}{R_3}$$

$$E_{P_2} = \frac{KQ_1}{r^2} ;$$

$$V_{P_2} = \frac{KQ_1}{r} + \frac{KQ_2}{R_2} + \frac{KQ_3}{R_3}$$

$$E_{P_3} = \frac{KQ_1}{r^2} + \frac{KQ_2}{r^2} ;$$

$$V_{P_3} = \frac{KQ_1}{r} + \frac{KQ_2}{r} + \frac{KQ_3}{R_3}$$

$$E_{P_4} = \frac{KQ_1}{r^2} + \frac{KQ_2}{r^2} + \frac{KQ_3}{r^2} ;$$

$$V_{P_4} = \frac{KQ_1}{r} + \frac{KQ_2}{r} + \frac{KQ_3}{r} //$$

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Gauss's Law

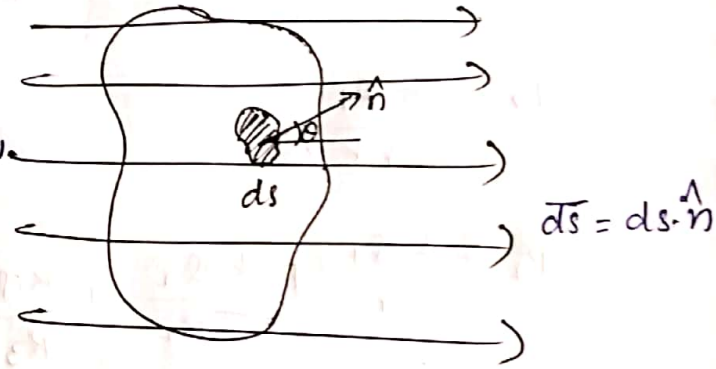
Electric flux (ϕ_E) :-

It is the no. of lines of force passing through an area is called electric flux.

$$\varepsilon = \frac{\phi}{\text{unit area}}$$

* Mathematically,

* Area is a vector. It's direction is perpendicular to it's surface.



* Flux through the small area,

$$d\phi_E = \vec{\varepsilon} \cdot d\vec{s}$$

$$d\phi_E = \varepsilon \, ds \cos\theta$$

* Total flux through entire area,

$$\phi_E = \oint_S d\phi_E = \oint_S \vec{\varepsilon} \cdot d\vec{s}$$

* ϕ_E is scalar.

* unit $\rightarrow \frac{N}{C} m^2 = N C^{-1} m^2$ (or) V-m.

* ϕ may be +ve, -ve or zero

$\theta \in [0°, 90°) \Rightarrow \phi = +ve$

$\theta \in (90°, 180°] \Rightarrow \phi = -ve$

$\theta = 90° \Rightarrow \phi = 0$.

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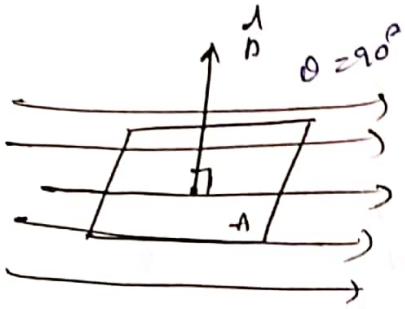
*

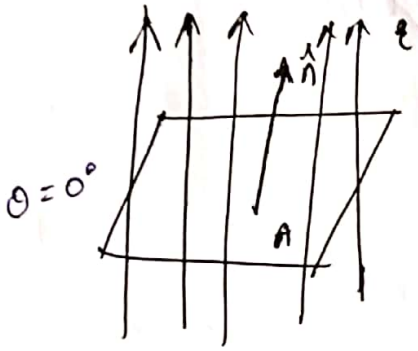


$\mathcal{E}$

$$\phi = \bar{E} \cdot \bar{A} = EA\cos 90° = 0$$

$$\boxed{\phi = 0}$$

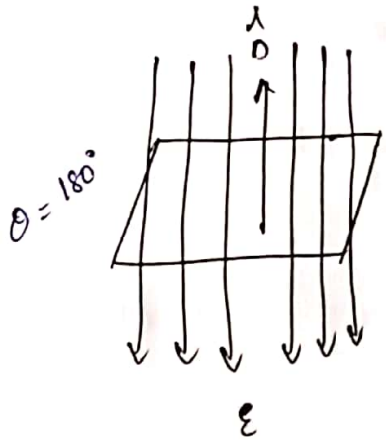
*



$$\phi = EA\cos 0°$$

$$\boxed{\phi = EA}$$

*



$$\phi = EA\cos 180°$$

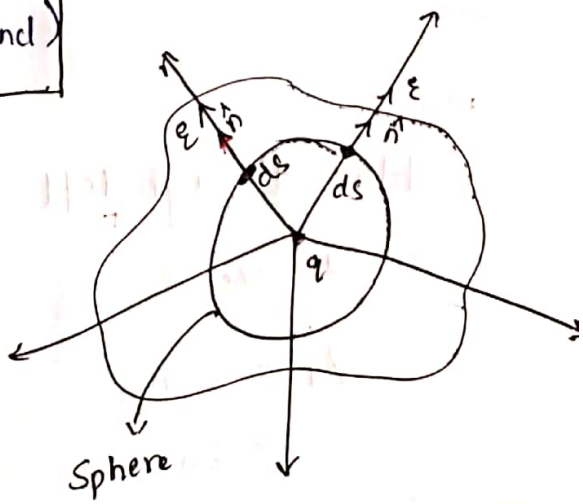
$$\boxed{\phi = -EA}$$

Gauss's Law :-

statement -

The total normal electric flux passed through a closed surface is equal $\frac{1}{\varepsilon_0}$ times the net charge enclosed by that surface.

$$\boxed{\phi_E = \oint_S \bar{E} \cdot d\bar{s} = \frac{1}{\varepsilon_0}(q_{net\ encl})}$$



proof :-

Flux through elementary area,

$$d\phi_E = \bar{E} \cdot d\bar{s} = E\,ds\cos 0°$$

$$d\phi_E = E\,ds$$

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Total flux, $\phi_\varepsilon = \oint_s d\phi_\varepsilon = \oint_s \varepsilon \cdot ds$

$$\phi_\varepsilon = \varepsilon \oint_s ds$$

$$\phi_\varepsilon = \frac{1}{4\pi\varepsilon_0} \frac{q}{r^2} (4\pi r^2)$$

$$\boxed{\phi_\varepsilon = \frac{q}{\varepsilon_0}}$$

Guass law: $\boxed{\phi_\varepsilon = \oint_s \bar{\varepsilon} \cdot d\bar{s} = \frac{q}{\varepsilon_0}}$.

★ Gravitation:

$$\phi_g = \oint_s \bar{\varepsilon} \cdot d\bar{s}$$

$$\phi_g = m(4\pi G)$$

$$\boxed{\phi_g = 4\pi G m}$$

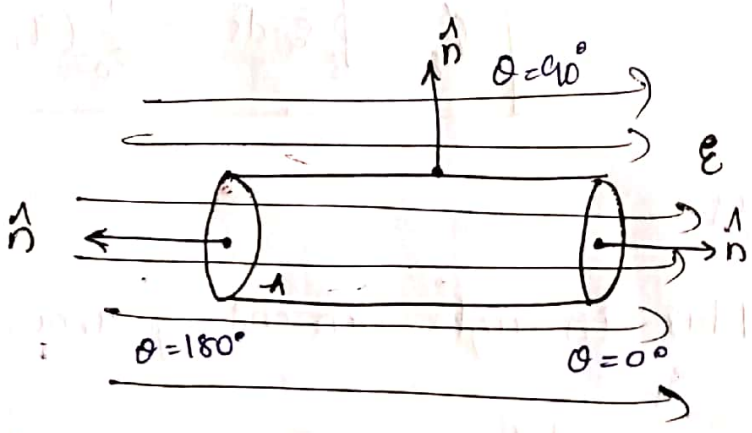
m = total mass enclosed by closed surface

Points:

① Flux through left surface,

$$\phi_1 = \bar{\varepsilon} \cdot \bar{A}$$

$$= \varepsilon A \cos 180^\circ = -\varepsilon A \text{ (entering flux)}$$



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flux through right surface,

$$\phi_2 = \vec{E} \cdot \vec{A} = EA\cos 0^\circ$$

$$\boxed{\phi_2 = EA}$$ (leaving flux)

flux through curved surface,

$$\phi_3 = \vec{E} \cdot \vec{A} = EA\cos 90^\circ = 0$$

∴ net flux through the closed surface,

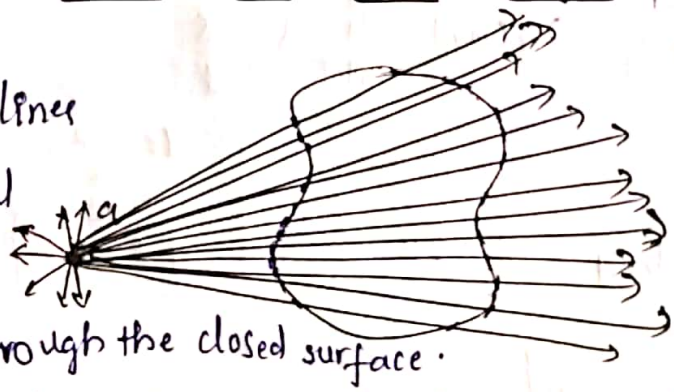
$$\phi = \phi_1 + \phi_2 + \phi_3$$

$$= -EA + EA + 0$$

$$\boxed{\phi = 0}$$

Note: net flux through any closed surface placed in external field is always zero.

② charge lies outside the closed surface:

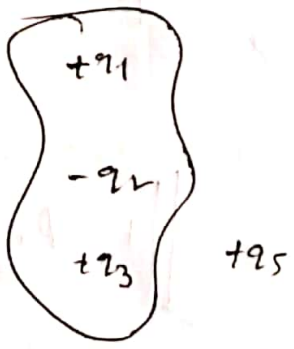


* No. of field lines entering is equal to No. of field lines leaving through the closed surface.

So, Net flux through the closed surface is zero.

③ net flux enclosed by closed surface,

$$\phi = \frac{q_{net}}{\varepsilon_0}$$



$$\phi = \frac{q_1 - q_2 + q_3}{\varepsilon_0}$$

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(4) In Guass's Law,

$$\oint_S \bar{E} \cdot d\bar{s} = \frac{q}{\varepsilon_0}$$

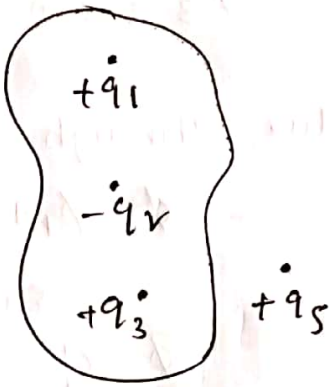
E = Net field at any point due to all charges that lie inside & outside the closed surface

q = net charge ~~enclosed~~ enclosed by closed surface only.

*.

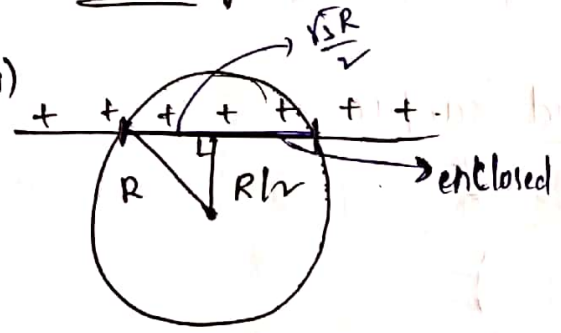
$$\oint_S (\bar{E_1} + \bar{E_2} + \bar{E_3} + \bar{E_4} + \bar{E_5}) \cdot ds = \frac{q_1 - q_2 + q_3}{\varepsilon_0}$$

*.



* Line charge :-

1)



$\lambda = \frac{Q}{l}$

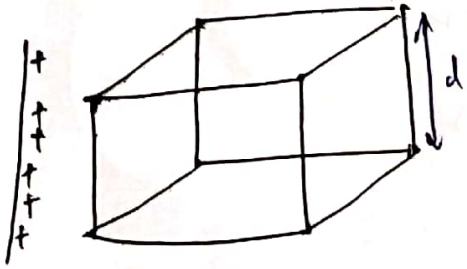
$l = \frac{\sqrt{3}R}{2} + \frac{\sqrt{3}R}{2}$

$q_{encl} = \lambda \sqrt{3} R$

$l = \sqrt{3}R$

$$\phi_E = \frac{\sqrt{3}\lambda R}{\varepsilon_0}$$

2)



$\phi_{min} = \frac{\lambda l}{\varepsilon_0}$ { line charge is placed along one of the face }

$\phi_{max} = \frac{\sqrt{3}\lambda l}{\varepsilon_0}$ { line charge is placed along body diagonal. }

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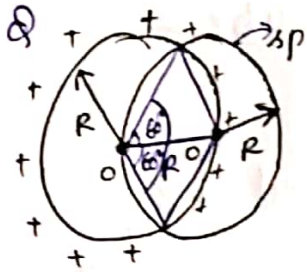
3) 

~~$\phi = \frac{Q}{\varepsilon_0}$~~

~~$\phi = 4\pi R^2$~~

flux enclosed by sphere,

$$\phi_{enclosed} = \frac{Q}{3\varepsilon_0}$$

$360° \rightarrow Q$

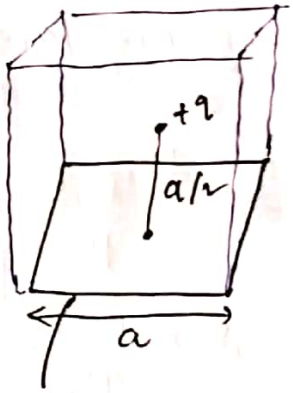
$120° \rightarrow \frac{Q}{360} \times 120 = \frac{Q}{3}$

Symmetry conditions

①

Flux through the cylinder

$$\phi = \frac{q}{2\varepsilon_0}$$

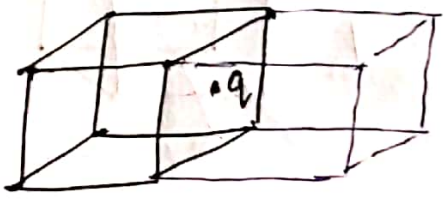
② 

square plate

Flux through each face (plate)

$$\phi = \frac{q}{6\varepsilon_0}$$

flux through cube $= \frac{q}{\varepsilon_0}$.

③ 

Flux through each cube,

$$\phi = \frac{q}{2\varepsilon_0}$$

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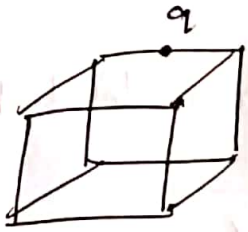
(4) Charge at midpt of edge of cube :-

we need total 4 cubes to enclosed charge.

So, flux through each cube is

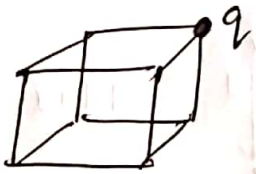
$$\boxed{\phi = \frac{q}{4\varepsilon_0}}$$



4 faces $\Rightarrow \frac{q}{4\varepsilon_0}$

(5) Charge at on corner of the cube :-

$$\boxed{\phi = \frac{q}{8\varepsilon_0}}$$



flux through the faces adjacent to the corner is zero.

Therefore, total flux through remaining three faces is

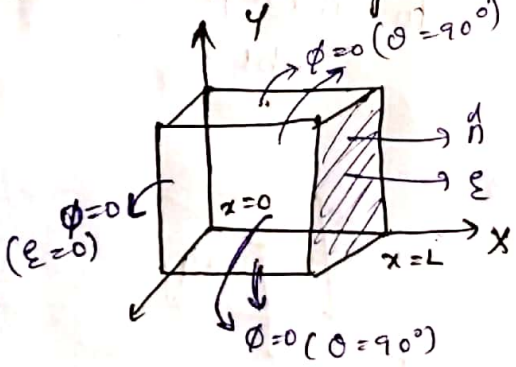
$$\boxed{\phi = \frac{q}{8\varepsilon_0}}$$

flux through each face is $\phi = \frac{q}{24\varepsilon_0}$.

Ex 6 For the func $\boxed{\vec{E} = E_0 x^2 \hat{i}}$; Find the net charge enclosed by the cube as shown.

$$\phi = \oint_S \vec{E} \cdot d\vec{s}$$

$$\phi = \vec{E} \cdot \vec{A}$$



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$$\phi = EA\cos 0^\circ$$

$$= (E_0 L^2)(L^2)$$

$$\boxed{\phi = E_0 L^4}$$

$$\phi = \frac{q_{en}}{\varepsilon_0}$$

$$\boxed{q_{en} = \varepsilon_0 E_0 L^4}$$

$q_{en} = \varepsilon_0 E_0 L^4$

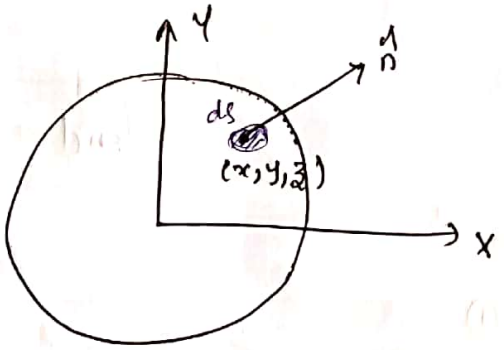
$\rho[L^3] = \varepsilon_0 E_0 L^4$

$$\boxed{\rho = \varepsilon_0 E_0 L}$$

Avg charge density

Ex: The Intensity of electric field depends only on position co-ordinates as follows $\varepsilon = \frac{a(x\hat{i}+y\hat{j})}{x^2+y^2}$ where a is const Find the charge with in the sphere of radius R with the centre at origin.

Sol:



we have,

$$\bar{R} = x\hat{i}+y\hat{j}+z\hat{k}$$

$$\hat{n} = \frac{\bar{R}}{|\bar{R}|} = \frac{x\hat{i}+y\hat{j}+z\hat{k}}{\sqrt{x^2+y^2+z^2}}$$

now, flux through the small area,

$$d\phi_\varepsilon = \bar{\varepsilon}\cdot d\bar{s}$$

$$= \frac{a(x\hat{i}+y\hat{j})}{x^2+y^2}\cdot ds\left(\frac{x\hat{i}+y\hat{j}+z\hat{k}}{R}\right)$$

$$= \frac{a(x^2+y^2)}{(x^2+y^2)R}\cdot ds$$

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$$d\phi_E = \frac{a}{R} ds$$

Total flux,

$$\phi_E = \oint_S d\phi_E = \oint \frac{a}{R} \cdot ds$$

$$\phi_E = \frac{a}{R} (4\pi R^2)$$

$$\boxed{\phi_E = 4\pi a R}$$

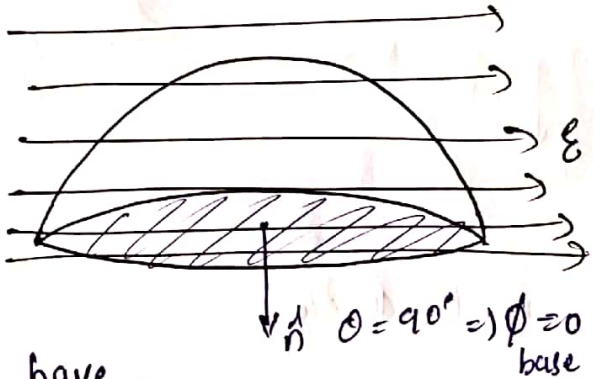
From gauss's law,

$$\phi_E = \frac{q_{encl}}{\varepsilon_0}$$

$$4\pi a R = \frac{q_{encl}}{\varepsilon_0}$$

$$\boxed{q_{encl} = 4\pi \varepsilon_0 a R}$$

Ex:6 ①



Find flux through curved surface of hemi sphere.

Sol:6 we have,

$\phi_E = 0$ (for hemi sphere)

$\phi_{base} + \phi_{curved\ surface} = 0$

$$\boxed{\phi_{curved\ surface} = 0}$$

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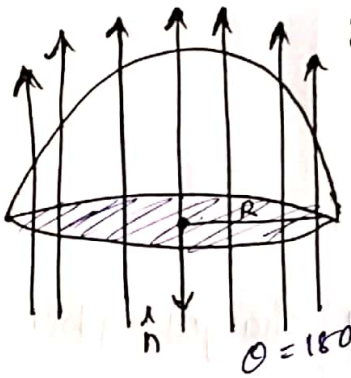
(2)



$\theta = 180^\circ \Rightarrow \phi_{base} = -E(\pi R^2)$

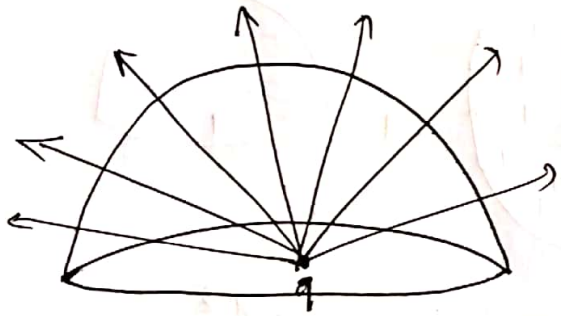
$\phi = 0$ (for hemi sphere)

$\phi_{base} + \phi_{curved} = 0$

$-E\pi R^2 + \phi_{curved} = 0$

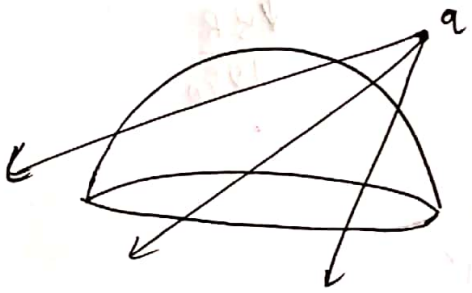
$$\boxed{\phi_{curved} = E\pi R^2}$$

(3)



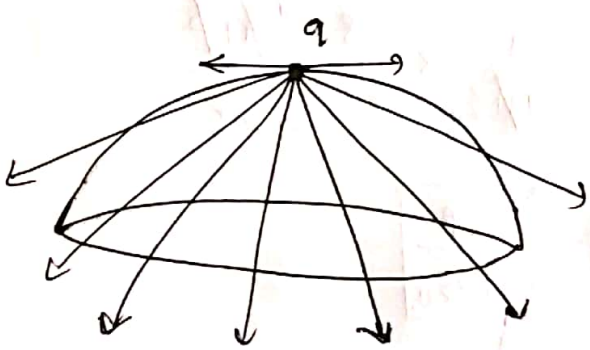
$$\boxed{\phi_{curved} = \frac{q}{2\varepsilon_0}}$$

(4)



$$\boxed{\phi = 0}$$

(5)



Flux through hemisphere $= \frac{q}{2\varepsilon_0} = \phi$

$\phi = \phi_{base} + \phi_{curved}$

flux through curved surface,

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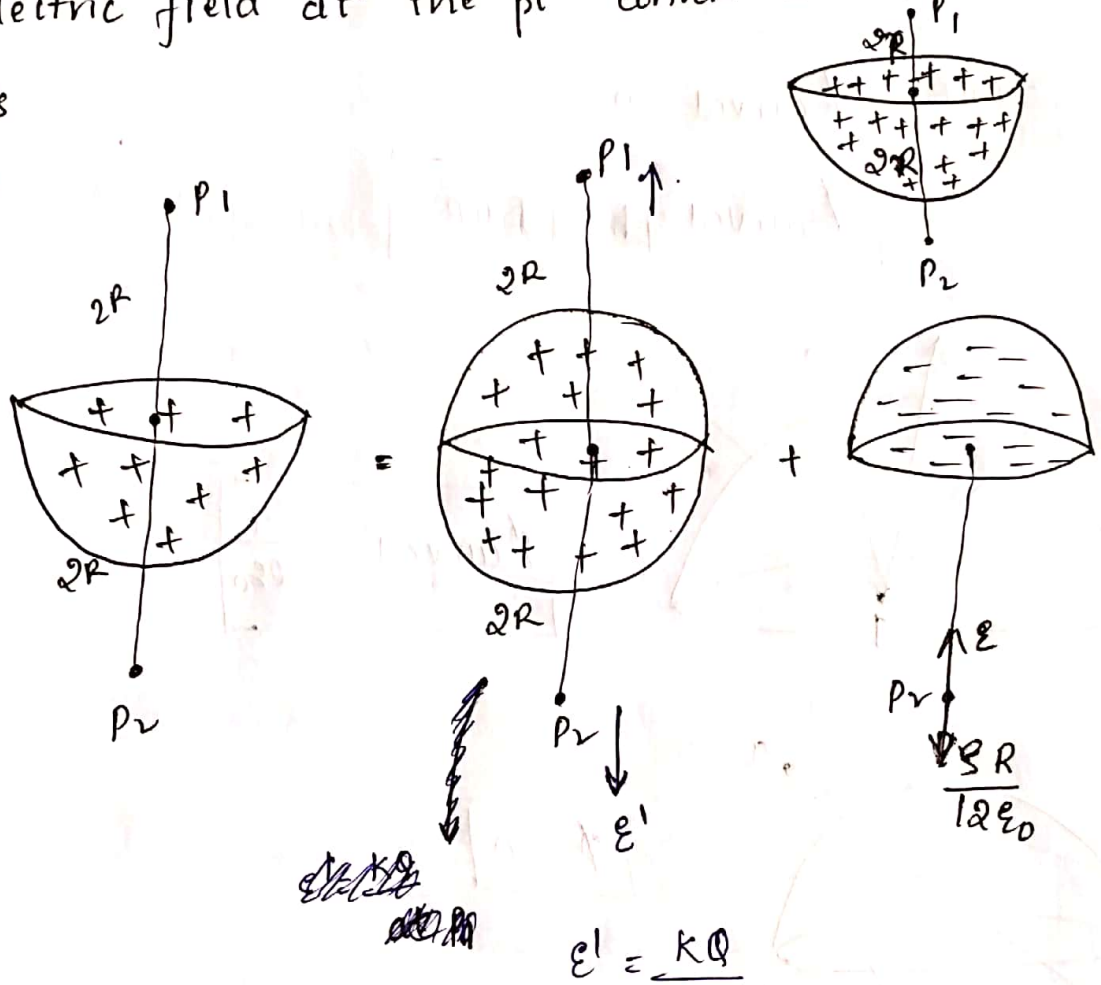
$$\Phi_{curved} = \frac{q}{2\varepsilon_0} - \Phi_{base}$$

Ex 6

The diagram shows a uniformly charged sphere of radius r. It has volume charge density ρ, if the electric field at pt 2R distance above centre is ξ. then the electric field at the pt which is 2R below its centre is

Sol 6



$$\varepsilon' = \frac{KQ}{(2R)^2}$$

$$= \frac{1}{4\pi\varepsilon_0} \times \frac{\rho \times \frac{4}{3}\pi R^3}{4R^2}$$

$$\boxed{\varepsilon' = \frac{\rho R}{12\varepsilon_0}}$$

field at $P_2 = \varepsilon' - \varepsilon = \frac{\rho R}{12\varepsilon_0} - \varepsilon$

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Ex 6

One micro coloumb charge is uniformly distributed on a spherical shell given by $x^2+y^2+z^2=25$. what will be the intensity of electric field at a point $(1,1,2)$.

Sol:

radius = 5

distance of given point from centre

$$=\sqrt{1^2+1^2+2^2} = \sqrt{6} < 5.$$

clearly point is inside the spherical shell.

$$\therefore \boxed{E = 0}$$

Ex:

Find flux through circular ring.

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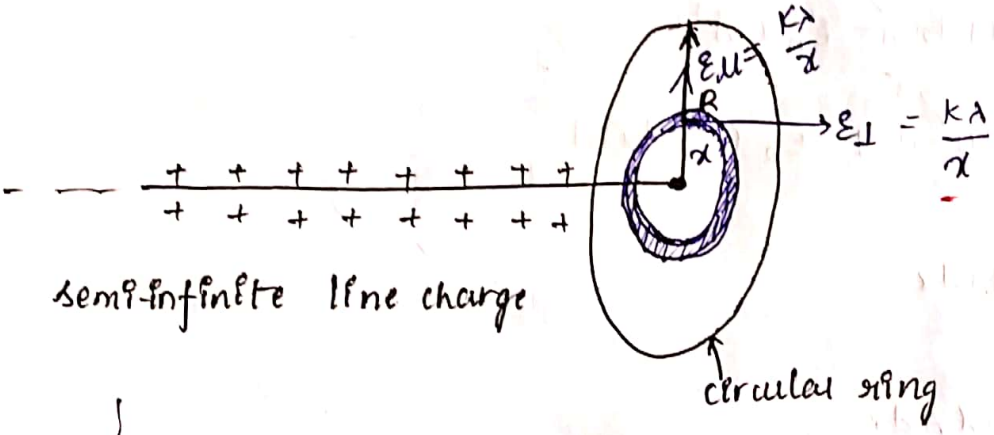
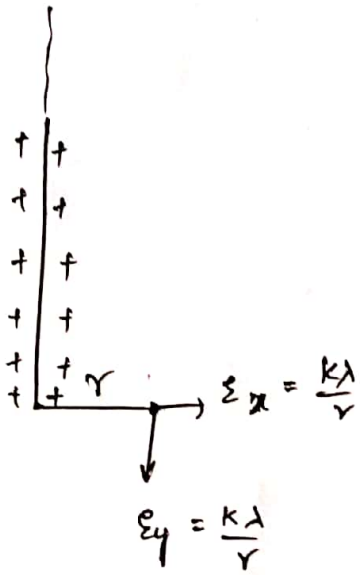


Sol:



Flux through elementary area,

$$d\phi = E_{\perp ler}\, dA$$

$$= \frac{K\lambda}{x}(2\pi x \cdot dx)$$

$$d\phi = 2\pi K\lambda\, dx$$

$$\text{Total flux} = \phi = \int d\phi = \int_{x=0}^{x=R} 2\pi K\lambda\, dx$$

$$\boxed{\phi = 2\pi K\lambda R}$$

$$\left(K = \frac{1}{4\pi\varepsilon_0}\right)$$

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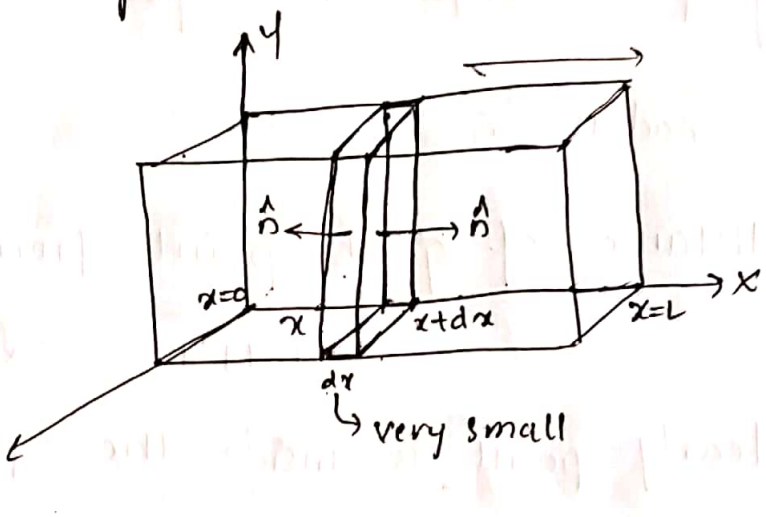
Calculation of charge density from field functions

① $\bar{E} = E_0 x^2 \hat{i}$. Find charge density as a function of 'x' in the cube.



M-I

$d\phi = \phi_{right\ face} + \phi_{left\ face}$

$d\phi = E_2 A - E_1 A$

$= E_0 (x+dx)^2 \hat{i} \cdot A - E_0 x^2 \hat{i} \cdot A$

$= E_0 (dx)^2 \cdot A + E_0 (2x\,dx) A$ ($(dx)^2 \to 0$)

$d\phi = 2E_0 x A\,dx$

$\frac{dq}{\varepsilon_0} = 2E_0 x A\,dx$

$\frac{\rho\,dv}{\varepsilon_0} = 2E_0 x A \cdot dx$

$\frac{\rho A dx}{\varepsilon_0} = 2E_0 x A dx$

$\boxed{\rho = 2\varepsilon_0 E_0 x}$

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M-II ρ_0

we have,

$$d\phi = \frac{dq}{\varepsilon_0}$$

$$d(EA) = \frac{\rho\, dV}{\varepsilon_0}$$

$$A\,dE = \frac{\rho A\, dx}{\varepsilon_0}$$

$$\boxed{\rho = \varepsilon_0 \frac{dE}{dx}} \text{ [Formula]}$$

$$\rho = \varepsilon_0 \frac{d}{dx}(E_0 x^2)$$

$$\boxed{\rho = 2\varepsilon_0 E_0 x}$$

* Average charge density in the cube,

$$\rho_{avg} = \frac{\text{Total charge}}{\text{Total volume}}$$

$$= \frac{\int \rho \cdot dv}{L^3}$$

$$= \frac{\int 2\varepsilon_0 E_0 x \cdot A\, dx}{L^3}$$

$$= \frac{\cancel{2}\varepsilon_0 E_0 \cancel{A} \frac{L^{\cancel{2}}}{\cancel{2}}}{L^3}$$

$$\boxed{\rho_{avg} = \varepsilon_0 E_0 L}$$

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* For 3D-vector,

i.e, $\bar{E} = E_x \hat{i} + E_y \hat{j} + E_z \hat{k}$

then $\rho_x = \varepsilon_0 \frac{\partial E_x}{\partial x}$

$\rho_y = \varepsilon_0 \frac{\partial E_y}{\partial y}$

$\rho_z = \varepsilon_0 \frac{\partial E_z}{\partial z}$

clearly charge density at a point is,

$$\rho = \rho_x + \rho_y + \rho_z$$

Ex 6 $\bar{E} = x^2 y \hat{i} + y^2 z x \hat{j} + z^3 \hat{k}$, Find charge density in the region at (1, 2, 1).

Sol 6

$\rho_x = \varepsilon_0 \times (2xy) = \varepsilon_0 (2(1)(2)) = 4\varepsilon_0$

$\rho_y = \varepsilon_0 (2xyz) = \varepsilon_0 (2 \times 1 \times 2 \times 1) = 4\varepsilon_0$

$\rho_z = \varepsilon_0 (3z^2) = \varepsilon_0 (3 \times 1) = 3\varepsilon_0$

$\rho = \rho_x + \rho_y + \rho_z$

$= 4\varepsilon_0 + 4\varepsilon_0 + 3\varepsilon_0$

$$\rho = 11\varepsilon_0$$

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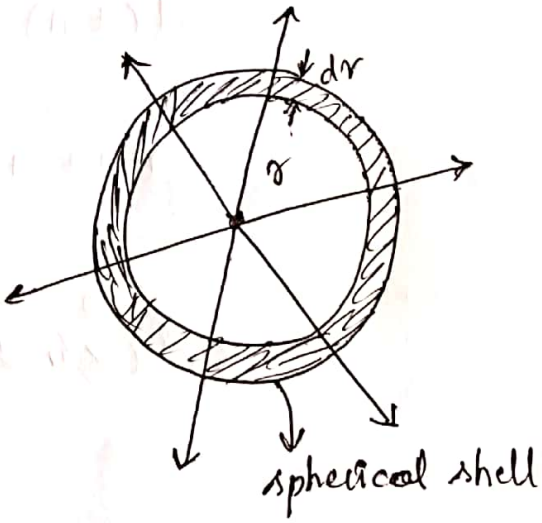
(Q) Radial Electric field :

Let $\boxed{E = cr^2}$. Find charge density as a function of r.



Here flux entering & leaving areas are different.

M-I $d\phi = E_2 A_2 - E_1 A_1$

$$\frac{dq}{\varepsilon_0} = c(r+dr)^2 (4\pi (r+dr)^2) - cr^2 (4\pi r^2)$$

$$= 4\pi c \left[(r+dr)^4 - r^4 \right]$$

$$= 4\pi c \left[((r+dr)^2 + r^2)((r+dr)^2 - r^2) \right]$$

$$= 4\pi c \left[2r^2 + 2rdr \right][2rdr]$$

$$= 8\pi c (2r^3 \cdot dr + 0)$$

$$\frac{dq}{\varepsilon_0} = 16\pi c r^3 \cdot dr$$

$$\frac{\rho dV}{\varepsilon_0} = 16\pi c r^3 \cdot dr$$

$$\frac{\rho \times 4\pi r^2 \cdot dr}{\varepsilon_0} = 16\pi c r^3 \cdot dr$$

$$\boxed{\rho = 4c\varepsilon_0 r}$$

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M-II 6

we have,

$$d\phi = \frac{dq}{\varepsilon_0}$$

$$d(EA) = \frac{\rho dV}{\varepsilon_0}$$

$$E dA + A dE = \frac{\rho (4\pi r^2 . dr)}{\varepsilon_0}$$

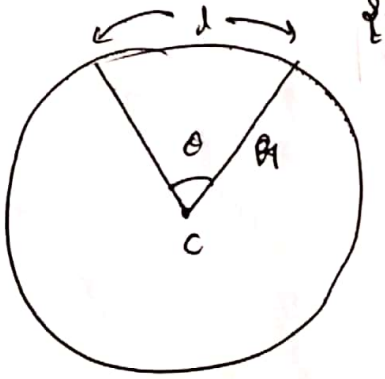
$$cr^2 (\rho 4\pi r \cdot dr) + 4\pi r^2 \cdot (2cr \cdot dr) = \frac{4\pi r^2 \rho}{\varepsilon_0} \cdot dr$$

$$2cr + 2cr = \frac{\rho}{\varepsilon_0}$$

$$\boxed{\rho = 4c\varepsilon_0 r}$$

Concept of Solid Angle (Ω)

→ Plane angle (θ):



{2D-angle}

$$\theta = \frac{l}{r}$$

radians.

For complete ring,

$l = 2\pi r$

$$\boxed{\theta = 2\pi} \text{ rad.}$$

→ Solid Angle (Ω): {3D-angle}

Angle blw the lines drawn from periphery of a surface at it's centre (or) angle measured by the lateral surface of the cone at it's centre is called solid angle.

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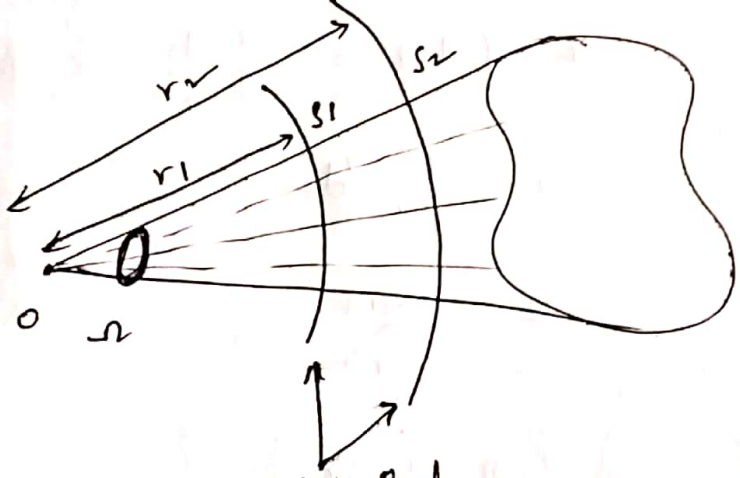
Mathematically,

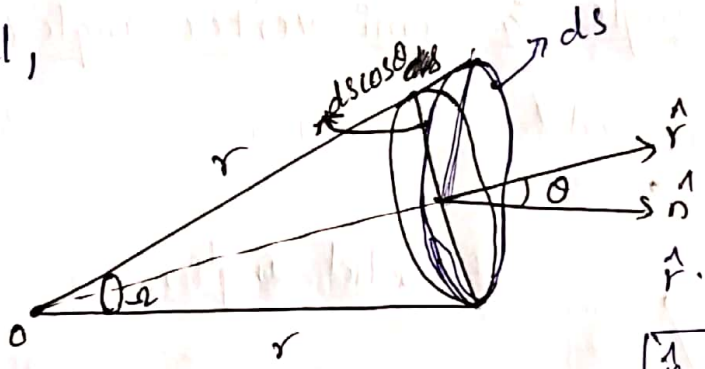
$$\Omega = \frac{S_1}{r_1^2} = \frac{S_2}{r_2^2} = \text{- - -}$$



unit : steradian (sr)

***

In general,



$\hat{r} \cdot \hat{n} = |\hat{r}| \cdot |\hat{n}| \cdot \cos\theta$

$$\hat{r} \cdot \hat{n} = \cos\theta$$

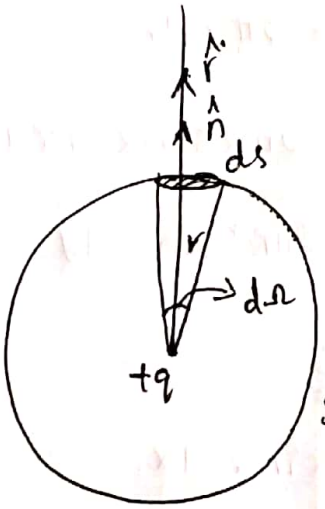
now, $\Omega = \frac{d\vec{s}}{r^2} \cdot \hat{r}$

$= \frac{ds\,\hat{n}}{r^2} \cdot \hat{r}$

$= \frac{ds}{r^2} (\hat{r} \cdot \hat{n})$

$$\Omega = \frac{ds\cos\theta}{r^2}$$

*



$$d\Omega = \frac{ds\cos\theta}{r^2} \quad \text{here } \theta = 0$$

$$d\Omega = \frac{ds}{r^2}$$

Total solid angle, subtended by sphere at its centre is,

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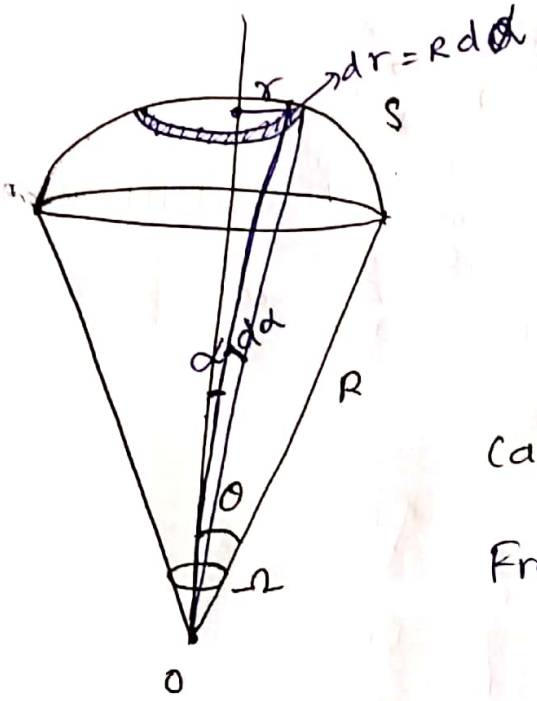
$$\Omega = \int d\Omega = \int \frac{ds}{r^2}$$

$$\Omega = \frac{1}{r^2} \cdot \int ds$$

$$\Omega = \frac{4\pi r^2}{r^2}$$

$$\boxed{\Omega = 4\pi sr}$$

Relation b/w solid angle & semi-vertex angle (θ) of cone



Solid angle,

$$\Omega = \frac{S}{R^2} \quad -(1)$$

Calculation of S:

From figure,

$$\sin\alpha = \frac{r}{R}$$

$$r = R\sin\alpha$$

$$ds = 2\pi r\, dr$$

$$= 2\pi R \sin\alpha \cdot R(d\alpha)$$

$$ds = 2\pi R^2 \sin\alpha \cdot d\alpha$$

∴ Total surface area,

$$S = \int_{\alpha=0}^{\alpha=\theta} ds = 2\pi R^2 \int_{\alpha=0}^{\alpha=\theta} \sin\alpha \cdot d\alpha$$

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$$S = 2\pi R^2 (-\cos\alpha)_0^\theta$$

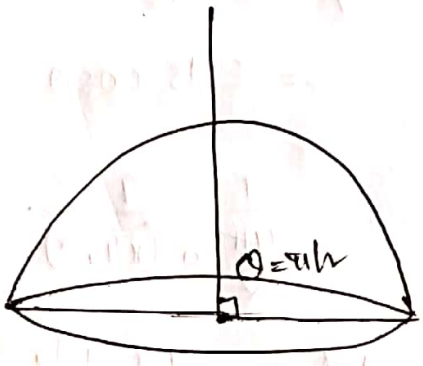
$$\boxed{S = 2\pi R^2 (1 - \cos\theta)}$$

$$\Omega = \frac{S}{R^2}$$

$$\boxed{\Omega = 2\pi(1-\cos\theta)}$$

Points:

For hemisphere



$$\Omega = 2\pi(1-\cos 90^\circ)$$

$$\boxed{\Omega = 2\pi \text{ sr}}$$

For complete sphere

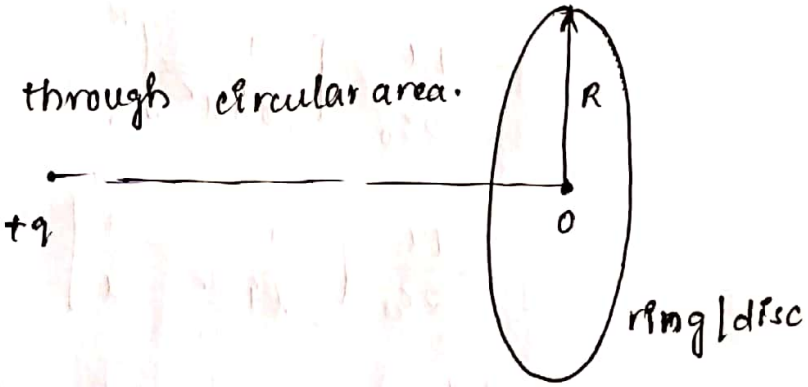
$$\Omega = 2\pi(1-\cos\pi)$$

$$\boxed{\Omega = 4\pi \text{ sr}}$$

7/04/25/Monday

Applications:

① Find flux through circular area.



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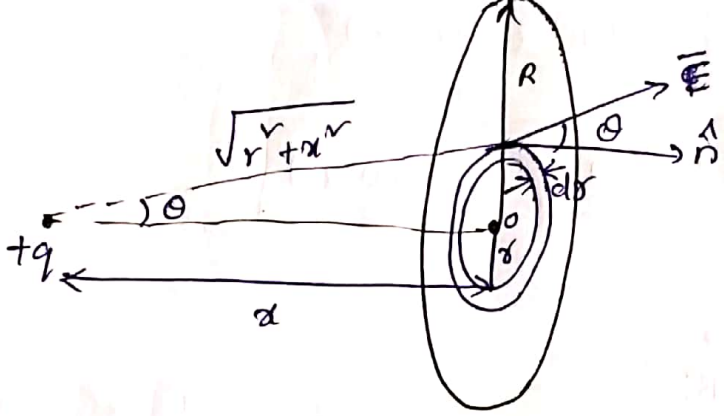
Sol⁶



M-86

Flux through elementary area,

$$d\phi = \bar{E} \cdot d\bar{s}$$

$$= E\,ds\cos\theta$$

$$= \frac{1}{4\pi\varepsilon_0}\frac{q}{(x^2+r^2)} \cdot 2\pi r\,dr \times \frac{x}{\sqrt{x^2+r^2}}$$

$$d\phi = \frac{q x r\,dr}{4\pi\varepsilon_0 (x^2+r^2)^{3/2}}$$

$$\phi = \int d\phi$$

$$= \frac{qx}{2\varepsilon_0}\int_{r=0}^{r=R} \frac{r\cdot dr}{(x^2+r^2)^{3/2}} \qquad x^2+r^2 = t^2$$

$$2r\cdot dr = 2t\cdot dt$$

$$= \frac{qx}{2\varepsilon_0}\int \frac{t\cdot dt}{t^3}$$

$$= -\frac{qx}{2\varepsilon_0}\left[\frac{1}{\sqrt{x^2+r^2}}\right]_0^R$$

$$= -\frac{qx}{2\varepsilon_0}\left[\frac{1}{\sqrt{x^2+R^2}} - \frac{1}{x}\right]$$

$$= \frac{qx}{2\varepsilon_0}\left[\frac{1}{x} - \frac{1}{\sqrt{x^2+R^2}}\right] = \frac{q}{2\varepsilon_0}\left[1 - \frac{x}{\sqrt{x^2+R^2}}\right]$$

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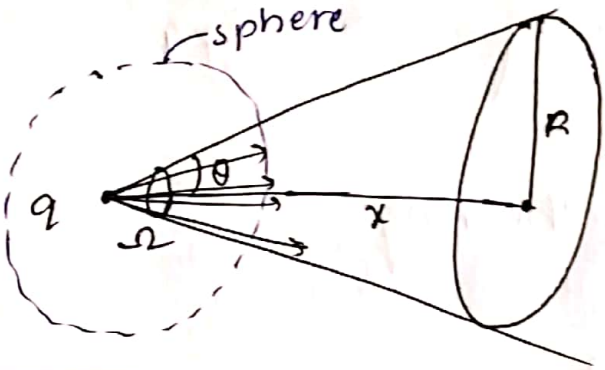
$$\boxed{\phi = \frac{q}{2\varepsilon_0}(1-\cos\theta)} \quad *$$

M-II 6



$$\boxed{\Omega = 2\pi(1-\cos\theta)}$$

Flux through the sphere $= \frac{q}{\varepsilon_0}$

Solid angle flux

$$4\pi \longrightarrow \frac{q}{\varepsilon_0}$$

$$1 \longrightarrow \frac{q}{4\pi\varepsilon_0}$$

$$\Omega \longrightarrow \frac{q\Omega}{4\pi\varepsilon_0}$$

$$\phi = \frac{q}{4\pi\varepsilon_0}(2\pi)(1-\cos\theta)$$

$$\phi = \frac{q}{2\varepsilon_0}(1-\cos\theta) \quad *$$

$$\boxed{\phi = \frac{q}{2\varepsilon_0}\left(1-\frac{x}{\sqrt{x^2+R^2}}\right)}$$

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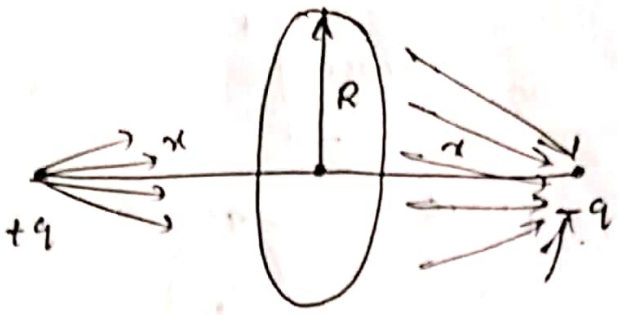
Ex 6



Flux through the ring ?

Sol6

$$\phi = \frac{q}{2\varepsilon_0}(1-\cos\theta) \times 2$$

$$\boxed{\phi = \frac{q}{\varepsilon_0}(1-\cos\theta)}$$

* If both charges are $+q$, $+q$ then $\boxed{\phi = 0}$.

Ex 6

$$\frac{q_1}{2\varepsilon_0}(1-\cos\alpha) = \frac{q_2}{2\varepsilon_0}(1-\cos\beta)$$

$$q_1(1-\cos\alpha) = q_2(1-\cos\beta)$$

$$\frac{q_1}{q_2} = \frac{2\sin^2\beta/2}{2\sin^2\alpha/2}$$

$$\boxed{\sqrt{\frac{q_1}{q_2}} = \frac{\sin\beta/2}{\sin\alpha/2}}$$

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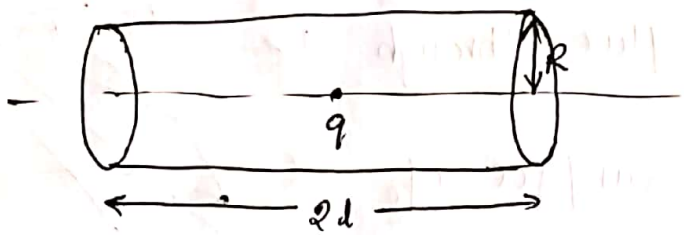
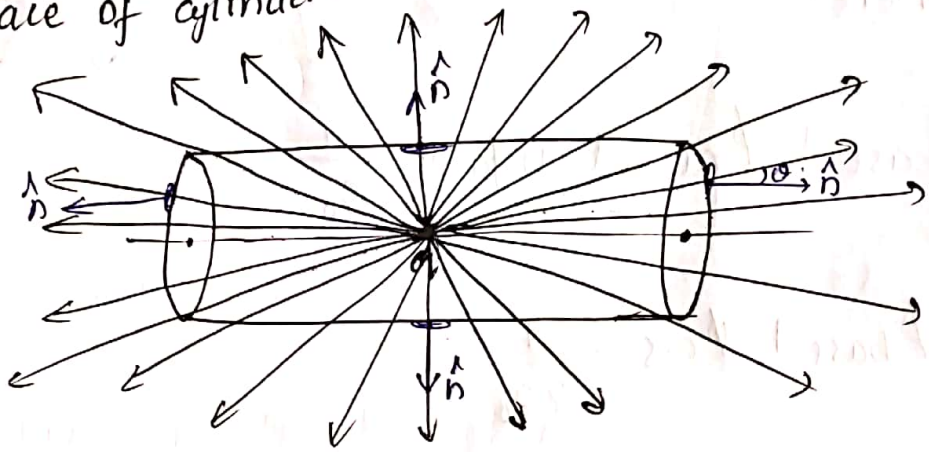
Ex 6

Find flux through curved surface of cylinder.



Sol 6



$$\phi_{total} = \frac{q}{\varepsilon_0}$$

$$\phi_{right} + \phi_{left} + \phi_{curved\ surface} = \frac{q}{\varepsilon_0}$$

$$\frac{q}{2\varepsilon_0}(1-\cos\theta) + \frac{q}{2\varepsilon_0}(1-\cos\theta) + \phi_{c.s} = \frac{q}{\varepsilon_0}$$

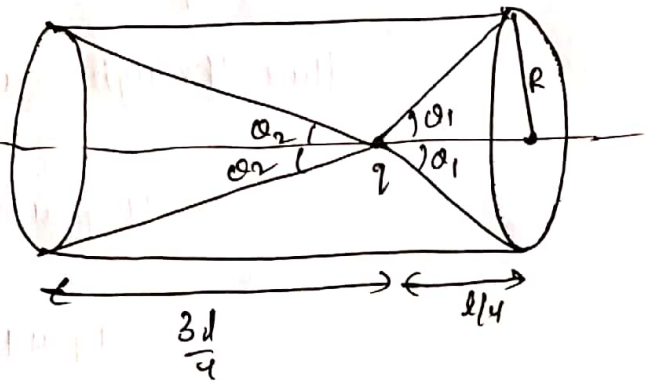
$$\frac{q}{\varepsilon_0}(1-\cos\theta) + \phi_{c.s} = \frac{q}{\varepsilon_0}$$

$$\boxed{\phi_{c.s} = \frac{q}{\varepsilon_0}\cos\theta}$$

Ex 6

$$\frac{q}{2\varepsilon_0}(1-\cos\theta_1) + \frac{q}{2\varepsilon_0}(1-\cos\theta_2) + \phi_{c.s} = \frac{q}{\varepsilon_0}$$

$$\boxed{\phi_{c.s} = \frac{q}{2\varepsilon_0}(\cos\theta_1 + \cos\theta_2)}$$



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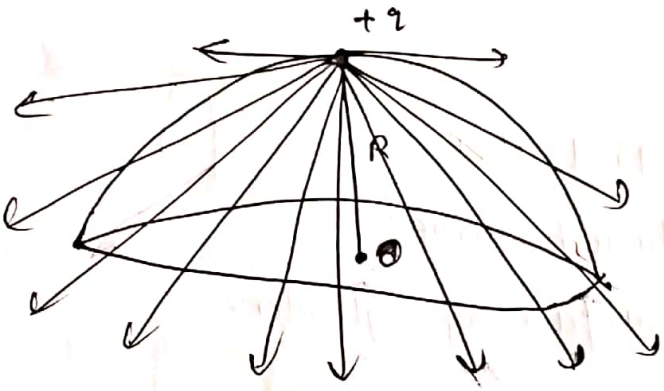
Ex 6

Find flux through curved surface of hemisphere.



Sol^n

$$\phi_{base} + \phi_{curved\ surface} = \frac{q}{2\varepsilon_0}$$

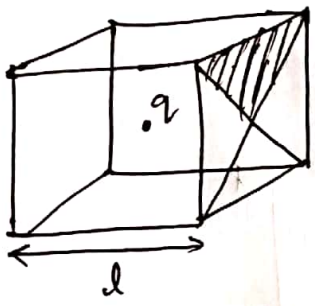
$$\phi_{base} + \phi_{c.s} = \frac{q}{2\varepsilon_0}$$

$$\frac{q}{2\varepsilon_0}(1-\cos\theta) + \phi_{c.s} = \frac{q}{2\varepsilon_0}$$

$$\phi_{c.s} = \frac{q}{2\varepsilon_0}\cos\theta \qquad (\because \theta = 45^\circ)$$

$$\boxed{\phi_{c.s} = \frac{q}{2\sqrt{1}\varepsilon_0}} \quad *$$

Ex 6



Find flux through shaded region.

Sol^n since charge is at centre of cube.

$$\text{flux through each face} = \frac{q}{6\varepsilon_0}$$

$$4 \text{ parts} - \frac{q}{6\varepsilon_0}$$

$$1 \text{ part} - \frac{q}{24\varepsilon_0}$$

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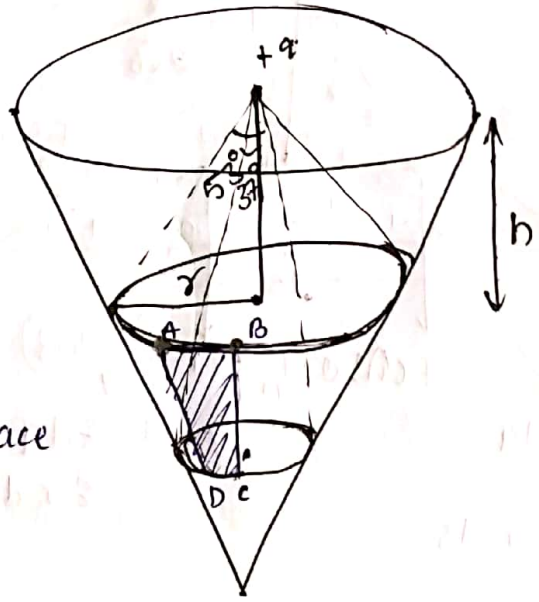
$$\phi_{shaded} = \frac{q}{2u\varepsilon_0}$$

Ex6.

Find flux through shaded portion ABCDA.

given AB = h/r



Sol6

flux through lateral surface containing ABCD.

$$\phi = \frac{q}{2\varepsilon_0}(1-\cos 53^\circ) - \frac{q}{2\varepsilon_0}(1-\cos 37^\circ)$$

$$\tan 53^\circ = \frac{r}{h}$$

$$= \frac{q}{2\varepsilon_0}\left(1-\frac{3}{5}\right) - \frac{q}{2\varepsilon_0}\left(1-\frac{4}{5}\right)$$

$$r = \frac{4h}{3}.$$

$$= \frac{q}{2\varepsilon_0}\left(\frac{2}{5}-\frac{1}{5}\right)$$

$$\boxed{\phi = \frac{q}{10\varepsilon_0}}$$

$2\pi r \longrightarrow \phi$

$1 \longrightarrow \frac{\phi}{2\pi r}$

$AB \longrightarrow \frac{\phi}{2\pi r}(AB)$

$$\phi' = \frac{q}{10\varepsilon_0} \times \frac{1}{2\pi\left(\frac{4h}{3}\right)}(h/2) = \frac{3q}{160\pi\varepsilon_0} \Rightarrow \boxed{\phi' = \frac{3q}{160\pi\varepsilon_0}}$$

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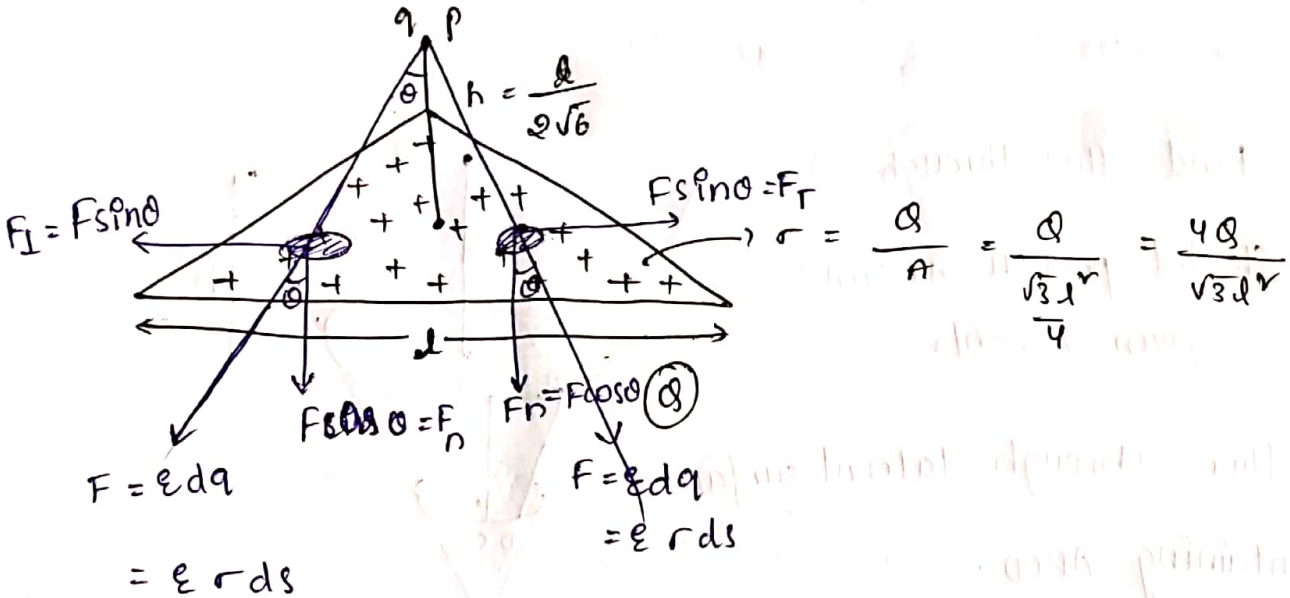
Force acting on the charged Sheet:



∴ net force on the charged sheet,

$$F_{net} = \int F\cos\theta$$

$$= \int E\sigma\, ds\cos\theta$$

$$= \sigma \int E ds\cos\theta$$

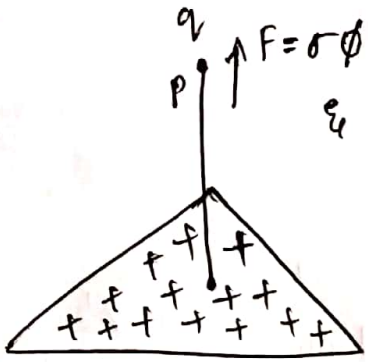
$$F_{net} = \sigma\phi$$

* $\boxed{F_n = \sigma\phi}$ (normal)

σ = charge density on sheet.

ϕ = flux through charged sheet due to q.

*



& $E = \frac{F}{q}$

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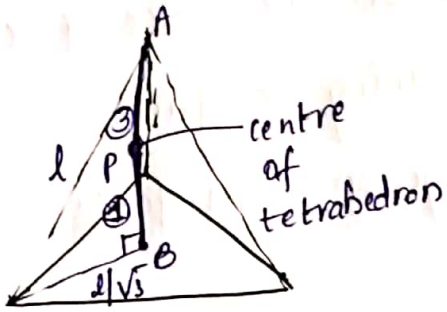
Here to find flux ϕ 6



$$AB = \sqrt{l^2 - \frac{l^2}{3}}$$

$$= \sqrt{\frac{2}{3}}\, l$$

$$3x + x = \sqrt{\frac{2}{3}}\, l$$

$$4x = \sqrt{\frac{2}{3}}\, l$$

$$\boxed{x = \frac{l}{2\sqrt{6}}}$$

$\therefore$ flux through tetrahedron $= \frac{q}{\varepsilon_0}$

flux through each triangular plate,

$$\phi = \frac{q}{4\varepsilon_0}$$

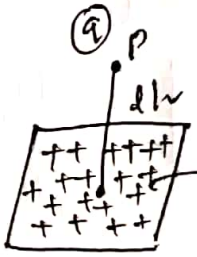
Force on sheet,

$$F = \sigma\phi$$

$$= \frac{\sigma q}{4\varepsilon_0}$$

$$\boxed{F = \frac{\sigma q}{\sqrt{3}\varepsilon_0 l^2}} \quad *$$

Exe 6



$\sigma = \frac{q}{l^2}$, $F_n = \sigma\phi$; $\phi = \frac{q}{6\varepsilon_0}$

$$F_n = \frac{\sigma q}{6\varepsilon_0} \Rightarrow \boxed{F_n = \frac{\sigma q}{6\varepsilon_0 l^2}}$$

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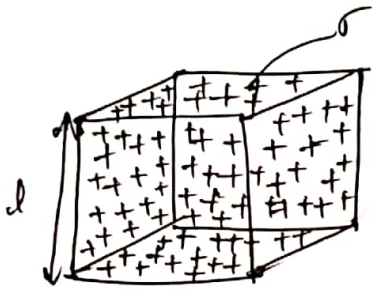
Ex 6



Find force on one face of cube due to remaining 5 faces.

Q is charge on each face.

Sol 6

$$\sigma = \frac{6Q}{6l^2} = \frac{Q}{l^2}$$

$$\boxed{F_n = \sigma \phi_5}$$

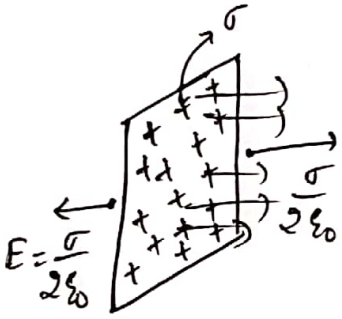
ϕ_5 = flux due to five faces.

To find ϕ_5 6-

Assume cubical surface having side length $(l - dl)$ where $dl \to 0$ inside the original cube.

Charge enclosed in that surface = 0

$$\phi_{net} = 0 \Rightarrow \phi_{5\,faces} + \phi_{6^{th}\,face} = 0$$



$$\phi_{5\,faces} = -\phi_{6^{th}\,face}$$

$$= -EA$$

$$= -\frac{\sigma}{2\varepsilon_0} l^2$$

(-ve sign indicates opposite direction of field)

$$\boxed{\phi_{5\,faces} = -\frac{\sigma}{2\varepsilon_0} l^2}$$

$$F = \sigma \phi_5$$

$$= \sigma \left(\frac{\sigma}{2\varepsilon_0} l^2\right) \Rightarrow \boxed{F = \frac{\sigma^2 l^2}{2\varepsilon_0}}$$

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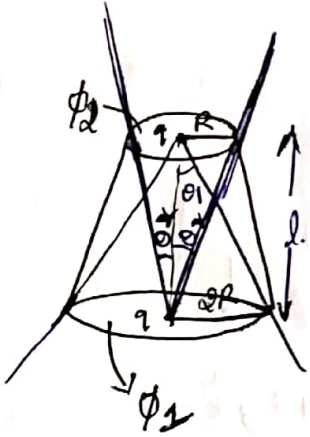
8/04/25 | Tuesday

Ex: Find ratio of ϕ_1 & ϕ_2.

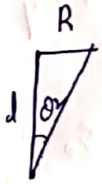


Sol:

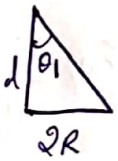
$$\phi_2 = \frac{q}{2\varepsilon_0}(1-\cos\theta_2)$$

$$\phi_1 = \frac{q}{2\varepsilon_0}(1-\cos\theta_1)$$

$$\boxed{\frac{\phi_1}{\phi_2} = \frac{1-\cos\theta_1}{1-\cos\theta_2}}$$



$$\cos\theta_2 = \frac{l}{\sqrt{R^2+l^2}}$$



$$\cos\theta_1 = \frac{l}{\sqrt{4R^2+l^2}}$$

** Guass law –

$$\phi_E = \oint_S \bar{E}\cdot d\bar{s} = \frac{q_{encl}}{\varepsilon_0}$$

where $q_{encl} = \int \lambda\, dl$

$$q_{encl} = \int \sigma\, dA$$

$$q_{encl} = \int \rho\, dv$$

we have, guass-divergence theorem,

$$\oint_S \bar{E}\cdot d\bar{s} = \oint_v \text{div}E\, dv$$

$$\therefore \oint_v \text{div}E\, dv = \frac{\oint_v \rho\, dv}{\varepsilon_0}$$

$$\text{div}E = \frac{\rho}{\varepsilon_0}$$

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$$\bar{\nabla} \bar{E} = \frac{\rho}{\varepsilon_0}$$

$$\bar{\nabla} = i \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} + k \frac{\partial}{\partial z}$$

$$\bar{E} = E_x \hat{i} + E_y \hat{j} + E_z \hat{k}$$

now, $\rho_x = \varepsilon_0 \frac{\partial E_x}{\partial x}$

$$\rho_y = \varepsilon_0 \frac{\partial E_y}{\partial y}$$

$$\rho_z = \varepsilon_0 \frac{\partial E_z}{\partial z}$$

Finally, $\boxed{\rho = \rho_x + \rho_y + \rho_z}$

* Again,

$$\frac{dE}{dr} = \frac{\rho}{\varepsilon_0}$$

& $E = -\frac{dV}{dr}$

$$-\frac{d^2V}{dr^2} = \frac{\rho}{\varepsilon_0}$$

$$\boxed{\frac{d^2V}{dr^2} = -\frac{\rho}{\varepsilon_0}} \quad *$$

*** While Using Guass Law:

Step-1: Construct a quassian surface such that angle between $\bar{E}$ and $d\bar{s}$ is either 0°, 90° (or) 180°.

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Step-2: Calculate net flux through the guassian surface

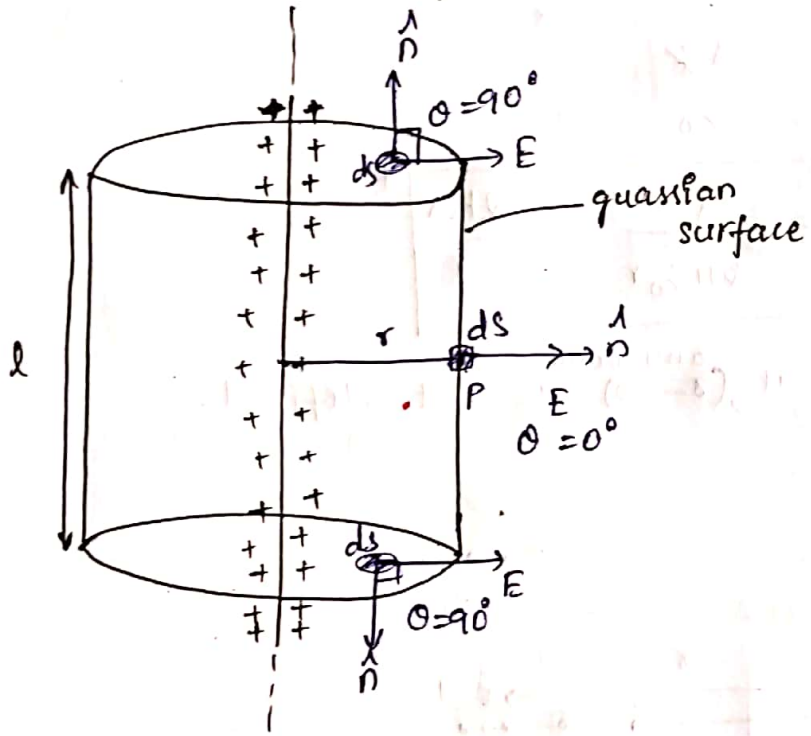
Step-3: Calculate net charge enclosed by guassian surface.

Step-4: use guass law, i.e., $\boxed{\phi_E = \frac{q_{encl}}{\varepsilon_0}}$

Step-5: To find potential at a point, use $\boxed{\int_i^f dV = -\int_P^f \bar{E} \cdot d\bar{r}}$

Applications:

① Infinite line charge:



Flux through guassian cylinder,

$$\phi_E = \phi_{c.s} + \cancel{\phi_{circular\ caps}}^{\,0}$$

$$\phi_E = \oint_s \bar{E} \cdot d\bar{s}$$

$$= \oint E \cdot ds \cos\theta = \oint E \cdot ds = E \int ds$$

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$$\phi_E = E \times 2\pi r l \quad -①$$

Net charge enclosed by quassian surface,

$$q_{encl} = \int \lambda dl$$

$$= \lambda \int dl$$

$$= \lambda l \quad -⑤$$

From quass law,

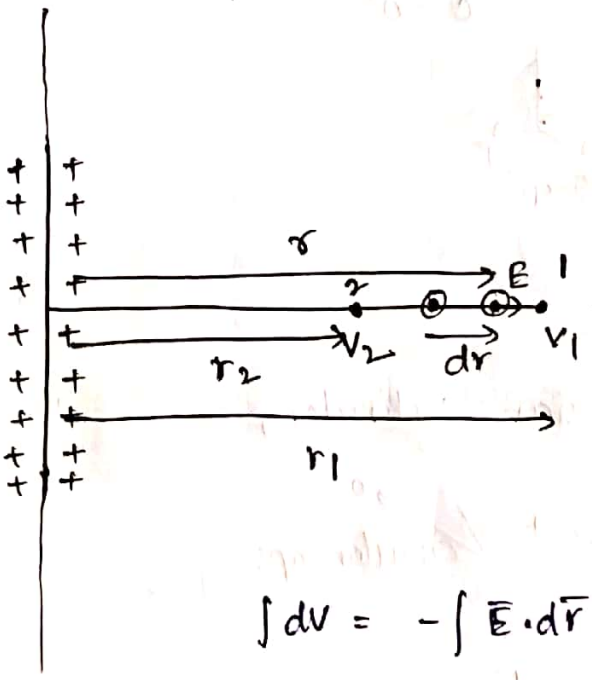
$$\phi_E = \frac{q_{encl}}{\varepsilon_0}$$

$$E \times 2\pi r l = \frac{\lambda l}{\varepsilon_0}$$

$$\boxed{E = \frac{\lambda}{2\pi\varepsilon_0 r} = \frac{2K\lambda}{r}} \quad *$$

* absolute potential at any point $(r \to \infty)$ is not defined.

*



$$\int dV = -\int \bar{E} \cdot d\bar{r}$$

$$\int_{V_2}^{V_1} dV = -\int_{r=r_2}^{r=r_1} E\,dr$$

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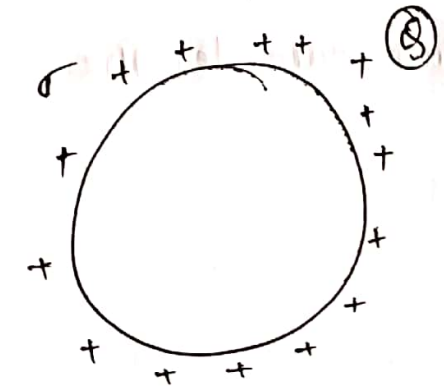
$$V_2 - V_1 = \int_{r_2}^{r_1} \frac{2K\lambda}{r}\,dr$$

$$\boxed{V_2 - V_1 = 2K\lambda \ln\left(r_1/r_2\right)}$$

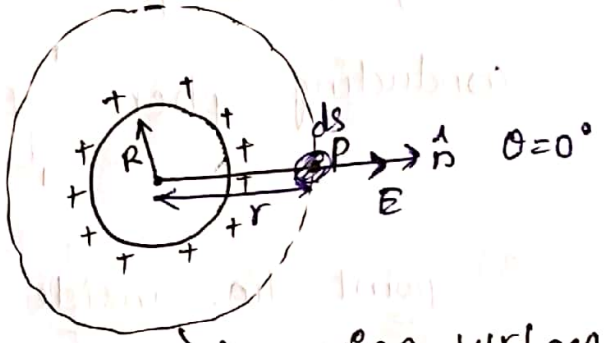
** for finite line charge we can not use guass law. use integration method.

② Charged Conducting Sphere / Shell :



metal sphere.

1) point lies outside :-



→ guassian surface

* $d\phi_E = \bar{E} \cdot d\bar{s} = E\,ds\cos 0 = E\,ds$

* $\phi_E = \oint_S E \cdot ds = E \oint ds$

$$\phi_E = E \times 4\pi r^2 \quad -①$$

* $q_{encl} = Q \quad -②$

* guass law, $\phi_E = \frac{q_{encl}}{\varepsilon_0}$

$$E_0 \times 4\pi r^2 = \frac{Q}{\varepsilon_0}$$

$$E_0 = \frac{Q}{4\pi\varepsilon_0 r^2}$$

$$\boxed{E_0 = \frac{kQ}{r^2}}$$

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(or)

$$E_0 = \frac{1}{4\pi\varepsilon_0 r^2}(\sigma \times 4\pi R^2)$$

$$\boxed{E_0 = \frac{\sigma R^2}{\varepsilon_0 r^2}}$$

2) point lies on the surface :-

$r = R$

$$\boxed{E_s = \frac{\sigma}{\varepsilon_0}}$$

Clearly electric field at a point just outside the conducting sphere $\boxed{E = \frac{\sigma}{\varepsilon_0}}$

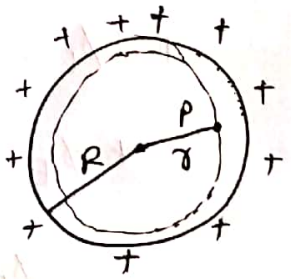
3) point lies inside the sphere :-

$r < R$

$\phi_E = 0$

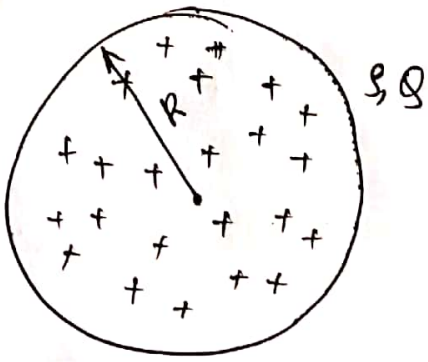
$q_{en} = 0$

$\boxed{E_i = 0}$



(3) Uniformly charged non-conducting sphere :

$$\rho = \frac{Q}{\frac{4}{3}\pi R^3} = \text{const}$$



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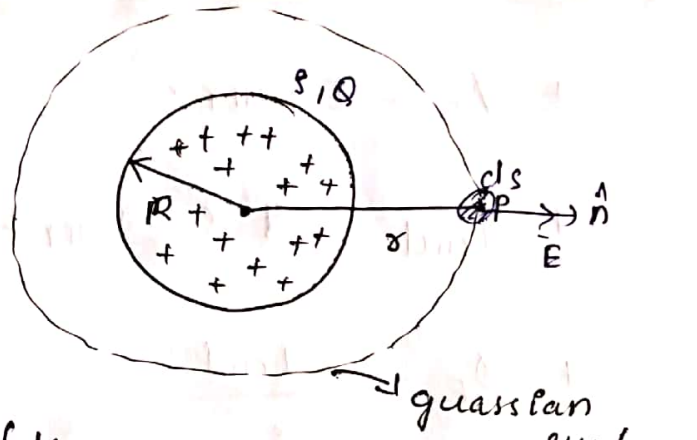
1) point lies outside :-



* $d\phi = \bar{E} \cdot d\bar{s}$

$= E\,ds\cos\theta$

$= E\,ds$

* $\phi_E = \oint_s d\phi_E = \oint_s E\,ds = E\int ds$

$\phi_E = E(4\pi r^2)$ — ①

* $q_{encl} = Q$

* $\phi_E = \frac{q_{encl}}{\varepsilon_0}$

$E(4\pi r^2) = \frac{Q}{\varepsilon_0}$

$E = \frac{Q}{4\pi\varepsilon_0 r^2}$

$$\boxed{E_o = \frac{KQ}{r^2}}$$

(or) $E_o = \frac{1}{4\pi\varepsilon_0 r^2} \times \frac{4}{3}\pi R^3 \rho$

$$\boxed{E_o = \frac{\rho R^3}{3\varepsilon_0 r^2}}$$

2) point lies on the surface :-

here $r = R$

$$\boxed{E_s = \frac{\rho R}{3\varepsilon_0}}$$

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3) point lies inside the body:

* $\phi_E = E \times 4\pi r^2$

* $q_{encl} = \rho \times \frac{4}{3}\pi r^3$

* $\phi_E = \frac{q_{encl}}{\varepsilon_0}$

$$4\pi r^2 \times E = \frac{\rho \times 4/3 \pi r^3}{\varepsilon_0}$$

$$\boxed{E_i = \frac{\rho r}{3\varepsilon_0}}$$

i.e, $$\boxed{\vec{E} = \frac{\rho \vec{r}}{3\varepsilon_0}}$$

(or)

$$E = \frac{Q}{4\pi R^3} \times \frac{r}{\varepsilon_0} = \frac{kQr}{R^3}$$

$$\boxed{E_i = \frac{kQr}{R^3}}$$

* $$V_p - V_\infty = -\left[\int_{r=\infty}^{r=R} E_1 \, dr + \int_{r=R}^{r=r} E_2 \, dr\right]$$

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Ex: For non-uniform charge density, $\rho = \rho_0\left(1-\frac{r}{R}\right)$ for $0 \le r \le R$ & $\rho = 0$ for $r > R$. Find electric field at (1) $r = R/2$ (2) $r = 2R$.

Sol:

(1) $$E \times 4\pi \left(\frac{R}{2}\right)^2 = \frac{\int \rho \cdot dv}{\varepsilon_0}$$

$$E \times \pi R^2 = \frac{\int_{r=0}^{r=R/2} \rho_0\left(1-\frac{r}{R}\right) \cdot 4\pi r^2 \cdot dr}{\varepsilon_0}$$

$$\varepsilon_0 E R^2 = 4\int_{r=0}^{r=R/2} \rho_0\left(r^2 - \frac{r^3}{R}\right) \cdot dr$$

$$E R^2 \varepsilon_0 = 4\rho_0 \left\{\left[\frac{r^3}{3}\right]_0^{R/2} - \left[\frac{r^4}{4R}\right]_0^{R/2}\right\}$$

$$E R^2 \varepsilon_0 = 4\rho_0 \left[\frac{1}{3}\times\frac{1}{8}\times R^3 - \frac{1}{4R}\times\frac{R^4}{16}\right]$$

$$E R^2 \varepsilon_0 = 4\rho_0 \left[\frac{R^3}{24} - \frac{R^3}{64}\right]$$

$$E = \frac{\rho_0}{\varepsilon_0}\left(\frac{10}{6\times 16}\right)$$

$$\boxed{E = \frac{5\rho_0}{48\varepsilon_0}}$$

(2) $$E \times 4\pi (2R)^2 = \frac{1}{\varepsilon_0}\int_{r=0}^{R} \rho_0\left(1-\frac{r}{R}\right) \cdot 4\pi r^2 \cdot dr$$

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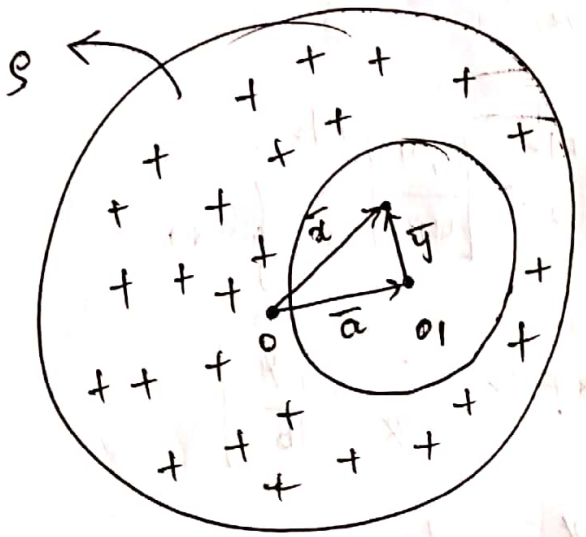
$$E \times 4R^2 = \frac{\rho_0}{\varepsilon_0} \left\{ \left[\frac{r^3}{3} - \frac{r^4}{4R} \right]_0^R \right\}$$

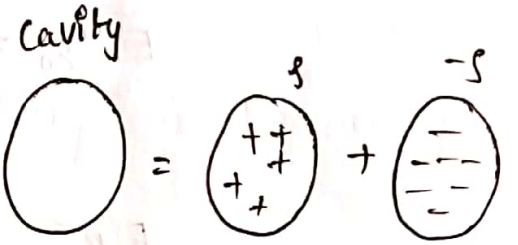
$$E \times 4R^2 = \frac{\rho_0}{\varepsilon_0} \left[\frac{R^3}{3} - \frac{R^3}{4} \right]$$

$$E \times 4R^2 = \frac{\rho_0}{\varepsilon_0} \left(\frac{R^3}{12} \right)$$

$$\boxed{E = \frac{\rho_0 R}{48 \varepsilon_0}}$$

Field inside the spherical cavity :-





we have,

$$\bar{E}_1 = \frac{\rho \bar{r}}{3\varepsilon_0}$$

$$\bar{E}_2 = \frac{-\rho \bar{a}}{3\varepsilon_0}$$

net field at P,

$$\bar{E}_P = \bar{E}_1 + \bar{E}_2$$

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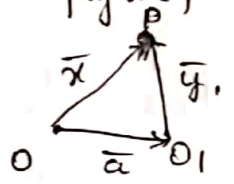
$$\bar{E}_P = \frac{\rho\bar{x}}{3\varepsilon_0} - \frac{\rho\bar{y}}{3\varepsilon_0}$$

$$= \frac{\rho(\bar{x}-\bar{y})}{3\varepsilon_0}$$

from figure,

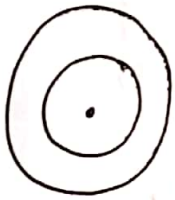


$\bar{x} = \bar{a} + \bar{y}$

$\bar{x} - \bar{y} = \bar{a}$

$$\boxed{\bar{E}_P = \frac{\rho\bar{a}}{3\varepsilon_0} = \text{uniform}}$$

* If $a = 0$ then $\boxed{E = 0}$



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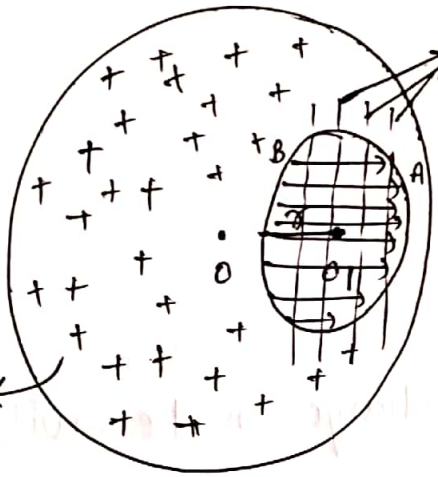
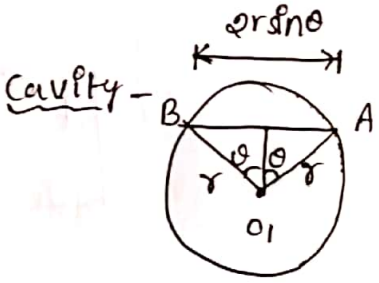


$$\boxed{E = \frac{\rho a}{3\varepsilon_0}}$$



r - radius of cavity.

* when the electron is released at A.

$F_e = qE$; A to B.

$$a_e = \frac{F_e}{m_e} = \frac{qE}{m_e} \quad ; \text{ A to B}$$

we have,

$$S = \cancel{ut}^{0} + \frac{1}{2}at^2$$

$$\rho r\sin\theta = \frac{1}{2} \times \frac{qE}{m_e} t^2$$

$$\boxed{t = \sqrt{\frac{4 r m_e \sin\theta}{qE}}}$$

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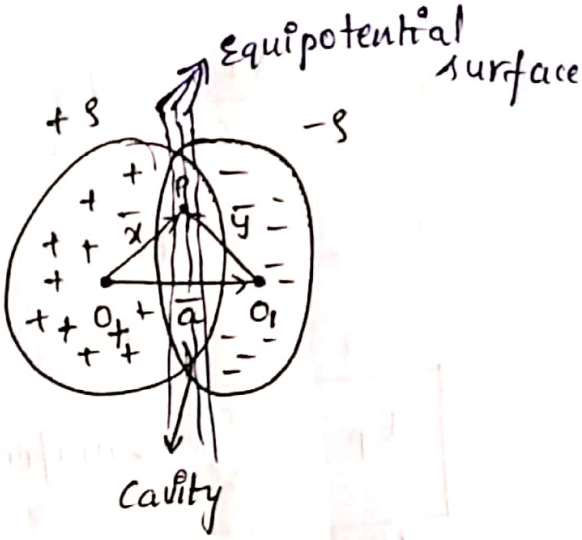
Intersecting Cylinders :-



$$\bar{E}_1 = \frac{\rho \bar{x}}{3\varepsilon_0}$$

$$\bar{E}_2 = -\frac{\rho \bar{y}}{3\varepsilon_0}$$

net field,

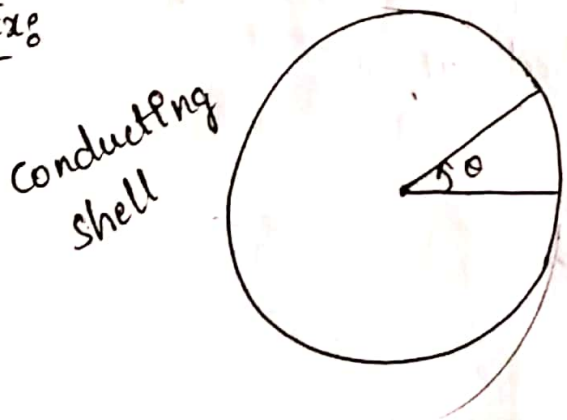
$$\bar{E} = \bar{E}_1 + \bar{E}_2$$

$$= \frac{\rho(\bar{x} - \bar{y})}{3\varepsilon_0}$$

$$\boxed{\bar{E} = \frac{\rho \bar{a}}{3\varepsilon_0}}$$

* workdone in moving a charge inside cavity is zero.

Ex:



$\sigma = \sigma_0 \cos\theta$

Find field at a point inside the shell.

Sol:

Consider a non-conducting equal and opposite charged spheres whose centres are sepelated by very small distance.

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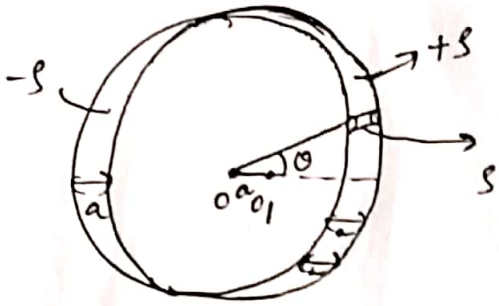


$\sigma \cong dA$ (∵ t is very small)

$\cos\theta = \frac{t}{a}$

$\boxed{t = a\cos\theta}$

* charge on both the elements is same,

$$\rho dV = \sigma dA$$

$$\rho \cancel{dA} \times t = \sigma \cancel{dA}$$

$$\boxed{\sigma = \rho a\cos\theta}$$

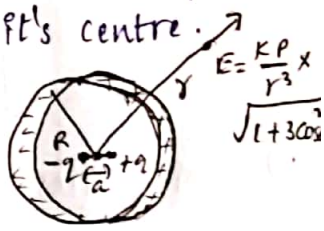
on comparing, $\boxed{\sigma_0 = \rho a}$

now, field inside the cavity,

$$E = \frac{\rho a}{3\varepsilon_0}$$

$$\boxed{E_i = \frac{\sigma_0}{3\varepsilon_0}}$$ (at any point inside the shell)

* To find field at outside points, Charge in the sphere can be treated as point charge at it's centre.



$\sigma_0 = \rho a$ $\quad \bar{P} = q\bar{a}$

$q = \rho \times \frac{4}{3}\pi R^3$

$E_\theta = \frac{KP}{r^3}\sqrt{1+3\cos^2\theta}$

$E_\theta = \frac{4K\pi R^3\sigma_0}{3r^3}\sqrt{1+3\cos^2\theta}$

(4) Conducting cylindrical shell :

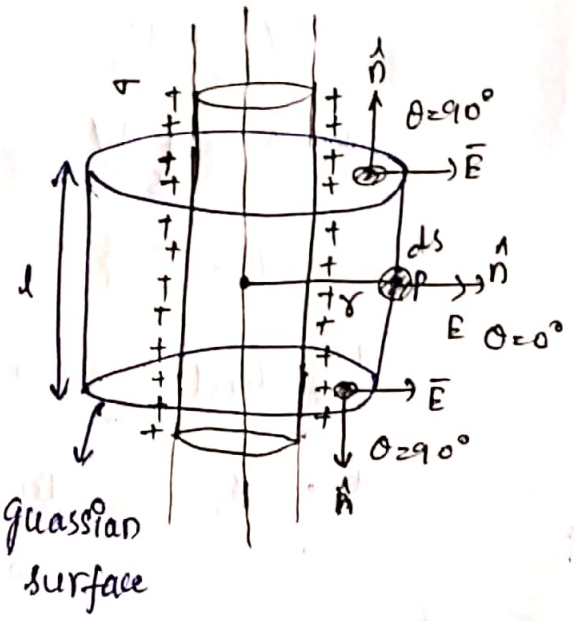
1) point lies outside :-

* $\oint d\phi = \oint \bar{E}.d\bar{s} = \oint E ds\cos\theta$

$$d\phi = E ds$$

$$\phi = E\oint ds$$

$$\phi = E \times 2\pi r l \quad \text{—①}$$



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* $q_{encl} = \sigma(2\pi R l)$ — (2)

* $\phi_E = \frac{q_{encl}}{\varepsilon_0}$

$E \times 2\pi r l = \frac{\sigma(2\pi R l)}{\varepsilon_0}$

$$\boxed{E = \frac{\sigma R}{r\varepsilon_0}} \quad *$$

2) point lies on the surface of

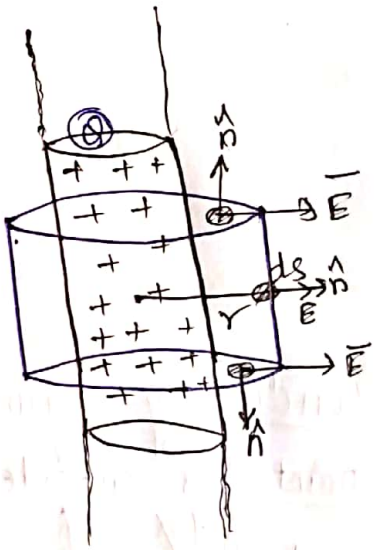
$r = R$

$$\boxed{E = \frac{\sigma}{\varepsilon_0}}$$

3) point lies inside the shell of

$$\boxed{E = 0}$$

(5) Non-conducting cylindrical shell of



1) point lies outside of

* $d\phi = \vec{E} \cdot d\vec{s} = E\,ds\cos 0°$

$d\phi = E\,ds$

* $\phi = \oint_S d\phi = \oint_S E\,ds$

$\phi = E \oint ds = E(2\pi r l)$ — (1)

* $q_{encl} = s(\pi R^2 l)$ — (2)

* $\phi_E = \frac{q_{encl}}{\varepsilon_0}$

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$$E \times 2\pi r l = \frac{\sigma(\pi R^2 l)}{\varepsilon_0}$$

$$\boxed{E = \frac{\sigma R^2}{2\varepsilon_0 r}} *$$

2) point lies on the surface &-

$$r = R$$

$$E = \frac{\sigma R^2}{2\varepsilon_0 R}$$

$$\boxed{E = \frac{\sigma R}{2\varepsilon_0}} *$$

3) point lies inside the surface &-

* $\phi_E = E(2\pi r l)$

* $q_{encl} = \sigma \times (\pi r^2 l)$

* $\phi_E = \frac{q_{encl}}{\varepsilon_0}$

$$E \times 2\pi r l = \frac{\sigma(\pi r^2 l)}{\varepsilon_0}$$

$$\boxed{E = \frac{\sigma r}{2\varepsilon_0}} *$$

(or)

$$E = \frac{\frac{Q}{\pi R^2 l} \times r}{2\varepsilon_0}$$

$$= \frac{1}{2\pi\varepsilon_0} \frac{Qr}{R^2 l}$$

$$\boxed{E = \frac{2KQr}{R^2 l}}$$

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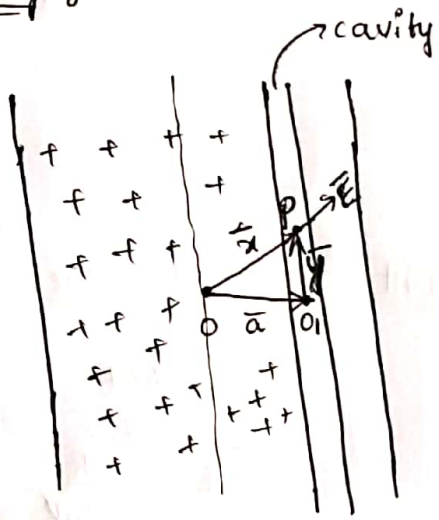
Field Inside the cylindrical Cavity :



$$\bar{E}_1 = \frac{\rho \bar{x}}{2\varepsilon_0}$$

$$\bar{E}_2 = -\frac{\rho \bar{y}}{2\varepsilon_0}$$

net field at P is,

$$\bar{E}_{net} = \bar{E}_1 + \bar{E}_2$$

$$\bar{E}_P = \frac{\rho(\bar{x} - \bar{y})}{2\varepsilon_0}$$

$$\boxed{\bar{E}_P = \frac{\rho \bar{a}}{2\varepsilon_0}}$$

Ex: a) charge density in a sphere is $\rho = \rho_0\left(1 - \frac{r}{R}\right)$; $0 \le r \le R$

$\rho = 0$; $r > R$

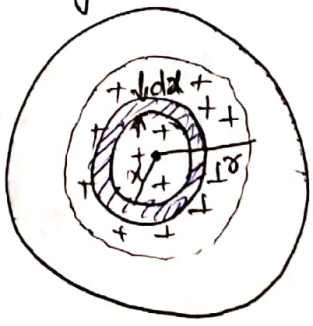
where ρ_0 = constant

r = distance of point from centre

R = radius of sphere.

i) Find Electric field as a function of distance r both inside and outside of sphere.

ii) Find max value of electric field strength.



Sol: i) For inside points :

$$\oint E\,ds = \frac{q_{end}}{\varepsilon_0}$$

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$$E \times 4\pi r^2 = \frac{q_{encl}}{\varepsilon_0}$$

$$E \times 4\pi r^2 = \frac{1}{\varepsilon_0}\left[4\pi \rho_0 r^3\left(\frac{1}{3}-\frac{r}{4R}\right)\right]$$

$$\boxed{E_q = \frac{\rho_0 r}{\varepsilon_0}\left(\frac{1}{3}-\frac{r}{4R}\right)} \quad *$$

Field becomes zero when,

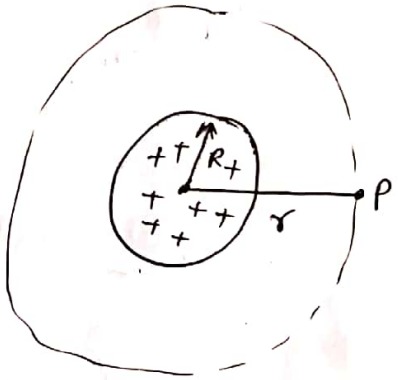
$$r\left(\frac{1}{3}-\frac{r}{4R}\right) = 0$$

$$r = 0 \;,\; 4R/3 \times$$

For Outside points :

$$\oint E\,ds = \frac{q_{encl}}{\varepsilon_0}$$

$$E \times 4\pi r^2 = \frac{q_{encl}}{\varepsilon_0} \quad - \quad (1)$$



To find q_{encl} :-

$$q = \int_{x=0}^{x=R} dq$$

$$q = 4\pi\rho_0\left(\frac{R^3}{3}-\frac{R^3}{4}\right) = \frac{4\pi\rho_0 R^3}{12}$$

(1) $\Rightarrow$ $E \times 4\pi r^2 = \dfrac{4\pi\rho_0 R^3}{12\varepsilon_0}$

$$\boxed{E_0 = \frac{\rho_0 R^3}{12\varepsilon_0 r^2}} \quad *$$

To find q :-

$$dq = \rho\, dv$$

$$dq = \rho_0\left(1-\frac{x}{R}\right)4\pi x^2\, dx$$

$$dq = 4\pi\rho_0\left(x^2-\frac{x^3}{R}\right)dx$$

$$q = \int_{x=0}^{x=r} dq$$

$$\boxed{q = 4\pi\rho_0\left[\frac{r^3}{3}-\frac{r^4}{4R}\right]}$$

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For $E = E_{max}$,

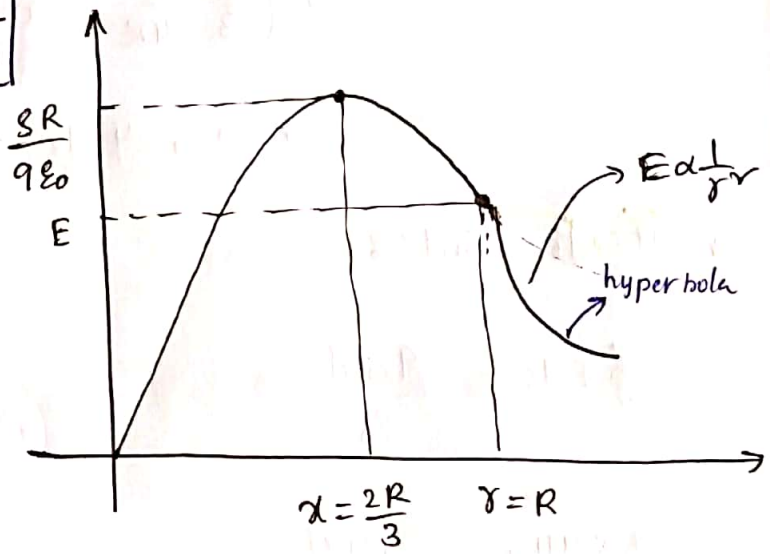
$$\frac{dE}{dr} = 0$$

$$E = \frac{\rho_0}{\varepsilon_0}\left(\frac{r}{3} - \frac{r^2}{4R}\right)$$

$$\frac{dE}{dr} = 0 \Rightarrow \frac{1}{3} - \frac{2r}{4R} = 0 \Rightarrow \boxed{r = \frac{2R}{3}}$$

$$E = \frac{\rho}{\varepsilon_0}\left(\frac{2R}{9} - \frac{4R^2}{936R}\right)$$

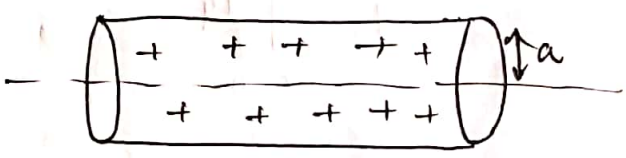
$$\boxed{E = \frac{\rho_0 R}{9\varepsilon_0}} \quad *$$



Ex! An infinite long cylinder having non uniform charge distribution such that volume charge density is directly proportional to distance of point from the cylinder.

charge density = 0 at $x = 0$.

$\rho = \rho_s$ at $x = a$.



Find electric field function in terms of x both inside & outside the cylinder.

Solⁿ Given $\rho \propto x$

$\rho = cx$

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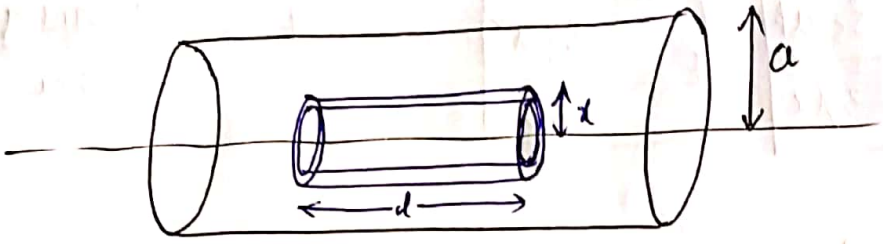
at $x = a$, $\rho = \rho_s$

$$\rho_s = ca$$

$$c = \frac{\rho_s}{a}$$

$$\therefore \boxed{\rho = \frac{\rho_s}{a} x}$$

Inside points :



$$\oint E \cdot ds = \frac{q_{encl}}{\varepsilon_0}$$

$$E \cdot 2\pi x l = \frac{q_{encl}}{\varepsilon_0}$$

$$E \cdot 2\pi x l = \frac{\rho_s (2\pi l) x^3}{3a\varepsilon_0}$$

$$\boxed{E = \frac{x^2 \rho_s}{3a\varepsilon_0}} \quad *$$

$$q = \int dq$$

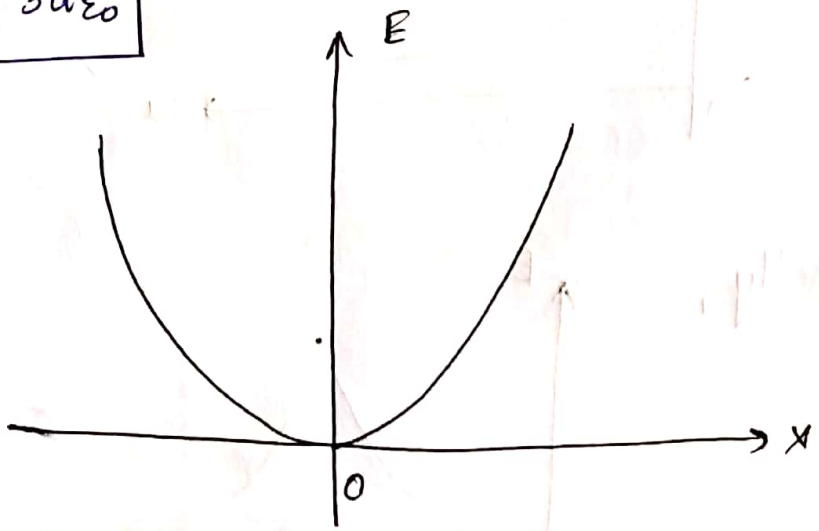
$$= \int_{x=x} \rho \, dV$$

$$= \int_{x=0} \frac{\rho_s}{a} x \, (2\pi x) l \, dx$$

$$= \frac{\rho_s}{a} 2\pi l \left[\frac{x^3}{3} \right]_0^x$$

$$q = \frac{\rho_s (2\pi l) x^3}{3a}$$

* $y = cx^n$ (parabola)



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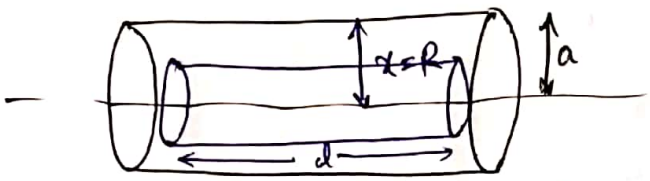
Outside points :



$$E \times 2\pi x l = \frac{\rho_s (2\pi l) R^3}{3a}$$

$$E = \frac{\rho_s R^3}{3x}$$

$$E = \frac{\rho_s a^3}{3x}$$

$$q = \int dq$$

$$= \frac{\rho_s 2\pi l}{a} \left[\frac{x^3}{3} \right]_0^R$$

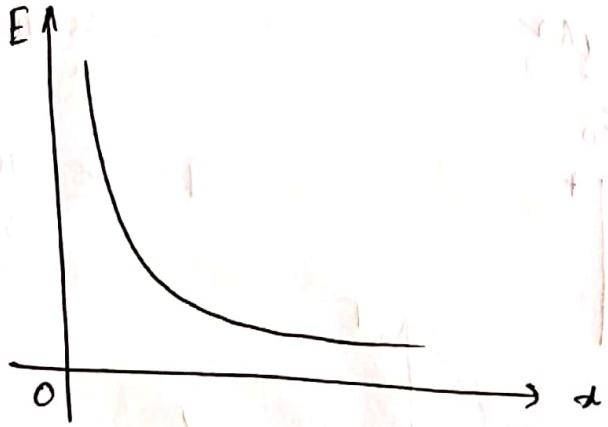
$$q = \frac{\rho_s 2\pi l a^3}{3a}$$

$$Ex = \frac{\rho_s a^2}{3a}$$

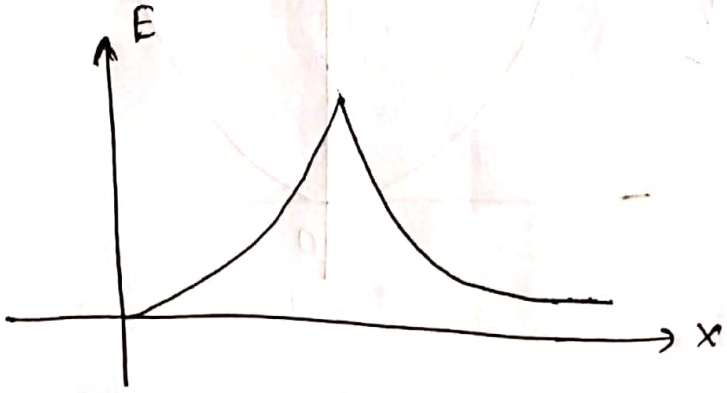
$\downarrow$

$xy = c^*$

(rectangular hyperbola)



Finally,



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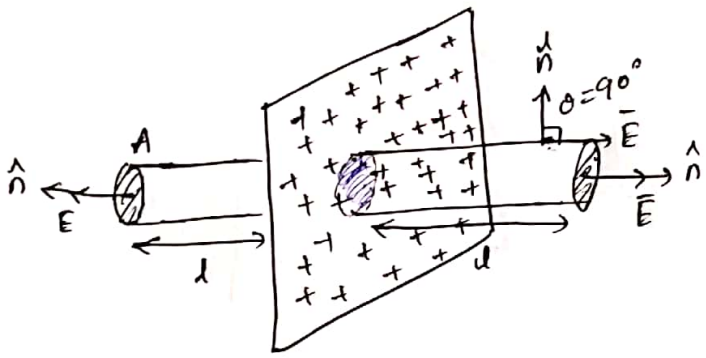
Infinite non-conducting charged sheet:



$$\phi = EA + EA$$

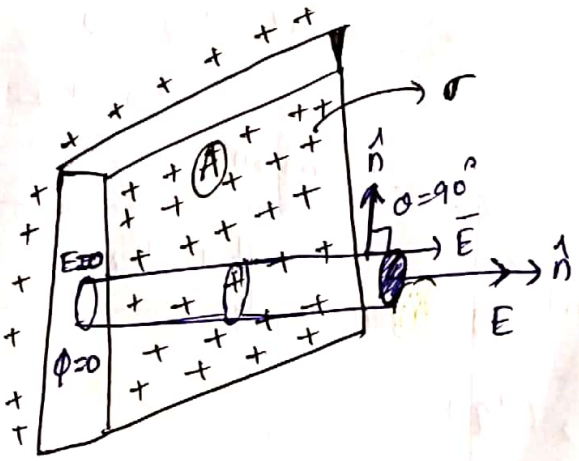
$$= 2EA$$

$$\phi = \frac{q_{encl}}{\varepsilon_0}$$

$$2EA = \frac{\sigma A}{\varepsilon_0}$$

$$\boxed{E = \frac{\sigma}{2\varepsilon_0}} \quad *.$$

Infinite conducting sheet: (combination of 2 non-conducting sheets.)



Total charge, $\boxed{q = 2\sigma A}$

Gaussian cylinder is considered such that one face lies inside the sheet & other face lies outside.

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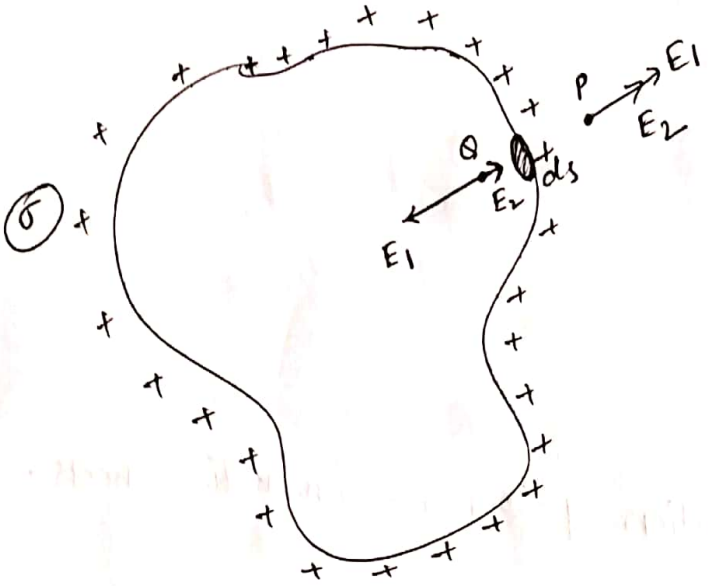
$$\phi = EA$$

$$\frac{q_{encl}}{\varepsilon_0} = EA$$

$$EA = \frac{\sigma A}{\varepsilon_0}$$

$$\boxed{E = \frac{\sigma}{\varepsilon_0}}$$

Electrostatic Pressure (P_E) :-



$$E_p = E_1 + E_2$$

we have,

$$\boxed{E_p = \frac{\sigma}{\varepsilon_0}}$$

i.e, $E_1 + E_2 = \frac{\sigma}{\varepsilon_0}$ — (1)

$$E_Q = E_1 - E_2$$

but $E_Q = 0$

$\Rightarrow E_1 = E_2$ — (2)

from (1) & (2)

$$\boxed{E_1 = E_2 = E = \frac{\sigma}{2\varepsilon_0}}$$

* Force on the element of area ds,

$$dF = E_2 (dq)$$

$$= E_2 (\sigma ds)$$

$$= \frac{\sigma}{2\varepsilon_0} (\sigma ds)$$

$$\boxed{dF = \frac{\sigma^2 ds}{2\varepsilon_0}}$$

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∴ Electrostatic pressure,

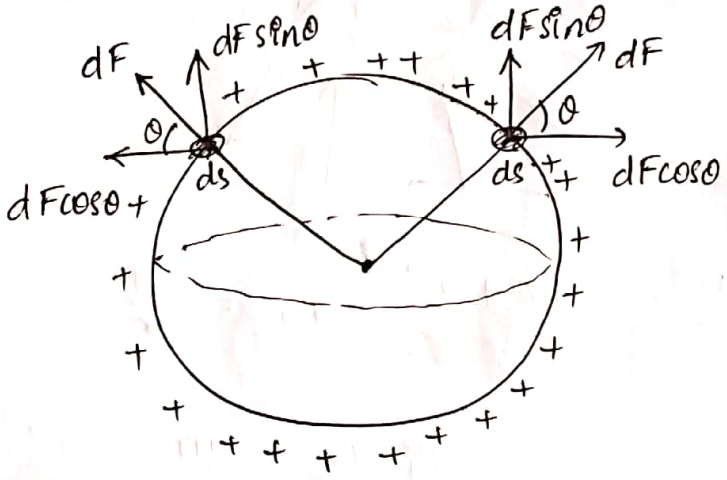
$$P_E = \frac{dF}{ds} = \frac{\sigma^2}{2\varepsilon_0} \quad *$$

we have, $E_{net} = \frac{\sigma}{\varepsilon_0}$

$$P_E = \frac{\sigma^2}{2\varepsilon_0} = \frac{1}{2}\varepsilon_0 E_{net}^2$$

Ex:

**Force on upper half of shell by lower half of shell:



$$F_{net} = \int dF\sin\theta$$

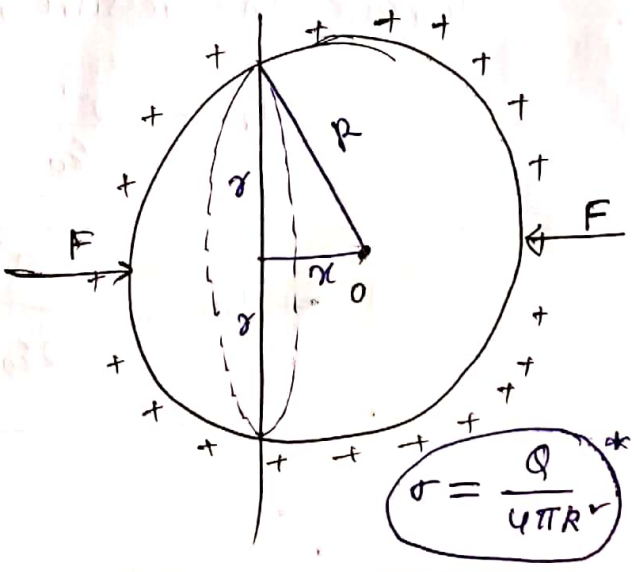
$$= \int (Edq)\sin\theta$$

$$= \int \frac{\sigma}{2\varepsilon_0}(\sigma ds)\sin\theta$$

$$F_{net} = \frac{\sigma^2}{2\varepsilon_0}\left(\int ds\sin\theta\right) \rightarrow \text{projection area}$$

$$F_{net} = \frac{\sigma^2}{2\varepsilon_0}(\pi R^2) \quad *$$

Ex: Find min force required to keep two parts together.



Solⁿ

$$F = \frac{\sigma^2}{2\varepsilon_0} \times \text{projection area}$$

$$= \frac{\sigma^2}{2\varepsilon_0}\pi r^2 = \frac{\sigma^2}{2\varepsilon_0}(R^2 - x^2)$$

$$\sigma = \frac{Q}{4\pi R^2} \quad *$$

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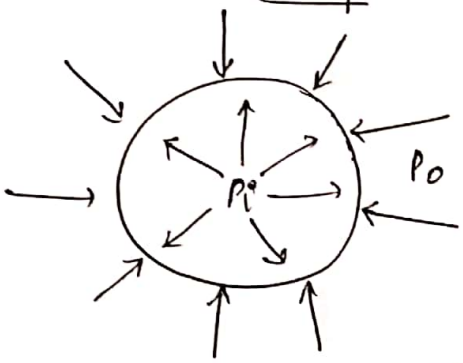
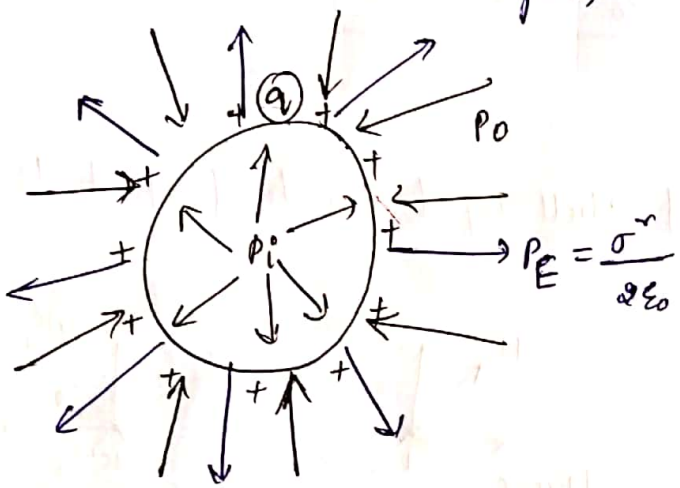
Charged Soap Bubble



* when there is no charge on the bubble,

$$P_i - P_o = \frac{4T}{r}$$

* when bubbled is charged,



$$P_i + \frac{\sigma^2}{2\varepsilon_0} - P_o = \frac{4T}{r}$$

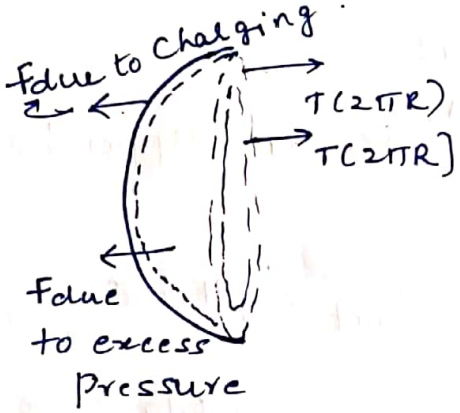
$$\boxed{P_i - P_o = \frac{4T}{r} - \frac{\sigma^2}{2\varepsilon_0}} \quad \text{——①}$$

Due to charging,

decrease in pressure,

$$-dp = \frac{\sigma^2}{2\varepsilon_0} = \frac{1}{2\varepsilon_0}\left(\frac{q}{4\pi r^2}\right)^2$$

$$\boxed{-dp = \frac{q^2}{32\varepsilon_0 \pi^2 r^4}}$$



$F_{ex} + F_{ch} = 4T(2\pi R)$

P_{ex} (proj. Area)

$+ \frac{\sigma^2}{2\varepsilon_0}$ [pro. Area]

$= 4T\pi R$

$P_{ex} A + \frac{\sigma^2}{2\varepsilon_0}(A)$

$= 4T\pi R$

$P_{ex} = \frac{4T\pi R}{A} \mp \frac{\sigma^2}{2\varepsilon_0}$

$P_{ex} = \frac{4T\pi R}{\pi R^2} - \frac{\sigma^2}{2\varepsilon_0}$

$P_{ex} = \frac{4T}{R} - \frac{\sigma^2}{2\varepsilon_0}$

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From Boyle's law,

$$PV = \text{const}$$

$$Pr^3 = \text{const}$$

$$\frac{dP}{P} = -3\frac{dr}{r}$$

* Increase in radius,

$$dr = -\frac{(dP)r}{3P}$$

$$\boxed{dr = \frac{q^2}{96\varepsilon_0\pi^2 r^3 P}}$$

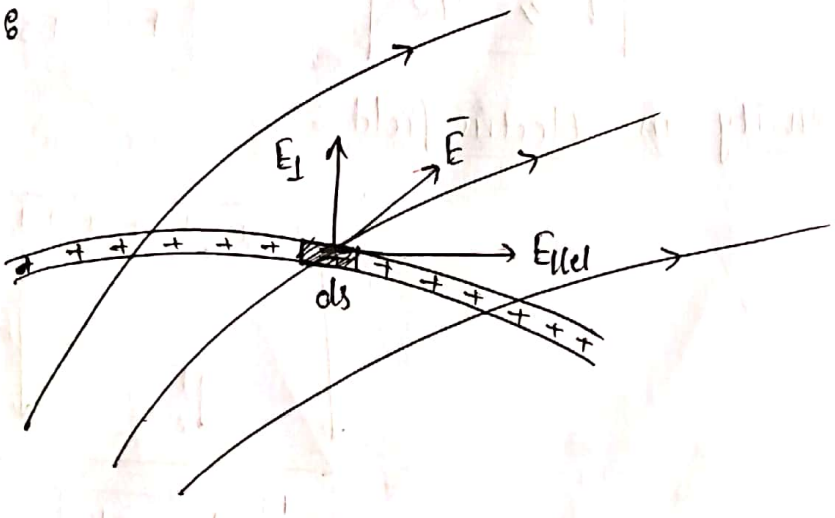
$$P_i - P_o = 0$$

when $\frac{4T}{r} = \frac{\sigma^2}{2\varepsilon_0}$

$$\boxed{\sigma = \sqrt{\frac{8T\varepsilon_0}{r}}}^{*}$$

$\Rightarrow$ Bubble will rapture.

Electric pressure when charged body is placed in external field &



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$E_{\parallel el} \longrightarrow$ tries to stretch the surface.

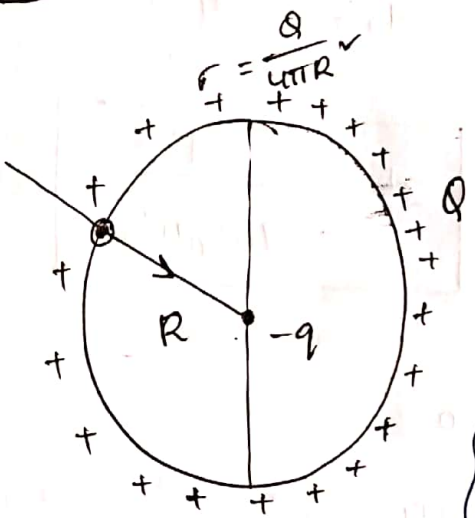
$E_{\perp er} \longrightarrow$ tries to push it outwards.

$$dF = E_{\perp er} \cdot dq$$

$$dF_\perp = \epsilon_\perp \sigma ds$$

$$\boxed{P_E = \frac{dF_\perp}{ds} = \sigma \epsilon_{\perp er}}$$

Ex:



* For two parts not to seperate,

$$(P_e)_f \geq (P_e)_p$$

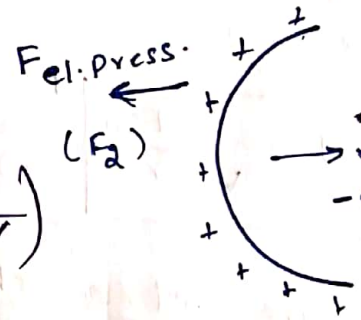
$$\not{\sigma}(E_\perp) \geq \frac{\sigma^{\not 2}}{2\epsilon_0}$$

$$\frac{1}{\not{4\pi}\epsilon_0} \frac{q}{R^{\not 2}} \geq \frac{1}{2\epsilon_0}\left(\frac{Q}{\not{4\pi} R^{\not 2}}\right)$$

$$\boxed{q \geq Q/2}^*$$

** F_1 on $-q$ = F_1 on hemisphere.

$$F_1 = E \cdot q \quad (\downarrow \frac{\sigma}{4\epsilon_0})$$

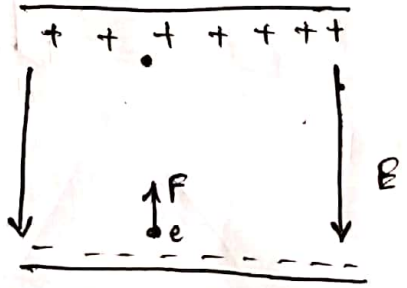


for not to seperate,

$F_1 = F_2$; $\frac{\sigma}{4\epsilon_0} \cdot 2 = \frac{\sigma^2}{2\epsilon_0} \cdot A$

9|04|25| wednesday

Energy density in Electric field :-



* $\boxed{\omega_e = \delta k}$

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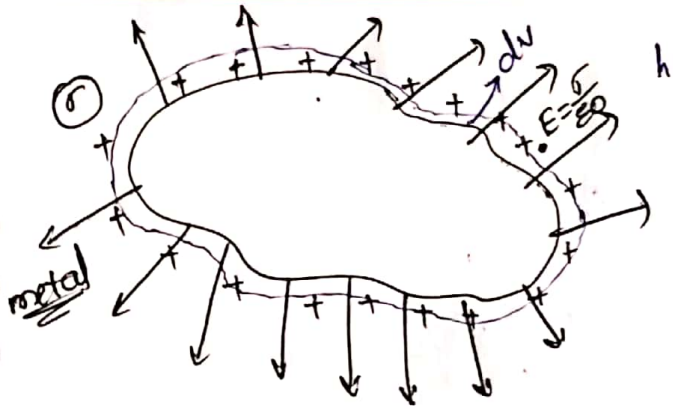


here decrease in field energy is used for expansion.

∴ stored field energy = work done.

decrease in energy,

$$dU = dW = P_E dV$$

Energy density,

$$\frac{dU}{dV} = P_E = \frac{\sigma^2}{2\varepsilon_0} = \frac{(\varepsilon_0 E)^2}{2\varepsilon_0}$$

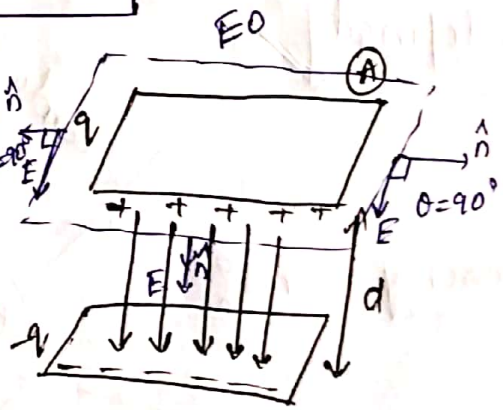
$$* \quad \boxed{u = \frac{1}{2}\varepsilon_0 E^2}$$

∴ Energy stored in a region of volume V,

$$\boxed{U = \int u \cdot dV}$$

$$* \quad \boxed{U = \int \frac{1}{2}\varepsilon_0 E^2 \cdot dV} \quad -①$$

Ex6 Find Energy stored in the region b/w the plates.



Solⁿ

$$U = \int \frac{1}{2}\varepsilon_0 E^2 dV$$

$$= \frac{1}{2}\varepsilon_0 E^2 V$$

$$\phi = EA$$

$$\frac{q}{\varepsilon_0} = EA$$

$$\boxed{E = \frac{q}{A\varepsilon_0}}$$

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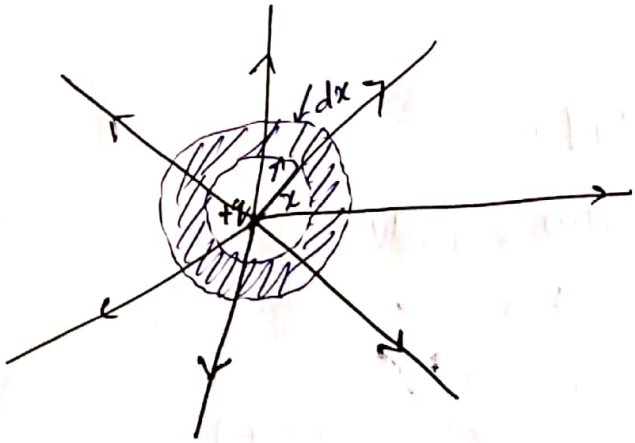
$$U = \frac{1}{2}\varepsilon_0 \left(\frac{q}{\varepsilon_0 A}\right)^2 \times (Ad)$$

$$\boxed{U = \frac{1}{2}\frac{q^2 d}{\varepsilon_0 A}}$$

Exp.



$$E \times 4\pi x^2 = \frac{q}{\varepsilon_0}$$

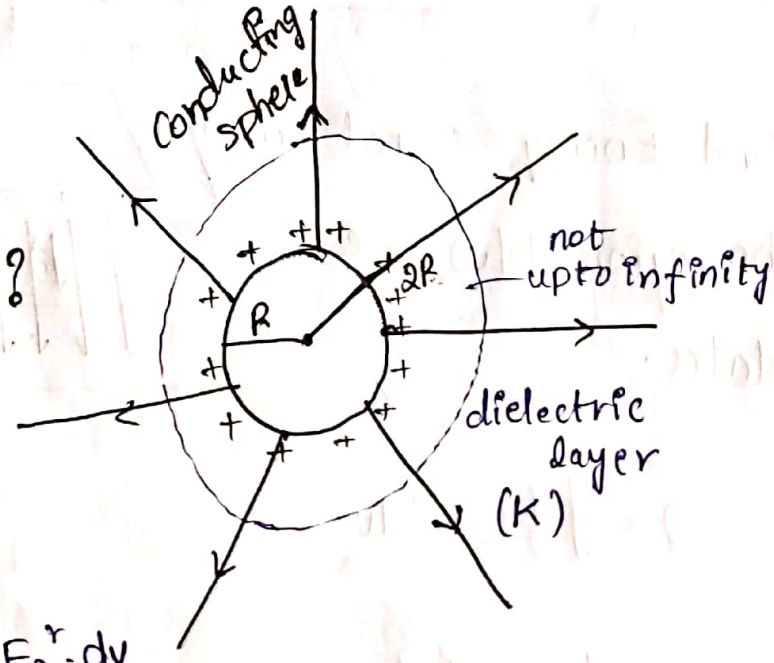
$$\boxed{E = \frac{q}{4\pi\varepsilon_0 x^2}}$$

$$U = \int_{x=0}^{x=\infty} \frac{1}{2}\varepsilon_0 \left(\frac{q}{4\pi\varepsilon_0 x^2}\right)^2 (4\pi x^2)\, dx$$

$$\boxed{U = \text{undefined}}$$

Exp.

Find energy stored?



Sol.

$$U = \int_{x=0}^{x=2R} \frac{1}{2}\varepsilon_0 K E_1^2 \cdot dv + \int_{x=2R}^{x=\infty} \frac{1}{2}\varepsilon_0 E_2^2 \cdot dv$$

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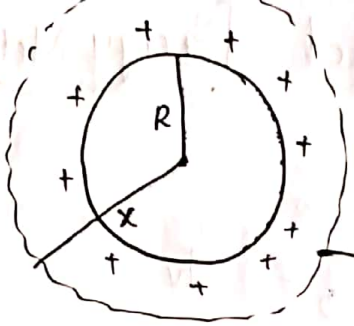
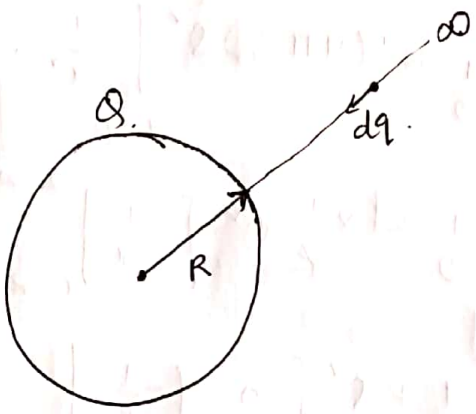
$$U = \int_{x=0}^{x=2R} \frac{1}{2}\varepsilon_0 K \left(\frac{q}{4\pi\varepsilon_0 K x^2}\right)^2 4\pi x^2 \cdot dx + \int_{x=2R}^{x=\infty} \frac{1}{2}\varepsilon_0 \left(\frac{q}{4\pi\varepsilon_0 x^2}\right)^2 4\pi x^2 \cdot dx$$



Self Energy :

① Conducting Sphere / Shell :

I-Method :



In Intermediate state,

potential at the surface of sphere, $V = \frac{KQ}{R}$

work done by an external agent in slowly bringing a charge from ∞ to surface of shell is,

$$dw = V\,dq$$

$$dw = \frac{KQ}{R}\,dq$$

∴ Total work done,

$$w = \int dw = \int \frac{KQ}{R} \cdot dq = \frac{K}{R}\int Q \cdot dq$$

$$\boxed{w = \frac{KQ^2}{2R}}$$ — electrostatic P.E (or) Self Energy.

$$\boxed{U_{self} = \frac{KQ^2}{2R}}$$ *

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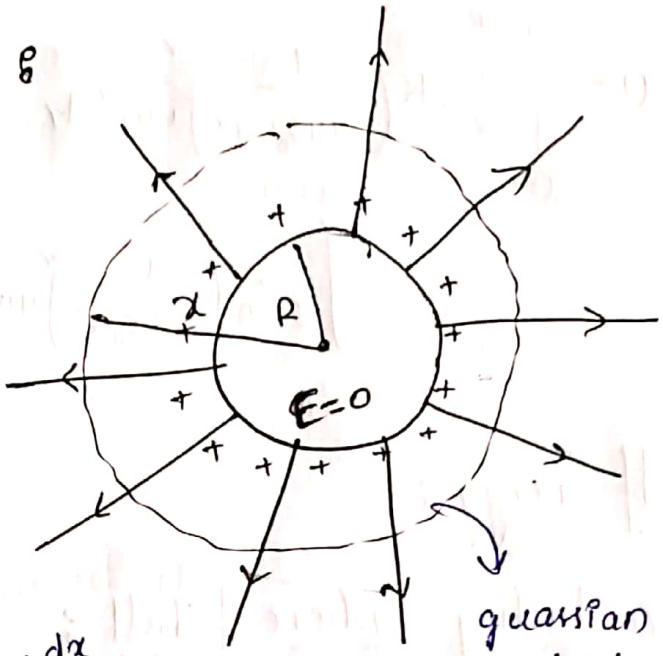
II- Method - [Energy Method] :



$$U = \int_{x=R}^{x=\infty} \frac{1}{2} \varepsilon_0 E^2 \, dv$$

$$= \int_{x=R}^{x=\infty} \frac{1}{2} \varepsilon_0 \left(\frac{KQ}{x^2}\right)^2 (4\pi x^2) \, dx$$

$$= \frac{1}{2} K^2 (4\pi \varepsilon_0) Q^2 \int_{x=R}^{x=\infty} \frac{1}{x^2} \cdot dx$$

$$= \frac{1}{2} K^2 \times \frac{1}{K} Q^2 \left[-\frac{1}{x}\right]_{x=R}^{x=\infty}$$

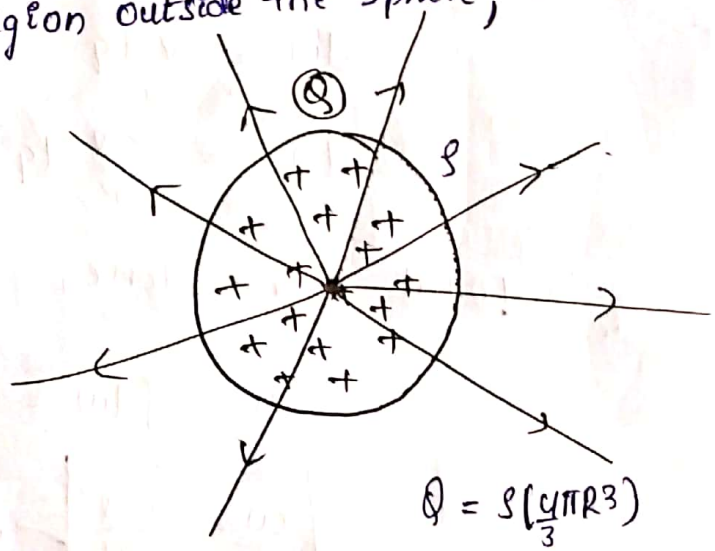
$$= \frac{1}{2} KQ^2 \left[0 - \left(-\frac{1}{R}\right)\right]$$

$$\boxed{U = \frac{1}{2} \frac{KQ^2}{R}}$$

$$\boxed{E = \frac{KQ^2}{x^2}}$$

(2) Non-Conducting charged sphere :

* Energy stored in the region outside the sphere,



$$U_0 = \int_{x=R}^{x=\infty} \frac{1}{2} \varepsilon_0 E_1^2 \cdot dv$$

$$= \frac{1}{2} \int \varepsilon_0 \left(\frac{KQ}{x^2}\right)^2 4\pi x^2 \cdot dx$$

$$\boxed{U_0 = \frac{KQ^2}{2R}}$$

* Energy stored inside the sphere,

$$U_i = \int_{x=0}^{x=R} \frac{1}{2} \varepsilon_0 E^2 \cdot dx$$

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$$= \int_{x=0}^{x=R} \frac{1}{2}\varepsilon_0 \left(\frac{KQx}{R^3}\right)^2 4\pi x^2 \cdot dx$$

$$= \frac{1}{2} \frac{KQ^2}{R^6} \int_{x=0}^{x=R} x^4 \cdot dx$$

$$= \frac{1}{2} \frac{KQ^2}{R^6} \times \frac{R^5}{5}$$

$$\boxed{U_0 = \frac{KQ^2}{10R}}$$

now, $\frac{U_i}{U_0} = \frac{\frac{KQ^2}{5(2R)}}{\frac{KQ^2}{2R}} \Rightarrow \frac{U_i}{U_0} = \frac{1}{5} \Rightarrow \boxed{U_0 = 5U_i}$

$$U_{Total} = U_i + U_0$$

$$= \frac{KQ^2}{10R} + \frac{KQ^2}{2R}$$

$$\boxed{U_{Total} = \frac{3KQ^2}{5R}} \leftarrow \text{self energy.}$$

Ex: Find work done by the electric field during the expansion of sphere from radius R to 2R.

Solⁿ

$$\Delta U = \frac{KQ^2}{2(2R)} - \frac{KQ^2}{2R} \quad (\because \Delta U = U_f - U_i)$$

$$\Delta U = -\frac{KQ^2}{4R}$$

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$$W = -\Delta U = \frac{KQ^2}{4R}$$

$$\boxed{W = \frac{KQ^2}{4R}} \quad *$$

Total Energy of System :

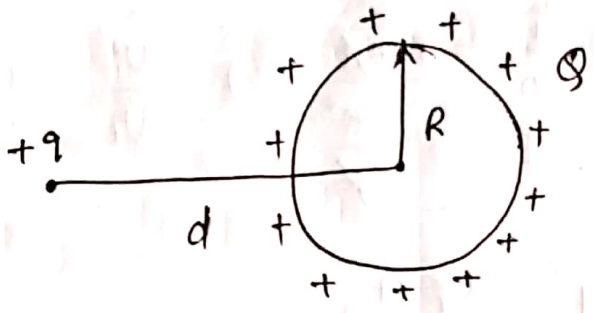
$$U = U_{self} + U_{interaction}$$

U_{self} : +ve

$U_{interaction}$: +ve / -ve

Ex:

Find total energy of system.

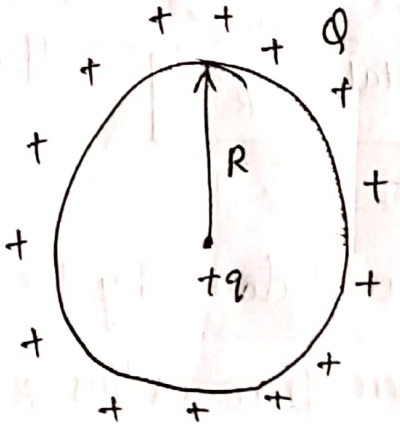


Sol:

$$\boxed{U = \frac{KQ^2}{2R} + \frac{KQq}{d}}$$

Ex:

Q) Find Potential Energy of system?



Sol: M-2:

$$\boxed{U = \frac{KQ^2}{2R} + \frac{KQq}{R}}$$

Q) Work done on expanding the shell from R to 3R. by electric field

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$$U_i = \frac{KQ^2}{2R} + \frac{KQq}{R}$$

$$U_f = \frac{KQ^2}{2(3R)} + \frac{KQq}{3R}$$

$$\boxed{W_e = U_i - U_f}$$

M-II:

$$U = \int_{x=0}^{x=R} \frac{1}{2}\varepsilon_0 \left(\frac{q}{4\pi\varepsilon_0 x^2}\right)^2 (4\pi x^2)\, dx + \int_{x=R}^{x=\infty} \frac{1}{2}\varepsilon_0 \left(\frac{(q+Q)}{4\pi\varepsilon_0 x^2}\right)^2 (4\pi x^2)\, dx$$

$$= \frac{1}{2}Kq^2\left[-\frac{1}{x}\right]_{x=0}^{x=R} + \frac{1}{2}K(q+Q)^2\left[-\frac{1}{x}\right]_{x=R}^{x=\infty}$$

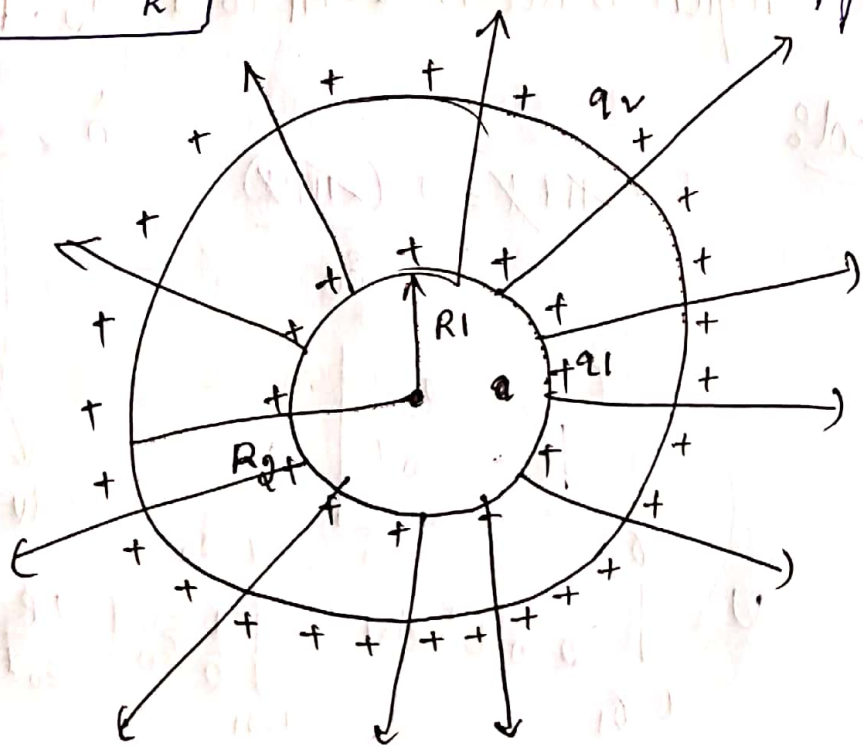
$$= -\frac{1}{2}\frac{Kq^2}{R} + \frac{1}{2}K(q^2+Q^2+2qQ)\left[\frac{1}{R}\right]$$

$$\boxed{U = \frac{KQ^2}{2R} + \frac{KQq}{R}}$$

** It is not valid as point charge has no self energy.

Ex6

Find Energy of the system



Sol: M-I:

$$U = \frac{Kq_1^2}{2R_1} + \frac{Kq_2^2}{2R_2} + \frac{Kq_1q_2}{R_2}$$

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M-II:

$$U = \int_{x=R_1}^{x=R_2} \frac{1}{2}\varepsilon_0 (E_1)^2 \cdot dV + \int_{x=R_2}^{x=\infty} \frac{1}{2}\varepsilon_0 (E_2)^2 \cdot dV$$

$$= \int_{x=R_1}^{x=R_2} \frac{1}{2}\varepsilon_0 \left(\frac{kq_1}{x^2}\right)^2 \cdot 4\pi x^2 \cdot dx + \int_{x=R_2}^{x=\infty} \frac{1}{2}\varepsilon_0 \left[\frac{k(q_1+q_2)}{x^2}\right]^2 4\pi x^2 \cdot dx$$

$$= \frac{1}{2} k q_1^2 \left[\frac{1}{R_1} - \frac{1}{R_2}\right] + \frac{1}{2} k (q_1+q_2)^2 \left[\frac{1}{R_2}\right]$$

$$= \frac{1}{2}\frac{kq_1^2}{R_1} - \frac{1}{2}\frac{kq_1^2}{R_2} + \frac{1}{2} k (q_1^2 + q_2^2 + 2q_1q_2)\left[\frac{1}{R_2}\right]$$

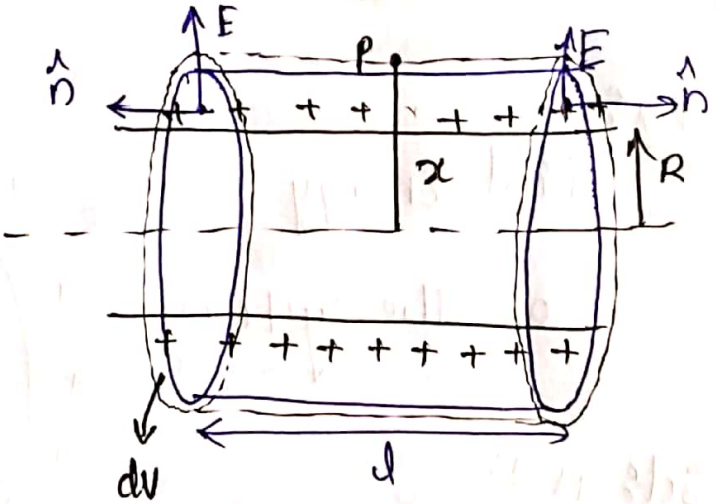
$$\boxed{U = \frac{1}{2}\frac{kq_1^2}{R_1} + \frac{1}{2}\frac{kq_2^2}{R_2} + \frac{kq_1q_2}{R_2}}$$

Ex: Find the energy stored per unit length in the region between $x = R_1$ to $x = R_2$. Here $r_2 > r_1 > R$.

Sol:

$$E \times 2\pi x l = \frac{\sigma (2\pi R l)}{\varepsilon_0}$$

$$\boxed{E = \frac{\sigma R}{\varepsilon_0 x}}$$



$$U = \int_{x=R_1}^{x=R_2} \frac{1}{2}\varepsilon_0 E^2 \cdot dV = \int_{x=r_1}^{x=r_2} \frac{1}{2}\varepsilon_0 \left(\frac{\sigma R}{\varepsilon_0 x}\right)^2 (2\pi x l) \cdot dx$$

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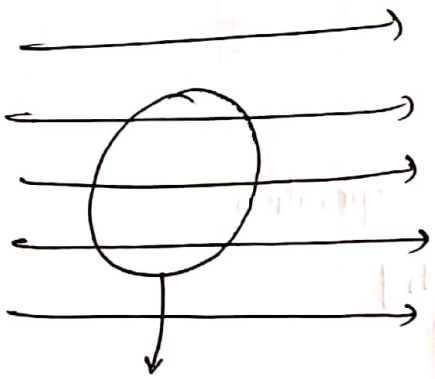
$$= \frac{1}{2} \varepsilon_0 \frac{\sigma^2 R^2}{\varepsilon_0^2} (2\pi l) \ln(r_2/r_1)$$

$$U = \frac{\pi \sigma^2 R^2 l}{\varepsilon_0} \ln(r_2/r_1)$$

10|04|25 | Thursday

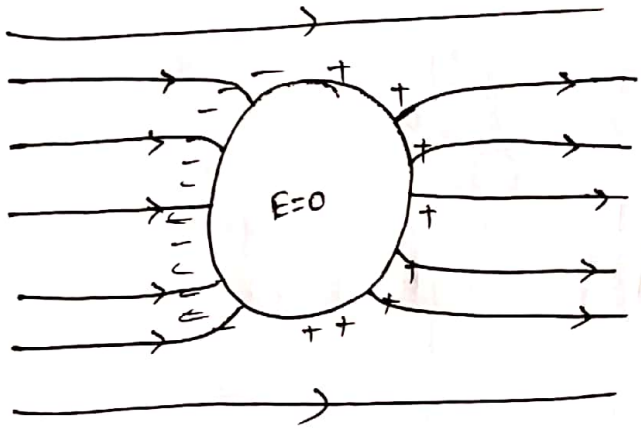
Properties of a Conductor :

①



before Induction

neutral metal sphere



After Induction

$$\bar{E} + \bar{E}_{ind} = 0$$

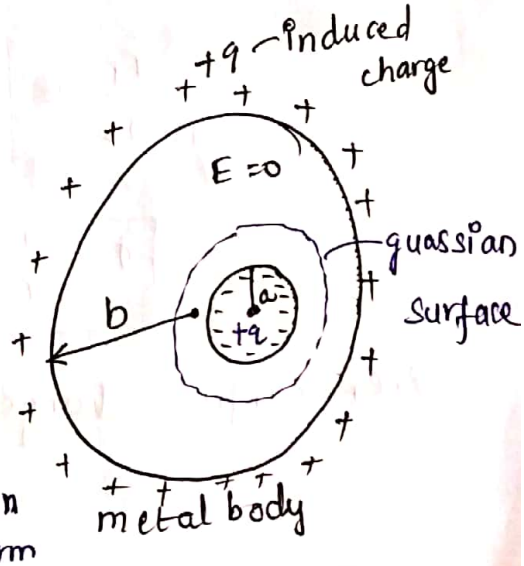
$$\boxed{\bar{E}_{ind} = -\bar{E}}$$

② Charge inside the metal body :

$E = 0$ (at any inside point)

$$\oint \bar{E} \cdot d\bar{s} = 0$$

$$\frac{q_{end}}{\varepsilon_0} = 0$$



* Charge distribution is uniform

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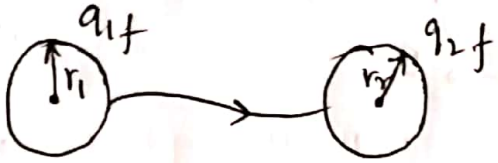
$$\boxed{q_{encl} = 0}^{*}$$

Induced charge inside the cavity is distributed to make net field and potential at every point outside the cavity is zero.

Charge distribution in metal bodies:

At very large seperation.



$$q_1 + q_2 = q_{1f} + q_{2f}$$

Also, $V_1 = V_2$

$$\frac{kq_{1f}}{r_1} = \frac{kq_{2f}}{r_2}$$

$$\frac{\sigma_1 4\pi r_1^2}{r_1} = \frac{\sigma_2 4\pi r_1^2}{r_2}$$

$$\frac{\sigma_1}{\sigma_2} = \frac{r_2}{r_1}$$

$$\boxed{\sigma \propto \frac{1}{r}}$$

i.e, surface charge density is inversly proportional

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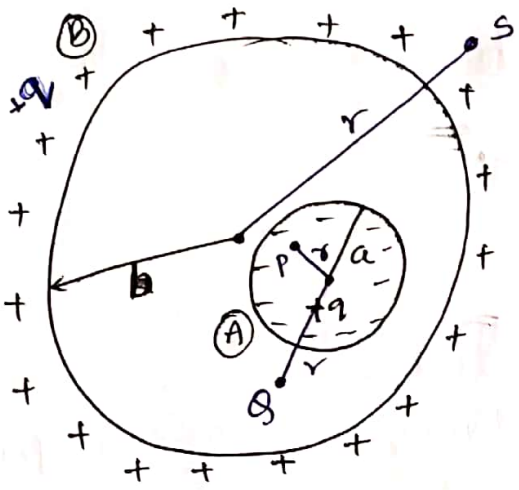


$$\sigma_A = \frac{-q}{4\pi a^2}, \text{ uniform}$$

$$\sigma_B = \frac{q}{4\pi b^2}, \text{ uniform}$$

At point P,

$$E_P = \frac{kq}{r^2} + 0 + 0 = \frac{kq}{r^2}$$

$$V_P = \frac{kq}{r} - \frac{kq}{a} + \frac{kq}{b}$$

At point Q,

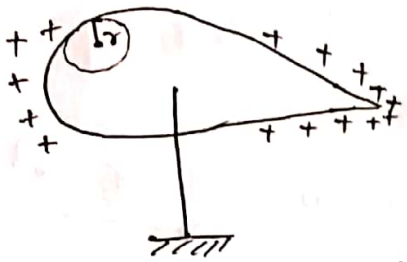
$$E_Q = 0$$

$$V_Q = \frac{kq}{b}$$

At points,

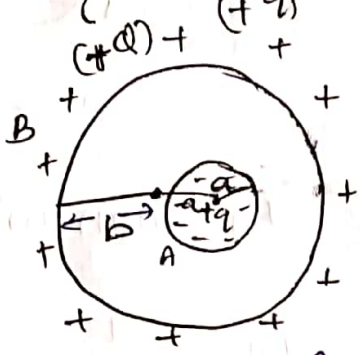
$$E_S = \frac{kq}{r^2}$$

$$V_S = \frac{kq}{r}$$



charge of body

induced charge



$$\sigma_A = \frac{-q}{4\pi a^2}, \text{ uniform}$$

$$\sigma_B = \frac{Q+q}{4\pi b^2}, \text{ uniform}$$

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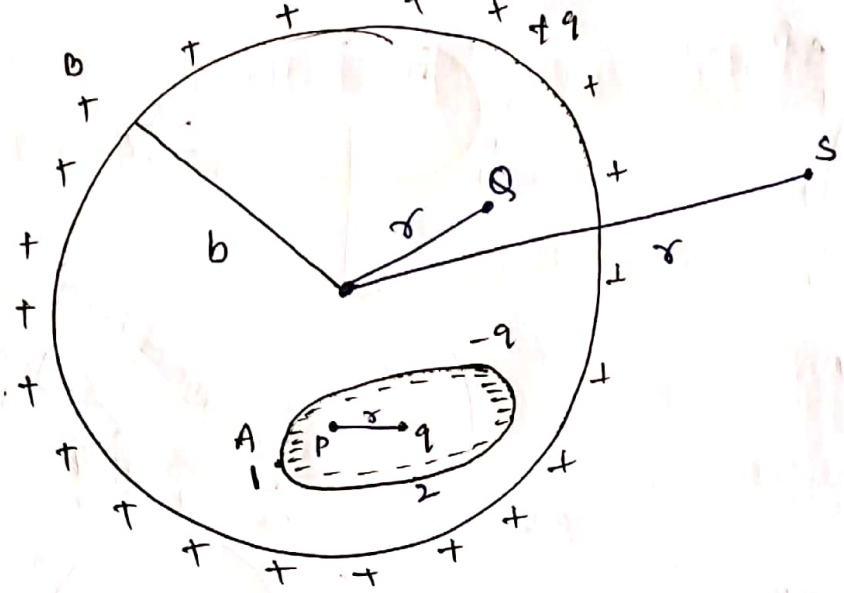
*



σ_A = non uniform

$\sigma_B = \frac{q}{4\pi b^2}$, uniform

* on the cavity surface,

$V_1 = V_2$ $\quad$ $\vec{E}.d\vec{r}$ → zero

$\sigma_1 \neq \sigma_2$ $\quad$ E → $\perp^{er}$ to Cavity surface.

$E_1 \neq E_2$ $\quad$ [∵ Metallic]

* we cannot determine potential at P.

* At S,

$$E_S = \frac{Kq}{r^2}$$

$$V_S = \frac{Kq}{r}$$

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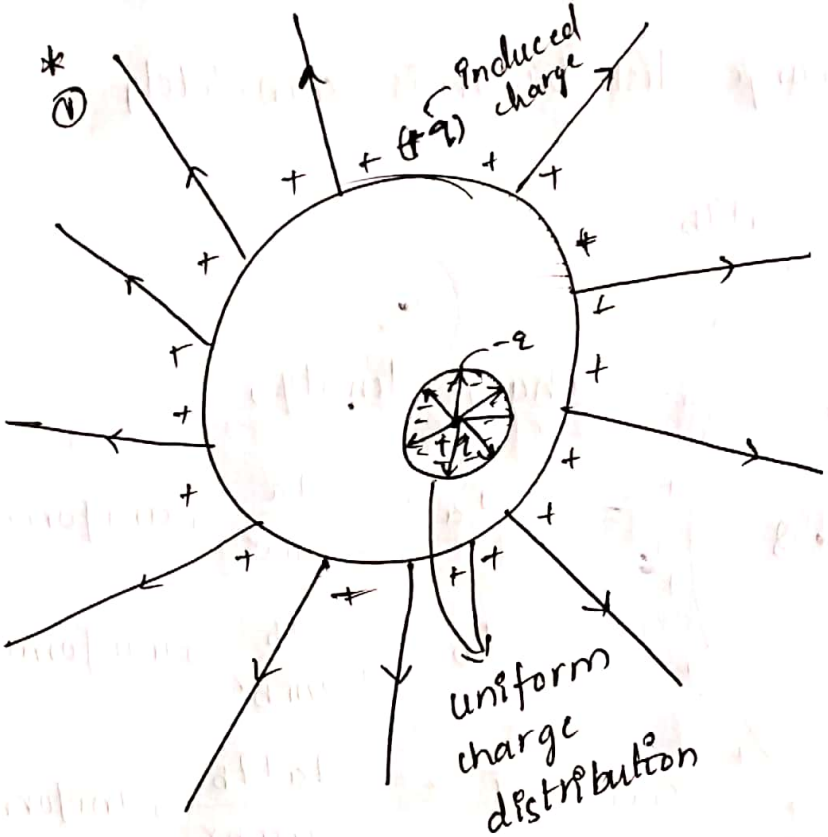
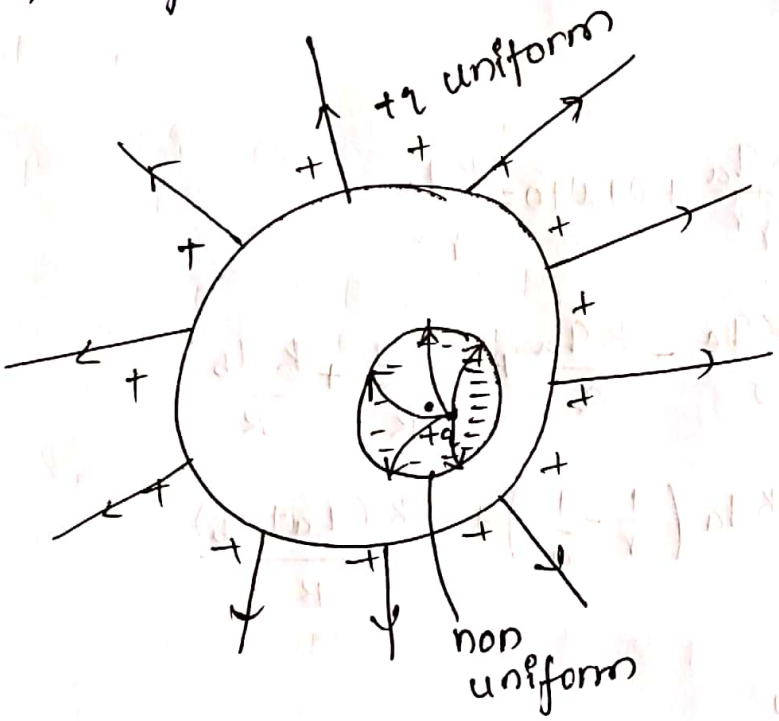
*

(1)

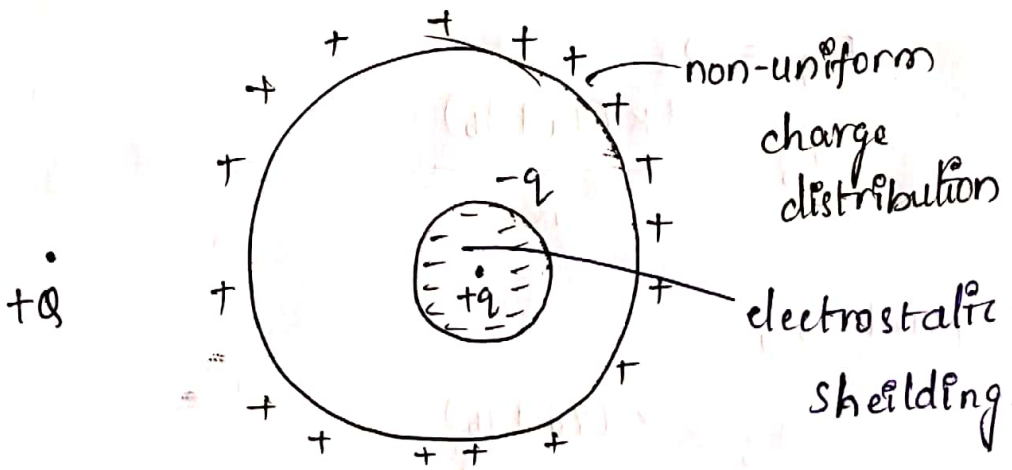


Now, charge at centre of cavity is slightly displaced.

(2)



(3)



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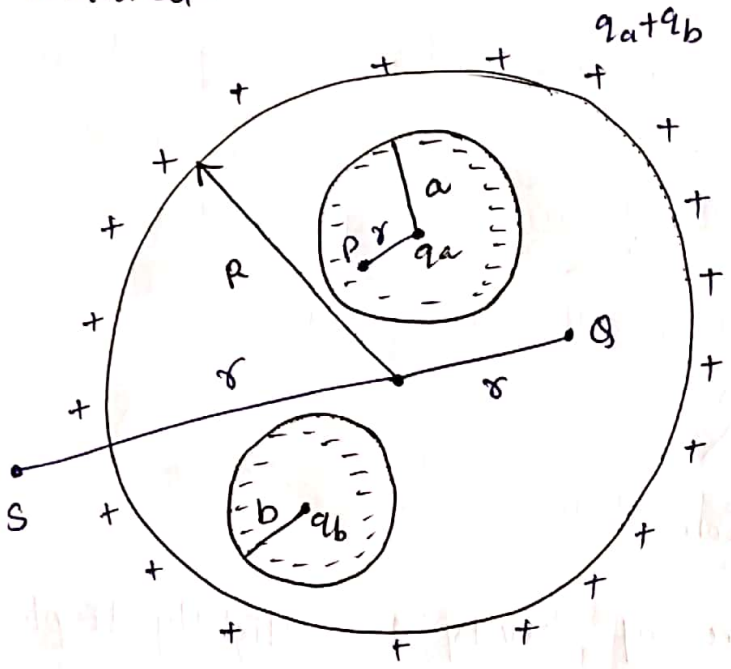
Here the effect of charge distribution is completely isolated.

*



charge density,

$$\sigma_a = \frac{-q_a}{4\pi a^2}, \text{ uniform}$$

$$\sigma_b = \frac{-q_b}{4\pi b^2}, \text{ uniform}$$

$$\sigma = \frac{q_a+q_b}{4\pi R^2}, \text{ uniform}$$

At Point P,

$$E_p = \frac{Kq_a}{r^2} + 0 + 0 + 0 = \frac{Kq_a}{r^2}$$

$$V_p = \frac{Kq_a}{r} - \frac{Kq_a}{a} + \frac{Kq_a}{R} + \frac{Kq_b}{R}$$

$$= Kq_a\left(\frac{1}{r} - \frac{1}{a}\right) + \frac{K(q_a+q_b)}{R}$$

At point Q,

$$E_Q = 0$$

$$V_Q = \frac{K(q_a+q_b)}{R}$$

At point S,

$$E_S = \frac{K(q_a+q_b)}{r^2}$$

$$V_S = \frac{K(q_a+q_b)}{r}$$

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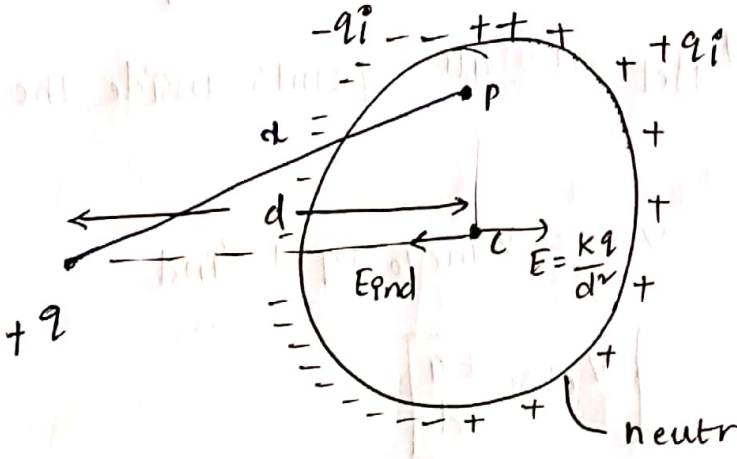
**Force on the charges q_a & q_b is zero.

$$E_{net} = 0$$

$$\Rightarrow F = Eq$$

$$\boxed{F = 0}$$

Ex 8



At centre c,

$$E_c = 0$$

$$\bar{E}_{due\ to\ +q} + \bar{E}_{pnd} = 0$$

$$\bar{E}_{pnd} = -\bar{E}_{due\ to\ +q}$$

$$= -\frac{kq}{d^2}$$

$$\boxed{\bar{E}_{pnd} = \frac{kq}{d^2}, \text{ towards } +q}$$

At any point P,

$$E_p = 0$$

$$\bar{E}_{due\ to\ +q} + \bar{E}_{pnd} = 0$$

$$\bar{E}_{pnd} = -\frac{kq}{x^2}$$

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$$E_{ind} = \frac{Kq}{x^2}, \text{ along } qP$$

Similarly, potential:

① At centre C,

Metal body is equipotential surface, volume is also equipotential vol.

V_{Net} at all points inside the body is same.

$$V_c = V_{due\ to\ +q} + V_{ind}$$

$$\boxed{V_c = \frac{Kq}{d}}$$

③ At point P,

$$V_p = V_{due\ to\ +q} + V_{ind}$$

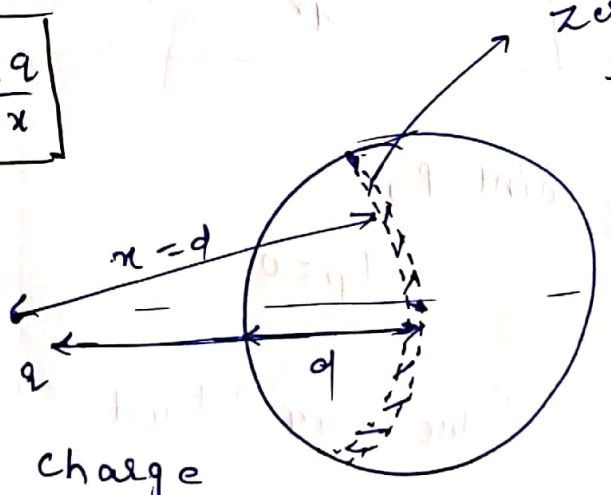
$$\Rightarrow V_p = \frac{Kq}{x} + V_{ind}$$

we have,

$$V_c = V_p$$

$$\frac{Kq}{d} + \cancel{\phi} = \frac{Kq}{x} + V_{ind}$$

$$\boxed{V_{ind} = \frac{Kq}{d} - \frac{Kq}{x}}$$



*, when $x = d$,

$V_{ind} = 0$.

Here, induced charge distribution is not uniform due to the presence of 'q'.

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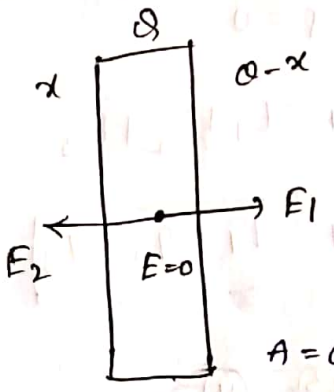
** Charge Distribution in metal plates:



A = area of face

we have,

$E = 0$

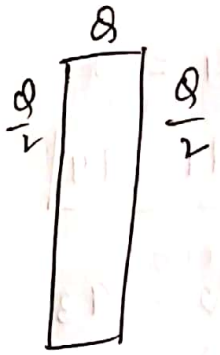
$E_1 - E_2 = 0$

$E_1 = E_2$

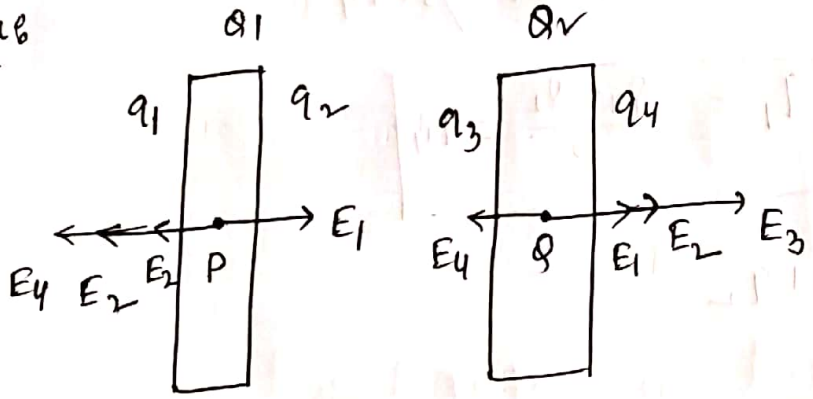
$$\frac{x}{2\varepsilon_0 A} = \frac{Q-x}{2\varepsilon_0 A}$$

$2x = Q$

$\boxed{x = Q/2}$



Ex 6



1) $Q_1 = q_1 + q_2$ — ①

2) $Q_2 = q_3 + q_4$ — ②

3) $E_P = 0$

$\Rightarrow E_1 - E_2 - E_3 - E_4 = 0$

$E_1 = E_2 + E_3 + E_4$

$$\frac{q_1}{2\varepsilon_0 A} = \frac{q_2}{2\varepsilon_0 A} + \frac{q_3}{2\varepsilon_0 A} + \frac{q_4}{2\varepsilon_0 A}$$

$$\boxed{q_1 = q_2 + q_3 + q_4} \quad -③$$

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Similarly,

$$E_Q = 0$$

$$E_4 = E_1 + E_2 + E_3$$

$$\boxed{q_4 = q_1 + q_2 + q_3} \quad -④$$

from ①, ②, ③, ④,

③, ④ ⇒ $q_1 - q_4 = 0$

$$\boxed{q_1 = q_4}$$

⇒ $q_2 = -q_3$

① + ② ⇒ $Q_1 + Q_2 = q_1 + q_2 + q_3 + q_4$

$$\boxed{q_1 = \frac{Q_1 + Q_2}{2} = q_4}$$

$$Q_1 = q_1 + q_2$$

$$Q_1 - \frac{Q_1 + Q_2}{2} = q_2$$

$$\boxed{q_2 = \frac{Q_1 - Q_2}{2}}$$

⇒ $$\boxed{q_3 = \frac{Q_2 - Q_1}{2}}$$

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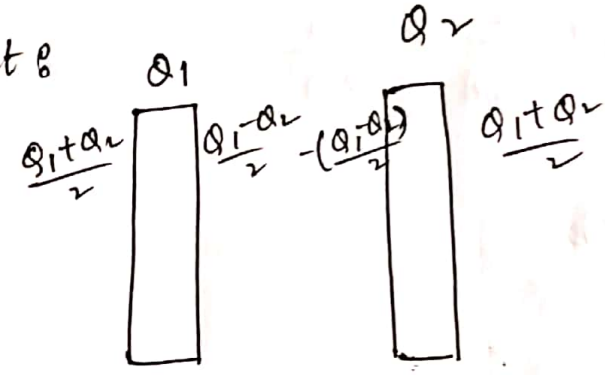
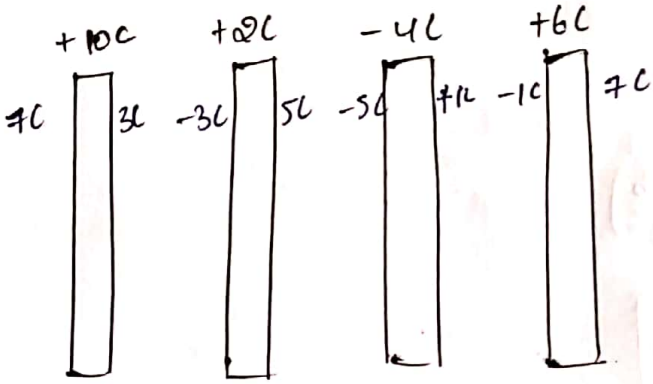
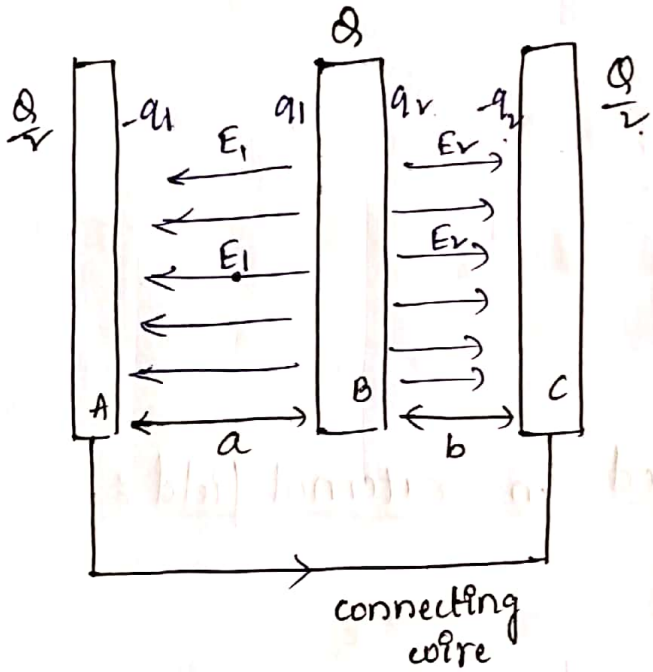
Short cut :



Ex :



Ex :



$$V_A = V_C$$

$$V_B - V_A = V_B - V_C$$

$$E_1 a = E_2 b$$

* where, $E_1 = \frac{q_1}{2\varepsilon_0 A} + \frac{q_1}{2\varepsilon_0 A} = \frac{q_1}{\varepsilon_0 A}$

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$$E_2 = \frac{q_2}{2\varepsilon_0 A} + \frac{q_2}{2\varepsilon_0 A} = \frac{q_2}{\varepsilon_0 A}$$

$$\frac{q_1}{\varepsilon_0 A}\, a = \frac{q_2}{\varepsilon_0 A}\, (b)$$

$$q_1 a = q_2 b \rightarrow (1)$$

and $Q = q_1 + q_2$

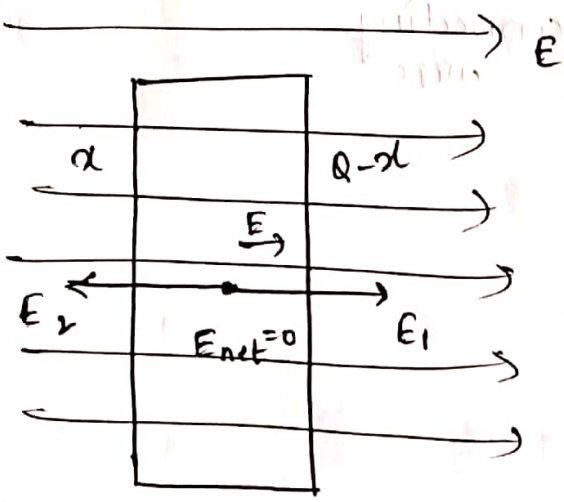
$$Q = q_2\left(\frac{a+b}{a}\right)$$

$$\boxed{q_2 = \frac{Qa}{a+b}}$$

$$\Rightarrow \boxed{q_1 = \frac{Qb}{a+b}}$$

*

Conducting sheet placed in external field E



$E_{net} = 0$

$E + E_1 = E_2$

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$$\frac{x}{2\varepsilon_0 A} + E = \frac{Q-x}{2\varepsilon_0 A}$$

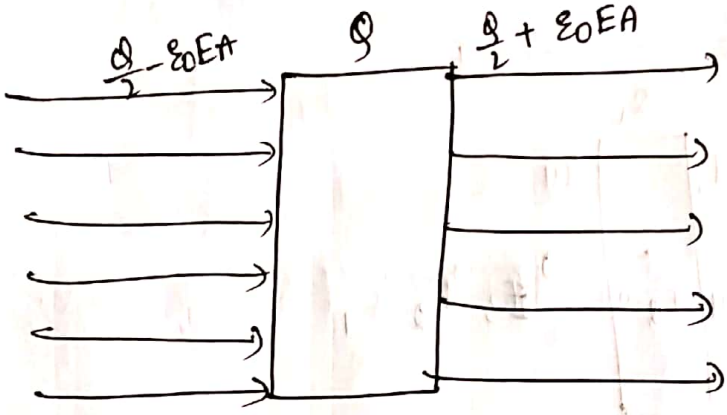
$$E = \frac{Q-2x}{2\varepsilon_0 A}$$

$$Q - 2x = 2\varepsilon_0 EA$$

$$x = \frac{Q}{2} - \underbrace{\varepsilon_0 EA}_{\text{Induced charge}}$$

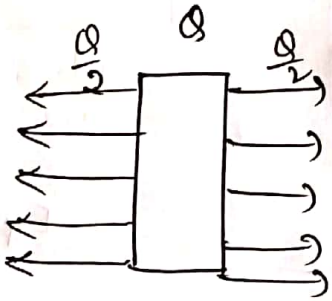
&

$$Q - x = \frac{Q}{2} + \underbrace{\varepsilon_0 EA}_{\text{Induced charge.}}$$



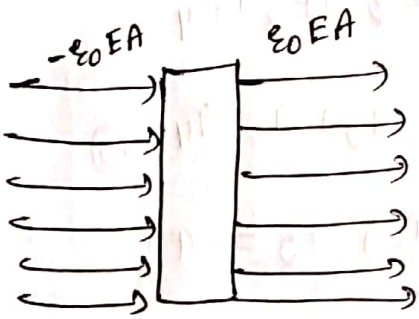
Points

① If $E = 0$



② If $Q = 0$ but $E \neq 0$

(neutral metal plate)



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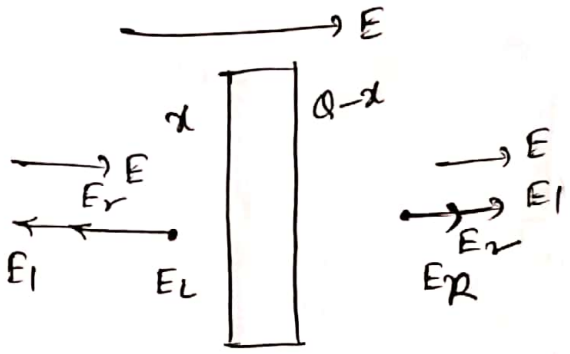
③

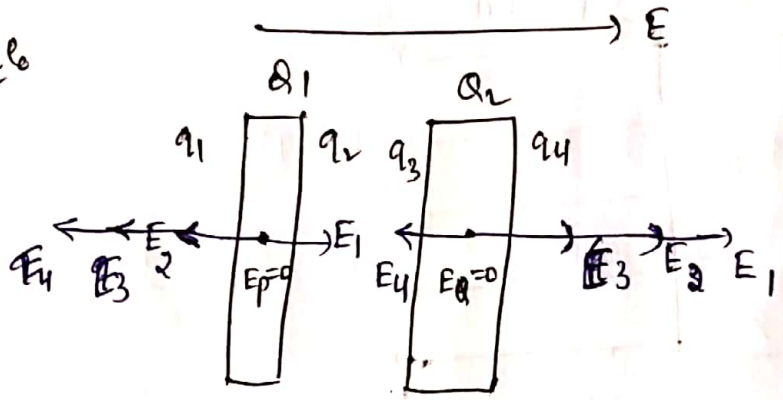


$$E_R = E_1 + E_2 + E$$

$$= E + \frac{x}{2\varepsilon_0 A} + \frac{Q-x}{2\varepsilon_0 A}$$

$$E_L = E_1 + E_2 - E$$

Ex 6



Sol 6

$$Q_1 = q_1 + q_2 \quad -①$$

$$Q_2 = q_3 + q_4 \quad -②$$

$E_P = 0$

$$E_1 + E = E_2 + E_3 + E_4$$

$$q_1 + 2\varepsilon_0 EA = q_2 + q_3 + q_4 \quad -③$$

$E_Q = 0$

$$E + E_1 + E_2 + E_3 = E_4$$

$$2\varepsilon_0 EA + q_1 + q_2 + q_3 = q_4 \quad -④$$

③, ④ $\Rightarrow$ $q_2 = -q_3$

③ + ④ $\Rightarrow$ $2q_1 + 4\varepsilon_0 EA = 2q_4$

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$$q_1 + 2\varepsilon_0 EA = q_4$$

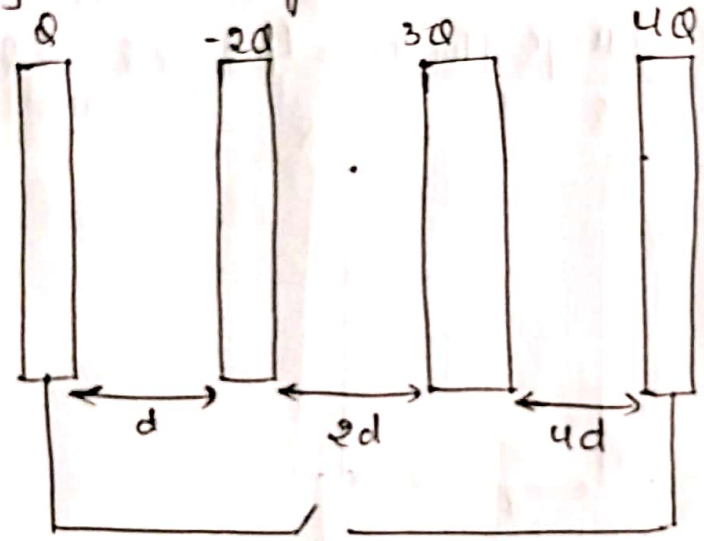
$$① + ② \Rightarrow Q_1 + Q_2 = q_1 + q_2 + q_3 + q_4$$

$$Q_1 + Q_2 = 2q_1 + 2\varepsilon_0 EA$$

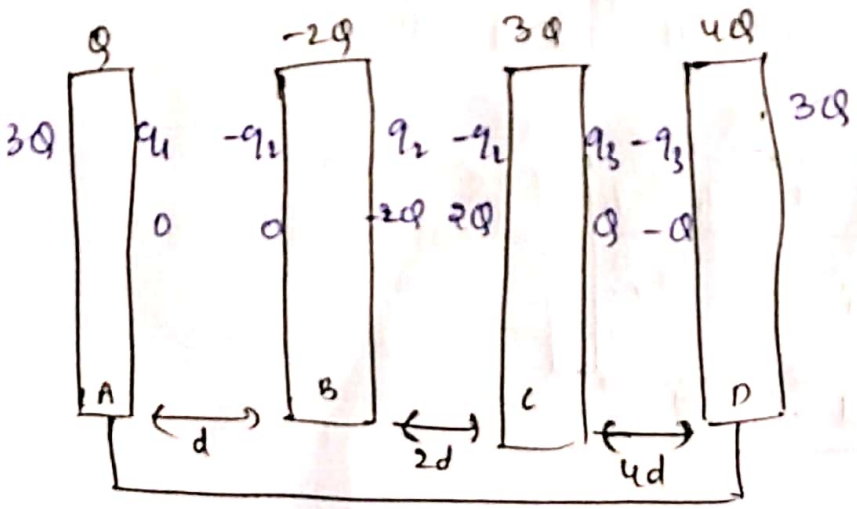
$$\boxed{q_1 = \frac{Q_1 + Q_2}{2} - \varepsilon_0 EA}$$

$$\boxed{q_4 = \frac{Q_1 + Q_2}{2} + \varepsilon_0 EA}$$

Ex: In the following diagram the plates are conducting and are held parallel, then amount of charge that flows through the switch after closing it is



Sol:



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$V_A = V_D$

$V_B - V_A = V_B - V_D$

$V_B - V_A = V_{BC} + V_{CD}$

$E_1(d) = E_2(2d) + E_3(4d)$

$E_1 = 2E_2 + E_3$

$\Rightarrow \quad q_1 = 2q_2 + q_3 \quad —①$

$-2Q = -q_1 + q_2$

$3Q = -q_2 + q_3$

$Q = -q_1 + q_3 — ③$

from ① & ② $\Rightarrow \boxed{q_2 = -2Q}$

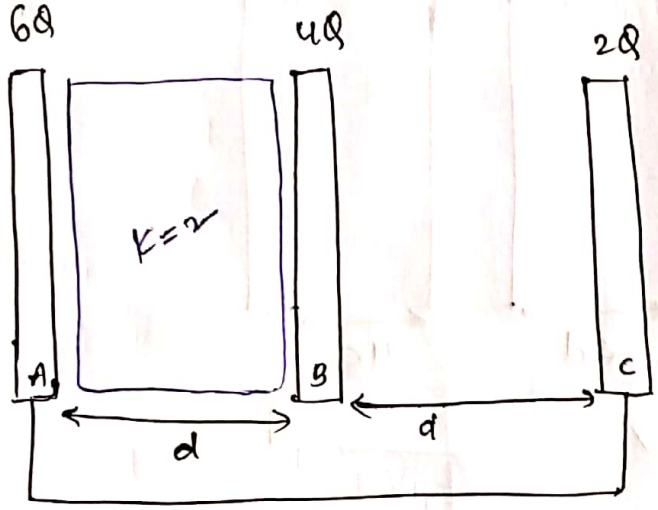
$\boxed{q_1 = 0}$

② $\Rightarrow \boxed{q_3 = Q}$

∴ Amount of charge flocon $= 4Q - 2Q = 2Q$.

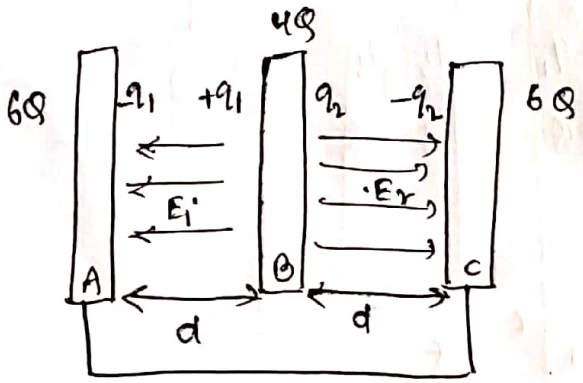
11|04|25 | Friday

Ex6



Find charges on all six faces.

Sol6



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$$E_1 = \frac{q_1}{2\varepsilon_0 KA} + \frac{q_1}{2\varepsilon_0 KA} = \frac{2q_1}{2\varepsilon_0 KA} = \frac{q_1}{\varepsilon_0 KA}$$

$$\& \quad E_2 = \frac{q_2}{2\varepsilon_0 A} + \frac{q_2}{2\varepsilon_0 A} = \frac{q_2}{\varepsilon_0 A}$$

$$V_A = V_C$$

$$V_B - V_C = V_B - V_C$$

$$E_1 d = E_2 d$$

$$\frac{q_1}{\varepsilon_0 KA} A = \frac{q_2}{\varepsilon_0 A} A$$

$$q_1 = q_2 K$$

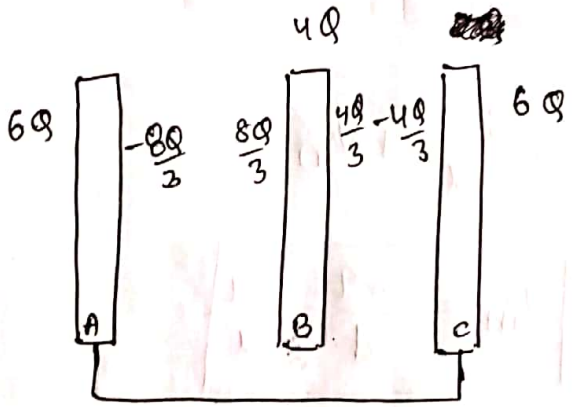
$$\boxed{q_1 = 2q_2}$$

$$q_1 + q_2 = 4Q$$

$$3q_2 = 4Q$$

$$\boxed{q_2 = \frac{4Q}{3}}$$

$$\therefore \boxed{q_1 = \frac{8Q}{3}}$$



Charge flow through the wire

$$= 6Q - \left(6Q - \frac{8Q}{3}\right)$$

$$= \frac{8Q}{3}.$$

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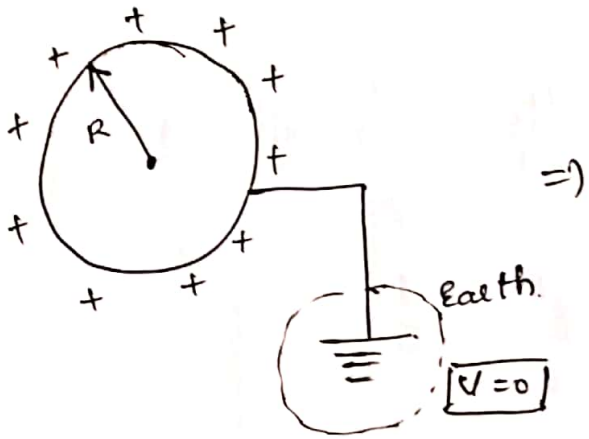
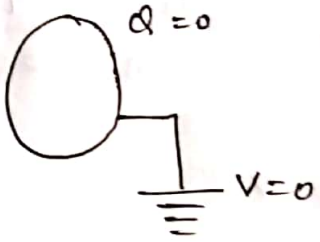
Earthening of a Conducting Body :

①



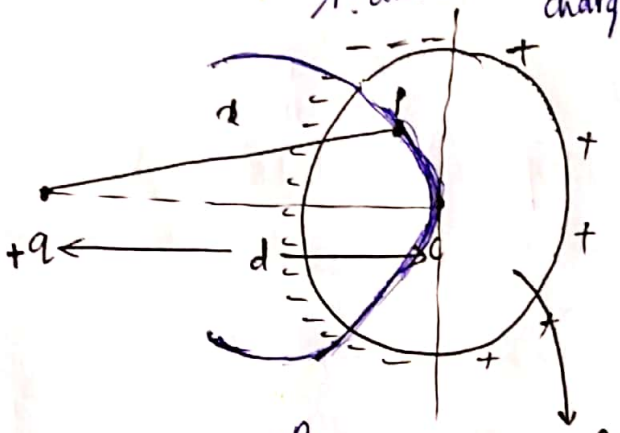
$\Rightarrow$



$$V_e = \frac{kQ_e}{R_e} = 0$$

$[\because R_e = 6400km]$

②



Sphere is in the field of +q.

$$V_c = \frac{kq}{d} + V_{in} = \frac{kq}{d}$$

$$V_p = \frac{kq}{x} + V_{in}$$

$$F = -\frac{dU}{dx}$$

$$U = \frac{1}{2}\varepsilon_0 E^2 V$$

$$U = \frac{1}{2}\varepsilon_0 \left(\frac{kq}{x^2}\right)^2 V$$

we have,

$V_c = V_p$

$$\Rightarrow \frac{kq}{d} = \frac{kq}{x} + V_{in}$$

$$\boxed{V_{in} = \frac{kq}{d} - \frac{kq}{x}}$$

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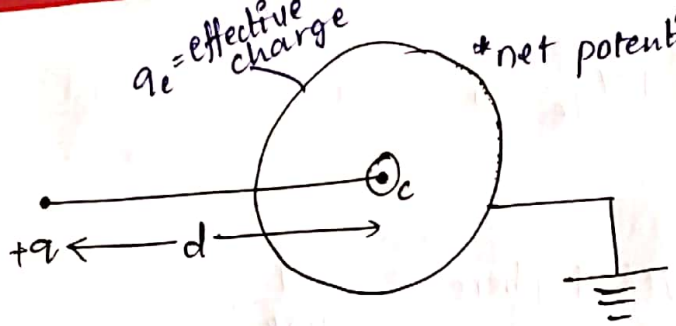
(3)



* net potential is zero when connected to earth.

* because of +q charge, some charge is flown from earth to sphere in order to make net potential zero.

* q_e & induced charges are distributed non-uniformly.

At centre,

$$V_C = \frac{Kq}{d} + \frac{K q_e}{R} + V_{ind}$$

($V_{ind} \to 0$; non-uniform)

here, $V_C = 0$ (earth connection)

$$\frac{Kq}{d} = -\frac{Kq_e}{R}$$

$$\boxed{q_e = -\frac{Rq}{d}} \Rightarrow [\text{electrons are flown from earth to sphere}].$$

$$E_C = \frac{Kq}{d^2} + E_{ind} + E_{q_e}.$$

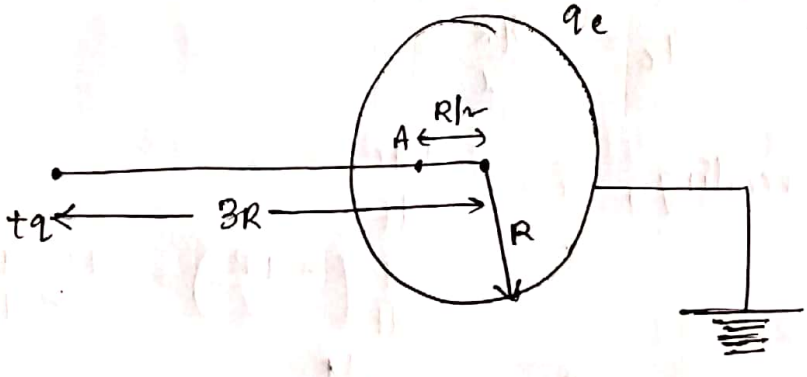
here, $E_C = 0$

$E_{due\ to\ charges\ on\ sphere} = E_{ind} + E_{q_e}$

we cannot determine seperately as they are non-uniform.

$$\boxed{E_{due\ to\ charges\ on\ sphere} = -\frac{Kq}{d^2}}$$

Ex:



Find potential ~~at~~ A due to all charges.

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Sol6

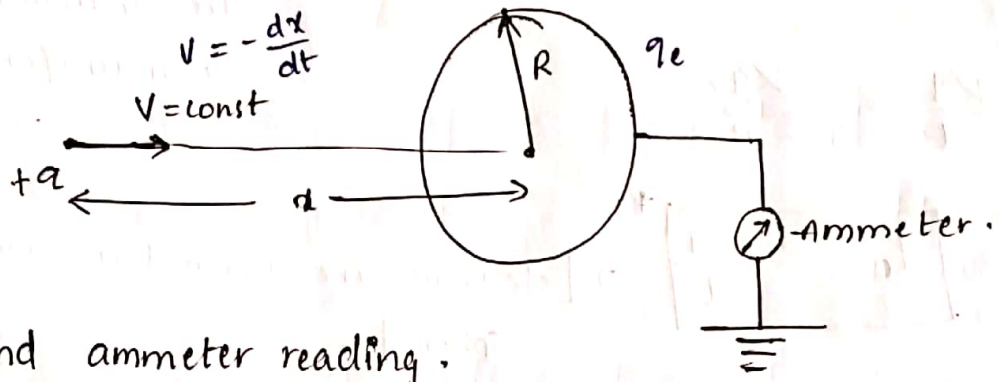
$$V_A = \frac{kq}{2.5R} + V_{ind} + V_{q_e} = 0$$

$V_{due\ to\ all\ charges\ of\ sphere} = -\frac{kq}{2.5R}$.

$$E_A = \frac{kq}{(2.5R)^2} + E_{ind} + E_{q_e} = 0$$

$E_{due\ to\ all\ charges} = \frac{-kq}{(2.5R)^2}$.

Eg6



Find ammeter reading.

Sol6

At any point, x = distance of $+q$ charge from centre to shell.

We have,

$$V_c = 0$$

$$\Rightarrow \frac{kq}{x} + \frac{kq_e}{r} + V_{ind}^{0} = 0$$

$$q_e = -\frac{qR}{x}$$

Ammeter reading, $i = \frac{dq_e}{dt}$

$$= -\frac{d}{dt}\left(+\frac{qR}{x}\right) = \frac{qR}{x^2}\cdot\frac{dx}{dt}$$

$$\boxed{i = \frac{-qRv}{x^2}}\ *$$

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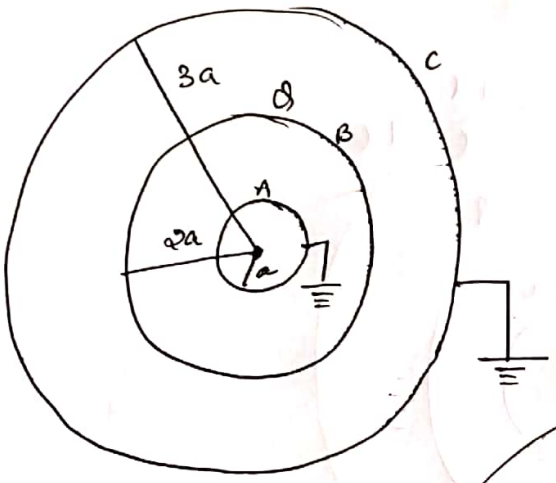
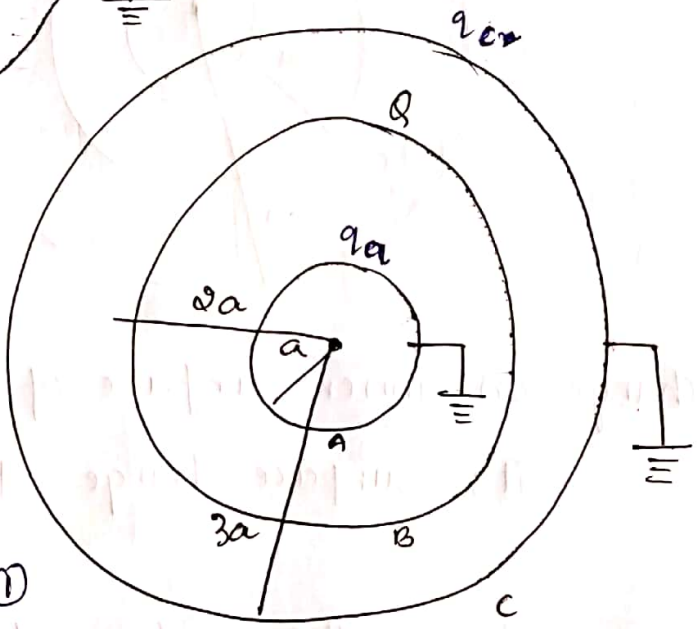
As $d(\downarrow) \Rightarrow \varphi(\uparrow)$.

Exe



Find final charge on spheres.

Sol



① $V_A = 0$

$$\frac{Kq_a}{a} + \frac{KQ}{2a} + \frac{Kq_c}{3a} = 0$$

$$q_a + \frac{Q}{2} + \frac{q_c}{3} = 0 \quad -①$$

② $V_C = 0$

$$\frac{Kq_a}{3a} + \frac{KQ}{3a} + \frac{Kq_c}{3a} = 0$$

$$q_a + Q + q_c = 0 \quad -②$$

Solving eqns ① & ②

$\Rightarrow$ ② − ① $\Rightarrow \frac{Q}{2} + \frac{2q_c}{3} = 0$

$$\boxed{q_c = -\frac{3Q}{4}}$$

② $\Rightarrow q_a = -Q + \frac{3Q}{4}$

$$\boxed{q_a = -Q/4}$$

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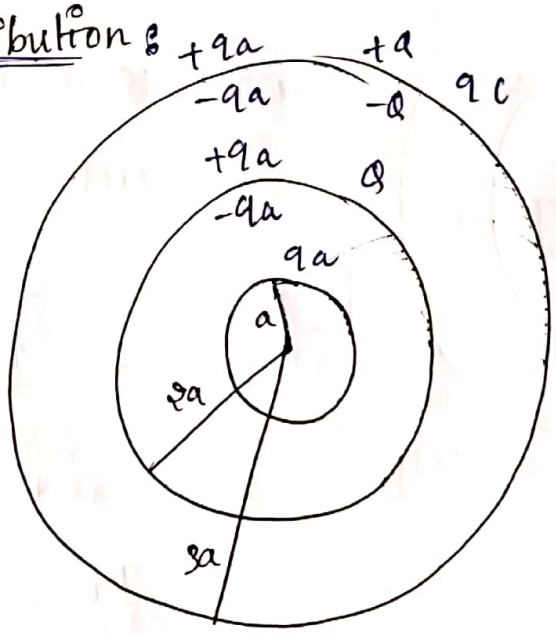
∴ final charges are ⇒ $q_a = -\frac{Q}{4}$, $q_b = Q$, $q_c = -\frac{3Q}{4}$.

*Charge distribution :



* charge on inner surface of middle sphere = $-q_a = +\frac{Q}{4}$.

it's surface charge density = $\frac{Q/4}{4\pi(2a)^2}$

* charge on outer surface of middle sphere = $+q_a + Q$

$= \frac{3Q}{4}$

it's surface charge density = $\frac{3Q/4}{4\pi(2a)^2}$

* charge on inner surface of outer sphere = $-q_a - Q$

$= +\frac{Q}{4} - Q$

$= -3Q/4$

it's surface charge density = $\frac{-3Q/4}{4\pi(3a)^2}$

* charge on outer surface of outer sphere = $q_a + Q + q_c$

$= -\frac{Q}{4} + Q - \frac{3Q}{4}$

$= 0$

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*potentials :

1) $V_A = 0$

2) $V_B = \frac{Kq_a}{2a} + \frac{KQ}{8a} + \frac{Kq_c}{3a}$

$= \frac{K}{a}\left(\frac{-Q}{8} + \frac{Q}{2} - \frac{Q}{4}\right)$

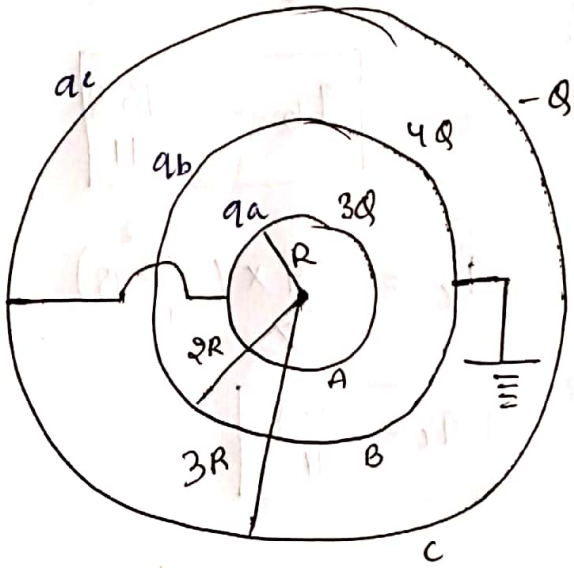
$= \frac{K}{a}\left(\frac{-3Q}{8} + \frac{4Q}{8}\right)$

$$\boxed{V_B = \frac{KQ}{8a}}$$

3) $V_C = 0$

Ex:

Find final charges q_a, q_b & q_c.



Sol:

1) $q_a + q_c = 3Q - Q = 2Q$

2) $V_A = V_c$

$$\frac{Kq_a}{R} + \frac{Kq_b}{2R} + \frac{Kq_c}{3R} = \frac{Kq_a}{3R} + \frac{Kq_b}{3R} + \frac{Kq_c}{3R}$$

$$q_a + \frac{q_b}{2} + \frac{q_c}{3} = \frac{q_a}{3} + \frac{q_b}{3} + \frac{q_c}{3}$$

$$\frac{2q_a}{3} = q_b\left(\frac{1}{3} - \frac{1}{2}\right) = -\frac{q_b}{6}$$

$$\boxed{q_a = -q_b/4}$$

3) $V_B = 0$

$$\frac{Kq_a}{2R} + \frac{Kq_b}{2R} + \frac{Kq_c}{3R} = 0$$

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$$\frac{q_a}{2} + \frac{q_b}{2} + \frac{q_c}{3} = 0$$

$$\frac{-q_b}{8} + \frac{q_b^{\times 4}}{2\times 4} + \frac{q_c}{3} = 0$$

$$\frac{3q_b}{8} + \frac{q_c}{3} = 0$$

$$q_c = -\frac{9q_b}{8}$$

$$q_a + q_c = 2Q$$

$$-\frac{2q_b}{8} - \frac{9q_b}{8} = 2Q$$

$$\boxed{q_b = -\frac{16Q}{11}}$$

$$q_c = -\frac{9}{8} \times \left(-\frac{16^{2}}{11} Q\right)$$

$$\boxed{q_c = \frac{18Q}{11}}$$

$$q_a = -\frac{1}{4} \times -\frac{16^{4}}{11} Q$$

$$\boxed{q_a = \frac{4Q}{11}}$$

∴ final charges q_a, q_b, q_c are $\frac{4Q}{11}, \frac{-16Q}{11}, \frac{18Q}{11}$ respectively

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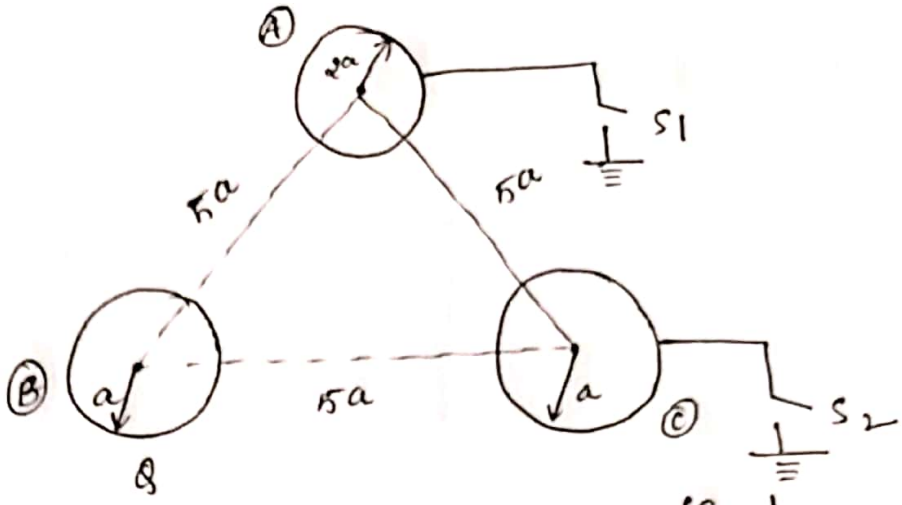
Ex 6



when both the switches are closed. find final charges on the spheres A & C and also find potential of B.

Sol 6

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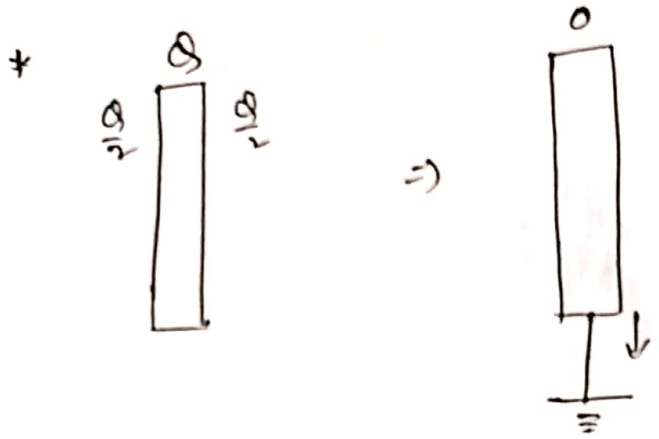
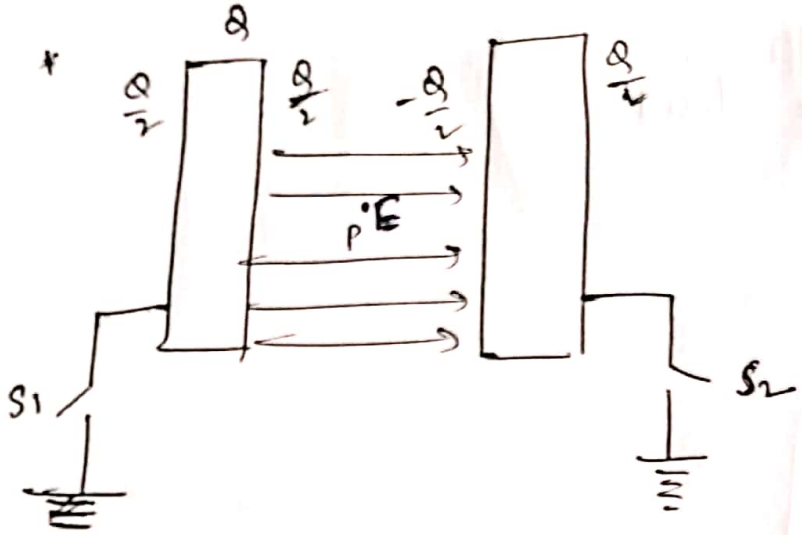
Conducting plates

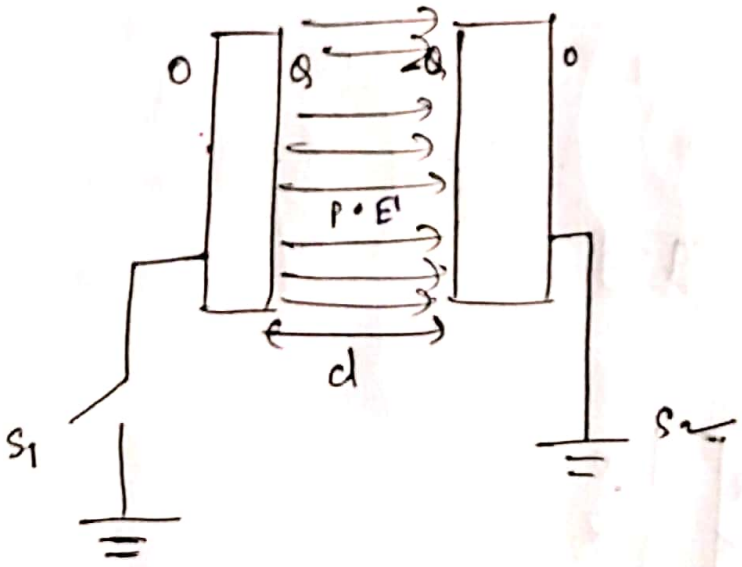




$$E = \frac{Q/2}{2\varepsilon_0 A} + \frac{Q/2}{2\varepsilon_0 A}$$

$$= \frac{Q}{2\varepsilon_0 A}$$

when S_2 is closed,



$$E' = \frac{Q}{2\varepsilon_0 A} + \frac{Q}{2\varepsilon_0 A} = \frac{Q}{\varepsilon_0 A}$$

clearly $\boxed{E' = 2E}$

Energy stored,

$$U = \frac{1}{2}\varepsilon_0 (E')^2 V \qquad (V = A \times d)$$

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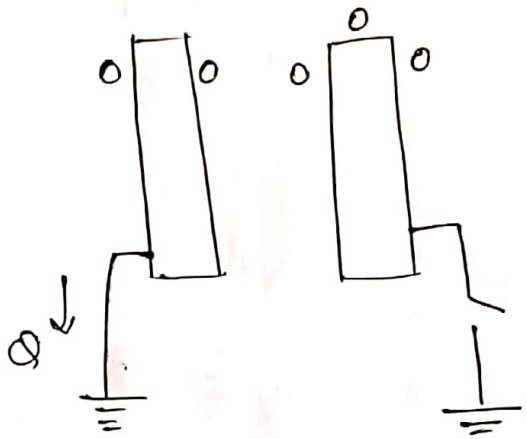
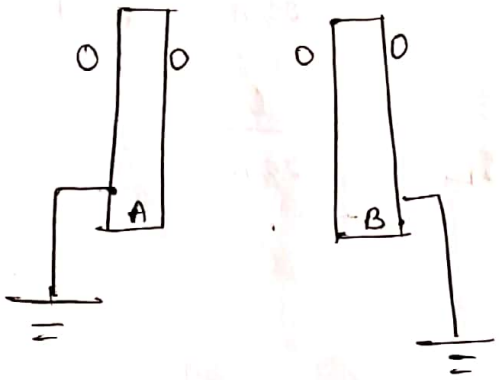
when only S_1 is closed



when both S_1 & S_2 are closed,



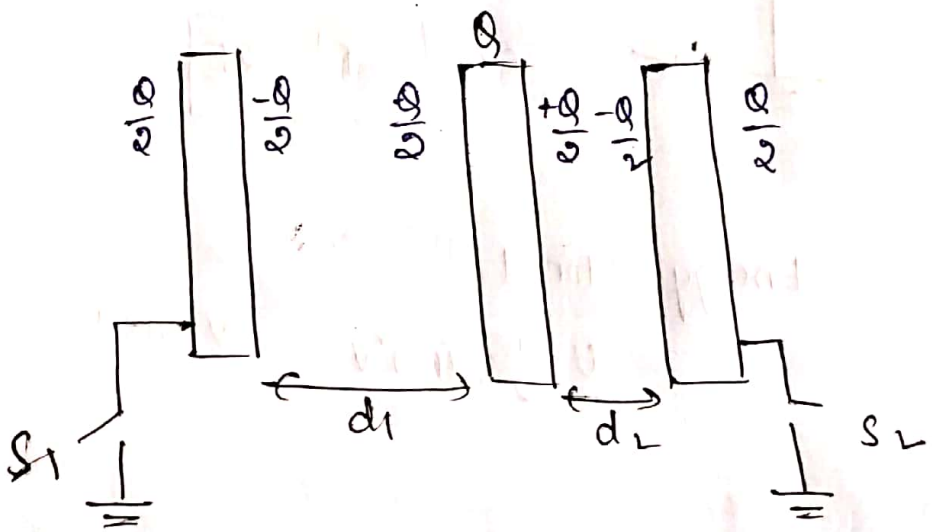
$V_A = 0$

$V_B = 0$

$V_A - V_B = 0$

$Ed = 0 \Rightarrow \boxed{E = 0} \Rightarrow \boxed{q = 0}$

Eg:



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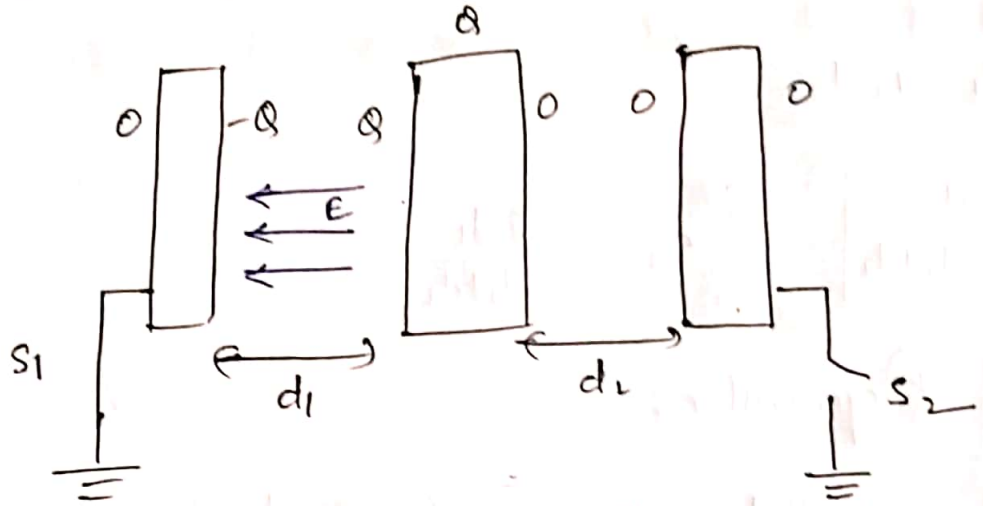
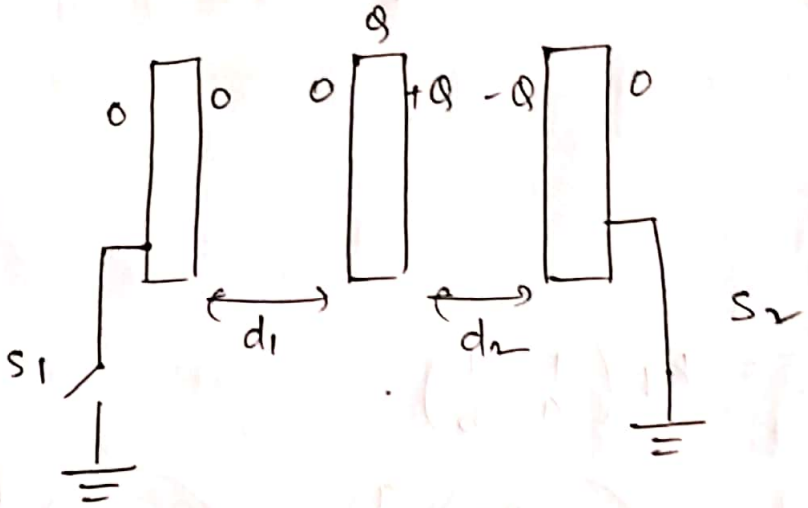
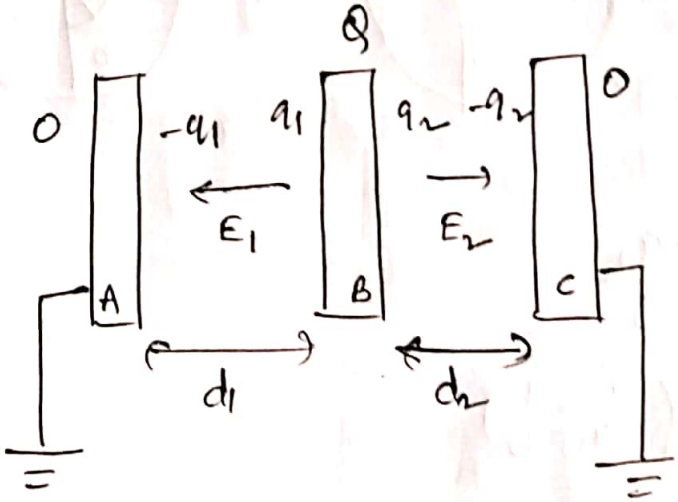
when only switch S_1 is closed,



when only switch S_2 is closed,



when both S_1 & S_2 are closed,



$V_A = V_c$

$V_B - V_A = V_B - V_c$

$E_1 d_1 = E_2 d_2$

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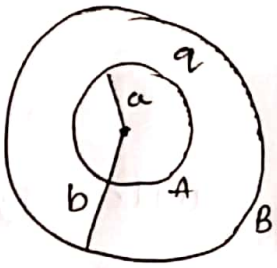
& $Q = q_1 + q_2$

$q_1 d_1 = q_2 d_2$

$$\boxed{q_1 = \frac{d_2 Q}{d_1 + d_2}} \qquad \boxed{q_2 = \frac{Q d_1}{d_1 + d_2}}$$

Vandegraff Generator

Ex: (1) Find potential difference between the shells.



Sol:

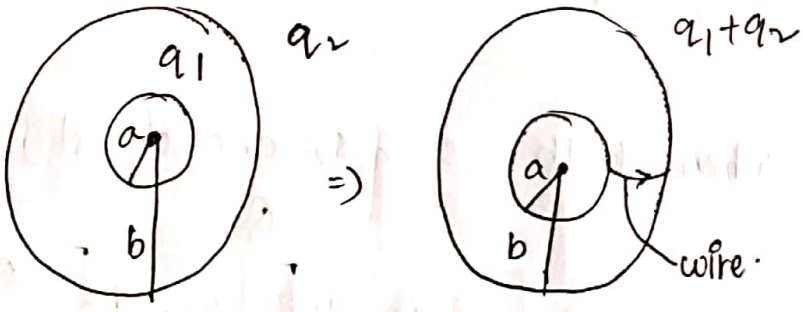
$$V_A = \frac{Kq}{a}$$

$$V_B = \frac{Kq}{b}$$

now,

$$V_A - V_B = Kq\left(\frac{1}{a} - \frac{1}{b}\right).$$

(2)



now,

$$V_A = \frac{Kq_1}{a} + \frac{Kq_2}{b}$$

& $V_B = \frac{Kq_1}{b} + \frac{Kq_2}{b}$

$$V_A - V_B = Kq_1\left(\frac{1}{a} - \frac{1}{b}\right)$$

Always, $V_A > V_B$

* when both the spheres are connected by wire, entire

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Charge from inner sphere will flow to outer sphere.
(It doesnot depends on charge on outer sphere).

Heat Generated in the wire,

$$H = U_f - U_i$$

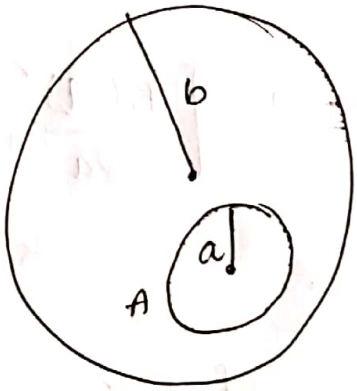
$$H = \frac{K(q_1+q_2)^2}{2b} - \left[\frac{Kq_1^2}{2a} + \frac{Kq_2^2}{2b} + \frac{Kq_1q_2}{b}\right]$$

$$\boxed{H = \frac{Kq_1^2}{2}\left(\frac{1}{b}-\frac{1}{a}\right)}$$

(we can write $H = ms\Delta\theta$)

Ex: A metal sphere (A) having radius 'a' is charged upto potential V. It is placed inside a conducting sphere of radius 'b'. (b > a) & both are connected with wire. Now find the potential of sphere A.

Sol:



Before connecting wire,

$$V = \frac{q}{4\pi\varepsilon_0 a}$$

$$\Rightarrow q = (4\pi\varepsilon_0)Va$$

After connecting wire,

charge flow from A to B.

$$V_B = \frac{q}{4\pi\varepsilon_0 b} \Rightarrow \boxed{V_B = \frac{Va}{b}}$$

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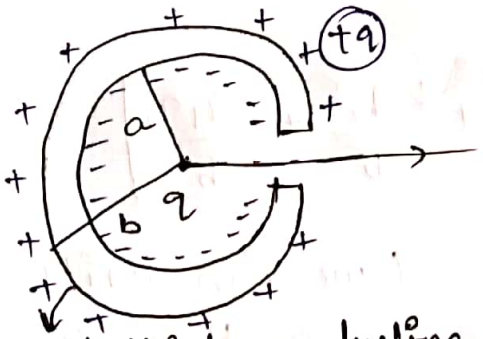
we have,

potential at all points inside the shell is same.

$$V_A = V_B = \left(\frac{a}{b}\right)V$$

Ex:



neutral thick conducting shell.

Find the work done in moving the charge from centre to infinity slowly.

Sol:

$$W = U_f - U_i$$

$$W = 0 - \left[\frac{K(-q)^2}{2a} + \frac{kq^2}{2b}\right.$$

$$\left. \frac{-kqq}{a} + \frac{kqq}{b} - \frac{kqq}{b}\right]$$

$$= -\left[\frac{kq^2}{2a} + \frac{kq^2}{2b} - \frac{kq^2}{a} + \frac{kq^2}{b} - \frac{kq^2}{b}\right]$$

$$= -\left[\frac{-kq^2}{2a} + \frac{kq^2}{2b}\right]$$

$$W = \frac{kq^2}{2}\left(\frac{1}{a} - \frac{1}{b}\right)$$

$$W_{ext} + W_{ele} = \Delta K^{0}$$

$$W_{ext} = -W_{ele}$$

$$= -(-\Delta U)$$

$$= \Delta U$$

$$= U_f - U_i$$

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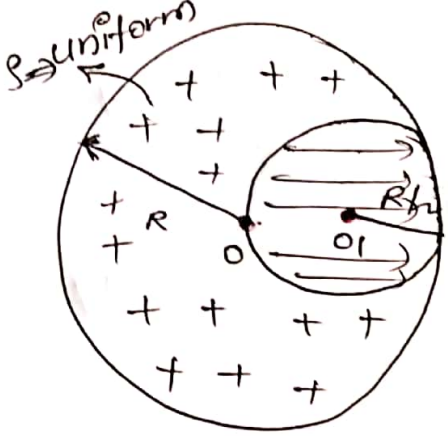
Ex 6 Find potential difference between centre of original sphere and centre of cavity.



Sol 6

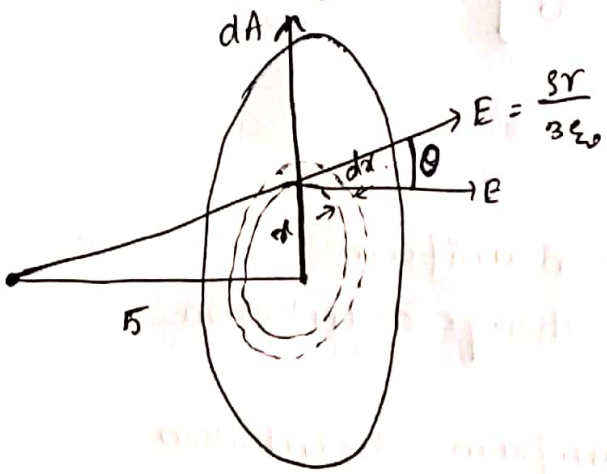
$$\bar{E} = \frac{\rho \bar{a}}{3\varepsilon_0} = \frac{\rho(R/2)}{3\varepsilon_0} = \frac{\rho R}{6\varepsilon_0} = \text{uniform}$$

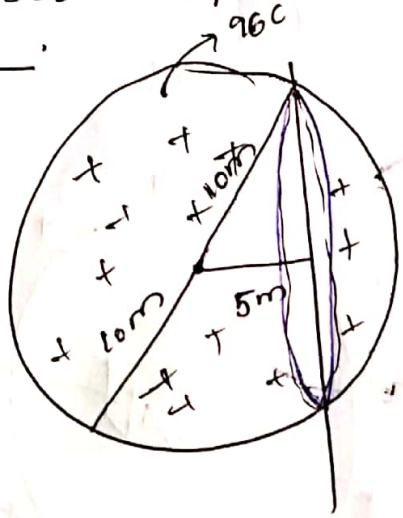
$$V_0 - V_{0_1} = Ed = \frac{\rho R}{6\varepsilon_0} \times \frac{R}{2} = \frac{\rho R^2}{12\varepsilon_0}$$

$$\boxed{P.d = \frac{\rho R^2}{12\varepsilon_0}}$$

Ex 7 A non-conducting sphere of radius 10m is uniformly distributed with a charge of 96C. Imagin a large plate sheet placed at a distance 5m from centre of sphere. The flux of electric field through the area of intersection of sheet with the sphere is $\frac{n}{\varepsilon_0}$ where n = 9.

Sol 7





$$d\phi = \frac{\rho r}{3\varepsilon_0} \times 2\pi x \cdot dx \cos\theta$$

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$$d\phi = \frac{\rho}{3\varepsilon_0} \times 2\pi x\, dx \times \frac{5}{x}$$

$$d\phi = \frac{10\pi\rho}{3\varepsilon_0} x\, dx$$

$$\phi = \int d\phi = \frac{10\pi\rho}{3\varepsilon_0} \left[\frac{x^2}{2}\right]_0^{x=5\sqrt{3}}$$

$$= \frac{5\pi\rho}{3\varepsilon_0} \times 25 \times 3$$

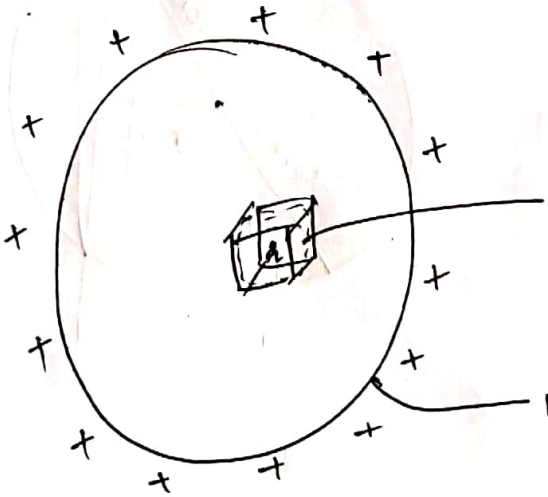
$$= \frac{125\pi\rho}{\varepsilon_0}$$

$$= \frac{125\pi \times \frac{24}{96}}{\frac{4}{3}\pi(10)^3} \Big/ \varepsilon_0$$

$$= \frac{125 \times 24 \times 3}{10 \times 10 \times 10} \Big/ \varepsilon_0$$

$$\boxed{\phi = \frac{q}{\varepsilon_0}}$$

Eqⁿ



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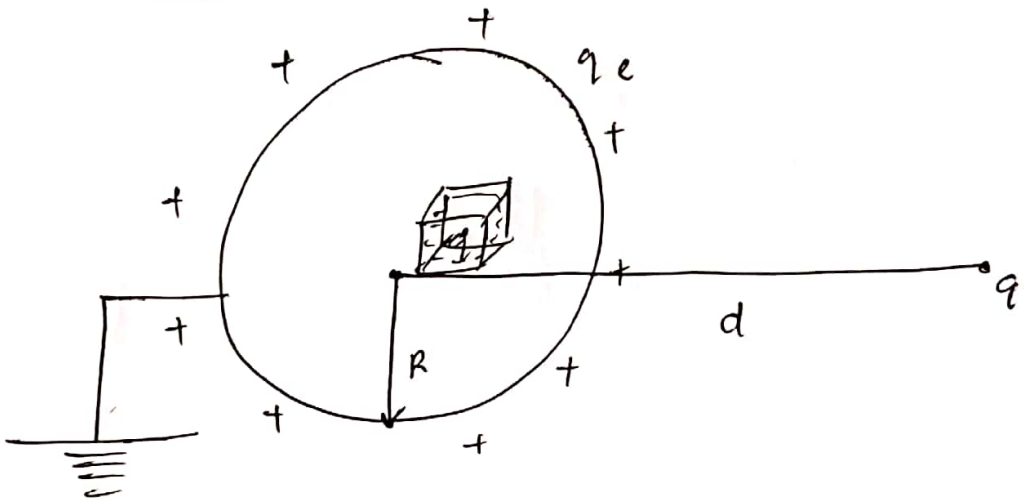
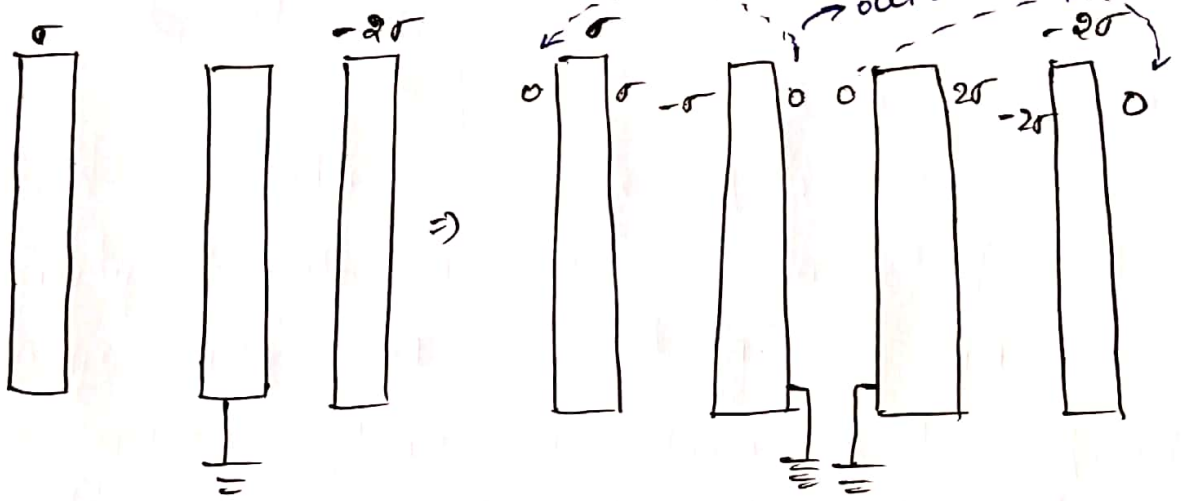
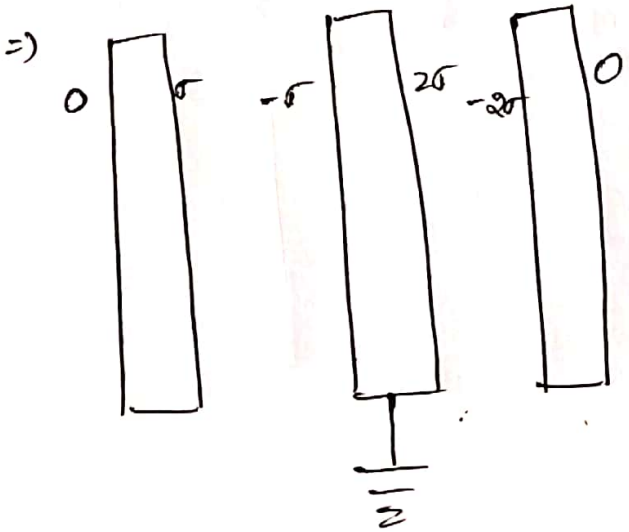


$$V_c = 0 \Rightarrow \frac{kq}{d} + \frac{kq_e}{R} + 0 = 0$$

$$\boxed{q_e = -\frac{qR}{d}}$$

Ex 6





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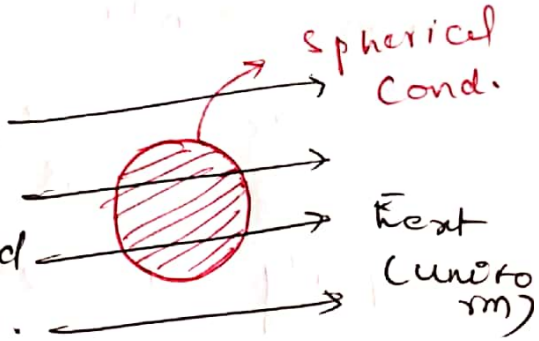
Properties of Conductor:-

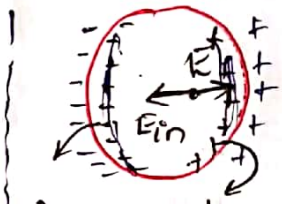
(1) Net Electric field, inside material of conductor in electrostatic condition is zero.

* e^- in the conductor move towards left, so, there will be some E. field induced in the conductor.



This E_{ind} will be equal and opposite to E_{ext} inside conductor.

$$\vec{E}_{ind} = -\vec{E}_{ext}$$



induced charges.

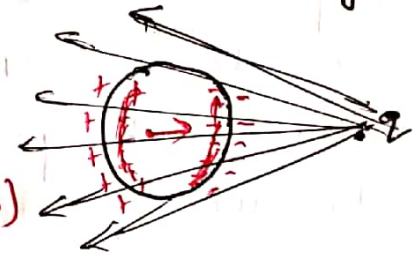
* So, always in electrostatic condition, inside E. field will be zero. [inside of the conducting material]

* Since, E_{inside} is zero everywhere, Then no, need to do any work to move a point charge from one point to another point

So, $W_{ext} = q[\Delta V]$

$0 = q[\Delta V]$; $\Delta V \neq 0$.

So, V is same every where [equipotential region]



in this case also $E_{inside} = 0$.

[∵ -ve induced charge = +ve induced charge]

[since conductor is initially neutral]

[$+\sigma_{ind} < \sigma_{2in}$]

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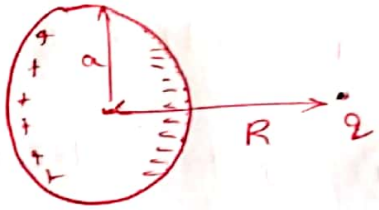
$V_c = V_q + V_{ind}$

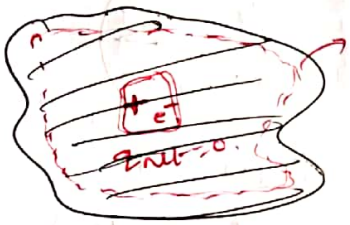


+ind, -ind charges are Equal and at Equal distance from centre. So, $V_{ind} = 0$,

$$V_c = \frac{k \cdot q}{R}$$

Property No:- 2:-

Extra Material Charge given to the conducting ~~surface~~, resides on surface only.



G.S is inside the Material so, E on G.S should be zero.

$$\phi = \int \vec{E} \cdot d\vec{s} = \frac{q_{in}}{\varepsilon_0} \quad ; \quad E = 0, \ \phi = 0$$

So, $q_{in} = 0$

so, if we give any Charge to this conductor, Then it will move to Surface only.

* That charge on the surface will distribute in such a way that,

E_{net} inside conductor $= 0$.

Property No:- 3:-

Surface of a Conductor is an Equipotential surface. inside region also Equipotential region. since, $E = 0$. inside

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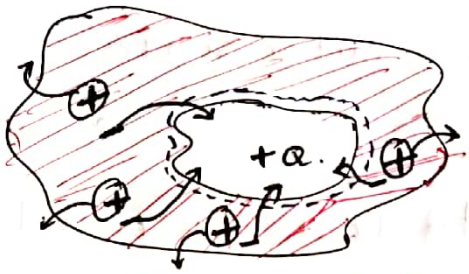
Property No:- 4 :- $\vec{E}_{net}$ lines are $\perp^{lr}$ to the surface of conductor.

Conductors with Cavity :-

Case 1) :- Q is placed inside cavity [contor is Neutral]



Consider a G.S just outside cavity, G.S is in the conducting material, so, 'E' must be zero, so, Negative charge will induce just inside the cavity.

$(Q + q_{ind}) = q_{inside\ The\ G.S} = 0$ $\quad [E=0,\ \phi=0,\ q_{in}=0]$

$$\boxed{q_{ind} = -Q}$$

-ve charges in the conducting material will come to the inner surface of cavity due to attraction of +Q. So, the Remaining +ve charges inside material will move to the Outer surface of conductor. and distribute themselves.

That -ve charge will induced on the inner surface of the cavity in such a way that, the electricfield due to the Cavity -ve induced charge and Q inside, will be zero at any point outside. the cavity.

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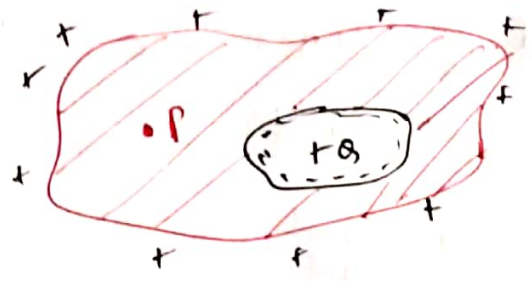
Now,



$\vec{E}_{P_{Net}} = 0.$

$\left(\vec{E}_{+Q\ cavi} + \vec{E}_{Q_{in}}\right) + \vec{E}_{Q_{out}}$

$(-Q)$ $(+Q)$

So, $\vec{E}_{outer\ Q}$ also zero. at 'P'. zero (since P is out-side point)

So, $+Q$ on the outer surface, will distribute in such a way that, E_{net} due to $+Q$ [outer surface] will be zero. ; $\vec{E}_{net}$ due to Q(cavity) & $-Q$ in anyways zero.

Case 2:- when a point charged is placed outside conductor.

This induced + and − charges cancel out the field due to $+Q$ inside so, that, $E_{inside} = 0$.

So, electric field ~~date to all charges~~ at a point inside a conductor due to outside charges [$+Q$ and Q_{ind} on surface] is zero. (every point → inside cavity also)

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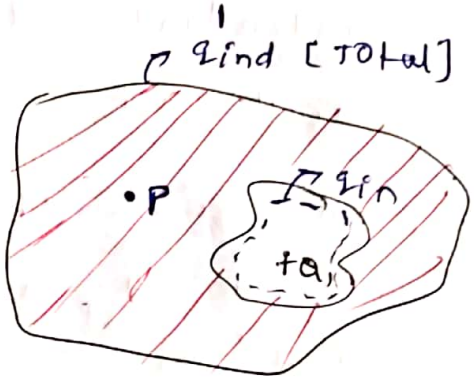
* E.F$_{net}$ due to outside charges can not able to enter the conducting material and cavity also.

* case 3:-

$\vec{E}_{net\,p} = (\vec{E}_{+q_1} + \vec{E}_{q_{in}}) + (\vec{E}_{q_{ind}} + \vec{E}_{Q_2})$.



$+q_1$ → zero. outside point.

q_{in} → zero. outside charges & inside point.

q_{ind} → $+Q_2$ to that charges.

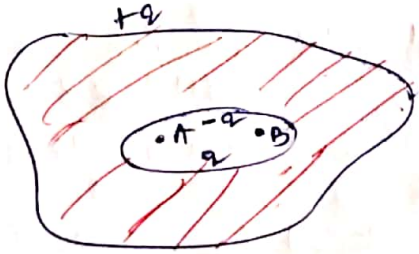
So,

here, The outside charge, on The surface of the conductor arrange themselves in such a way that, The electric field inside conductor and cavity is zero due to outside charge Q_2 and charge on outer surface.

$\vec{E}_{outer} + \vec{E}_{+q} = 0$ for any inside points

outside charges

*



$+Q$.

if we move '+q' which is present out-side, Then +q on surface will also change in such a way that $\vec{E}_{+q} + \vec{E}_{+q} \Rightarrow$ Inside zero. [at every point]

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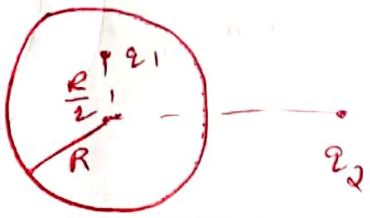
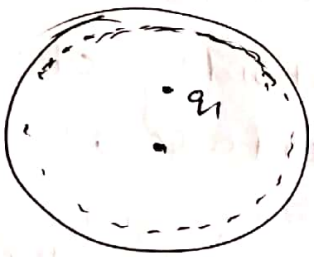
so, electric field in the cavity which is due to 'q' [inside cavity] and -q induced, will not change. So, Potential diff. blw A and B will not change.

* A point charge q_1 is placed inside a neutral conducting shell at a distance $\frac{R}{2}$ from The centre of Shell. Another point charge q_2 is placed at a distance $\frac{\sqrt{7}}{2}R$ from The centre of Shell. Then



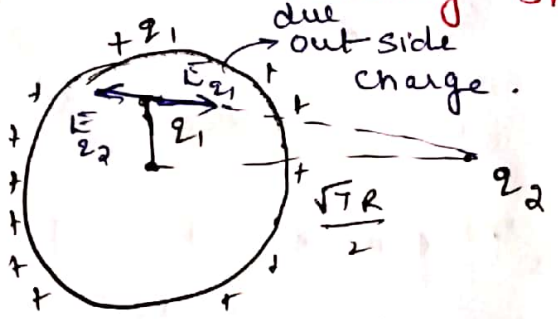
(i) The magnitude of force on q_1 due to induced charges at the inner surface of the conducting shell is zero. [True or false]

Sol:-



Since, -q induced charge distribution is not, uniform, -q ind will apply force on q_1.

(ii) The mag. of force on q_1 due to induced charges at the outer surface of the conducting shell is $\frac{q_1 q_2}{8\pi\varepsilon_0 R^2}$.



We know that-

$\vec{E}_{q_2} + \vec{E}_{q_1} = 0$ [inside at every point]

($\vec{E}_{q_2}$: outer charges)

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$|\vec{E}_{\text{due } q_2}| = \frac{k \cdot q_2}{(\sqrt{2}R)^2}$.

force on q_1 due to $q_2 = \frac{q_1 q_2}{8\pi\varepsilon_0 R^2}$.

This force will cancel out by $+q_1$ (ind) [outside of the surface]. So, That- fnet on q_1 zero due to outside charges. [$\because$ $\vec{E}$net due to outside charges zero]

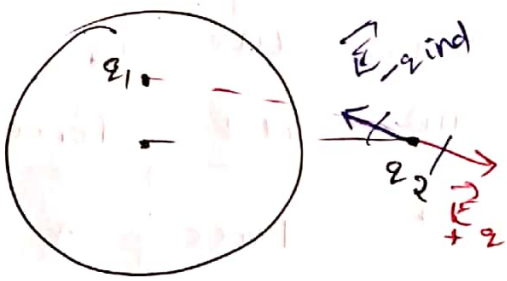
(iii) The mag. of force on q_2 due to ind. charges at the inner surface of the shell is —

$\vec{E}_{+q \text{ (inside cavity)}} + \vec{E}_{-q \text{ (induced)}} = 0$ [for outside points]

so, $|\vec{E}_{+q}| = |\vec{E}_{-q \text{ [ind]}}| = \frac{q_1 q_2}{8\pi\varepsilon_0 R^2}$.

So, force applied by $-q$ ind on q_2

$= \frac{q_1 q_2}{8\pi\varepsilon_0 R^2}$.



* in a neutral conducting hallow sphere of inner and outer radii 5cm, 10cm respectively, a point charge $Q = 1C$ is placed at a point A, that is 3cm from the centre 'c' of the hallow sphere. An external uniform electric field of magnitude 20N/C is also applied. Net electric field on this charge is 15N, away from the centre of the sphere as shown. Then

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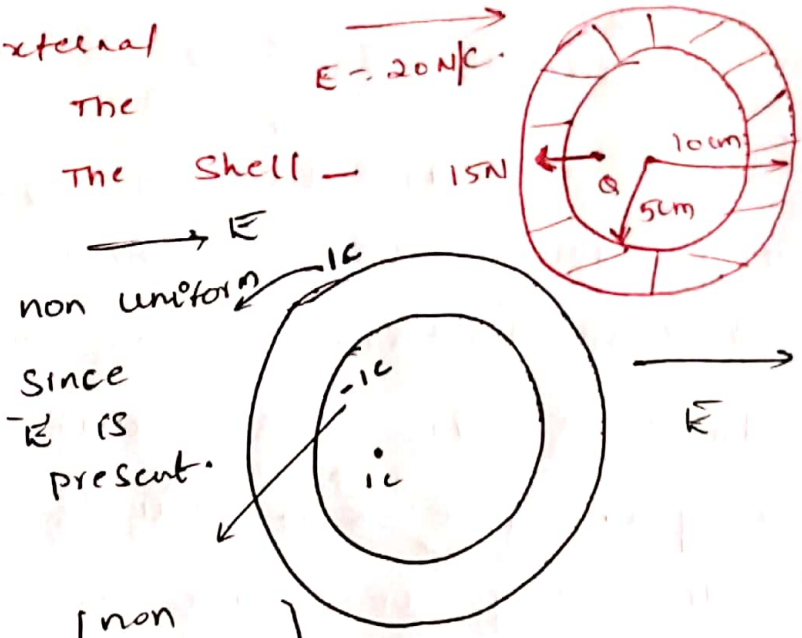
i) force due to external electric field on the outer surface of the shell —



*

Since $\vec{E}$ is uniform,

$\vec{f} = q\vec{E}$

q_T

$q_{Total\ on\ outer\ charge} = 1C.$

So, $\vec{F} = q_T\vec{E}$

$\boxed{\vec{F} = 20N}$ [right]

[non uniform since $\vec{E}$ is present.]

[since 1C is not at centre]

ii) Force on outer surface is —

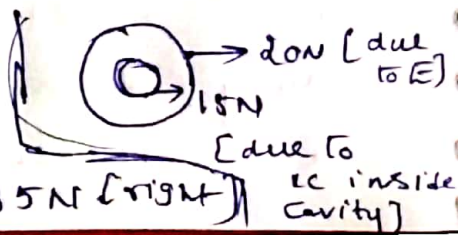
Force is nothing but 20N only since, for outer points, 1C, -1C effect will be zero. So, the only force on outer surface is due to $\vec{E}$ only.

(iii) Net force on shell :-

* Force on 1C [inside cavity] is 15 [left] [Given]

The only force on -1C is due to 1C [inside cavity] only. force due to $\vec{E}$ and 1C [outside the surface] on -1C is zero since -1C is inside. for inside points $E_{net} = 0$ for outside charges, so, $\vec{F}_{on\ 1C} = -\vec{F}_{on\ -1C}$

So, $F_{on\ -1C} = 15$ [right]



So, Total force on shell = 15N + 20N = 35N [right]

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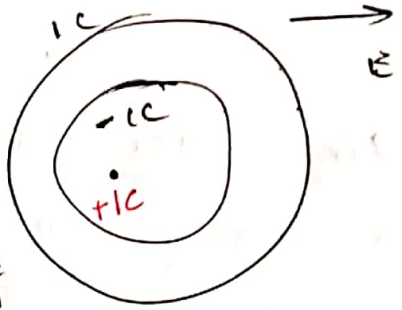
(iii) Net force on point charge due to shell :-

Force on 1C due to -1C is 15 ←.



Now, Force due to 1C on 1C is nothing
(outer) (inside cavity)

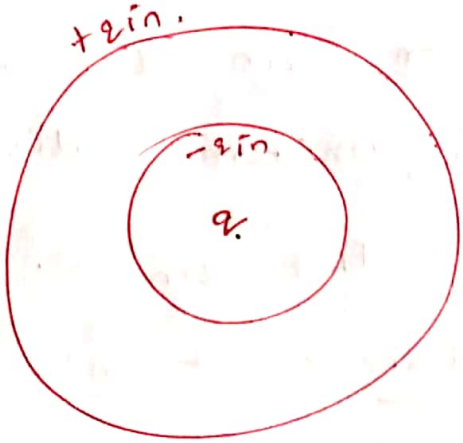
but, equal and opposite to force on 1C due to $\vec{E}$.

Force on 1C due to $\vec{E}$ = 20 × 1 = 20N [right]

So, force due to 1C [outside] on 1C [cavity] is 20N [left]

So, total force on 1C (cavity) due to shell = $\vec{F}_{-1C}$ + $\vec{F}_{1C}$ = 15 ← + 20 ←

= 35 N ←.



$-q_2$

1. q is placed at centre, so, -qin dis. is uniform and +qin dis. is non uniform since $-q_2$ is present.

* Net force on q is zero. since, -qin dis. is uniform so, field due to

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$-q_{in}$ at centre is zero. and, effect of $-q_2$ and $+q_{in}$ on q inside cavity is zero. ∵ inside point for $-q_2$ and $+q_{in}$. for outside charges inside field is zero. So, force due to $+q_{in}$, $-q_2$ is zero.

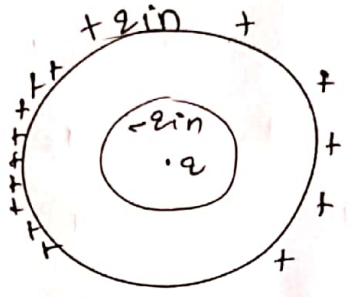
So, 'q' will be at rest.

* Net force on shell :-

Net force on shell is nothing but net force on $-q_{in}$ and $+q_{in}$.

* Force on $-q_{in}$:-

force due to $-q_2$ and $+q_{in}$ on $-q_{in}$ is zero (inside $\vec{E}$ due to outer charges is zero)



force on $-q_{in}$ due to q is also zero [∵ uniform distribution]

So, $\vec{F}_{net}$ $-q_{in}$ = 0.

* Force on $+q_{in}$:-

Force on $+q_{in}$ due to 'q' and $-q_{in}$ is zero. since, For outside point E_{net} due to q, $-q_{in}$ is zero.

Force on $+q_{in}$ due to $-q_2$ will be left direction because more +ve is are closer to $-q_2$. So, shell moves towards left.

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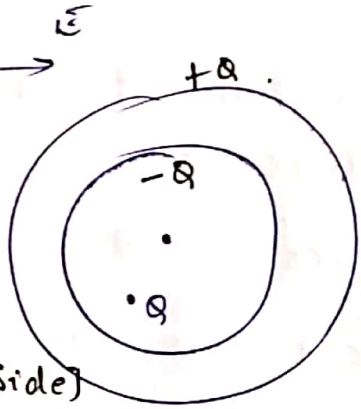
(i) fnet on Q due to E is not zero



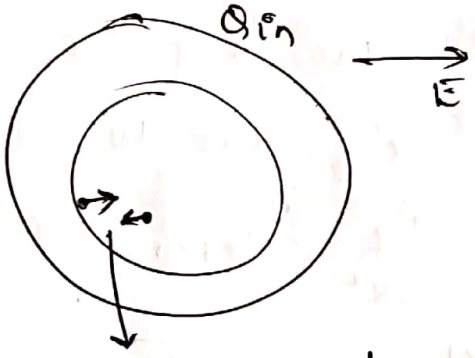
Net force on Q due to E and +Q (outside) is zero.

* Net force on 'Q' is also, not zero. force on 'Q' due to E and +Q is zero. But, $-Q_{in}$ apply force on 'Q' Since, $-Q_{in}$ distribution is non uniform.

* $\vec{F}_{on\ Q\ due\ to\ -Q_{in}} = \vec{F}_{on\ -Q_{in}\ due\ to\ Q}$.

But There will be some force on outer surface due to E' That force is

$F = Q_{in} \cdot E$ [right]



[cancel out Eaach Other].

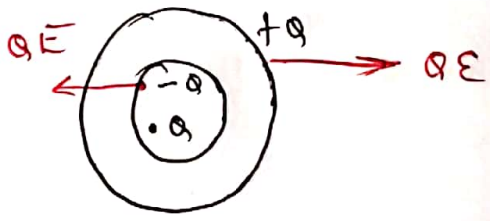
So,

Q and shell as a system, fnet on system is not zero.

** "Fnet on Q and inner shell is zero."

force on shell due to E :-

fnet on shell due to E (only) = zero.



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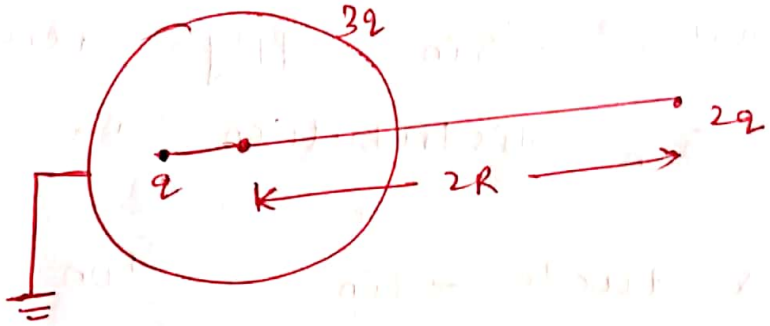
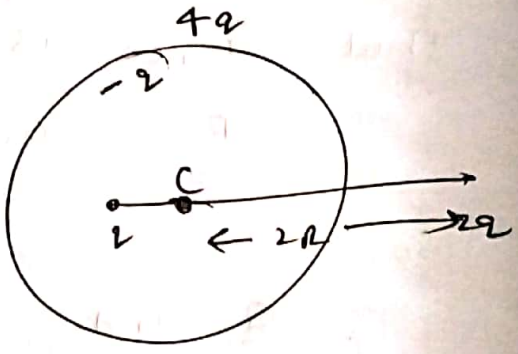
A thin conducting spherical shell of radius R is given charge 3q on its surface and a point charge 'q' is placed at a distance $\frac{R}{2}$ from its centre 'C'. Also there is a charge 2q placed outside the shell at a distance of 2R as shown. Then $\left[k = \frac{1}{4\pi\varepsilon_0}\right]$



1) The mag. of electric field at the centre, 'C' due to outer surface of shell is $\frac{kq}{2R^2}$. Before closing switch.

Sol: $\vec{E}_{4q} + \vec{E}_{2q} = 0$ at centre.



[∵ inside point for 4q & 2q both]

So, $|\vec{E}_{4q}| = |\vec{E}_{2q}| = \frac{k\cdot(2q)}{(2R)^2} = \frac{kq}{2R^2}$.

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2) electric potential at the centre 'c' due to outer surface of shell is ___ (Before closing the switch)

Sol:-

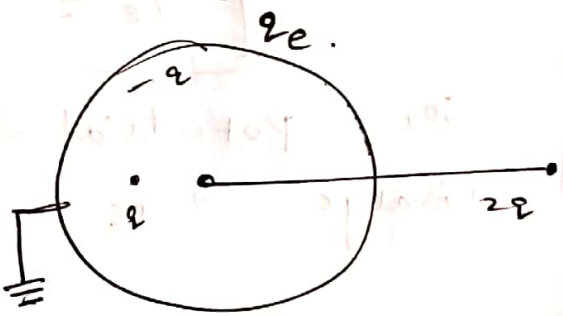
Charge on outer surface = +4q.

So, potential at centre due to outer surface = $\frac{k(4q)}{R}$

[Before closing the switch]

*. Potential at the centre 'c' due to charges on outer surface of shell is $-\frac{kq}{R}$ after closing the switch 's'

Sol:- let q_e be the charge on the outer surface after closing the switch.



→ Since outer surface connected to ground,

$V_{outersur} = 0$.

$V_q + V_{-q} + V_{qe} + V_{2q} = 0$ (individually)

($V_q + V_{-q}$ = zero.

(counterpoint for $q, -q$).

$V_{2e} = \frac{k \cdot q_e}{R}$. [on the surface]

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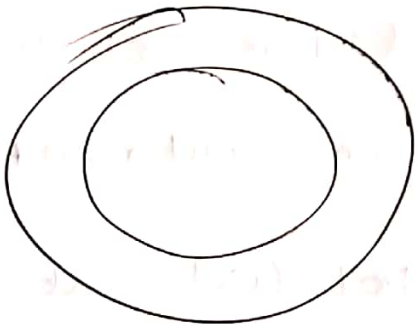
V due to q_2 :-

* inside field is zero due to q_2.

So, The region is Equipotential [only due to $2q_2$].



So, V at centre = V on surface.

$$\frac{kq(2)}{2R} = V \text{ on surface} = \frac{kq}{R}.$$

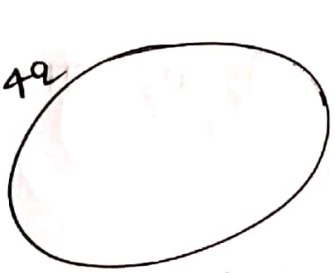
So,

$$V_{Net \text{ on surface}} = \frac{kq_e}{R} + \frac{kq}{R} = 0$$

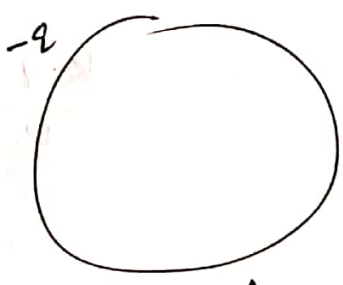
$$\boxed{q_e = -q}$$

* So, Potential at centre, due to outside charge is $\frac{-kq}{R}$.

* Charge flown through the switch into the earth after closing the switch is $5q$



(Initial)



(final)

So, transferred "5q".

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